



HomeSend API Suite 2.3



Product	HomeSend
Document Version	10.00
Status	Final
Reference	HOS_EIS_HomeSend-Ws-API-Suite-2.3_EN

Version Control

This is a managed document, and if any changes are made a complete replacement will be issued. The current release number is shown at the bottom of each page.

Author	Version	Comments	Date
Sebastien Lefevre	01.00	First public release.	2011-01-17
Sebastien Lefevre	02.00	Added Airtime transfer use cases (Airtime Transfers).	2012-05-04
Sebastien Lefevre	03.00	Fixed Airtime transfer wording: "designation" renamed to "denomination".	2012-05-25
Sebastien Lefevre	04.00	Fixed transactionId length (the XSD was correct, not the documentation). Added Error Type (Final or Transient Failure) for each HWS error code, additional HWS implementation guidelines about error handling. Added Reverse Advice of Charge call flow.	2013-10-16
Gaël Saulnier	05.00	Updated with HomeSend documentation template.	2014-03-24
Sebastien Lefevre	06.00	Fixed incorrect bullet list formatting. Added documentation for ER0034 on HWS/adviceOfCharge.	2014-12-19
Sebastien Lefevre	07.00	Added documentation for new URI scheme: pan.	2015-10-14
Sebastien Lefevre	08.00	Added documentation for token-like schemes. Aligned to HWS 2.3.1 that brings relaxed requirements on HWS/remittance sourceUri and destinationUri elements.	2016-06-13
Sebastien Lefevre	09.00	E-Recharge service deprecation: related call flows removed; related operations and types have been flagged as not to be used anymore. Airtime transfer deprecation: airtime-related attributes flagged as not to be used anymore. Added characters constraints on hsTransactionId (generated by HomeSend – no impact on existing implementations).	2017-03-22
Sebastien Lefevre	10.00	Aligned to HWS 2.3.2 that brings relaxed requirements on remittanceCbsType, which is now a string instead of an enumeration. This allows to use new service types, whose list is maintained in the Use Cases documentation.	2017-05-31

Document Purpose

This document describes the HomeSend Web Service API suite 2.3 (HWS, OEWS, TEWS, R-TEWS) and its usages as needed when implementing a connector for an E-Money/E-Wallet system to interconnect with HomeSend.

Audience

This document is intended for development engineers from the E-Wallet System providers that have to enhance their system to connect to the HomeSend hub.

Knowledge of Web Service concepts and technologies is required.

Reference documents

Reference	Document Name
-----------	---------------

[XMLSCHEMA]	W3C Recommendation (2 May 2001): "XML Schema Part 2: Datatypes". Available at: http://www.w3.org/TR/2001/REC-xmlschema-2-20010502/ .
[XMLSIG]	W3C Recommendation (12 February 2002): "XML-Signature Syntax and Processing". Available at: http://www.w3.org/TR/2002/REC-xmlsig-core-20020212/ .
[WSI]	WS-I Basic Profile Version 1.0: "Final Material". Available at: http://www.ws-i.org/Profiles/BasicProfile-1.0-2004-04-16.html .
[WSDL]	W3C Note (15 March 2001): "Web Services Description Language (WSDL) 1.1". Available at: http://www.w3.org/TR/2001/NOTE-wsdl-20010315
[WSS]	OASIS Standard 200401 (March 2004): "Web Services Security: SOAP Message Security 1.0 (WS-Security 2004)". Available at: http://docs.oasis-open.org/wss/2004/01/oasis-200401-wss-soap-message-security-1.0.pdf
[ISO4217]	ISO 4217: "Codes for the representation of currencies and funds".
[DS]	Digital Signature usage for HomeSend Ws API Suite eSGE_EIS_HomeSend-Ws-API-Digital-Signature, version 01.00 or later.

Copyright

This document contains proprietary, confidential, or customer-sensitive information, and may not be copied or circulated without prior approval.

Table of Contents

TABLE OF CONTENTS.....	4
CONVENTIONS AND TERMINOLOGY	10
1 WHAT'S NEW IN THE WS API SUITE 2.3.....	12
1.1 OVERVIEW	12
1.2 COMPATIBILITY	12
1.3 HWS 2.3.....	12
1.3.1 New Operations	12
1.3.2 Updated Operations.....	12
1.3.3 Removed or Deprecated Operations	13
1.4 OEWS 2.3	13
1.4.1 New Operations	13
1.4.2 Updated Operations.....	13
1.4.3 Removed or Deprecated Operations.....	13
1.5 TEWS 2.3.....	13
1.5.1 New Operations	13
1.5.2 Updated Operations.....	14
1.5.3 Removed or Deprecated Operations.....	14
1.6 R-TEWS 2.3.....	14
1.6.1 New Operations	14
1.6.2 Updated Operations.....	14
1.6.3 Removed or Deprecated Operations.....	14
2 INTRODUCTION	15
2.1 HOMESEND OVERVIEW	15
2.2 E-WALLET SYSTEMS INTERCONNECTIONS	15
2.3 CALL FLOWS OVERVIEW.....	15
2.3.1 Synchronous Call Flow	15
2.3.2 Asynchronous Call Flow	16
2.4 INTRODUCTION TO HOMESEND ROUTING	17
2.4.1 HomeSend Routing Principles.....	17

2.4.2	End-Users URIs	18
2.5	PERFORMANCE EXPECTATIONS	20
3	USE OF WEB SERVICE TECHNOLOGIES	21
3.1	WEB SERVICE MESSAGE CONTENT.....	21
3.1.1	SOAP	21
3.1.2	XML.....	21
3.1.3	HTTPS	21
3.2	WEB SERVICE INTERFACE DEFINITIONS.....	21
3.3	SECURITY	21
3.3.1	Transport-Level Security.....	21
3.3.2	Message-Level Security	22
4	DETAILED SERVICE DESCRIPTION & SEQUENCE DIAGRAMS	23
4.1	GENERAL MECHANISMS.....	23
4.1.1	Transactions Structure.....	23
4.1.2	Retransmissions	23
4.1.3	Real-Time Retry Mechanism	24
4.1.4	Automatic Reconciliation	28
4.2	ONE-SHOT REMITTANCE	29
4.2.1	Synchronous Model	30
4.2.2	Asynchronous Model	32
4.3	REMITTANCE WITH ADVICE OF CHARGE.....	38
4.3.1	Advice Of Charge Timeout	40
4.3.2	Advice Of Charge Cancellation	41
4.3.3	Correlations	41
4.3.4	Asynchronous Variant.....	42
4.3.5	Reverse Advice of Charge.....	43
4.4	ASYNCHRONOUS CREDIT PROCESSING	45
4.4.1	Use Cases	45
4.4.2	Principle	45
4.4.3	“Server-side” Notification Model	46
4.4.4	Client-side Polling Model	48
4.4.5	Pre-Credit Notification.....	48

4.4.6	Cancelling a Pending Transaction	49
4.4.7	Amending a Pending Transaction	53
4.5	GETTING A TRANSACTION STATUS EXPLICITLY	57
4.5.1	Sending Side: Getting a Transaction Status from HomeSend	57
4.5.2	Receiving Side: Getting a Transaction Status	58
4.6	LINKS MONITORING: WATCHDOG MECHANISM	60
4.6.1	HWS Monitoring.....	60
4.6.2	OEWS Monitoring	60
4.6.3	TEWS Monitoring.....	61
4.6.4	R-TEWS Monitoring.....	61
5	IMPLEMENTATION GUIDELINES	62
5.1	SENDING SIDE	62
5.1.1	Operations to Implement	62
5.1.2	Mandatory Inconclusive Request Detection	62
5.1.3	Optional Request-level Retransmission Mechanism	63
5.1.4	A Typical HWS Client with Advice of Charge Support	66
5.1.5	Indicative End-User Feedbacks.....	68
5.2	RECEIVING SIDE	82
5.2.1	Operations to Implement	82
6	NAMESPACES	84
7	COMMON DATA TYPES DEFINITION.....	85
7.1	SIMPLE TYPES	85
7.2	SENDERINFORMATION STRUCTURE.....	86
7.3	AMOUNTINFORMATION STRUCTURE	86
7.4	SECURITY STRUCTURE.....	86
7.5	VENDORSPECIFICFIELD STRUCTURE [2.1].....	86
7.6	VENDORSPECIFICFIELDS STRUCTURE [2.1].....	87
7.7	BASEEXCEPTION STRUCTURE	87
8	HOMESEND WEB SERVICE (HWS).....	89
8.1	SPECIFIC TYPES	89
8.1.1	Payer Indicator.....	89
8.1.2	RemittanceCbsType Enumeration.....	89

8.1.3	ERechargeCbsType Enumeration	89
8.1.4	AocValidity	89
8.1.5	TransactionSummary Structure	90
8.1.6	ProvisionalResponse Structure [2.2]	90
8.1.7	TransactionStatus Choice [2.2]	91
8.1.8	PendingTransactionStatus Structure [2.2]	91
8.1.9	CompleteTransactionStatus Structure [2.2]	92
8.1.10	RejectedTransactionStatus Structure [2.2]	92
8.1.11	HomeSendServiceException Structure	93
8.2	OPERATIONS	93
8.2.1	HWS/adviceOfCharge	93
8.2.2	HWS/remittance	96
8.2.3	HWS/eRecharge	101
8.2.4	HWS/cancellation	101
8.2.5	HWS/status [2.2]	104
8.2.6	HWS/amend [2.3]	106
8.2.7	HWS/deviceWatchdog [2.2]	108
9	ORIGINATING ENDPOINT WEB SERVICE (OEWS)	109
9.1	SPECIFIC TYPES	109
9.1.1	Reservation Identifier	109
9.1.2	Chargeback Reason	109
9.1.3	OriginatingServiceException Structure	109
9.2	OPERATIONS	110
9.2.1	OEWS/reserve	110
9.2.2	OEWS/charge	112
9.2.3	OEWS/release	114
9.2.4	OEWS/chargeback [2.2]	115
9.2.5	OEWS/deviceWatchdog [2.2]	117
10	TERMINATING ENDPOINT WEB SERVICES (TEWS & R-TEWS)	119
10.1	TEWS AND R-TEWS TECHNICAL INTERFACES	119
10.2	SPECIFIC TYPES	119
10.2.1	State Enumeration (TEWS only)	119

10.2.2	AccountInformation Structure (TEWS only).....	119
10.2.3	ProvisionalResponseSupport Structure [2.2]	120
10.2.4	ProvisionalResponse Structure [2.2]	120
10.2.5	TransactionStatus Choice [2.2]	121
10.2.6	PendingTransactionStatus Structure [2.2].....	121
10.2.7	CompleteTransactionStatus Structure [2.2].....	122
10.2.8	RejectedTransactionStatus Structure [2.2].....	122
10.2.9	CancellationReason Structure [2.3].....	124
10.2.10	HsProvidedTransactionStatus Choice [2.3].....	124
10.2.11	HsProvidedPendingTransactionStatus Structure [2.3]	124
10.2.12	HsProvidedCompleteTransactionStatus Structure [2.3]	125
10.2.13	HsProvidedRejectedTransactionStatus Structure [2.3]	125
10.2.14	TerminatingServiceException Structure [2.2]	125
10.3	OPERATIONS	126
10.3.1	TEWS/accountInformation.....	126
10.3.2	TEWS/credit.....	128
10.3.3	TEWS/status.....	131
10.3.4	TEWS/cancellation [2.3]	133
10.3.5	TEWS/amend [2.3]	135
10.3.6	TEWS/statusReport [2.3].....	137
10.3.7	TEWS/deviceWatchdog [2.2].....	140
10.3.8	R-TEWS/statusReport	140
10.3.9	R-TEWS/deviceWatchdog.....	142
11	ANNEXES	143
11.1	CALL FLOWS.....	143
11.1.1	Simple Remittance Call Flows	143
11.1.2	Advice of Charge Call Flows	145
11.1.3	Automatic Reconciliation Call Flows.....	147
11.2	WSDL FILES	156
11.2.1	HomeSend Web Service (HWS) 2.3	156
11.2.2	Originating Endpoint Web Service (OEWS) 2.3	165
11.2.3	Terminating Endpoint Web Service (TEWS and R-TEWS) 2.3.....	170

11.3	SAMPLE REMITTANCE TRACES.....	180
11.3.1	Synchronous Credit Model	180
11.3.2	Asynchronous Credit Model – Successful Transaction	187
11.3.3	Asynchronous Credit Model – Failed Transaction.....	198
11.3.4	Sample Custom Faults	201

Conventions and Terminology

Icons

The following icons are used as visual cues to draw attention to important information.

**Note:**

Indicates useful and complementary information. Explanation, comment, or short expansion of the text object that is intended to catch your attention.

**Tip:**

Indicates practical but nonessential information that makes the solution easier to use or operate (e.g. keyboard shortcut, alternative way to perform a step in a procedure, etc.)

**Warning :**

The Warning icon indicates a caution. Would this information be ignored, it could cause possible and irreversible damage to the equipment, data or software.

Terminology

Definitions

For the purposes of the present document, the following terms and definitions apply:

- **Application:** Computer program that accesses a Web Service.
- **Physical Interconnection Point (PIP):** A system connected to HomeSend, handling end-user balances and payments and providing an interface to the served end-users. A PIP may offer multiple logical interfaces, declared as Functional Interconnection Points (FIPs) in HomeSend, each of them acting for a particular Service Provider in case of a shared PIP. Common PIPs implementations include E-Wallet systems, E-Money systems.
- **Web Service:** Software system designed to support interoperable machine to machine interaction over a network
- **Web Service Provider:** Entity that provides Web Services interfaces to capabilities offered
- **Web Service Client:** Entity that wants to make use of providers' web services to execute services.
- **Service Provider (SP):** an organization interfacing a HomeSend service to its end-users. Could be a MNO, MVNO, an e-wallet provider, a bank or any other financial institution.
- **Corridor:** a logical link between a source Service Provider and a destination Service Provider, enabling an end-user managed by the source Service Provider to perform a remittance or a recharge towards an end-user managed by the destination Service Provider. A corridor is directional: forming a corridor from SPA to SPB does not allow SPB's end-users to send anything to SPA's end-users. A corridor contains specific service charges and AML (Anti Money Laundering) mechanisms and only applies for a particular cross-border service (remittance or recharge).

Abbreviations

For the purposes of the present document, the following abbreviations will be used:

Name	Quick Description
AML	Anti-Money Laundering
FIP	Functional Interconnection Point
HWS	HomeSend Web Service
IT	Information Technology
MNO	Mobile Network Operator
MTO	Money Transfer Operator
MVNO	Mobile Virtual Network Operator
OASIS	Organization for the Advancement of Structured Information Standards
OEWS	Originating Endpoint Web Service
PIP	Physical Interconnection Point
PP	Platform Provider
PSP	Payment Service Provider
R-TEWS	Reverse Terminating Endpoint Web Service
SP	Service Provider
SPA	Service Provider A (sending service provider)
SPB	Service Provider B (receiving service provider)
TEWS	Terminating Endpoint Web Service
W3C	World Wide Web Consortium
WS	Web Service
WSDL	Web Service Description Language
WS-I	Web Services Interoperability Organization
XML	Extensible Markup Language

1 What's New in the Ws API Suite 2.3

This section provides an overview of the differences between the Ws API Suite 2.2 and 2.3.

1.1 Overview

The Ws API Suite 2.3 brings the following features:

- Ability to cancel asynchronous, pending transactions: a sending system can now attempt to cancel a transaction that is in "Credit In Progress" state, i.e. which was submitted successfully to a receiving system, but not completely processed yet. Transactions that are being authorized may also be cancelled.
- Ability to amend selected fields in a pending transactions: if both sending and receiving systems support this feature, it is possible for a sender to update one of several fields for a submitted transactions, for instance to fix a misspelled name, to allow the transaction completion. The receiving system can optionally notify the sending system that the transaction needs amending.
- HomeSend can now push a transaction status to the receiving system. This is a generic framework that will first be used to notify a receiving system that a transaction has been blocked by the sanction list screening subsystem and is waiting for an investigation from a receiving service provider's compliance officer.

1.2 Compatibility

The Ws API Suite 2.3 is fully backward compatible with 2.2: messages sent using HWS, OEWS, TEWS, or R-TEWS 2.2 can be fully read without any modification by a server implementing the 2.3 counterpart. However, the reverse is not guaranteed, as a 2.3 message may contain some new fields that cannot be understood by a 2.2 implementation.

All elements defined in HWS, OEWS, TEWS, and R-TEWS 2.2 are still valid and fully used in 2.3, which is just a superset of 2.2.

1.3 HWS 2.3

This section highlights the differences between HWS 2.2 and 2.3.

1.3.1 New Operations

HWS 2.3 adds the following new operation:

- HWS/amend: this operation enables to update one or several end-user related information elements for a pending transaction. For instance, this can be used to fix a typo in a beneficiary name, preventing the completion of a cash-out transaction. Both solicited and unsolicited amend requests are supported.

1.3.2 Updated Operations

The HWS/*adviceOfCharge* operation has been improved:

- HWS/*adviceOfCharge* requests now support an optional `reverseQuote` element to request HomeSend a quote based on the amount to be received, instead of the amount to be sent.
- HWS/*adviceOfCharge* responses have been extended to contain the initial/base amount used for service charges computations. This is also the amount to send when confirming a reverse quote.

- In addition, HWS/*adviceOfCharge* responses now include an expiration date that is more absolute than the expiration delay, which is still provided for compatibility with existing implementation. Implementers are suggested to use this date instead of the delay to know when the advice of charge proposal expires.
- In HWS/*remittance* requests following a HWS/*adviceOfCharge* request, *sourceUri* and *destinationUri* fields can now be omitted. If provided, they still need to be equal to the ones provided in the HWS/*adviceOfCharge* request. *This has been introduced in a fully backward-compatible WSDL file referenced with the version HWS 2.3.1. This still remains a fully compatible HWS 2.3 WSDL.*
- The *cbsType* field in HWS/*adviceOfCharge* and HWS/*remittance* requests is now an open enumeration (string) instead of a closed ones (with only specific values). This allows for more service type values to meet business needs (such as B2B, P2P, etc).

The HWS/*cancellation* request has been extended to include an optional cancellation reason. This request can now be used to cancel an asynchronous transaction which is in a “credit in progress” state, in addition to canceling running advices of charge.

In addition, a HWS client using HWS 2.3 is now guaranteed never to receive a ER0009 to a HWS/*remittance*. Instead, a HWS/*remittance* provisional response will be sent. This simplifies HWS implementations, and makes them transparent to synchronous and asynchronous transactions, which can now be handled exactly the same way.

1.3.3 Removed or Deprecated Operations

No operations were removed or marked as deprecated from HWS 2.2.

1.4 OEWS 2.3

This section highlights the differences between OEWS 2.2 and 2.3.

1.4.1 New Operations

No operations were added.

1.4.2 Updated Operations

Existing operations have not been updated.

1.4.3 Removed or Deprecated Operations

No operations were removed or marked as deprecated from OEWS 2.2.

1.5 TEWS 2.3

This section highlights the differences between TEWS 2.2 and 2.3.

1.5.1 New Operations

TEWS 2.3 adds the following new operations:

- **TEWS/amend:** this operation enables to update one or several end-user related information elements for a pending transaction. For instance, this can be used to fix a typo in a beneficiary name, preventing the completion of a cash-out transaction.
- **TEWS/cancellation:** this operation enables to request the destination system to cancel a pending transaction. Even if the cancellation request is accepted, it does not mean that the transaction will be actually cancelled.
- **TEWS/statusReport:** this operation is used by HomeSend to notify a destination system that one of its transactions needs a particular action from its side before being submitted. This is typically used to notify the receiving systems that a transaction needs a sanction list screening investigation by one of its compliance officers, in the case where such a screening service has been delegated to the HomeSend platform provider.

1.5.2 Updated Operations

TEWS/accountInformation responses now support an optional list of secondary currencies. A default, primary currency for the destination account must still be provided in the `currency` element. If this account can be credited using other currencies, however, the receiving system is now able to provide them to HomeSend.

TEWS/status requests now contain an optional `routingTag` element.

1.5.3 Removed or Deprecated Operations

No operations were removed or marked as deprecated from TEWS 2.2.

1.6 R-TEWS 2.3

This section highlights the differences between R-TEWS 2.2 and 2.3.

1.6.1 New Operations

No operations were added.

1.6.2 Updated Operations

Existing operations have not been updated.

1.6.3 Removed or Deprecated Operations

No operations were removed or marked as deprecated from R-TEWS 2.2.

2 Introduction

2.1 HomeSend Overview

HomeSend enables and facilitates money and/or credit transfers between E-Wallet Systems. Such transfers are possible on a peer to peer basis, but the benefit of hub interconnection is to simplify agreement setups and operational procedures, especially to implement international remittances and e-recharge operations.

HomeSend applies cross-border commissions and taxes according to the corridor's agreements, and the source and destination E-Wallet systems shall not add additional service charges, to make sure HomeSend's advice of charge capabilities work correctly.

2.2 E-Wallet Systems Interconnections

HomeSend interconnects multiple E-Wallet systems through a set of Web Services APIs:

- The Source E-Wallet system, on which HomeSend requests reservation and charge operations to charge the sending end-user, publishes a web service called OEWS (Originating Endpoint Web Service). HomeSend implements an OEWS client to access it.
- The Destination E-Wallet system, on which HomeSend requests additional end-user information and credits the receiving end-user, publishes a web service called TEWS (Terminating Endpoint Web Service). HomeSend implements a TEWS client to access it. For asynchronous responses, TEWS is completed by a technical Replier API, dubbed R-TEWS, published by HomeSend.
- The system that requests the remittance or the electronic recharge operation towards HomeSend implements a client to access a web service called HWS (HomeSend Web Service), published by HomeSend. Such a system is usually the same as the Source E-Wallet. That's why both systems are referred under the single name of "Source E-Wallet system" in this document – but they can be split on different logical or physical systems if needed.

HWS, OEWS, and TEWS (with its R-TEWS technical service) compose the HomeSend WS API suite. This document describes their versions 2.3.

2.3 Call Flows Overview

HomeSend supports two kinds of transaction call flows, called "synchronous" and "asynchronous".

Detailed call flows and explanations are provided in the following chapters.

2.3.1 Synchronous Call Flow

Here is a example of a typical transaction that could be performed through HomeSend: a one-shot remittance (i.e. without advice of charge), here implemented using the "Synchronous" model:

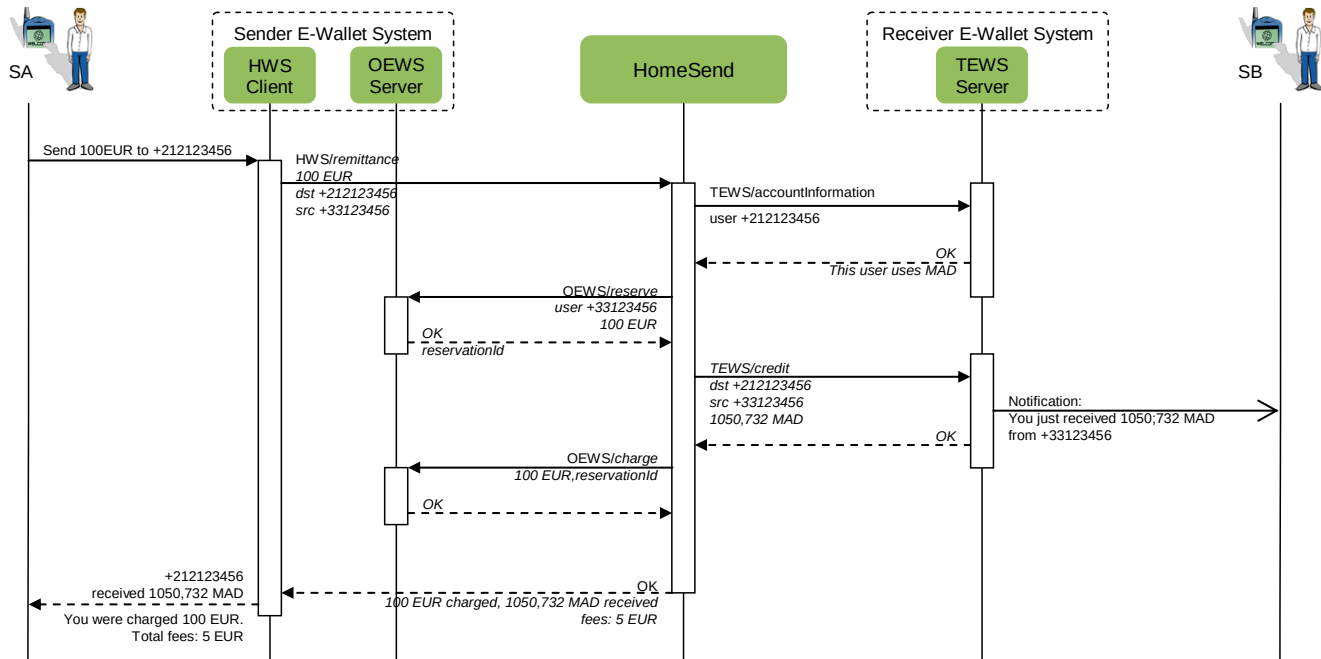


Figure 1 - A typical one-shot remittance, synchronous model

This synchronous model is well suited for destination E-Wallet systems that are able to commit a credit in real-time. Destination systems that require minutes, days, or weeks of processing before being able to accept definitively a transfer should implement the asynchronous model as described below.

2.3.2 Asynchronous Call Flow

The same transaction as above, i.e. a one-shot remittance, can also be implemented using the asynchronous model. It is well suited for destination systems that are not able to commit a credit in real-time (banks, MTOs, cash-out systems).

The optimum call flow for this case introduces a status report notification message sent by the destination systems when the credit operation has been fully processed:

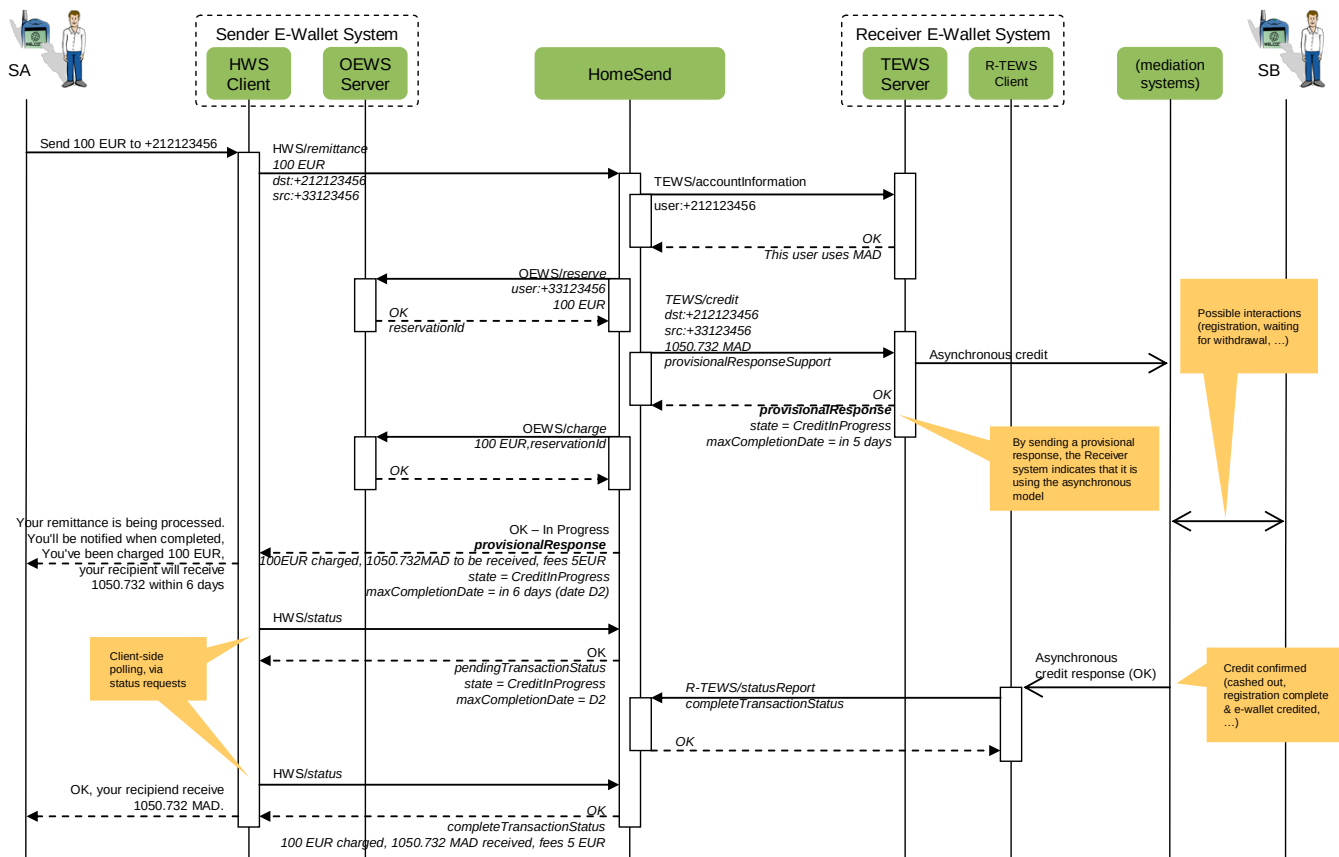


Figure 2 - A typical one-shot remittance, asynchronous model

HomeSend may also use a client-side polling mechanism to retrieve an updated status for the submitted credit operation; this may prove useful if the server-pushed status report notification has been missed, but it is less efficient as the transaction completion time then depends on HomeSend's polling interval configuration.

Please see [Asynchronous Credit Processing](#) for further details.

2.4 Introduction to HomeSend Routing

HomeSend uses a routing engine that enables to finding the destination E-Wallet system to perform a transaction based on source, destination, and requested service criteria.

This introduction to HomeSend's routing should help the reader understand the way some API fields may be used during the implementation of a HomeSend connector on an E-Wallet system.

2.4.1 HomeSend Routing Principles

HomeSend performs a routing at two levels:

- Service level: it selects the corridor the transaction is routed through and the appropriate service charges to apply, according to the agreements between the involved service providers.
- Access level: HomeSend selects the receiver E-Wallet systems that may handle the transaction, in accordance with both technical properties and the selected destination service provider from the service-level routing above.

Applying these levels of routing leads to a list of candidate corridors that are reachable through a set of different destination E-Wallet systems. Based on this list, HomeSend performs a dynamic routing to locate the actual destination system that will handle the transaction by sending *TEWS/accountInformation* requests.

Such requests contain the identifier of the end-user to serve and an optional indication of the corridor/service to apply. Based on this information, the targeted destination E-Wallet system can accept or reject the *TEWS/accountInformation* request, completing the routing processing.

Typical reasons to reject such a request:

- The destination E-Wallet system does not serve this end-user: unsupported end-user type, not available on this system, or ported-out.
- The destination E-Wallet system can serve this end-user, but not through the requested corridor or service – for instance, when the end-users can be served by this E-Wallet system only if they subscribed to a specific service.
- Service temporarily unavailable; this includes connection/response timeouts as well.

Depending on the routing configuration and the rejection reason, HomeSend may try the next destination E-Wallet system candidate or retry the same system with a different requested corridor/service.

2.4.2 End-Users URIs

End-users are uniquely identified by URIs. They are transmitted as strings on the wire, following the RFC2396 format. HomeSend defines a collection of supported URI schemes:

Scheme	Format	Description
tel	+E164	E164 number identifying a Telephony Subscriber. The number (usually a MSISDN) must be expressed as an E164 number (starting with a country code). The '+' sign is mandatory (it is compliant with RFC3966 format / international number only). However HomeSend does not support hyphenation). Example: <code>tel:+3312345678</code>
iban	ISO 13616:2007, packed (i.e. no blank characters)	International Bank Account Number as a ISO 13616:2007 identifier, case sensitive. This scheme is provided for convenience. You may prefer use the BAN scheme instead. A receiver system that supports IBAN-based identification of its managed accounts should also support BAN-based identification for those same accounts. Example: <code>iban:CH5108686001256515001</code>
ewallet	<alphanumeric string> URI parameters are also supported	E-Wallet account identifier. As there is no standard to identifier an e-wallet account globally, the HomeSend Platform Provider must decide with its SP what the exact format of such an ID will be. To help keep the format simple and avoid collisions between SP's local identification convention, an additional URI parameter named 'sp' may be used, identifying the the e-wallet account provider. With such a model, each SP can keep its local e-wallet identification scheme and transpose it transparently to HomeSend. Examples: Using the sp URI parameter: <code>ewallet:user011;sp=paypal</code> Using a naming convention (not transparent for the SP): <code>ewallet:paypal_user011</code>

ban	<alphanumeric string>;bic=<bic> With <bic> matching the ISO 9362 standard, in uppercase	A bank account number (BAN) identifier, formed by the account number (its format depends on the bank involved) and the bank identifier code (BIC), as an URI parameter. This scheme is suitable to identify most bank accounts worldwide, contrary to the IBAN scheme, which is mostly limited to European bank accounts. Notice that an IBAN identifier can also be replaced by a BAN identifier using this model. The BIC is formed by 8 uppercase alpha numeric characters, optionally followed by a 3 alpha numeric character branch code. The account number usually contains alphanumeric characters, but the exact format and length depends on the bank that holds the account. HomeSend does not syntax-check the BIC or the account number value. Example: ban:30056001140114000251817;bic=CCFRFRPP
pan	[0-9]+	A Private Account Number, i.e. a payment or credit card number. Depending on the card brand and issuer, the number of digits varies between 12 and 16 typically. This URI scheme is only usable as a destination URI in HWS requests - it won't be sent by HomeSend in OEWS or TEWS requests or responses. This allows to directly send a remittance to a card-backed account (could it be a banking or ewallet account or other kinds of accounts that are associated to a card). Source URIs identified by a pan are not supported yet. Using this URI scheme requires the sending system to be fully PCI-DSS certified and onboarding of such systems will require additional steps (including but not limited to a proof of this PCI-DSS compliance, and the use of a dedicated HomeSend HWS URL to follow a PCI-DSS certified path). Example: pan:5555123412345678
mpan	'xxxxxxxxxxx' [0-9]{4}	Masked PAN, i.e. a payment or credit card number that is partially masked and usable for end-user display or data capture without requiring a PCI-DSS compliance. Only the last 4 digits of the card are displayed, prefixed by 12 'x' characters (regardless of the actual number of digits in the original PAN). This URI scheme can only be used as a source URI in TEWS requests, so exclusively sent by HomeSend, to help identify the card the remittance was sent from. In particular, TEWS implementers might expect this URI scheme in a source URI in TEWS/credit requests. Example: pan:xxxxxxxxxxxx5678
*token	Token-dependent	<i>In addition to the standard URI schemes above, HomeSend also supports vendor-specific tokens to be used instead of PANs, for destination service providers that can either publish a tokenization API to HomeSend, or re-use tokens as retrieved by HomeSend through one of its tokenization providers.</i> <i>In any case, the token format and scheme is discussed with the service providers willing to use such identifiers during IOT preparation.</i>

URI formats to use depend on the kind of end-users or end-user accounts managed by the E-Wallet system.

For ewallet URIs, additional discussions with the HomeSend platform provider should be held to establish a namespace (usually using the ; sp= URI parameter or a particular naming convention) to avoid colliding with other e-wallet's identifiers managed by other providers connected to the hub.

When an E-Wallet system receives a request related to an URI that uses a scheme that is not provided here, it should reply with an "unknown subscriber" error, instead of "invalid format", as supported schemes may be added without breaking the API.

2.5 Performance Expectations

To guarantee a good end-user experience, at least 90% of OEWS and TEWS requests (respectively sent to the sender E-Wallet systems and receiver E-Wallet systems) are expected to reply in less than **500ms**.

3 Use of Web Service Technologies

All HWS, OEWS, TEWS, R-TEWS Web Services are built over the same Web Service protocol stack.

3.1 Web Service Message Content

3.1.1 SOAP

All Web Service messages **MUST** send and accept messages that conform to the SOAP usage defined in the WS-I Basic Profile [WSI], using the document/literal encoding style.

The supported SOAP version is 1.1.

3.1.2 XML

All Web Service messages **MUST** send and accept messages that conform to the XML usage defined in the WS-I Basic Profile 1.1 [WSI].

3.1.3 HTTPS

All Web Service messages **MUST** send and accept messages that conform to the HTTPS usage defined in the WS-I Basic Profile 1.1 [WSI].

OEWS and TEWS HTTPS servers may require a client-side certificate. Their HTTPS server certificate, however, must be known by HomeSend for both TEWS and OEWS interfaces to avoid connecting to an uncontrolled E-Wallet system.

It is up to the integrator to decide what HTTPS server certificate should be used (and checked) to access HomeSend on its HWS or R-TEWS interface (HomeSend is usually deployed behind a reversed proxy). Those interfaces do not require a HTTPS client certificate.

3.2 Web Service Interface Definitions

HWS, OEWS, TEWS, R-TEWS are formally defined using a WSDL 1.1 file, following the WSDL specification [WSDL] and compliant with the WSDL usage made in WS-I Basic Profile 1.1 [WSI].

These WSDL files can be found in this document (see [WSDL Files](#)) and may be used to develop HWS clients as well as OEWS and TEWS servers (and R-TEWS client) using the different Web Service toolkits & frameworks available on the market.

**Tip:**

Relevant WSDL files are also embedded into this PDF document, as attached files.

The TEWS WSDL file also contains the R-TEWS service definition.

3.3 Security

3.3.1 Transport-Level Security

HomeSend requires that all SOAP messages are sent over HTTPS (TLS/SSLv3).

The HTTPS server's certificate must be checked by the client side upon connection, to avoid connecting to an unidentified OEWS, TEWS, HWS, or R-TEWS server (avoiding the man-in-the-middle attack).

In addition, HomeSend can send a client-side HTTPS certificate when connecting to OEWS or TEWS servers for further transport-level security.

HomeSend does not currently require HTTPS clients certificates for HWS and R-TEWS links.

In production, the traffic between E-Wallet systems and HomeSend are usually sent over dedicated links or virtual private networks, according to the IP interconnection rules imposed by the HomeSend platform provider.

3.3.2 Message-Level Security

HomeSend also supports message digital signature based on WS-Security X.509 Certificate Token Profile 1.1.

For more information regarding this mechanism, please see [DS].

4 Detailed Service Description & Sequence Diagrams

This section discusses the main HomeSend scenarios.

4.1 General Mechanisms

4.1.1 Transactions Structure

HomeSend processes its high-level operations (one-shot remittance, remittance with advice of charge, and e-recharge) by creating a transaction between 3 entities:

- The system requesting the remittance or e-recharge (usually the sender E-Wallet System) and HomeSend, using the HWS API,
- The receiver E-Wallet System and HomeSend, using the TEWS API, as well as R-TEWS API for receiver system-pushed status reports,
- The sender E-Wallet System and HomeSend, using the OEWS API (optional)

A transaction is uniquely identified by a transaction identifier generated by HomeSend as soon as the request starting a new transaction is received. This transaction identifier is globally unique.

Transactions contain request/response couples to perform local operations between the different systems at stake.

All of them are described in the next sections.

4.1.2 Retransmissions

Requests can be retransmitted at any time by the initial sending entity. Such retransmissions could occur in different cases/contexts:

- For reliability purposes, a request that was too long to respond may be retransmitted, as it could have been dropped due to a network or software error between the sender and the recipient. This is used to implement what HomeSend calls a “Real-time retry mechanism”, and can be used by the sender E-Wallet system to retransmit its requests, as well as HomeSend when sending the different requests to the sender and receiver E-Wallet systems.

Interleaving retransmitted requests for the same transactions is not allowed: only the last received request may be retransmitted. For instance, a sending E-Wallet system may receive an OEWS/reserve request, then a OEWS/charge request, then a OEWS/charge retransmission: but a subsequent OEWS/reserve retransmission should reply with an error.

Requests retransmitted in this context are exactly the same as the initial ones (same SOAP payload). There is no way, from a syntactic point of view at SOAP level, to distinguish a retransmission from the initial request.

Sender and receiver E-Wallet systems must support incoming request retransmissions (i.e. detect duplicate requests) for at least **36 hours** after receiving the request for the first time (OEWS/reserve, OEWS/charge, OEWS/chargeback, TEWS/credit). HomeSend also detects incoming HWS/adviceOfCharge, HWS/remittance, and HWS/cancellation duplicate requests for at least 36 hours.

This 36-hour delay is called the *maximum automatic reconciliation timeframe*, noted *MAX_{reconciliation}*. It enables to repair the transaction automatically in case of ambiguous/inconclusive requests (see below). After this delay, no retransmissions related to the transaction must be sent, as they would be considered as new requests instead of retransmissions.


Warning :

The source E-Wallet system shall not send a duplicate request after this delay – HomeSend would consider the incoming request as a new request, not a retransmitted one, thus starting a new transaction (with possible remittance or e-recharge).

In return, HomeSend guarantees never to send a duplicate requests to any E-wallet systems after this delay, too.

- A system may also retransmit a request even after receiving a response: this mechanism is called “status query”. It is mainly used when the response was an `ER0009` (indicating a case where the final transaction response cannot be provided immediately) and retries are attempted to get a more final status, as it may still evolve (see “Automatic Reconciliation” below).

While the retransmitted request is exactly the same as for real-time retry, it can be retransmitted longer after receiving the first response, over a long period (the maximum reconciliation timeframe $MAX_{reconciliation}$). Sender E-Wallet system implementing a HWS client MUST guarantee that no retransmission will be performed after this maximum delay, otherwise the retransmitted request can be considered as a brand new request, thus executing the same operation a second time. HomeSend, as a OEWS and TEWS client, guarantees it.

The number of status queries N_Q and the delay between them R_Q are configured independently from the parameters governing the real-time retry mode, and are usually much greater. However, $(N_Q \times R_Q)$ is always lower or equal to $MAX_{reconciliation}$.

Each status query may be itself retransmitted “locally”, using the real-time retry mechanism above.

On HWS side, while this mechanism is fully supported, it is highly recommended that “status queries” after receiving a `HWS/remittance` or `HWS/adviceOfCharge` response from HomeSend are implemented using the `HWS/status` operation. This is a safer model that won’t risk remittance re-processing even after $MAX_{reconciliation}$. Please refer to [Getting a Transaction Status from HomeSend](#) for further details.

These use cases are described in more details in the next sections.

4.1.3 Real-Time Retry Mechanism

Requests sent within a transaction may be retransmitted by the initial sender. This retransmission mechanism improves the transaction reliability without relying on the sole TCP layer, which is usually not reactive enough for near real-time operations.

This local mechanism enables to decide if a request was in error, inconclusive (ambiguous - no way to know, if the recipient processed it), or correctly responded, *in a limited timeframe* (usually $\leq 9000\text{ms}$). Depending on this status, the sender may trigger different behaviors.

This mechanism is parameterized by two variables:

- N_R : the number of request attempts (advised value: 3), i.e. the total number of transmissions
- D_R : the delay between two transmissions attempts (advised value: 3000ms)

On HomeSend side, the values for N_R and D_R are configurable independently for each E-Wallet system. Typical values are 2 and 2000ms respectively, but are usually customized for each system.

4.1.3.1 On Request Initiator's Side

A request initiator may retransmit a request in the following cases:

- Connection error at TCP level (refused or timed out) – the TCP connection timeout must be lower or equal to D_R
- Sending error (once the TCP connection is established)
- No response received after a configurable timeout (D_R)
- Incomplete response received (usually due to a TCP disconnection)
- The SOAP Fault `ER0003 Request already in progress` was received. The next attempt may provide an updated status.

Any other event should stop the retransmission mechanism, as a final request status will be known.

Each retransmission must be sent into a new TCP connection, and the previous attempt's TCP connection must be closed first.

The request is considered **in error** if and only if:

- All TCP connection attempts failed, or
- A SOAP fault was received, different from `ER0003 (Request in progress)`

The request is considered **OK** if and only if:

- An attempt provided a valid response (non-SOAP fault)

The request is considered **inconclusive** in all other cases, in particular:

- If at least one TCP connection succeeded (i.e. not all attempts led to a TCP connection error) but no explicit SOAP Fault or OK response was received
- The last received SOAP faults were `ER0003`

Usually, the request initiator that decided that the last request attempt was timed out, closes the TCP connection. If not, a late response may be received. In this case, it must be discarded by the initiator.

4.1.3.2 On Request Recipient's Side

If a system received a retransmitted request (at least, what it considers to be a retransmitted request):

- If the request was already processed and replied (and was the last replied request for the transaction), send back the same response message (unless the status evolved from `ER0009 Ambiguous` to a new status; in this case, the new status must be used in response),
- If the request is still being processed, reply with a `ER0003 Request already in progress` response message.

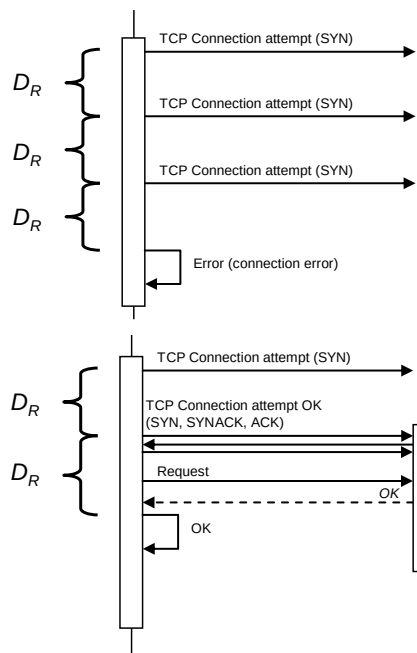
This behavior shall be guaranteed for a minimal duration after receiving the initial request, which corresponds to the maximum reconciliation timeframe $MAX_{reconciliation}$, as discussed above, as this retransmission mechanism may also be used in a “status queries” context (see [Automatic Reconciliation](#)). The duplicate detection engine on the recipient’s side must then be able to detect this retransmission during that long.

Depending on the implementation of the recipient system, a retransmitted request after this delay may be considered as an invalid message or a new request.

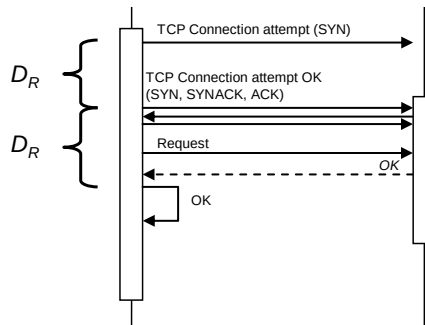
4.1.3.3 Examples

In these examples, $N_R=3$.

Connection error:
(final request result: error)

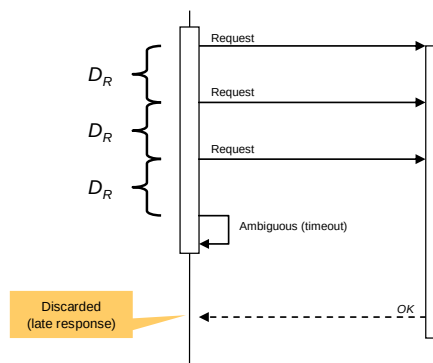


Connection error, recovered on second attempt:
(final request result: OK)

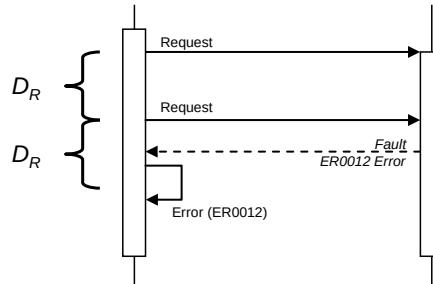


In the following call flows, the TCP level is not represented for clarity.

TCP connection OK, but request timeout
(if the last TCP connection was not closed by the sender, any late response must be discarded):
(final request result: inconclusive)

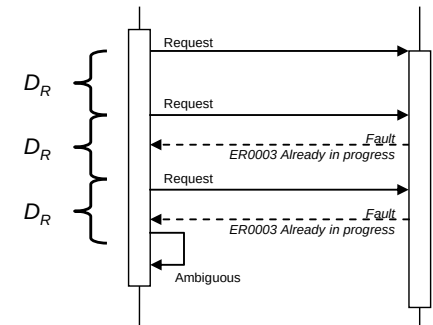


A final response (here, an error) was retrieved on second attempt:
(final request result: error)

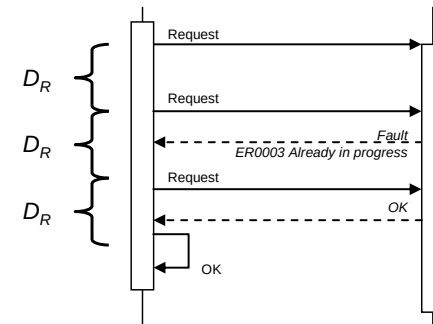


ER0003 Already in Progress

Ambiguous response on timeout, due to the destination being explicitly too long to process the request:
(final request result: inconclusive)

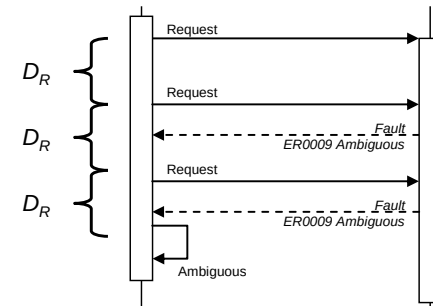


Previous case, but recovered to OK on 3rd attempt:
(final request result: ok)

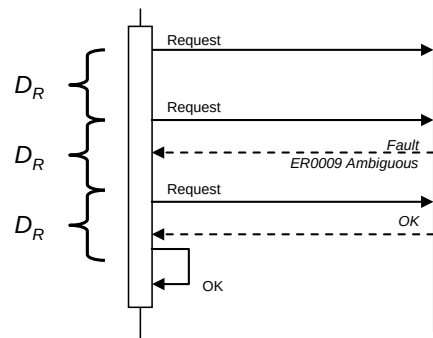


ER0009 Ambiguous

Ambiguous response that was not resolved within the given timeframe, leading to an ambiguous status:
(final request result: inconclusive)



Previous case, but recovered to OK on 3rd attempt:
(final request result: ok)



4.1.4 Automatic Reconciliation

During a transaction, some failed or non-replied requests towards E-Wallet systems may result in the transaction being “ambiguous”, i.e. it already performed some operations partially, but due to ambiguous response on credit, the final transaction’s status cannot be well known.

In some other cases, the transaction’s global status is already known (OK or NOK) but the standard call flow could not be completed. This transaction is said to be “incomplete”. In such cases, the request opening the transaction is replied to the sender E-Wallet system with a provisional response so that a feedback can be provided to the end-user, and the transaction is completed “in the background” to make sure that all systems are synchronized with the transaction’s final status.

This mechanism may take a long time and is called “reconciliation”. It can be run for several days (up to $MAX_{reconciliation}$, though a manual reconciliation may take longer), and may require additional manual, administrative interventions to provide operations status that could not have been obtained automatically.

4.1.4.1 Resolving Inconclusive Transactions

When a TEWS/*credit* request is inconclusive, HomeSend replies to the source E-Wallet system with a provisional response. In parallel, it keeps trying to get a TEWS/*credit* response from the destination system.

Behavior on sender side (HWS Client)

In this case, the sender E-Wallet system should start polling regularly HomeSend to get an updated status on this transaction, while HomeSend tries to complete the transaction automatically.

This polling can be implemented using two methods:

- Status requests: this mode consists in using the HWS/*status* operation to get an updated status on the transaction. However, it is only possible to use this mode if the HWS client was notified of the HomeSend transaction ID to use in such a HWS/*status* request; this is only possible after a HWS/*adviceOfCharge* response,
- A legacy method is also supported, sending duplicate requests: this mode consists in retransmitting a duplicate HWS/*remittance* request as the one that was responded with the provisional response. Such duplicate requests are said to be “status query” requests, as they won’t trigger any remittance or e-recharge again, but just used to get an updated status once available. Any kind of valid HWS/*remittance* response is possible, including a provisional one.

The sender E-Wallet system may retransmit duplicate requests up to $MAX_{reconciliation}$ after the initial request. It is important NOT to send them later, as HomeSend may consider them as starting a new remittance or e-recharge transaction.

This is a legacy model that is only supported to ease porting from older HWS API versions to HWS 2.2 and later. Instead of using duplicate retransmissions, the HWS/*status*-based model is now preferred (see above).

As soon as a final response is provided to the sender E-Wallet system, it should stop sending duplicate or status requests (and usually provides a feedback to the sending end-user).

How often these status queries (duplicate requests or HWS/*status* requests) are sent is up to the sender E-Wallet system implementer. However, the interval between status query requests may be advised by the HomeSend platform provider, in accordance with the reconciliation intervals used by HomeSend itself.

A 1-hour interval is the minimal recommended value, through shorter polling interval may be used for testing.

When HomeSend is unable to resolve the transaction automatically, because no conclusive TEWS/*credit* response was received before $MAX_{reconciliation}$, a manual action is required to unblock the transaction. The service provider responsible for the sender E-Wallet system will be notified by the platform provider of such an abnormally long transaction. Notifying the end-user in such a case is out of the scope of this API.

Behavior on HomeSend side

In the meanwhile, while the sender E-Wallet system sends status query requests to get an updated status, HomeSend tries to resolve the inconclusive TEWS/*credit* request by retransmitting it regularly, at larger interval than D_r , and more times than N_r . However, HomeSend gives up when reaching $MAX_{reconciliation}$ after the initial credit attempt.

As soon as the TEWS/*credit* request outcome is known, next status queries will be responded either with an OK (if the credit succeeded), a provisional response, or an appropriate error (if the credit failed), so that a final end-user response can be provided.

HomeSend will then complete the transaction (OEWS/*charge* or OEWS/*release*, if applicable) in the background.

Message flows associated to this Ambiguous transactions resolution mechanism are available in annex ([Recovering from Ambiguous Credit Request](#)).

4.1.4.2 Completing Incomplete Transactions

The automatic reconciliation process used to complete incomplete transactions, i.e. transactions whose OEWS/*reserve*, OEWS/*charge*, OEWS/*release* or OEWS/*chargeback* operation failed, is completely driven by HomeSend, and requires the following behavior on the sender and receiver E-Wallet system:

- The sender E-Wallet system must implement the requirement regarding the retransmission mechanism: being able to identify a retransmitted/duplicate OEWS/*reserve*, OEWS/*charge*, OEWS/*release* or OEWS/*chargeback* request, for a at least $MAX_{reconciliation}$ after receiving the first request.
- The receiver E-Wallet system must implement the requirement regarding the retransmission mechanism: being able to identify a retransmitted/duplicate TEWS/*credit* request for at least $MAX_{reconciliation}$ time after receiving the first TEWS/*credit*. Notice that detecting retransmitted TEWS/*accountInformation* request is not mandatory, as this is an idempotent operation. Detecting TEWS/*status* duplicates is also useless as they can be reprocessed safely.
- In addition, if the (sender or receiver) E-Wallet system replied an ER0009 *Ambiguous* to any of these requests, it must be able to return the latest known status, when receiving the retransmitted request (so that, after several attempts, HomeSend gets a clear request status instead of the initial ER0009 response).

For reference, call flows related to automatic reconciliation are provided in annex ([Automatic Reconciliation Call Flows](#)).

4.2 One-Shot Remittance

One-shot remittance allows a subscriber to transfer an amount of money from his/her account managed by the source E-Wallet System to another account in another E-Wallet System. The two E-Wallet Systems can be located in different countries, but may also be located in the same country, and may manage different kinds of accounts (for instance, the source system could interface e-wallet accounts,

identified by a telephone number or a particular e-wallet ID, while the destination system could interface bank accounts, identified by an IBAN number).

Additionally, it enables to indicate which participant (sender or receiver) should assume the service charges. If the receiver assumes the charges, applied taxes and commissions are deducted from transferred amount. If the sender assumes them, the sender balance is deducted from the transferred amount plus applied taxes and commissions.

4.2.1 Synchronous Model

In a typical scenario, a prepaid mobile subscriber wishes to transfer money to his/her family. The subscriber uses his/her mobile phone or other landline phone to interact with an IVR system (for example). In order to transfer money, once identified, the subscriber enters the MSISDN of the receiver, and the amount to transfer. The E-Wallet system behind the IVR then triggers HomeSend's remittance logic with a new *HWS/remittance* request:

1. HomeSend first locates the address of the service managing the receiver, using one or multiple *TEWS/accountInformation* requests. Once located, HomeSend computes the service charges that apply to the transfer,
2. An optional reservation is done on the sender's account balance (*OEWS/reserve*),
3. The receiver's account balance is then increased by the received amount (*TEWS/credit*),
4. The sender's reservation is confirmed (*OEWS/charge*) (optional, only if a reservation was performed),
5. HomeSend finally replies to the *HWS/remittance* request with a summary of what was debited/credited.

This sequence is shown below.

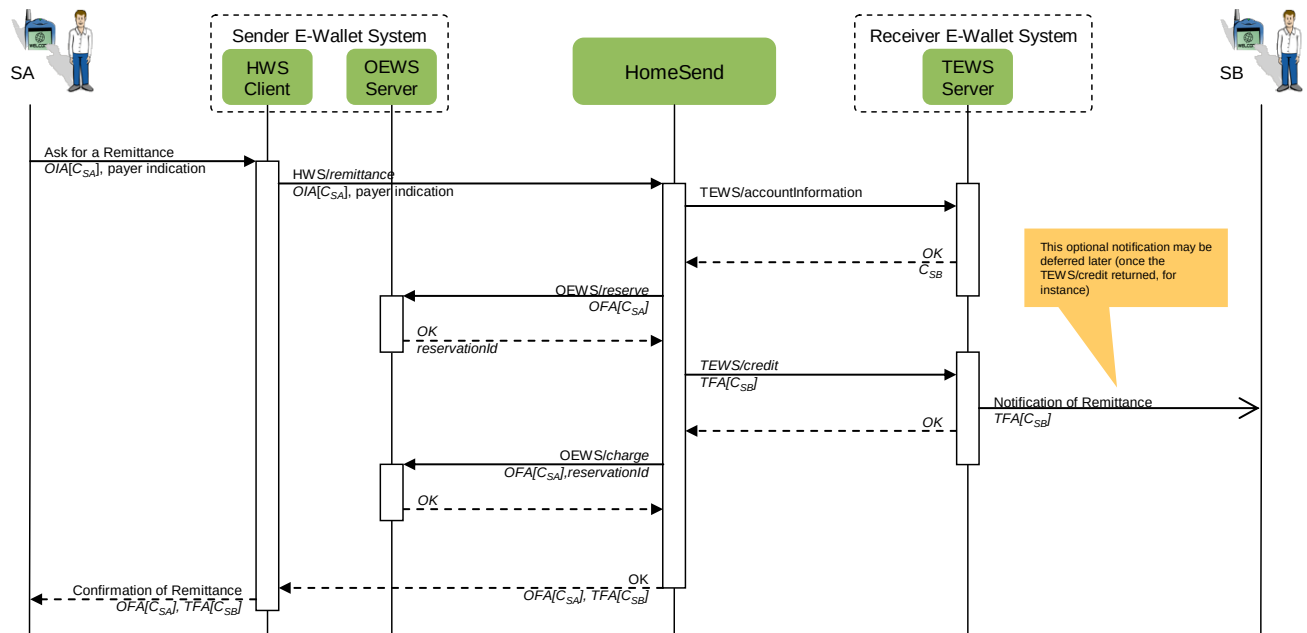


Figure 3 - One Shot Remittance sequence diagram

The following amounts are considered:

- **OIA:** Originating Initial Amount: the amount the sending end-user wants to send.

- **OFA**: Originating Final Amount: the amount the sending end-user has been charged. Could be equal to OIA, if the receiver pays the fees (default), or equal to $OIA + \text{transaction fees}$ if the sender pays them. Who pays the fees is indicated in the HWS/remittance payer indication.
- **TFA**: Terminating Final Amount: the amount actually received by the receiving end-user.

$A[C_{SA}]$ and $A[C_{SB}]$ are used to denote an amount A in sending end-user currency (C_{SA}) or in receiving end-user currency (C_{SB}).

The transactions identifiers at stake are stored into `transactionId` and `hsTransactionId` fields. The following sequence focuses on the way they are managed:

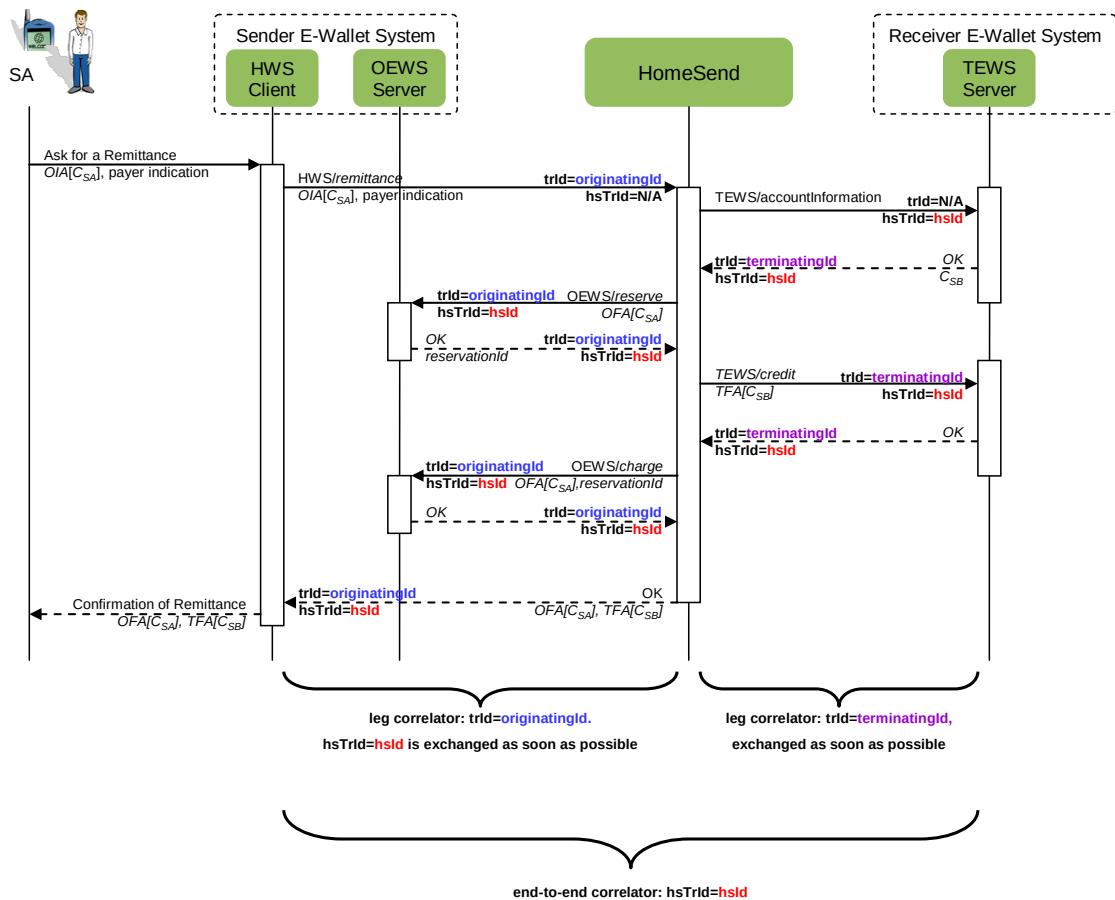


Figure 4 - One Shot Remittance requests correlation
(the actual fields names are `transactionId` and `hsTransactionId`)

The reservation/charge requests are actually optional. It depends on the `reservationRequired` flag in the HWS/remittance requests. By default, it is `true`, leading to the call flows above. When set to `false`, the call flow becomes:

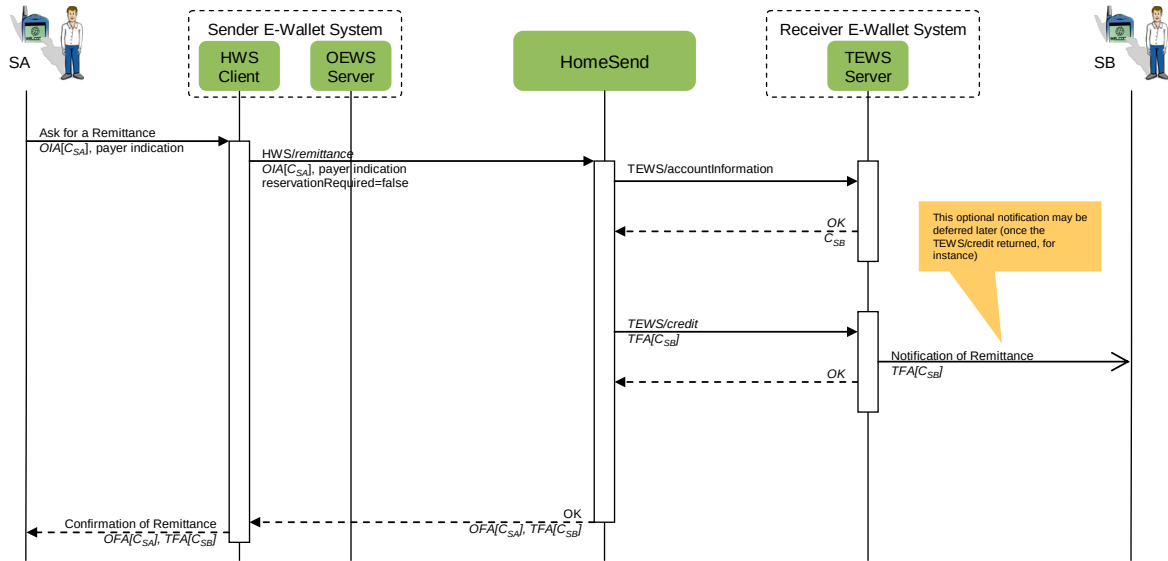


Figure 5 – One-shot remittance with reservationRequired=false

This model is well adapted to sender system implementations that do not want HomeSend to orchestrate the reservation for them. In this case, the sender system is responsible for reserving the amount to charge before sending the HWS/remittance request, and commit or release this reservation once the final HWS/remittance response is received.

As it may not be possible to know how much should be charged before receiving the HWS/remittance response (as the charged amount may include the transaction fees), this simplified call flow is typically used with an advice of charge (see [Remittance with Advice of Charge](#)).

This is also the preferred implementation for a “cash in”-like service easily, where an operator is able to verify manually that the sender has the amount he/she wants to send, including the additional fees, if the sender pays them. In this case, there is no need for an actual E-Wallet system on source side, but only a way to trigger the remittance through HomeSend.

This call flow is logically equivalent to reserve and charge requests being always replied successfully and thus all call flows related to remittance and remittance with advice of charge also apply to it.

4.2.2 Asynchronous Model

One-shot remittances can be implemented using an asynchronous credit model.

Switching to the asynchronous model is up to the destination system: if a provisional response is provided to the TEWS/credit request, the transaction is then said to be asynchronous, as the final TEWS/credit status will be retrieved later. If a final response is provided to this request, the transaction is said to be synchronous, and the call flows above apply.

From the sender’s point of view, such an asynchronous transaction begins the same way as a synchronous one:

1. The sender system triggers HomeSend’s remittance logic with a new HWS/remittance request,

2. HomeSend locates the address of the service managing the receiver, using one or multiple TEWS/*accountInformation* requests. Once located, HomeSend computes the service charges that apply to the transfer,
3. An optional reservation is done on the sender's account balance (OEWS/*reserve*),
4. HomeSend submits the credit to the receiver system (TEWS/*credit*),

Then what follows is specific to this model:

5. The receiver system replies with a TEWS/*credit* provisional response, providing a maximum completion date *D1* (which can be typically days or weeks in the future, as such credit/payment processing may take a long time),
6. HomeSend does a "preliminary charge" to confirm the sender's reservation (OEWS/*charge*) (optional, only if a reservation was performed),
7. Then replies to the HWS/*remittance* request with a provisional response, indicating how much was preliminary charged, how much is about to be received, and a maximum completion date *D2* (= *D1*+*Grace*), which is an adjusted deadline when the transaction is assumed to be completed by. The Grace is an additional period HomeSend gives itself to retrieve the final transaction status if it was not provided on time (i.e. before *D1*) by the destination system; it is usually set to a few days.

The sender system may then notify the sending end-user that the remittance was correctly taken into account, and he/she should expect a final response by *D2*.

At this point, two parallel processes are executed:

- On the sending side (HWS), the sender system starts polling HomeSend to get an updated status for the transaction. HWS/*status* requests are regularly sent (for instance, 2-4 times a day, to discuss with the HomeSend platform provider). They are replied with HWS/*status* response indicating the current pending state of the transaction (if not completed), or with a final transaction status, either OK or rejected. Once such a final status has been received, the sender system shall stop polling HomeSend, and the sender end-user can be notified of the final outcome of the transaction.

If at *D2*, no final status was retrieved, the polling rate should be reduced (for instance, to once a day or less) to avoid network load. Being in this state is not considered as a normal situation, but may happen and must be supported.

- On the receiving side (TEWS/R-TEWS), HomeSend has to retrieve the outcome of the credit operation:
 - First, it waits until *D1* (the maximum completion date as advertised by the receiver system) for a receiver system-initiated status report that provides the credit's final response (R-TEWS/*statusReport*).
 - If at *D1*, the receiver system did not send a final status report, HomeSend enters a "grace" period until *D2*. During this timeframe:
 - HomeSend actively polls the receiver system for an update, using TEWS/*status* requests, until a final status is known,
 - The receiver system may still send a "late" status report that provides the credit's final response (R-TEWS/*statusReport*),
 - A manual credit confirmation or rejection can be performed through the HomeSend's Web Customer Care GUI.

- If at D2, HomeSend was unable to get a final status, the transaction is stuck in a “Ambiguous” (inconclusive) state that requires a manual credit confirmation or rejection through the HomeSend’s Web Customer Care GUI, though a late status report will still be accepted (R-TEWS/*statusReport*).

Once the credit's final status is known:

- If it is OK, HomeSend just waits for a new HWS/*status* request from the sender system to provide the final status.
- If it is rejected, HomeSend first reverts the preliminary charge operation (step 6) using a OEWS/*chargeback* request (optional, only performed if a reservation was done), then waits for a new HWS/*status* request from the sender system to provide the final status.

Reaching *D1* (the initially advertised maximum completion date) is considered as a non-optimum case. The grace period between *D1* and *D2* provides a margin (typically about several days) to fall back to a client-side polling if for whatever reasons a R-TEWS/*statusReport* was missed. Reaching *D2* should be considered as an abnormal situation and must be avoided.

Notice that from a HWS client's point of view, a synchronous transaction's whose TEWS/*credit* request was inconclusive, requesting an automated reconciliation between HomeSend and the destination system, will be seen as an asynchronous transaction.

4.2.2.1 Standard Asynchronous Call Flows with Reservation

The following sequence diagrams correspond to the cases where the reservation is orchestrated by HomeSend, and a response is asynchronously provided by the receiver systems before *D1*:

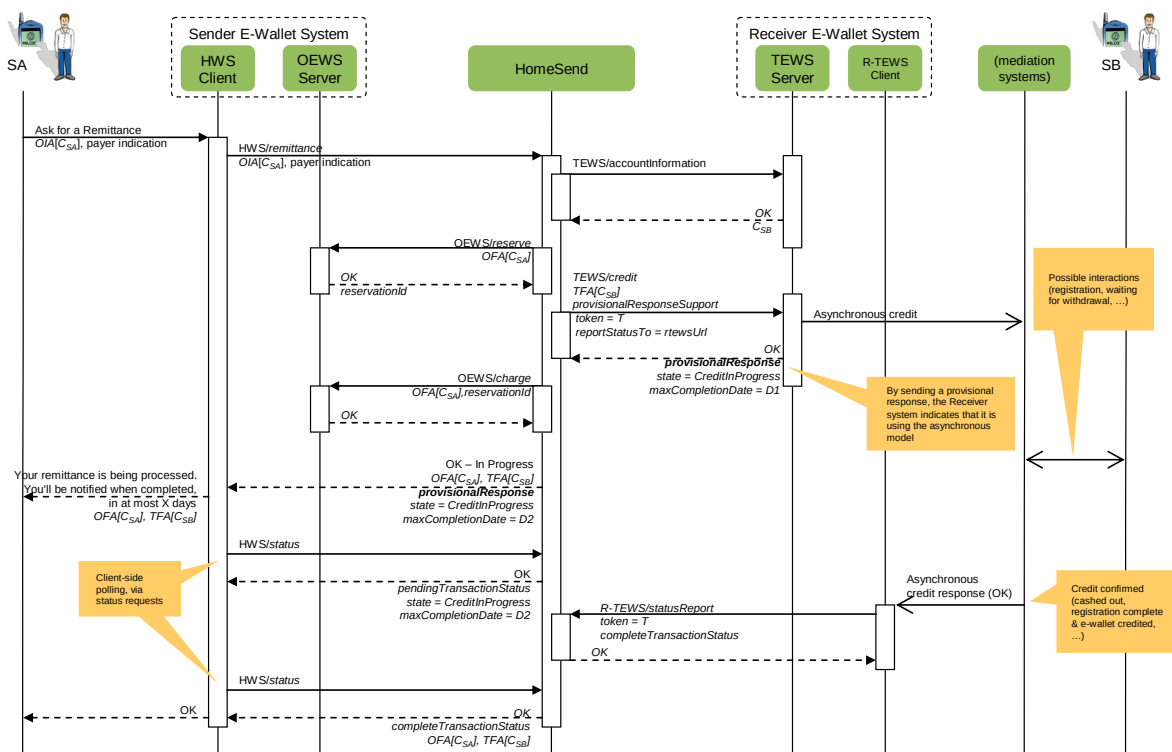


Figure 6 - One-shot remittance, asynchronous model, OK

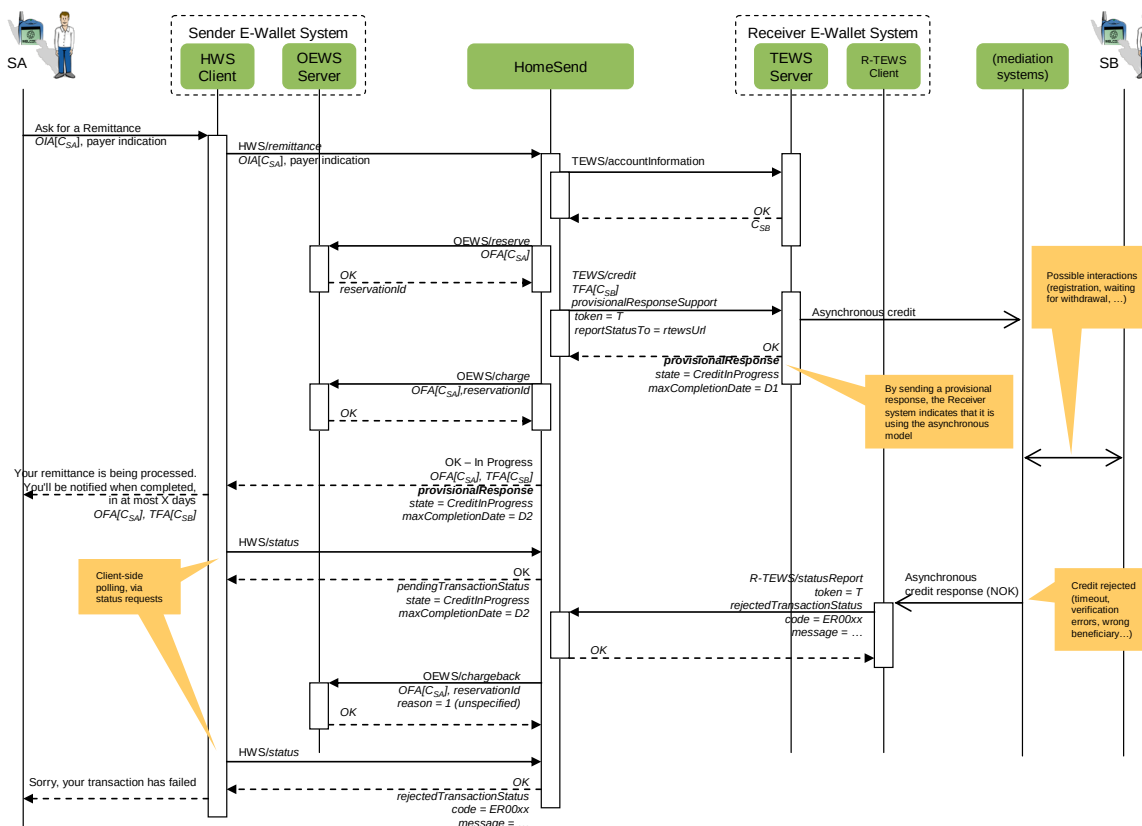


Figure 7 - One-shot remittance, asynchronous model, NOK

4.2.2.2 Standard Asynchronous Call Flows without Reservation

As for the synchronous model, asynchronous remittances can disable the reservation orchestration, simplifying the call flows:

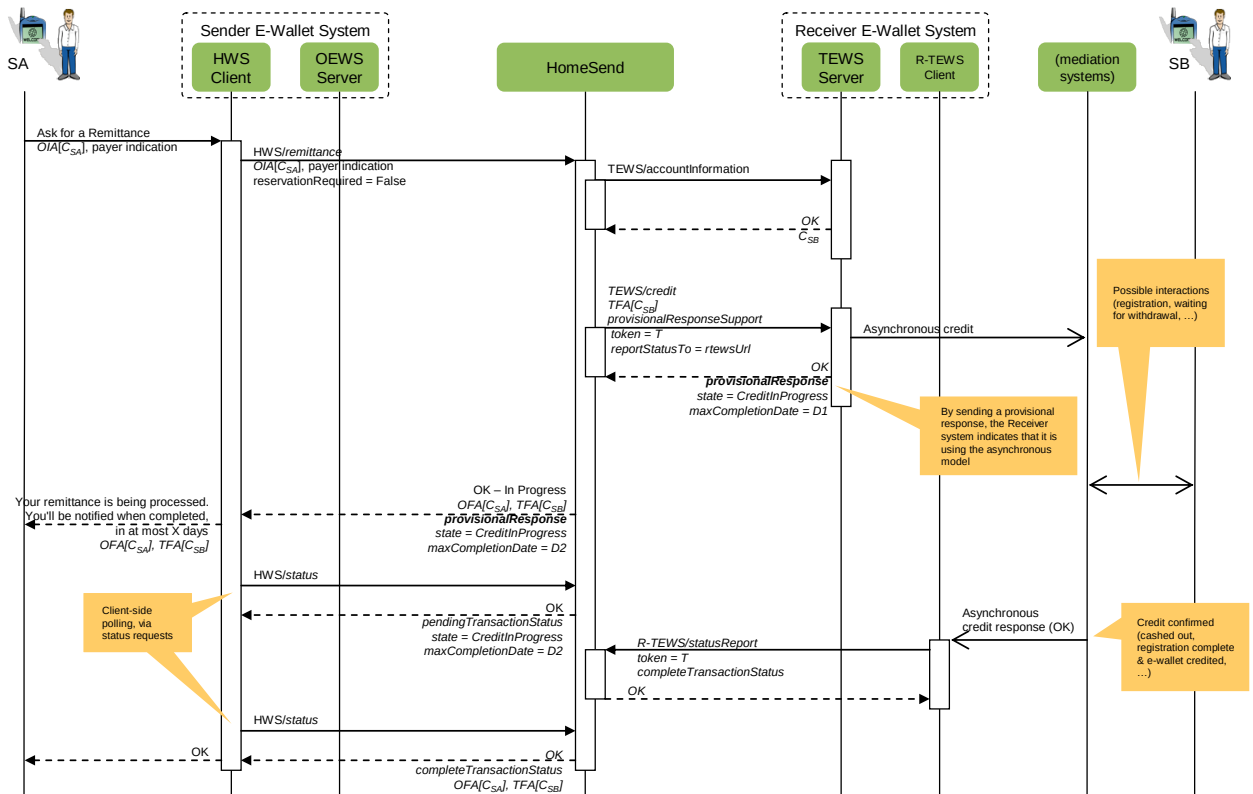


Figure 8 - One-shot remittance with reservationRequired=false, asynchronous, OK

In this case, it is recommended that the sender's balance is charged upon the HWS/*remittance* provisional response. This response contains the amount to charge with regards to the transaction's fees and the HWS/*remittance* payer indication.

In case of a final transaction rejection, a reservation-less call flow becomes :

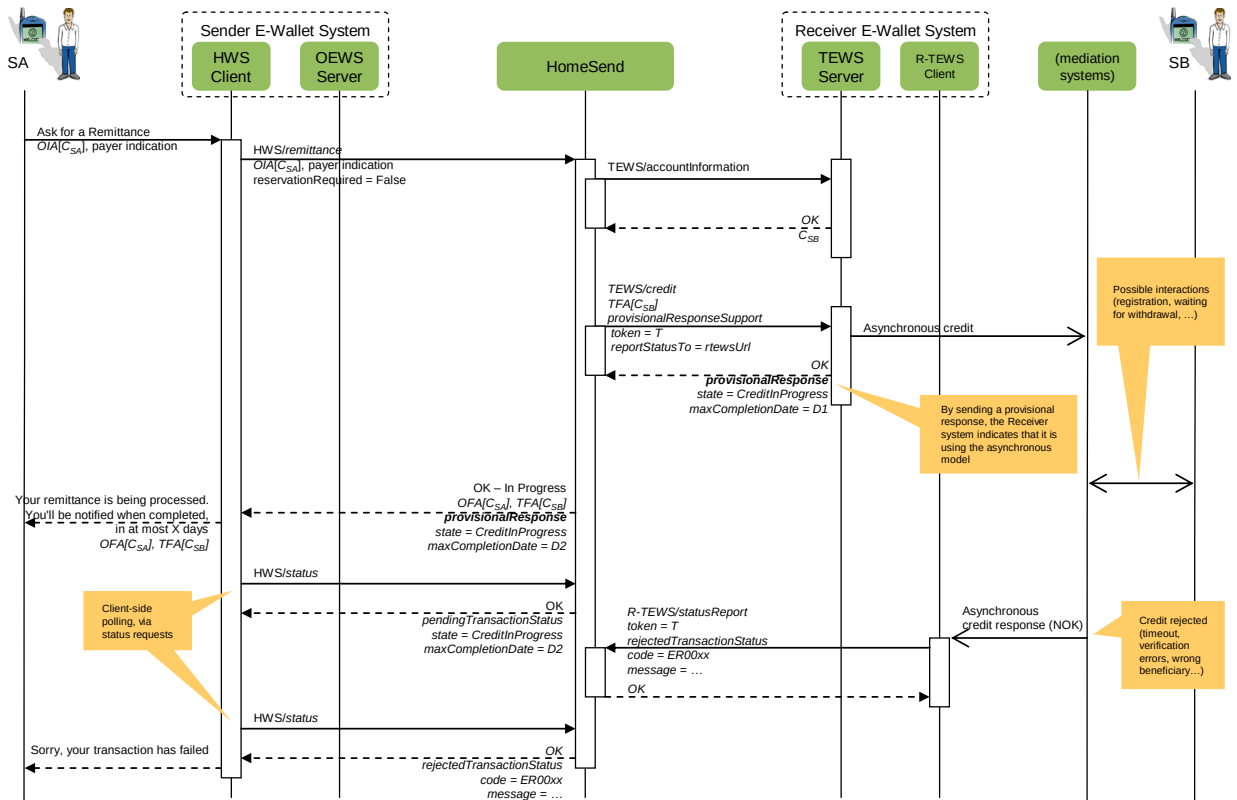


Figure 9 - One-shot remittance with reservationRequired=false, asynchronous, NOK

Notice that there is no OEWS/chargeback in this case. The sender system will be responsible for reverting the preliminary charge (if any was performed) when receiving the final HWS/status response indicating the transaction's failure.

4.2.2.3 Correlations

Transaction identifiers are basically managed as for a synchronous transactions:

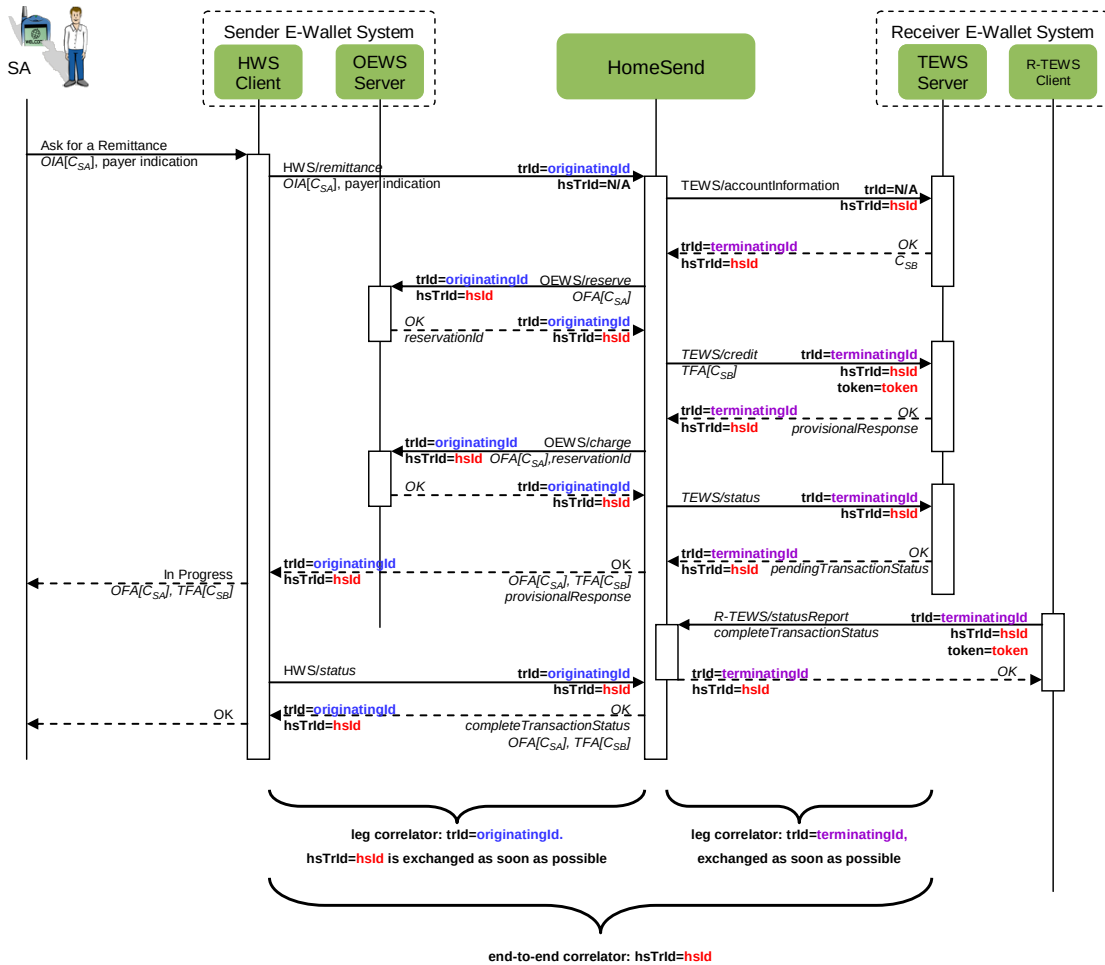


Figure 10 - Correlations in an asynchronous one-shot remittance

Notes:

- All HWS/status requests reuse the same transaction identifiers, as they are part of the same dialogue.
- R-TEWS/statusReport requests must also include the token that was provided in the TEWS/credit provisional response support advertisement.

4.3 Remittance with Advice of Charge

This type of transaction is an enhancement over the one-shot remittance.

Such a transaction starts with the HWS/adviceOfCharge request. HomeSend routes the transaction and, if OK, compute two proposals for a possible associated remittance. These proposals are returned into the HWS/adviceOfCharge response and the sender E-Wallet system usually announces them to the end-user:

- One proposal contains OFA[C_{SA}] (the amount the sender will be charged for, in his/her currency) and TFA[C_{SB}] (the amount the receiver will receive) if the sender pays the transaction fees,
- The other proposal, when present, contains the same amounts in the case where the receiver pays the fees. This proposal is not provided if the amount the receiver would receive is negative in this case.

- Both proposals also contain the transaction fees, in the sender's currency (they are the same for both options).

Based on this information, the end-user may choose one or the other option, and the E-Wallet system shall then send a `HWS/remittance` request with the payer indicator with regard to the selected option.

This `HWS/remittance` contains the HomeSend transaction ID, as provided in the `HWS/adviceOfCharge` response, so that the same exchange rates and service charges are applied for the remittance as those that were valid when computing the advices of charge.

This remittance request is then processed as a one-shot remittance with the usual reservation/charge operations (if `reservationRequired=true`, which is the default) and the credit operation. The routing processing, however, was already performed and is skipped here.

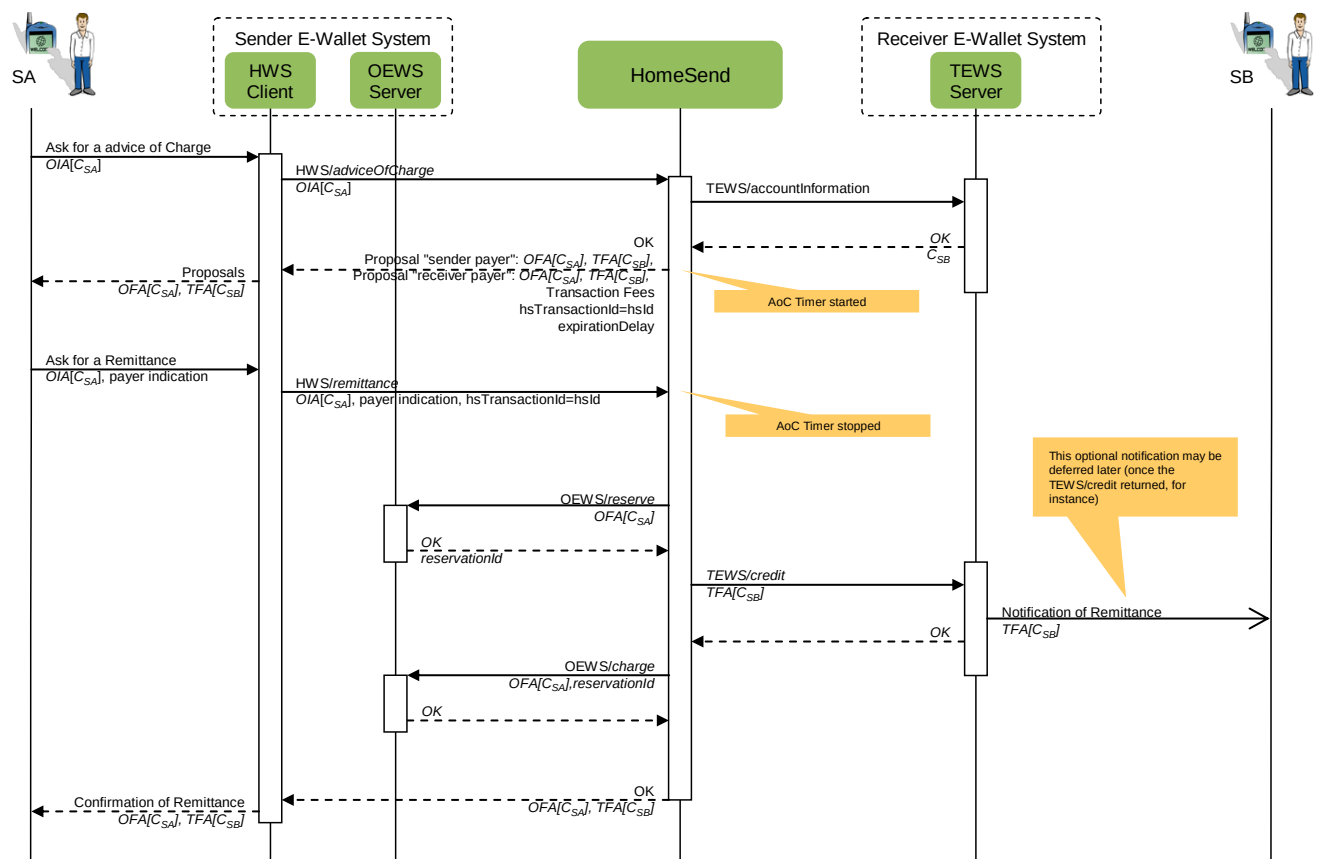


Figure 11 - Remittance with Advice Of Charge

The advice of charge proposals, however, have an expiration delay (usually about 1 hour, but fully configurable per interconnected service provider, depending on its service logic towards its end-users).

However, an advice of charge has a limited validity. It can be cancelled:

- On timeout: HomeSend keeps the advices of charge opened for a remittance only for a defined time. On timeout, the transaction is closed, and the sender E-Wallet system is not notified of it. However, it is able to know when it expires, thanks to the `expirationDelay` parameter in the `HWS/adviceOfCharge` response.
- Administratively: Service providers or the HomeSend platform provider agents may cancel a proposal administratively. The sender E-Wallet system is not notified in such a case.

- Explicitly: the end-user and/or the sender E-Wallet system may cancel an unsatisfactory proposal, using the dedicated HWS/*cancellation* request (see below). If possible, when the end-user decided not to accept any option, the sender E-Wallet system should send this request to avoid unnecessary load on the HomeSend system.

Once an advice of charge is cancelled or expired, the subsequent associated HWS/*remittance* requests are rejected with the specific fault `ER0033 Transaction Cancelled`.

Upon receiving such an error, the sender E-Wallet should then restart a new advice of charge transaction. It shall not try a new one-shot remittance (without any `hsTransactionId`), because it may lead to the use of different exchange rates and service charges thus rendering the previous advice of charge not correct.

4.3.1 Advice Of Charge Timeout

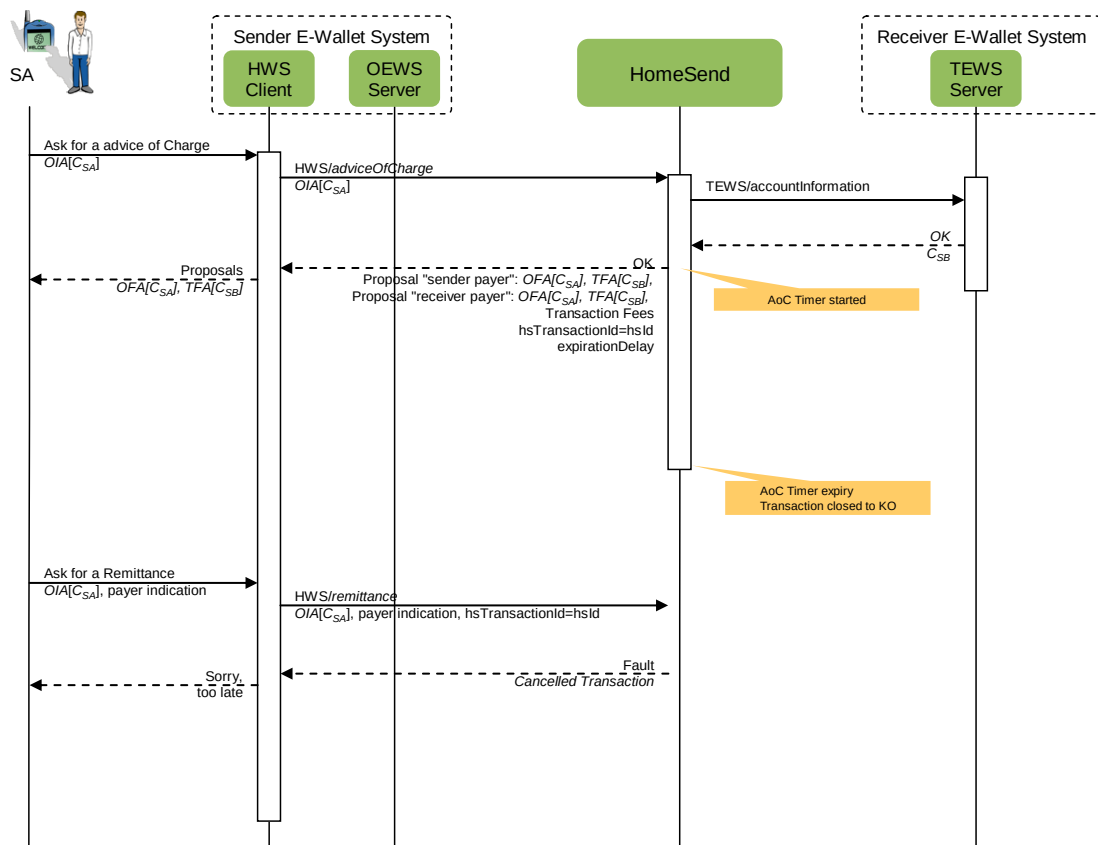


Figure 12 - Late HWS/*remittance* rejected after an advice of charge timeout

As explained above, a remittance may be rejected if an advice of charge was accepted too late. However, the specific `ER0033 Transaction Cancelled` fault is generated, only if the late remittance arrives within the HomeSend's transaction lingering timeout (equal to `MAXreconciliation`). After this delay, a `ER0017 Correlation Error` is sent instead, as no open transaction corresponding to the provided `hsTransactionId` could be found.

Anyway, the sender E-Wallet can know when the advice of charge expires, and can avoid sending the remittance for nothing. It is responsible for notifying the user asynchronously of the expiry, though this is fully optional.

4.3.2 Advice Of Charge Cancellation

If the end-user explicitly rejects the advice of charge proposals, the sender E-Wallet should send a *HWS/cancellation* request to close the transaction as soon as possible on HomeSend's side.

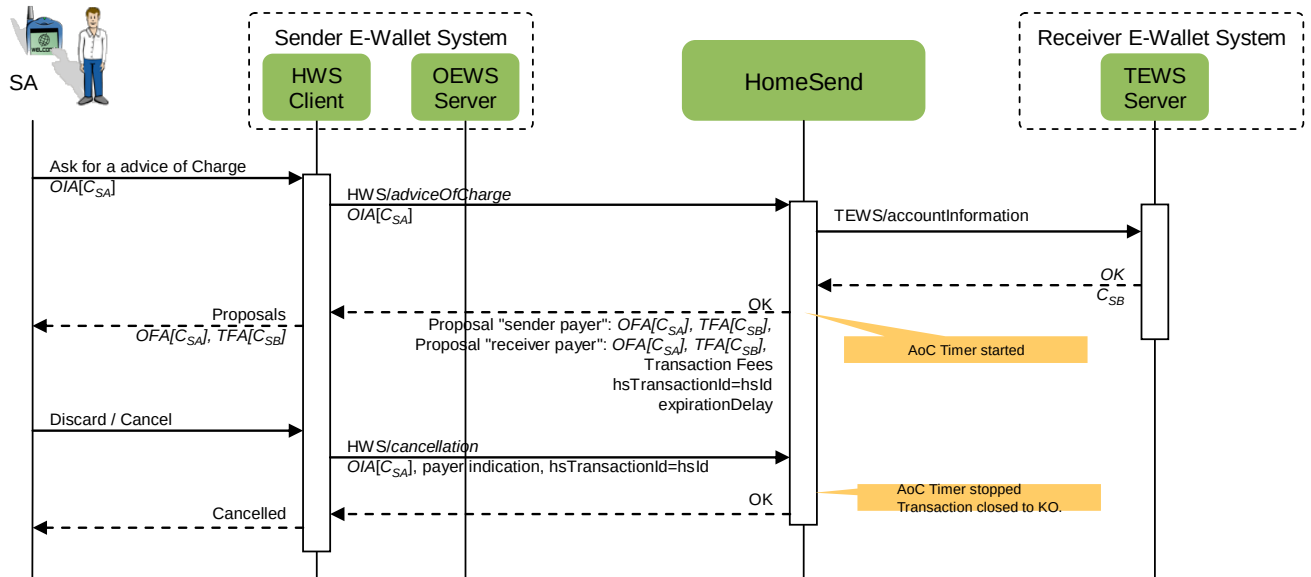


Figure 13 - Advice of Charge cancellation

Once the transaction has been cancelled, *HWS/remittance* requests are handled as described above.

Notice that further *HWS/cancellation* requests on an already cancelled transaction, results in the **ER0033 Transaction Cancelled** fault. From the sender E-Wallet system's point of view, this should not be considered as an error, as the transaction is already correctly cancelled.

4.3.3 Correlations

For reference, the ways transaction identifiers are exchanged are described below.

Notice that the `hsTransactionId` is the main correlation between the *HWS/adviceOfCharge* request and *HWS/remittance* (or *HWS/cancellation*) request.

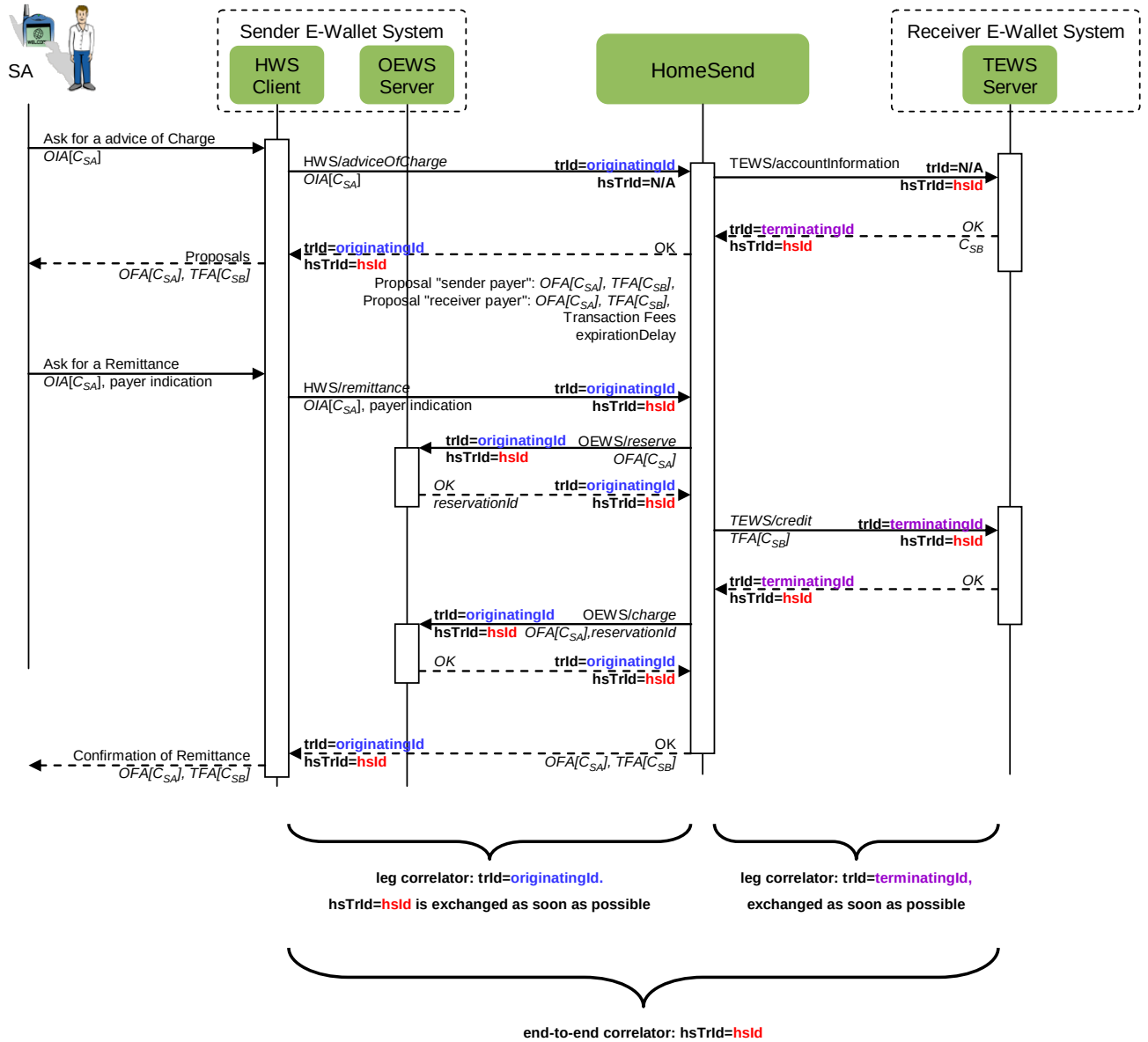


Figure 14 - Remittance with Advice of Charge requests correlation (the actual fields names are transactionId and hsTransactionId)

4.3.4 Asynchronous Variant

As for a one-shot remittance, a remittance with advice of charge can be executed asynchronously.

The preliminary HWS/adviceOfCharge request is handled the same way, and only the remittance part of the call flow varies, following the same logic as for a one-shot remittance.

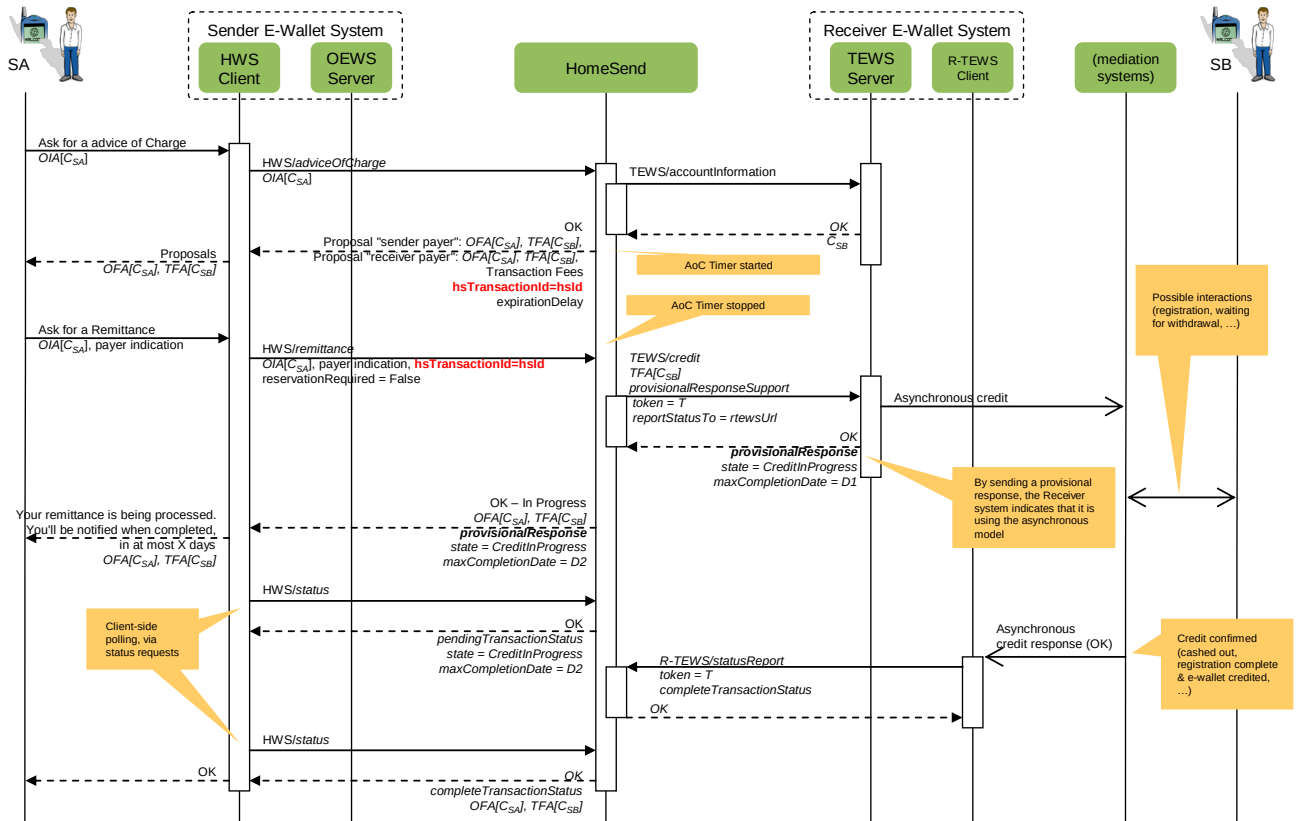


Figure 15 - A typical Remittance with Advice Of Charge, asynchronous, with reservationRequired=false

For more details about the asynchronous model, please see [Asynchronous Model](#).

4.3.5 Reverse Advice of Charge

A reverse advice of charge is a way to use an Advice of Charge request to know how much the sender must send/will be charged so that the beneficiary receives a specified amount. Calls flows are exactly the same as for a standard (also known as a forward) advice of charge transaction; the initial Advice of Charge is also the same, but with specific fields filled that way:

- The amount A in the Advice of Charge must be expressed in one of the currencies assumed to be supported by the beneficiary (noted C_{SB}). For instance, if sending a remittance to a beneficiary in Philippines, it is safe to assume that the local currency, PHP, will be supported. We call this amount the “target received amount”.
- In addition, to flag this Advice of Charge request as a reverse Advice of Charge, a specific “reverse quote” indication must be set, along with the currency of the sending account C_{SA} .

This way, HomeSend will be able to calculate a quote replying to the question “How much, in C_{SA} , will the sender be charged, so that the beneficiary receives A C_{SB} ?”.

The response associated to such a reverse Advice of Charge is expressed as a single Advice of Charge proposal, marked as being “sender payer”, indicating:

- The principal amount that must be sent, in C_{SA} , (named `initialAmount` in the response)

- The amount that will be charged on the sending side, in C_{SA}
- The actual amount that will be received, in C_{SB} . This amount may be slightly greater than the amount A provided by the sending user, due to various currency rounding rules and banding effects that could occur with certain service charge models.

If the user-provided currency C_{SB} is not supported by the beneficiary, the reverse Advice of Charge request fails with the specific error code `ER0045`. Currently this API does not provide a way to get the supported currencies for a beneficiary.

If HomeSend can't get a proposal that would lead to a received amount closed enough (within a configurable margin that can be provided by the HomeSend platform provider), the reverse Advice of Charge fails with the error code `ER0016`.

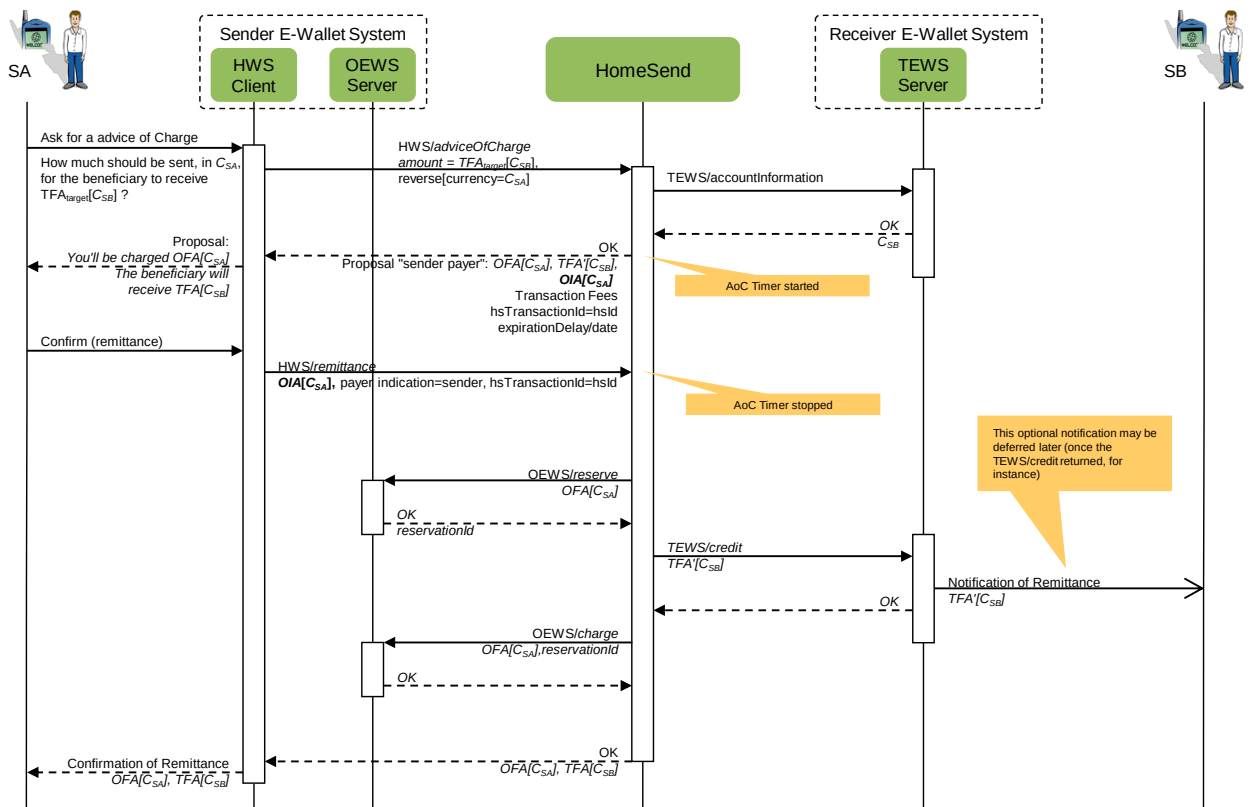


Figure 16 - A nominal Remittance with Reverse Advice of Charge

Upon receiving a reverse Advice of Charge response, the sending system should present the quote to the end-user, clearly presenting the charged amount and the adjusted received amount. If confirmed, the sending system should send a Remittance request with the same `hsTransactionId` as the one received in the Advice of Charge response, and with the exact same amount as received in the proposal's `initialAmount`. The payer flag must also be set to "sender".

The transaction is sent processed and completed as usual, as any other Remittance with Advice of Charge transaction (i.e. with support for asynchronous transactions/provisional responses, optional reservation, etc).

4.4 Asynchronous Credit Processing

This mechanism only applies to the remittance service, i.e. for one-shot remittances and remittances with advice of charge transactions. Electronic recharges do not support it.

4.4.1 Use Cases

Processing a credit operation asynchronously is well adapted in the following use cases:

- The destination system is a bank, able to commit the transfer only after daily or batched operations (or any financial institution based on batched processing),
- The destination system requires a mandatory beneficiary registration prior to being able to receive the funds (e-wallet creations may require such a registration step),
- The destination system requires additional KYC elements or verifications, involving the beneficiary, an agent, or another manual intervention,
- The destination system only confirms the credit/payment upon actual payout,
- And generally in all cases where the destination system cannot guarantee that the credit will be taken into account in real time (i.e. in less than several seconds).

When it is possible to confirm a credit in real time, however, the synchronous model must be preferred, for the sake of simplicity and efficiency.

4.4.2 Principle

A destination system may decide to handle a credit operation asynchronously on a per-transaction basis, depending on its own business logic. HomeSend has no way to prevent such an asynchronous handling, and the sending system must be prepared to handle such cases as well.

When deciding such a processing model, the destination system shall respond with a credit “provisional response” to HomeSend, indicating that the credit request was valid (low-level, technical verifications, such as syntax validations, are expected to be performed in real-time, though it is not forbidden to defer them) and was correctly taken into account by the system.

Such a provisional response basically contains a hint about how long the processing may last (hours, days, weeks), which is the normal period during which the sending end-user should wait for a final transaction status. During this period, HomeSend waits for an asynchronous credit completion, providing the final status for the transaction: successfully completed, or rejected, with an associated reason.

This nominal credit completion model is called a “server-side” notification model, or push-model, as the destination system is responsible for pushing the completion up to HomeSend.

If after this delay, no credit completion was received, HomeSend enters a grace period during which it will try to get the status by itself, by polling the destination system regularly. This fallback mechanism, which should be avoided due to its inefficiency, is called a “client-side” polling model, or pull-model.

Both those strategies enable HomeSend to get a final status for the asynchronous transaction. On sending side, however, only a client-side polling model is available, and the sending system should regularly poll HomeSend after receiving a provisional response until it gets a final transaction status.

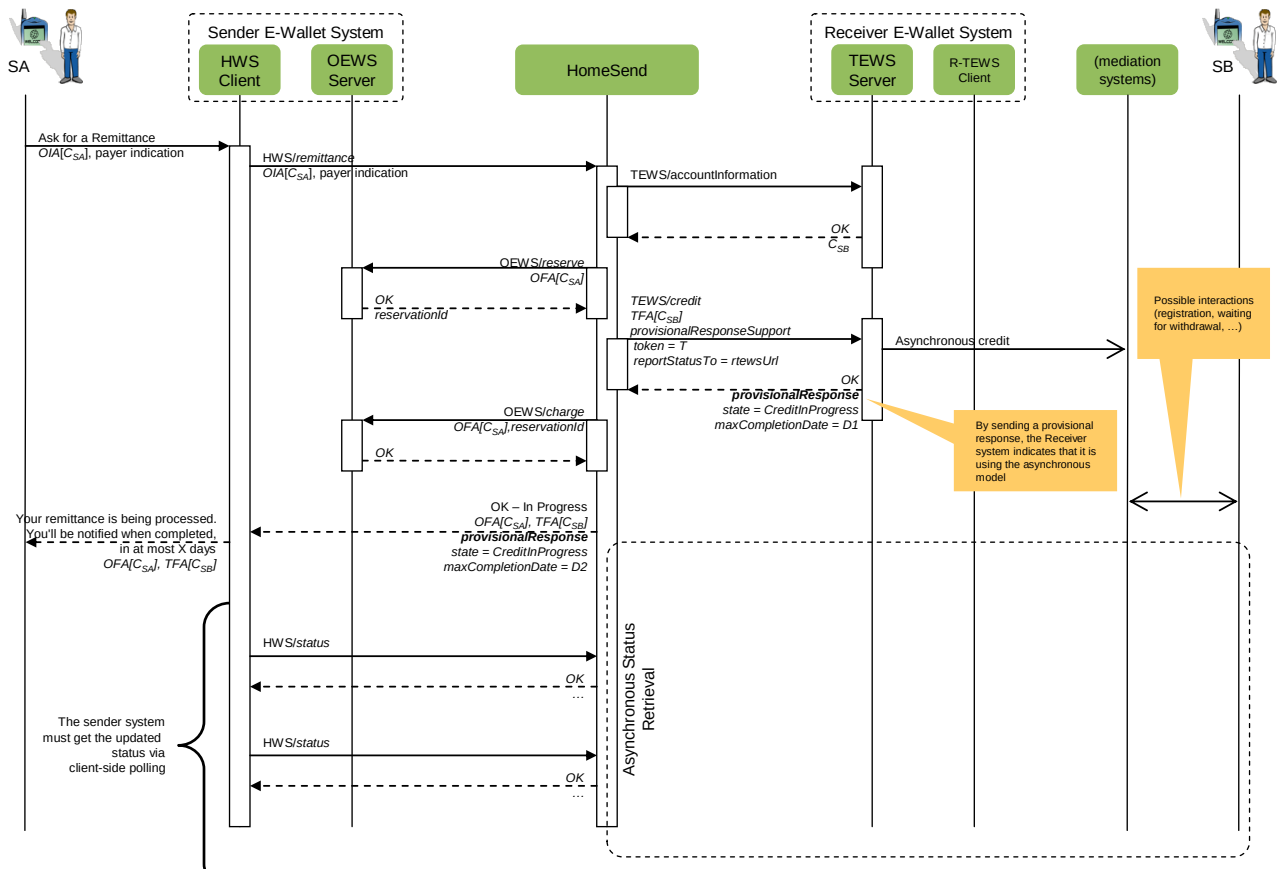


Figure 17 - Asynchronous Credit Processing, general model [1]. Possible completion methods are described below.

4.4.3 “Server-side” Notification Model

This mechanism, which is the preferred one to complete an asynchronous credit operation, relies on a technical interfaced named R-TEWS, which could be considered as a **R**everse web service for TEWS.

Its URL is dynamically discovered thanks to what is advertised in the TEWS/*credit* request, and must be used to report the status of a submitted transaction using the R-TEWS/*statusReport* operation, along with a security token that was also provided in the TEWS/*credit* request.

As soon as the credit (or payment) is known to be fully processed and confirmed on the destination side, the receiver system should send a R-TEWS/*statusReport* containing a `complete` element, with possible optional Vendor Specific Fields, as if it was a synchronous (but deferred) TEWS/*credit* response.

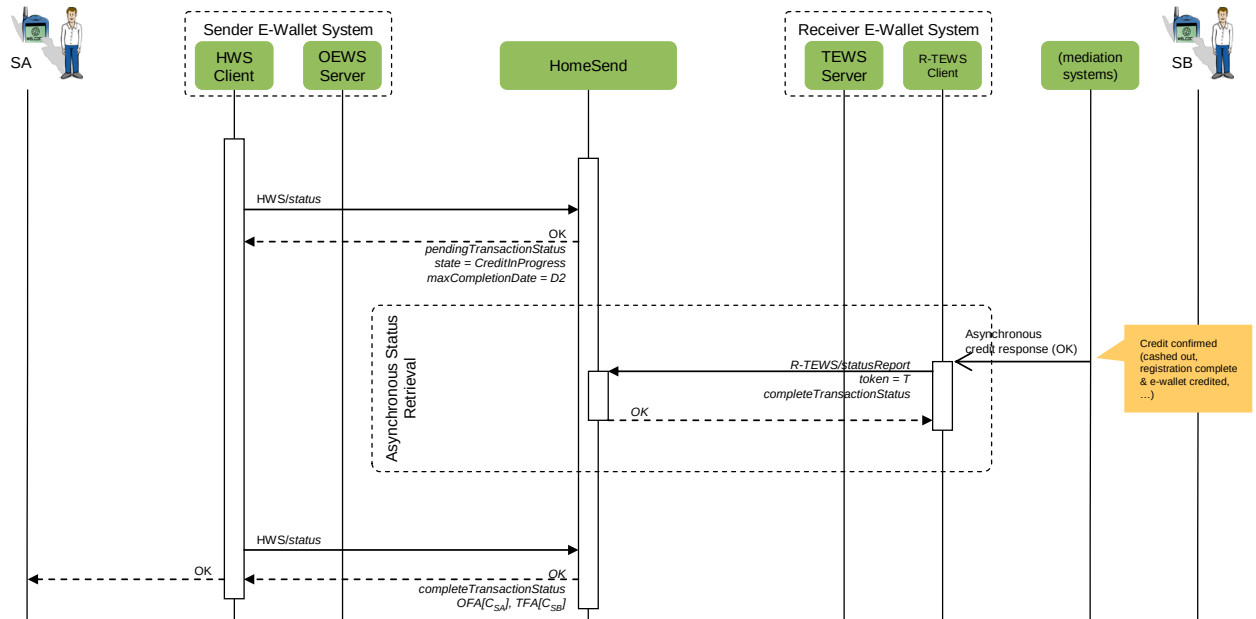


Figure 18 - Call flow continued from [1], successful transaction, server-side notification model

If the credit (or payment) is known to be rejected (or expired), , the receiver system should send a R-TEWS/statusReport containing a rejected element, containing the error cause, as if it was a synchronous (but deferred) TEWS/credit Fault. A specific error cause is provided to indicate a transaction expiry.

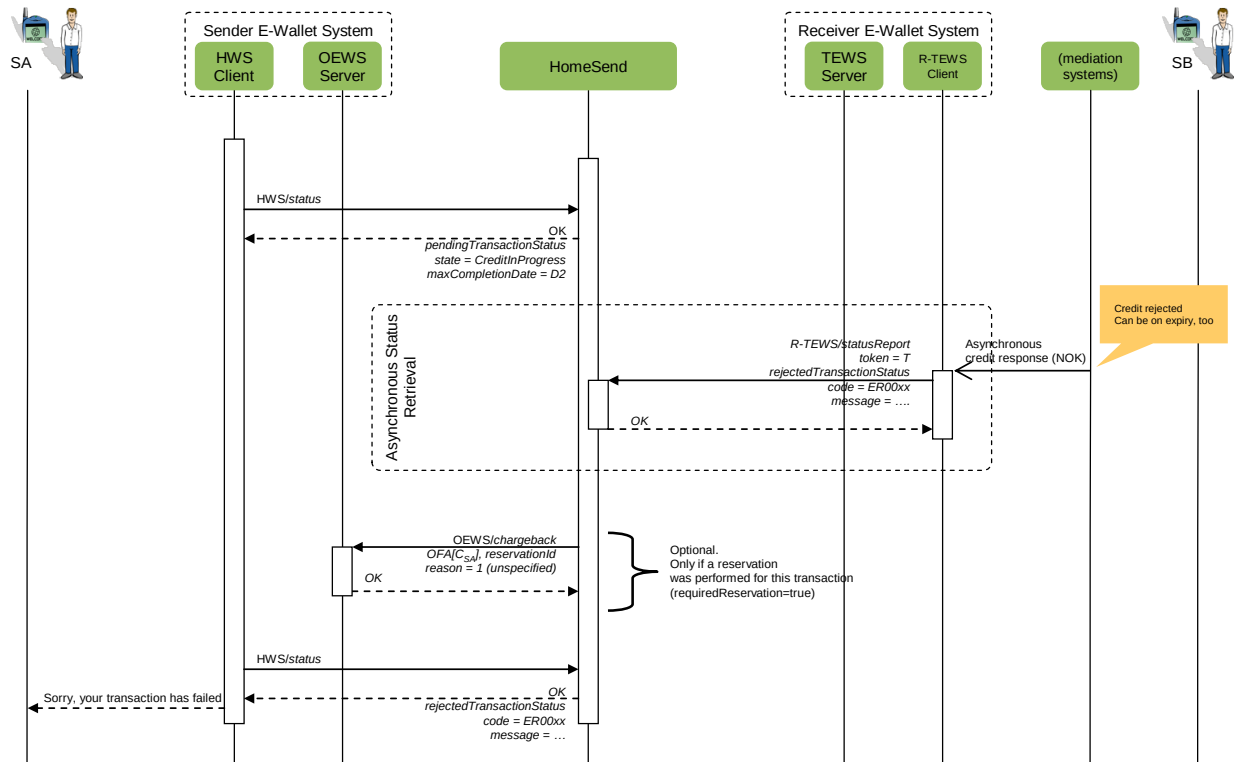


Figure 19 - Call flow continued from [1], rejected transaction, server-side notification model

Once the transaction has been completed (as succeeded or rejected), further notifications, if any, will be ignored (the R-TEWS/statusReport request will be rejected with a Fault), unless this is a duplicate of the initial R-TEWS/statusReport that completed the transaction.

This status report model may also be used to track the evolution of the transaction on the destination side. In this case, multiple R-TEWS/statusReport requests transporting a pending element may be accepted for a transaction before getting a final R-TEWS/statusReport complete or rejected request, which will close the transaction.

4.4.4 Client-side Polling Model

This fallback model relies on the TEWS/status operation. This is detailed in [Receiving Side: Getting a Transaction Status](#).

4.4.5 Pre-Credit Notification

In some particular cases, when the destination service provider delegates to HomeSend the handling of sanction list screening verification, the destination system may receive a specific TEWS/statusReport request to notify it that a transaction is about to be sent to its system, but not sent yet as it requires a manual action from a compliance officer.

The receiver system should be prepared to receive such a request after a TEWS/accountInformation and before a TEWS/credit request (that may never come if the transaction is cancelled in the meantime) only if the delegated sanction list screening feature is used. Otherwise, it is advised not to implement this operation (i.e. send a ER0044 Operation not supported when receiving a TEWS/statusReport request).

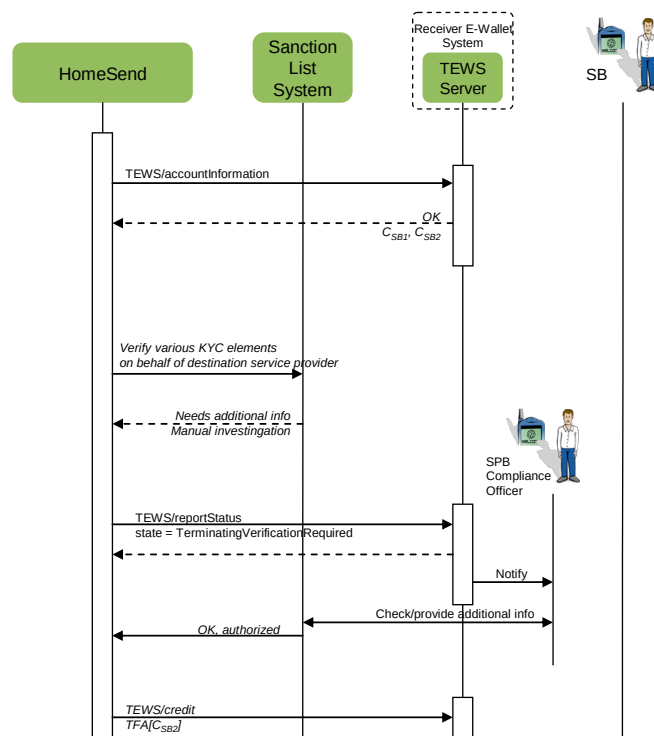


Figure 20 - A typical use of TEWS/reportStatus for Sanction List Screening manual verification

When receiving such a notification, the receiver system is likely to trigger some additional business operations depending on the reported transaction state. Possible cases are indicated in [TEWS/statusReport request](#) section.

4.4.6 Cancelling a Pending Transaction

The remittance initiator may try to cancel a transaction in several states:

- After receiving a HWS/*adviceOfCharge* response, the initiator may send a HWS/*cancellation* request to HomeSend to explicitly forget the proposal. This is called “Advice of Charge cancellation”, and is described in [Advice Of Charge Cancellation](#). This use case only involves the sender system and HomeSend:

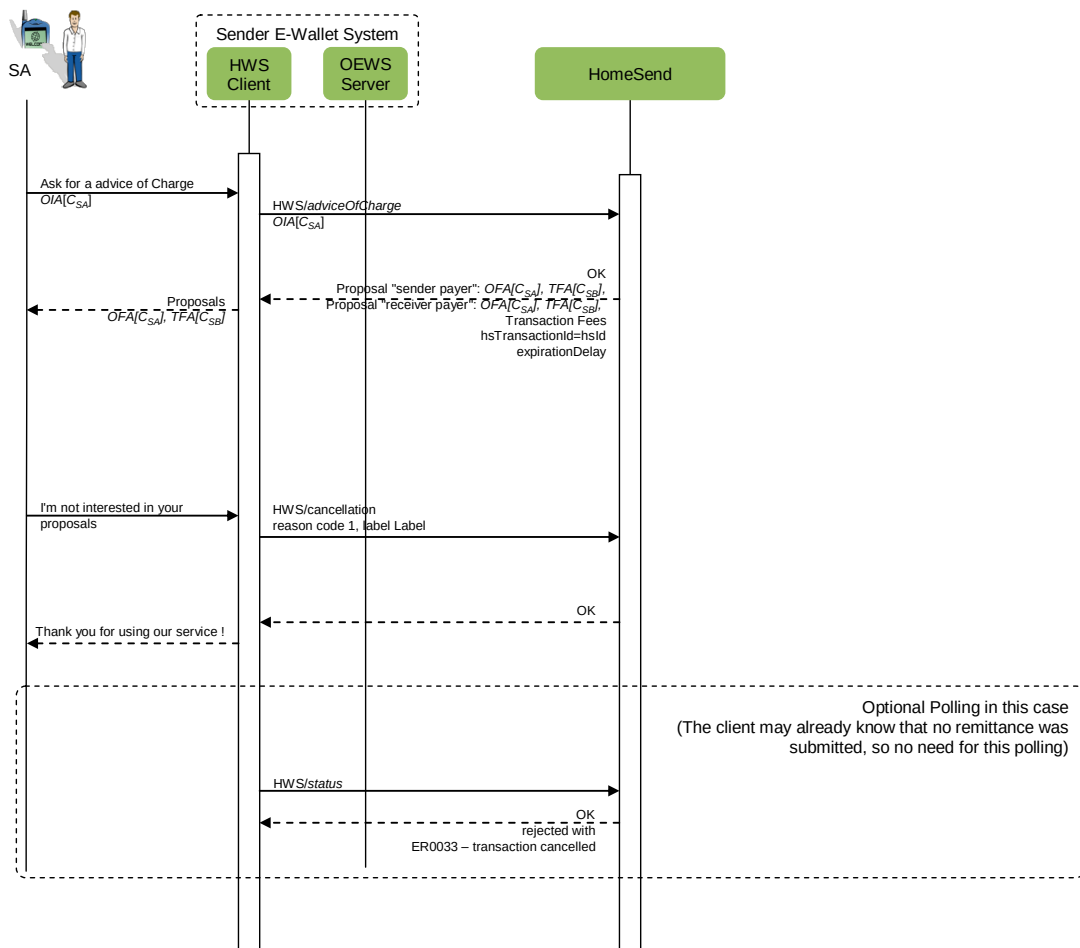


Figure 21 - Explicit cancellation of an Advice of Charge proposal

- After a HWS/*remittance* has been replied with a HWS/*remittance* provisional response, the initiator sees the transaction as “asynchronous”. In this case, it is possible to *attempt* a transaction cancellation. This is what this chapter describes.

The following sequence diagram summarizes a successful cancellation of an asynchronous transaction:

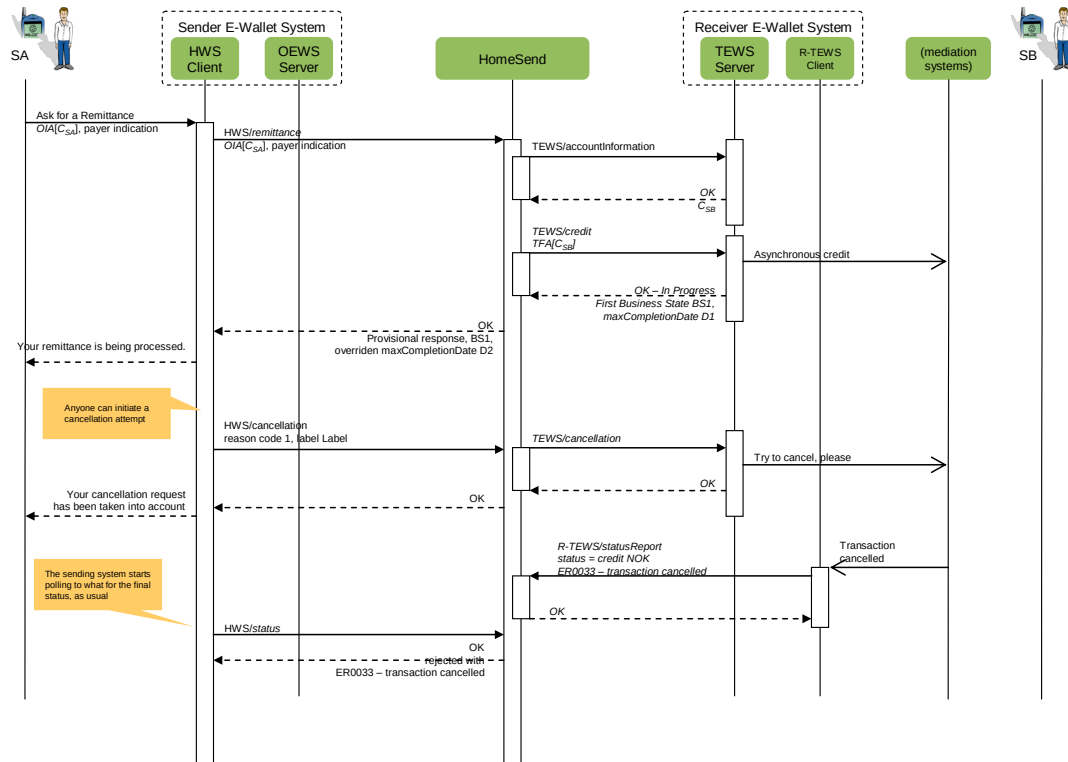


Figure 22 - End-to-end, asynchronous cancellation sequence diagram

4.4.6.1 On Sender System Side

The remittance initiator may send a *HWS/cancellation* request anytime after receiving a *HWS/remittance* provisional response. In that case, however, there is no guarantee that the actual cancellation will be processed, this is just an attempt. A cancellation in such a state may indeed require HomeSend to forward a cancellation request to the destination system, which in return may need to process the cancellation asynchronously if the transaction was already pushed to another external system (such as a banking network).

In such cases, the actual cancellation may not be achieved as the transaction may have been already completed remotely. Some destination systems may also be unable to process such a cancellation request as they may not have the required business logic to do it.



Note:

The cancellation operation is used to cancel a transaction that is known to be still pending. Once the transaction is completed, if you need to “cancel” it, you will have to refund the transaction instead. The refund operation is not part of this HWS API, which is dedicated to remittance initiation, but is a Customer Care operation.

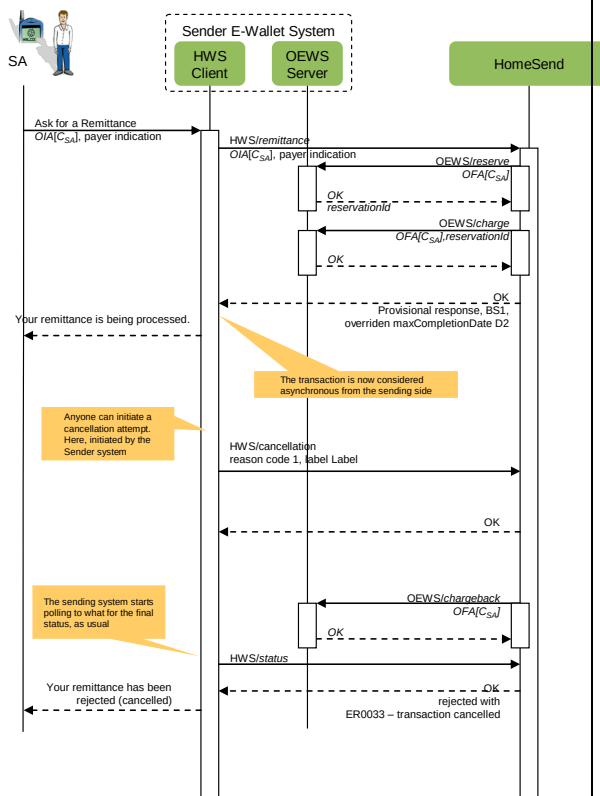
Practically, a sending system may try to cancel a remittance as soon as it has received a *HWS/remittance* provisional response. If a final *HWS/remittance* response is received instead (an final error or a successful completion), any cancellation attempt will fail with an error.

This can either be a user-initiated cancellation request, or an administrative or technical cancellation triggered by the system itself – this is transparent for the API and HomeSend.

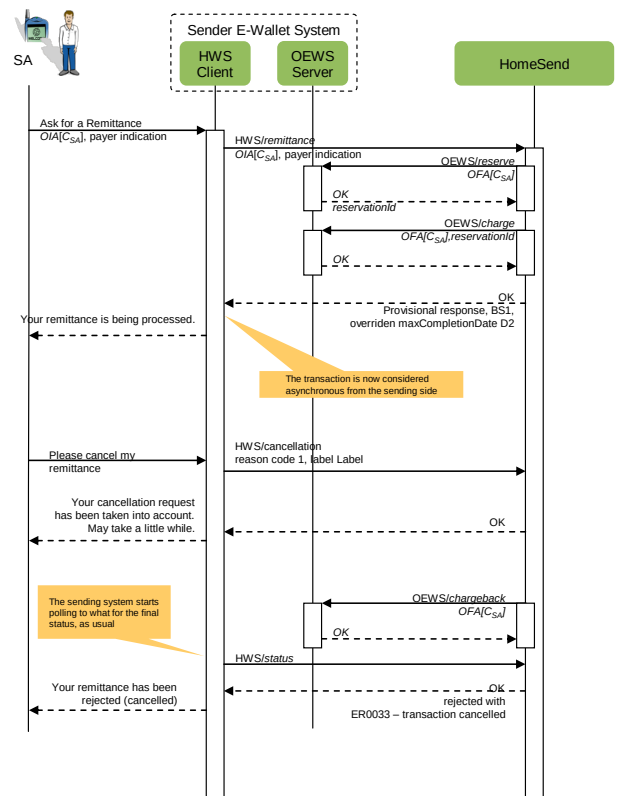
The cancellation request may then be replied with:

- A fault, indicating that the cancellation was not taken into account. The errors `ER0034` and `ER0036` are non-permanent errors and the sending system may attempt to cancel the transaction later. All other faults means that it won't be possible to cancel the transaction.
- A OK response, indicating that the cancellation request was correctly taken into account, but the transaction is not cancelled yet. After receiving such a response, the sending system should continue polling for a completion, as it started doing after receiving the `HWS/remittance` provisional response. If the transaction is successfully cancelled, a next polling attempt will lead to a “transaction rejected” status, with `ER0033 Transaction cancelled` error code. Even if a cancellation request was sent, the transaction may still be completed successfully, typically due to message cross-over somewhere in the transaction processing chain.

The diagram below describes the interaction between the sending system and HomeSend for a sending-system initiated cancellation. In this example, the cancellation works as expected, i.e. the transaction is eventually rejected with a `ER0033 Transaction cancelled` error code:



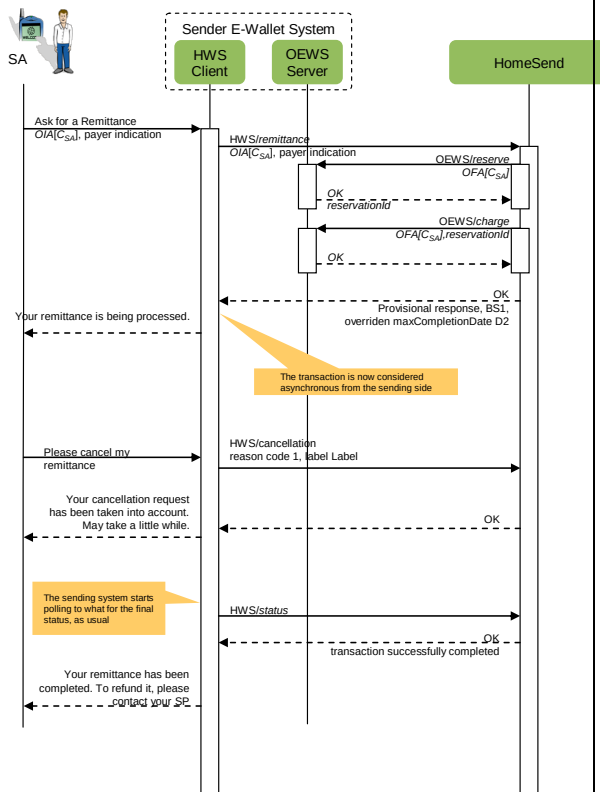
The same case for a user-initiated cancellation. From an API point of view, this is strictly equivalent to the previous case:



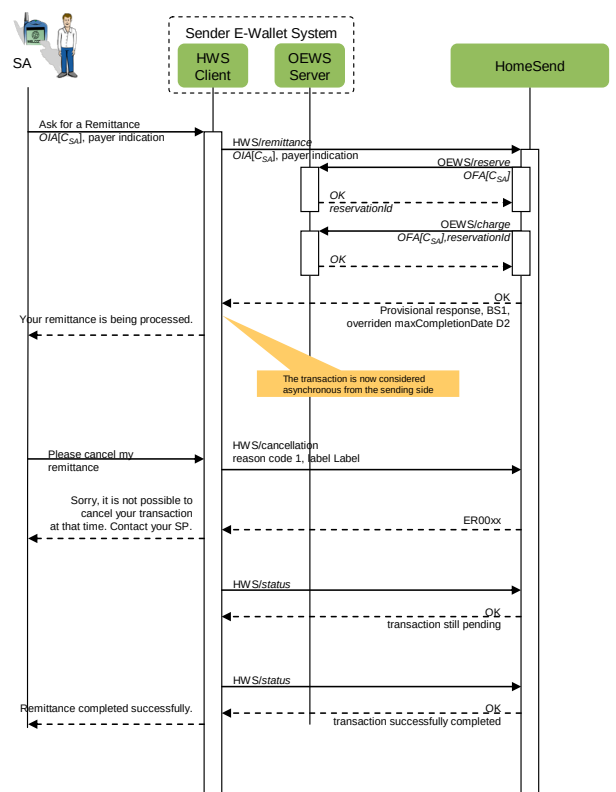
In the following case, the (user-initiated) cancellation request was correctly taken into

A case where the `HWS/cancellation` request is explicitly rejected; it may not be possible to

account, but the transaction was eventually completed successfully. The cancellation order, which is likely to be handled asynchronously by the destination system, could not abort a transaction that was in a near-completion state already:



cancel the transaction at all (for instance, the destination system may not support this operation), and the standard status polling mechanism must still continue as usual:



Notice that the transaction may still be rejected for other reasons, even if the cancellation order was not taken into account.

4.4.6.2 On Receiving System Side

A receiving system that is able to cancel a submitted, asynchronous transaction (i.e. a transaction whose TEWS/credit was responded with a TEWS/credit provisional response) should implement the TEWS/cancellation operation this way:

- The system must be ready to accept a TEWS/cancellation request right after sending a TEWS/credit provisional response. HomeSend will not send a TEWS/cancellation request before (otherwise, should be replied with a ER0034 Unexpected request error).
- The system must respond with a TEWS/cancellation OK response if and only if it is able to take the cancellation order into account, or forward it to a mediation system that supports pending transactions cancellations.
- As for any other TEWS requests, the system may receive duplicate TEWS/cancellation requests. It is supposed to respond with the same response as the first one.

- If the cancellation operation is not supported by the destination system, it must reply with a ER0044 Operation not supported error.

Other errors may apply as usual.

Once a TEWS/*cancellation* request has been accepted (i.e. responded with a TEWS/*cancellation* OK response), the system is assumed to try to abort the pending transaction, and wait for a feedback from its mediation server indicating that the transaction was rejected (typically with due to a cancellation). As a consequence, it will send a R-TEWS/*statusRequest* request “rejected” to HomeSend or reply TEWS/*status* response “rejected” to HomeSend-initiated TEWS/*status* requests to complete the transaction.

If the transaction was rejected due to this cancellation, the expected error code is ER0033 Transaction cancelled. However, using other error codes is not a problem in this context.

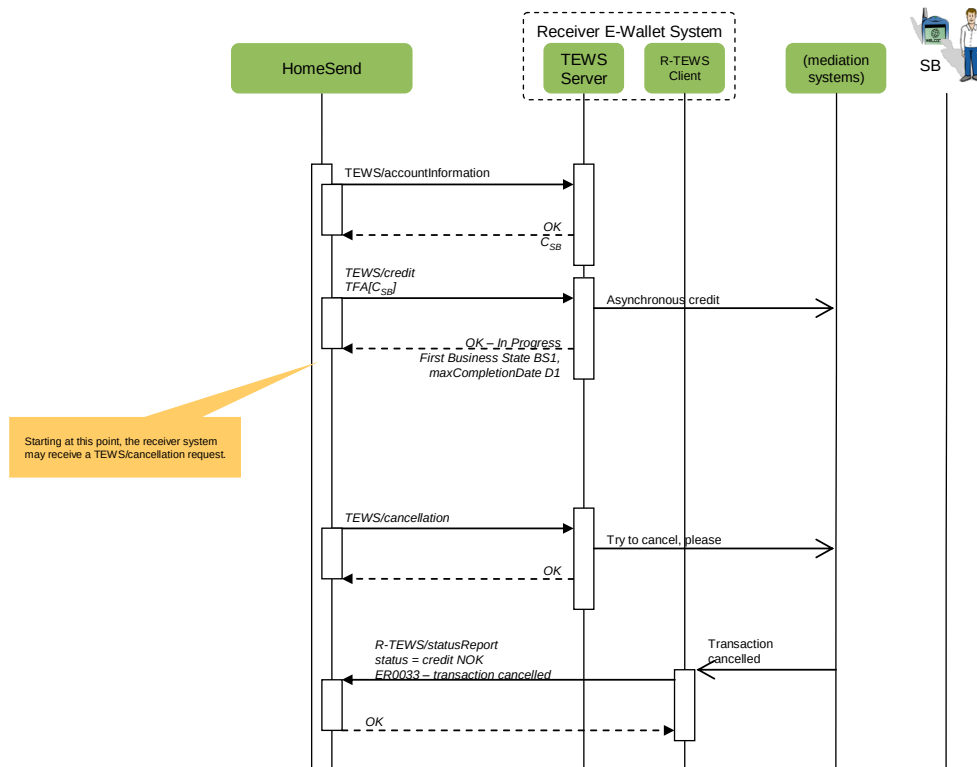


Figure 23 - Canceling a pending transaction, from the receiving system's point of view

4.4.7 Amending a Pending Transaction

Some destination service providers may be unable to distribute sent money to their final users in case of some inconsistencies in beneficiary information. For instance, in a Mobile-to-Cash use case, where the money is collected via a cashier, the cashier may need to check the beneficiary's name against the beneficiary's name as provided by the sender in the remittance transaction. In case of a mismatch, typically a typo, the transaction may be rejected.

In such cases, it is possible for the destination service provider to ask for a transaction update, so that the beneficiary's name, for instance, could be fixed.

Notifying the sender of the required update can be achieved two ways:

- The beneficiary may need to contact the sender by his/her own means and explain the reason why he/she could not withdraw the cash; in this case, HomeSend won't be notified that an update is required for the transaction. This case is called "Unsolicited Amend".
- The destination system may also be able to notify the sender through HomeSend that there is an update expected on one of several business fields; in this case, HomeSend will be aware that an update is required. This second case is called "Solicited Amend".

In both cases, it requires both receiver and sender systems to support the Amend operation (*HWS/amend* on the sender side, *TEWS/amend* on the receiver side).

4.4.7.1 General Amend Model

On the Sending Side

In all cases, regardless of the solicited/unsolicited amend use case, the sender system is responsible for sending a *HWS/amend* request to HomeSend, containing a list of elements to amend.

This request will be rejected if not sent while the transaction is in a "Credit in progress" state, as only pending, asynchronous transactions may be amended.

Multiple elements may be amended at the same time. However, all of them must be flagged as amendable by HomeSend and the destination system. If at least one of them is not amendable, the request is fully rejected and nothing is updated.

An element to amend is expressed as a Vendor Specific Field, as they represent non-core, non-essential information elements for transaction routing or service charge calculations. They typically convey KYC-related information such as the sender's or receiver's name, address, birth date, etc. Source or destination URI and transaction amounts, in particular, are not considered as VSFs and cannot be updated.

Finally, a *HWS/amend* request must contain at least one field to update.

When a *HWS/amend* is correctly processed by HomeSend and the destination system, it is replied with a *HWS/amend OK* response, indicating that the update was successfully performed.

Amending a transaction does not change its main state; it won't abort or complete the transaction successfully; it will remain in a "Credit in progress" state until its normal completion. The amend operation is here to help its successful completion, such like the cancellation operation helps its rejection.

On the Receiving Side

When HomeSend accepts a sender-initiated amend request, it sends a *TEWS/amend* request to the receiver system. This can only occur after a *TEWS/credit* request responded with a *TEWS/credit* provisional response, and while the transaction is not known as completed by HomeSend (no final status received).

The *TEWS/amend* request contains a non-empty list of fields to update, with their new values only. Only KYC-related elements are updatable; everything used for transaction routing and amount calculations won't be amendable.

Anyway, if a field is considered as not amendable by the destination system, it must reject the whole request and must not attempt to update anything.

If the TEWS/*amend* operation is not supported, because such a transaction update feature is not needed or not supported by the receiver system, it should be replied with a ER0044 Operation not supported.

Replying OK to a TEWS/*amend* means that all the provided fields were updated.

4.4.7.2 Unsolicited Amend

In this case, the HWS/*amend* request is sent by the receiving system “out of the blue”, and HomeSend forwards it as a TEWS/*amend* request to the destination request when possible (correct transaction states, updatable fields only).

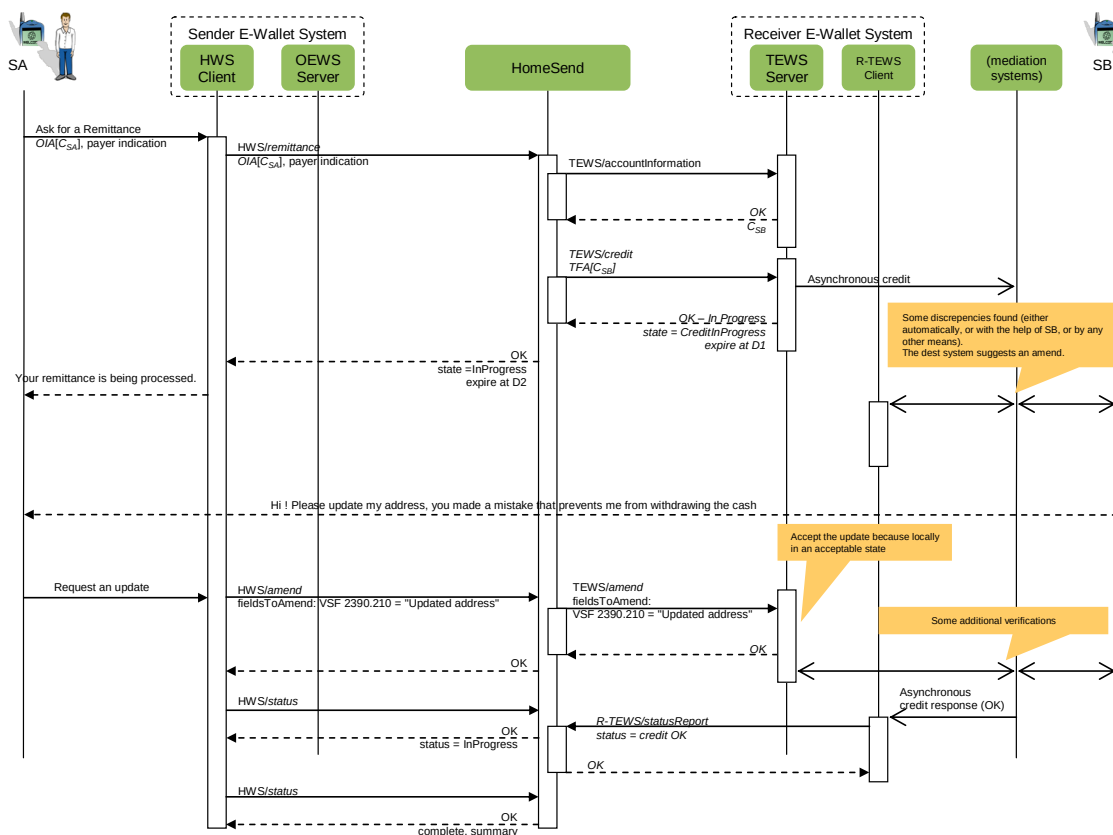


Figure 24 – A typical Unsolicited Amend

No additional feedback is provided to either party. The amend is assumed to be sufficient to help complete the transaction successfully.

4.4.7.3 Solicited Amend

In this case, the destination system that is requesting an update on a pending transaction notifies HomeSend of the required action: it sends a R-TEWS/*statusRequest* pending transaction, with the state set to the literal `AmendRequested`.



Note:

It is also possible to notify HomeSend of the `AmendRequested` state in a `TEWS/status` response. However this is much less efficient as it will require to wait for HomeSend to start polling the system for a transaction status.

In this case, HomeSend updates its internal transaction status so that this `AmendRequested` state can be forwarded back to the sender system on next `HWS/status` polling attempt. This is the asynchronous way the sender system can be notified that it needs to amend the transaction (and as a consequence, probably ask the sending user to provide the amended fields).

It is advised that the status update containing the `AmendRequested` state also contains a list of the fields to update, with their current (incorrect) values, as state-specific fields. They are encoded using the same structure as the Vendor Specific Fields, again to be consistent with the fact that only non-core elements may be updated. If such state-specific fields are provided by the receiver system, they are transparently forwarded to the sender system.

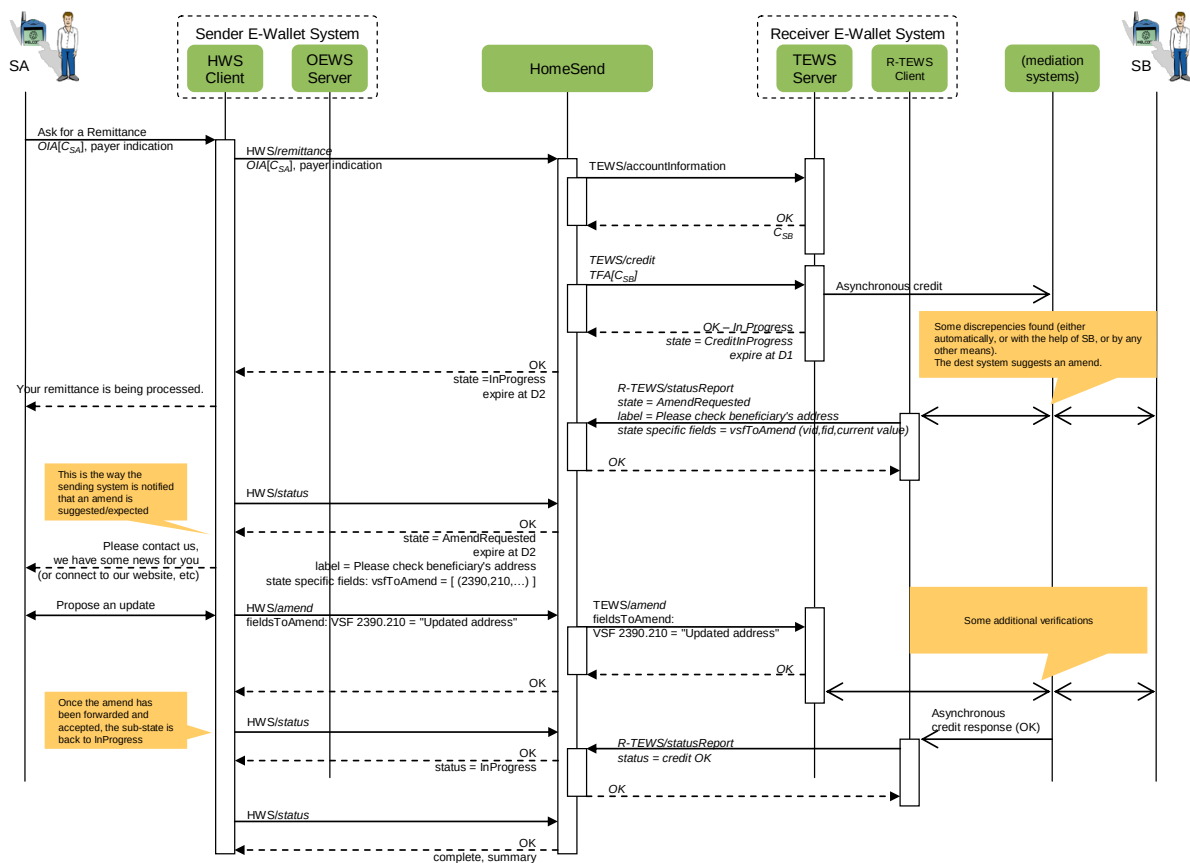


Figure 25 - A typical Solicited Amend

Upon receiving the `AmendRequested` state update, the sender system is expected to send a `HWS/amend` request containing the amended elements. As in the unsolicited amend case, HomeSend forwards it as a `TEWS/amend` request if all internal checks are passed.

Once accepted by the receiver system, a `HWS/amend OK` response is sent back to the sender side, and the receiver system does not need to send a `RTEWS/statusReport` update to HomeSend: the transaction's internal state is automatically reverted to its state before switching to the `AmendRequested` state. As a consequence, the sender system won't get any more `AmendRequested` state when polling HomeSend, so that no other amend is needed.

It is possible to ask for different fields to amend in different `AmendRequested` “sessions”. However, to limit the impact on the user experience, it is preferred to ask for all updates at the same time.

**Note:**

If the sender system does not support the HWS/amend operation, the transaction may stay in the `AmendRequested` state until it is cancelled or rejected on timeout on destination side..

4.5 Getting a Transaction Status Explicitly

Explicit *status* operations are made available so that a HWS client (sender system) or a TEWS client (HomeSend) can explicitly query the status of transactions submitted to HomeSend or to a receiver system respectively. This mechanism completes the retransmission-based status query (which is still supported but limited to `MAXreconciliation` after the first request was received to lower the cost of the duplicate detection engine design) and provides a safe way to get such a status without the risk of re-processing a retransmitted request.

However, for scalability reasons, it may not be possible to get the status of transactions that were submitted too long ago:

- HomeSend won't be able to provide the status of a transaction that was completed more than 2 weeks in the past – this delay, however, can be discussed with the HomeSend platform provider and may be increased,
- Receiver systems may also have such a limit. For systems that will reply synchronously to TEWS/*credit* requests, 3 days is a recommended minimum. Systems implementing the asynchronous credit model must be able to provide their transactions' status for at least their credit expiration time (could be weeks) plus 3 days.

These durations may be discussed with the HomeSend platform provider, too.

4.5.1 Sending Side: Getting a Transaction Status from HomeSend

As soon as the Sender E-Wallet System is aware of the HomeSend transaction ID, it may query the transaction status regularly, until HomeSend replies with an Unknown transaction error.

The transaction's status is obtained by sending a HWS/*status* request with consistent `transactionId` and `hsTransactionId` fields, and is replied with a HWS/*status* response that contains the state of the transaction with associated parameters:

- If the transaction is still being processed (`pending`), the response contains an indicative date hinting about how long the transaction may be processed (`maxCompletionDate`). The sending end-user should consider it is not abnormal not to complete the transaction before this date, as it takes a long time on the destination side to perform an end-user registration or to wait for the end-user to cash out. In addition, this response may contain Vendor Specific Fields, such as a secret code that may be required for the payout, as it would have been provided in a HWS/*remittance* provisional response.

A detailed state is also provided, but unless specified otherwise a sending E-Wallet System provider should not construct a business logic on it, as its possible values depend on the destination service provider.

- If the transaction has been successfully `completed`, the response contains a transaction summary (providing the charged and credited amounts as well as the transaction's total fees), and optional Vendor Specific Fields; this is the same level of information as in a HWS/*remittance* response.

- If the transaction has been *rejected* for whatever reasons, the response contains an error code and a non-normative label/message. The possible error codes are a subset of those in a *HWS/remittance* fault.

This mechanism is not limited to asynchronous transactions. It also works on closed/completed transaction during a limited time (typically several days), regardless of whether a *HWS/remittance* provisional response was sent or not.

4.5.2 Receiving Side: Getting a Transaction Status

A Receiver E-Wallet System must implement the *TEWS/status* operation to enable HomeSend to get a submitted transaction status in a particular case following a provisional *TEWS/credit* response.

Such a provisional response contains a *maxCompletionDate* field, which basically indicates how long the credit operation may take, at most – this depends on the associated service logic, that may include end-user registration (weeks) or daily processing through multiple sub-systems (days), or even waiting for the beneficiary to actually cash out (weeks, before the transaction expires).

Until this date, HomeSend waits for the Receiver E-Wallet System to complete the transaction by itself, sending a *R-TEWS/statusReport* with a *complete* or *rejected* status (server-side notification/push model). If no status report is received at this date, HomeSend enters a grace period during which it polls the system to get an updated status, regularly sending *TEWS/status* requests. As soon as the transaction is known to be completed (succeeded or failed), the polling stops.

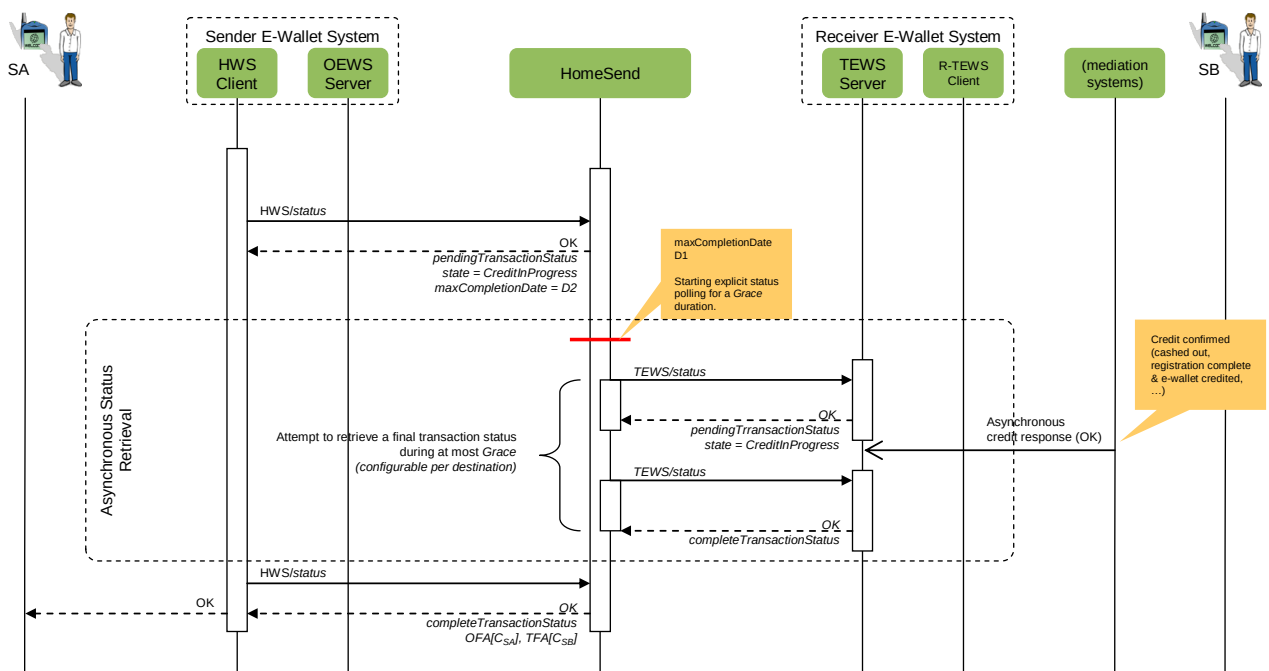


Figure 26 - Possible continuation after [1], resolving an asynchronous credit response to OK via *TEWS/status* requests

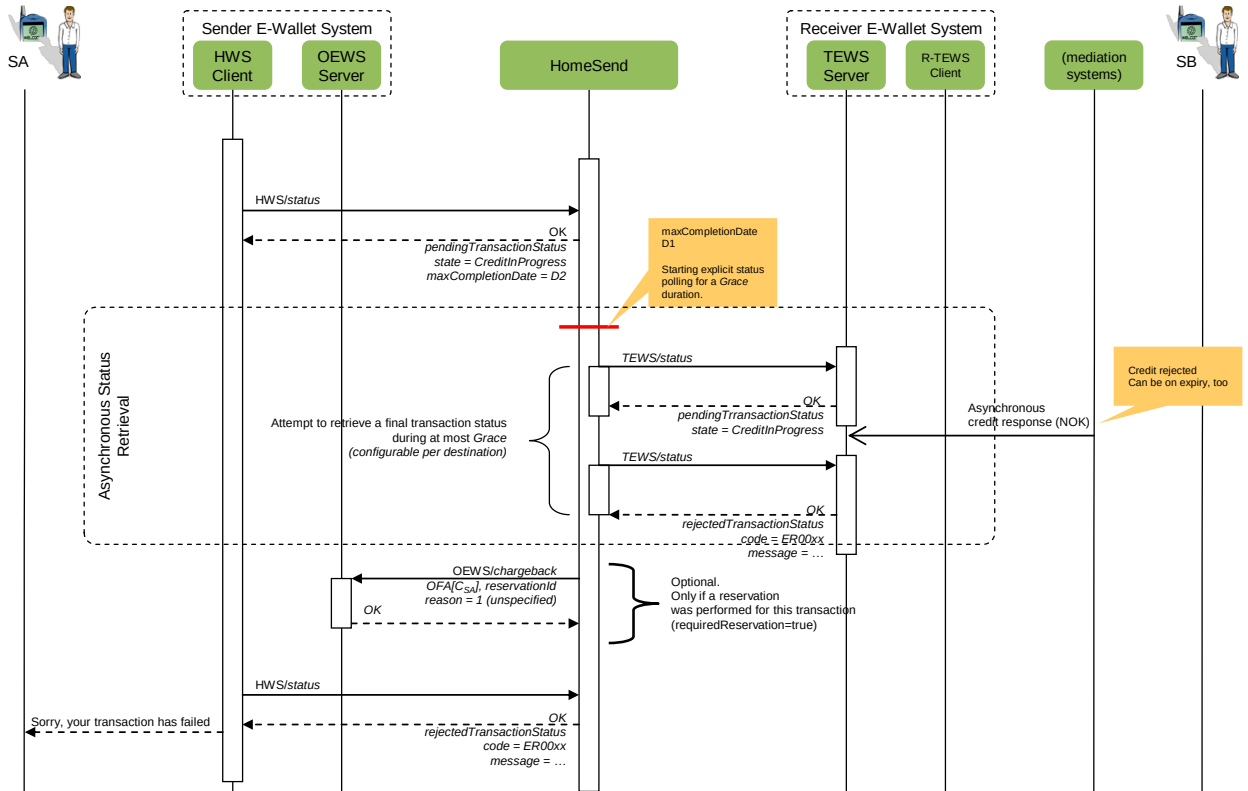


Figure 27 - Possible continuation after [1], resolving an asynchronous credit response to NOK via TEWS/status requests

The polling frequency is typically one attempt every 6 hours, over a configurable grace period (1-3 days). These values may be adjusted to the needs.

If no final status has been retrieved after the grace period, a manual action is required to complete the transaction, which was abnormally too long to process.



Note:

HomeSend will still accept a valid R-TEWS/statusReport request to complete the transaction even if it is received during or after the *grace* period.

When sending TEWS/status requests, HomeSend expects one of these kinds of TEWS/status responses:

- If the transaction is still being processed or waiting for the user on the destination side, a `pending` element is returned, with a state indication that will be used to track the transaction ("CreditInProgress", "ReadyToBePaid", "WaitingForCustomerRegistration", etc). These are custom states that will be discussed and formalized when integrating the system against HomeSend, according to the business needs. Additional, optional Vendor Specific Fields may also be provided.
- If the transaction has been successfully completed (paid, credited), a `complete` element shall be returned, with possible optional Vendor Specific Fields that would have made sense in a TEWS/credit final response.
- If the transaction has been rejected or expired, a `rejected` element shall be returned, containing a formal error code and a non-normative label, as if it were a TEWS/credit Fault.

4.6 Links Monitoring: Watchdog Mechanism

Each technical link between HomeSend and an external system can be proactively monitored by the client side, without waiting for an actual activity to detect a failure. This applies to:

- HWS: a sender E-Wallet system can monitor if the HomeSend service is still up and running,
- OEWS: HomeSend may be configured to verify that a given OEWS server hosted on a sender E-Wallet system is still available,
- TEWS: HomeSend may be configured to verify that a given TEWS server hosted on a receiver E-Wallet system is still available,
- R-TEWS: a receiver E-Wallet implementing the asynchronous mechanism may check that the HomeSend replier TEWS service will be available when needed. The R-TEWS link may be different per transaction (though, typically, the same URL is used).

A dedicated operation, called `deviceWatchdog`, is used to implement this client-side proactive monitoring. This is a stateless operation that is aimed to verifying both the network link and the presence of a valid Web Service running on the other side.

4.6.1 HWS Monitoring

This monitoring can be optionally implemented by a HWS client. If implemented, the following model is suggested:

- The HWS client should regularly send a `HWS/deviceWatchdog` request to HomeSend's HWS interface,
- Then wait several seconds for a `HWS/deviceWatchdog` response,
- If a connection error occurs, nothing is received, a fault is received, or an invalid `HWS/deviceWatchdog` response is received, the HWS client may consider the HWS link as down, and may notify the sending service provider's monitoring system about the event. The link may be considered down only after several failed retries in a row to avoid false positive due to network latency.
- Regular `HWS/deviceWatchdog` requests should be still sent after such a failure detection to monitor the link as up again as soon as a valid `HWS/deviceWatchdog` response is received on time,
- Detecting a HWS link as down should not prevent the HWS client from trying to place new transactions (as it may be up again before a new device watchdog probe detected it up). If the initial transaction's request is responded on time (either with a HomeSend Service Exception fault or a response), the HWS link may be considered up again, without waiting for the device watchdog mechanism to detect it as is.

The advised interval between two `HWS/deviceWatchdog` probe attempts is at least 15 minutes. The advised timeout to wait for a `HWS/deviceWatchdog` response is 5 seconds.

4.6.2 OEWS Monitoring

The HomeSend's platform provider may configure HomeSend to monitor an OEWS link regularly.

In this case, the OEWS server shall expect `OEWS/deviceWatchdog` requests and must reply with a valid `OEWS/deviceWatchdog` response as soon as possible.

A typical probing rate is one request every 15 minutes, but may be increased depending on the observed network link quality for improved network failure detections.

Implementing the *OEWS/deviceWatchdog* operation is mandatory when providing a OEWS server.

4.6.3 TEWS Monitoring

The HomeSend's platform provider may configure HomeSend to monitor a TEWS link regularly.

In this case, the TEWS server shall expect *TEWS/deviceWatchdog* requests and must reply with a valid *TEWS/deviceWatchdog* response as soon as possible.

A typical probing rate is one request every 15 minutes, but may be increased depending on the observed network link quality for improved network failure detections.

Implementing the *TEWS/deviceWatchdog* operation is mandatory when providing a TEWS server.

4.6.4 R-TEWS Monitoring

This monitoring can be optionally implemented by a R-TEWS client. If implemented, the following model is suggested:

- The R-TEWS client should start regularly send a *R-TEWS/deviceWatchdog* request to HomeSend's R-TEWS interface once its URL is known, i.e. only after receiving a *TEWS/credit* request,
- The R-TEWS client should not monitor the link if no *TEWS/credit* provisional response was sent, as the R-TEWS link won't be used for the transaction,
- The R-TEWS client should not monitor the same URL that is already monitored in the context of another submitted transaction waiting for a completion,
- The R-TEWS client should wait several seconds for a *R-TEWS/deviceWatchdog* response after sending the request,
- If a connection error occurs, nothing is received, a fault is received, or an invalid *R-TEWS/deviceWatchdog* response is received, the HWS client may consider the HWS link as down, and may notify the sending service provider's monitoring system about the event. The link may be considered down only after several failed retries in a row to avoid false positive due to network latency.
- Regular *R-TEWS/deviceWatchdog* requests should be still sent after such a failure detection to monitor the link as up again as soon as a valid *R-TEWS/deviceWatchdog* response is received on time,
- Detecting a R-TEWS link as down should not prevent the R-TEWS client from trying to send a *R-TEWS/statusReport* request (as it may be up again before a new device watchdog probe detected it up),
- The R-TEWS client should stop monitoring the R-TEWS link once the transaction has been completed on terminating side.

The advised interval between two *R-TEWS/deviceWatchdog* probe attempts is at least 15 minutes. The advised timeout to wait for a *R-TEWS/deviceWatchdog* response is 5 seconds.

5 Implementation Guidelines

This section provides some guidelines when it comes to implementing a HomeSend adapter, either on the sending side or the receiving side.

They are guidelines and may not be followed by the implementer, or may be freely adapted according to more specific needs. They are not dependent on a particular implementation middleware or web service stack. More middleware-specific notes & hints may be provided during integration testing with the platform provider's teams.

5.1 Sending Side

5.1.1 Operations to Implement

The following table summarizes the operations a sending system should implement, depending on the kind of service it wants to offer:

Operation	Remittance
HWS/adviceOfCharge	O
HWS/remittance	M
HWS/status	M
HWS/cancellation	O
HWS/amend	O
HWS/deviceWatchdog	O

M: Supporting the operation is mandatory

O: Implementing the operation is optional – may depend on the service the sending service provider wants to offer

When the sending system wants HomeSend to orchestrate a reservation process against a source E/M-wallet account, OEWS may be used:

Operation	Remittance
OEWS/reserve	M
OEWS/charge	M
OEWS/release	M
OEWS/chargeback	M
OEWS/deviceWatchdog	M

Using the OEWS interface is useless in the other cases.

5.1.2 Mandatory Inconclusive Request Detection

The sending system must be able to detect if a HWS/*remittance* request has been correctly processed (with a positive or a negative result) or if the request processing was *inconclusive*, i.e. it is unsure whether HomeSend processed the request or not. Section [Result Classification](#) provides some hints to help detect inconclusive cases.

In that case, the sending system MUST NOT assume that the request was rejected by HomeSend. Instead, it must tell the sending end-user that the transaction has been taken into account, and starts polling HomeSend for a transaction update, as if the HWS/*remittance* were replied with a provisional response. However, since no actual provisional response was received, there is no given `maxCompletionDate`, which can be then be set to `now + 3 weeks`

Detecting inconclusive cases on other requests is optional. All inconclusive cases for these other requests could be considered as a failed request.

5.1.3 Optional Request-level Retransmission Mechanism

It is highly recommended to implement a request-level retransmission mechanism (“Local retry mechanism”, aka “Real-time retry mechanism”, as described in [Real-Time Retry Mechanism](#) section) in order to improve the sending end-user experience, for all requests. Such a mechanism is meant to maximize the chances of success of a request, compensating for short network failures/delays. For instance, instead of having an Advice of charge being immediately rejected due to a connection error (turning the end-user to another remittance service), it is better to retry the request and get a valid Advice of Charge proposal within the next 2-3 seconds.

Keep in mind that while the request-level retransmission mechanism is optional, correct result classification for at least HWS/*remittance* requests is mandatory to be able to detect inconclusive cases.

Here is a typical, suggested workflow for a standard HWS request handling, including both request-level retransmission mechanism and inconclusive request detections.

In this section, we will make a distinction between:

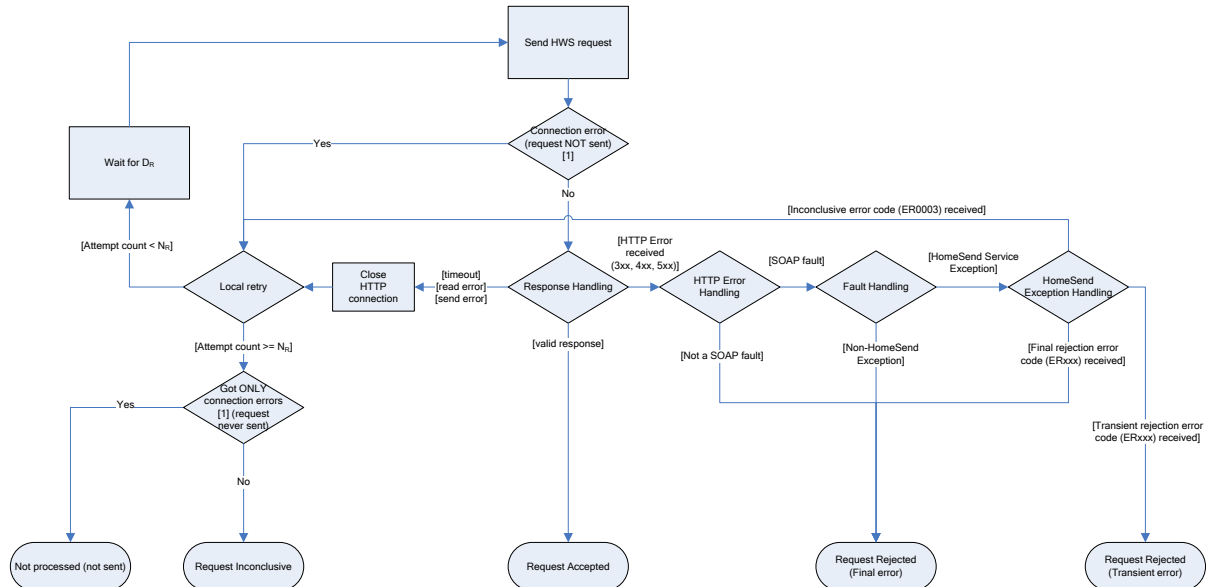
- The operation, which is a logical action triggered by the business logic, provides a global status: inconclusive, failed, success.
- Requests, which are actual SOAP requests sent over the network. A request can have the following results/status: not processed, inconclusive, final response received (success or failure). An operation is typically implemented as a loop of requests attempts (retries), where the number of maximum retries in the loop is configurable. The interval between two requests in the loop is also typically configurable to let the HWS server or the network recover from temporarily failures when applicable.

5.1.3.1 Workflow

This method can apply to all HWS requests, even if it may be sub-optimal in some cases – implementers are free to use their own optimizations.

Typical, suggested workflow for HWS request processing
(including a « Local retry mechanism », aka « Real-time retry mechanism »)

Applies to all HWS requests, and leads to 5 possible outcomes



Local retry parameters:
 D_R : Retry interval, between two HWS requests attempts. Suggested value: 3000ms
 N_R : The maximum number of attempts before giving up. Suggested value: 3 (i.e. one initial attempt and 2 possible retries).

Typical connection errors that prevent sending the request (even partially):
- tcp connection failure
- unresolvable host
- SSL handshake error (untrusted server certificates, hostname mismatch...)

Send error example:
- connection closed prematurely while writing the request

Read error examples:
- connection closed prematurely while reading the response
- invalid SOAP payload (missing SOAP headers or invalid XML, ...)
- not a SOAP payload (html instead of XML, ...)
- invalid HTTP payload (missing chunks, data, etc)

It is not mandatory to implement this exact work flow. However, it may help implementers design their own solution.

5.1.3.2 Result Classification

For each SOAP request attempt, implementers may use the following table to classify the request result into something suitable for the algorithm above:

Event when sending the request	Request local status	Typical Action	Comments
<i>Transport-level events</i>			
TCP connection refused	Not processed (not sent)	Continue local retries	
TCP connection timeout	Not processed (not sent)	Continue local retries	Depending on your implementation model, you may need or prefer to close the TCP connection explicitly to avoid late response events handling.
SSL connection/negotiation failed (including failed certificate validations, untrusted hosts, etc)	Not processed (not sent)	Continue local retries	Implementers are free to apply a more or less strict policy regarding certificate & hostname validations.

<i>HTTP-level events</i>			
read timeout/response timeout	Inconclusive	Continue local retries	
HTTP 3xx errors	Rejected (Final failure)	Stop local retries, global operation status: rejected	
HTTP 4xx errors	Rejected (Final failure)	Stop local retries, global operation status: rejected	408 is "Request Timeout", meaning that this should be considered as an inconclusive response. However, HomeSend will never send this error code.
HTTP 5xx errors that are not SOAP faults	Rejected (Final failure)	Stop local retries, global operation status: rejected	
Non-SOAP response received	Inconclusive	Continue local retries (Could be optimized by stopping the local retries loop)	
<i>SOAP-level events</i>			
Valid response received according to the WSDL/XSD	Accepted	Stop local retries, global operation status: accepted (may be a provisional response though)	
Valid SOAP response, not matching the WSDL/XSD	Inconclusive	Stop local retries, global operation status: inconclusive	
Generic SOAP Fault	Inconclusive	Stop local retries, global operation status: rejected	
HomeSendService Exception SOAP Fault with an error type "Inconclusive"	Inconclusive	Stop local retries, global operation status: inconclusive	
HomeSendService Exception SOAP Fault with an error type "Transient error"	Rejected (Transient failure)	Stop local retries, global operation status: rejected. A new transaction (same amounts, destination)/request may be retried later (in a few seconds, minutes) with a good chance of success.	
HomeSendService Exception SOAP Fault with an error type "Final error"	Rejected (Final failure)	Stop local retries, global operation status: rejected.	
Other cases (not explicitly described in this table)	Inconclusive	Continue local retries	It is very important NOT to consider that the request was failed if the sender has no clue about its actual processing status. The inconclusive status is a safe bet.

Global operation result from request local status:

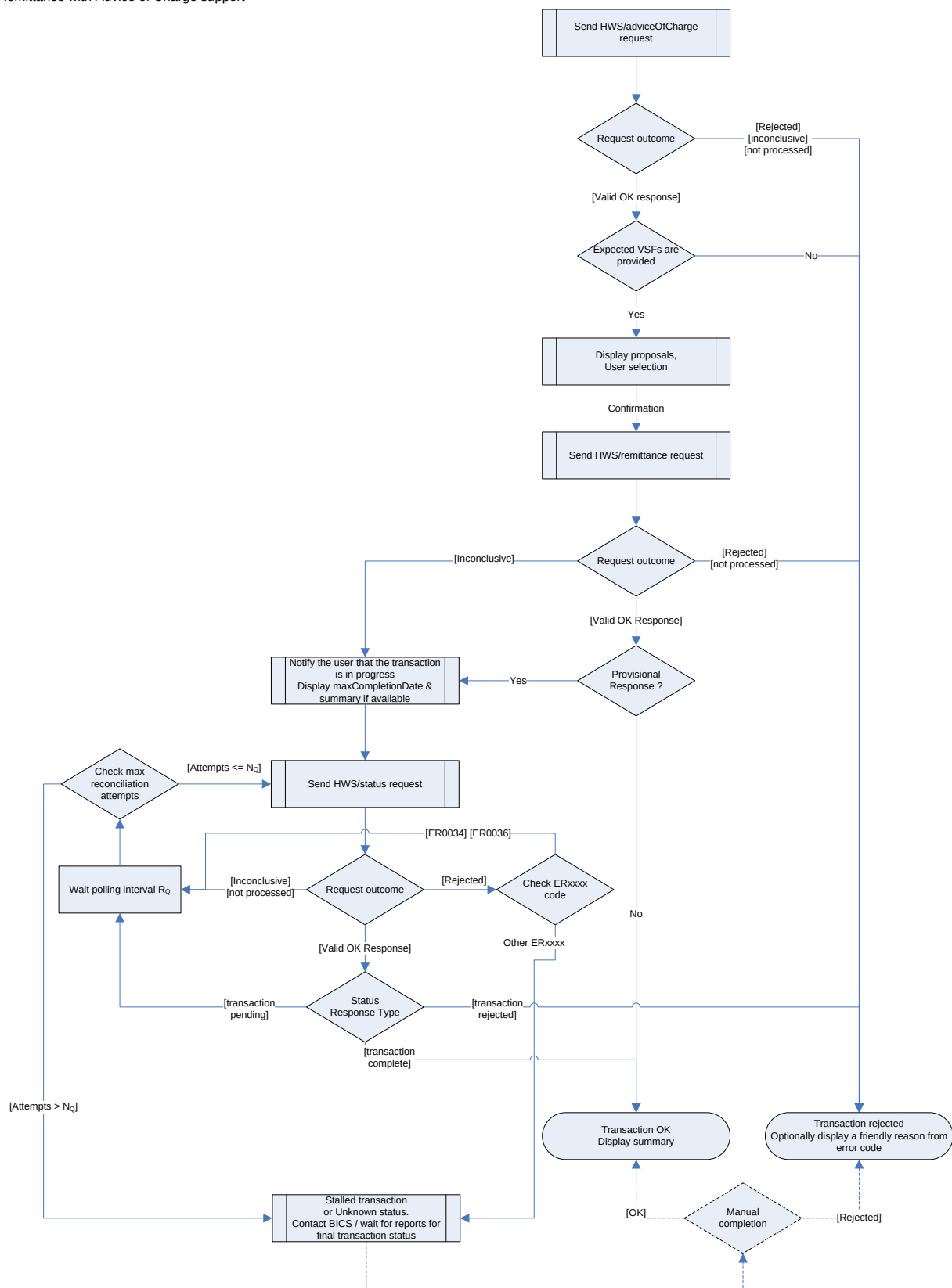
- If a final response was received during the local retries loop: the loop is stopped and the final response is used to get the global operation result (**accepted**, or **rejected**)

- Or, if at least once inconclusive request result occurred during the loop, the global operation status is **inconclusive**
- Or, if all request results were “not processed”, guaranteeing that the request could never be processed during the retry loop, the global operation status is **not processed**.

5.1.4 A Typical HWS Client with Advice of Charge Support

This workflow is a suggestion of implementation to implement a standard HWS client with advice of charge support.

Typical, suggested workflow for a HWS 2.2+ client
Remittance with Advice of Charge support



Status polling parameters:

R_Q : Polling interval, between two HWS/status attempts. Suggested value: 1h/6h/24h depending on the corridor

N_Q : The maximum number of polling attempts before giving up. N_Q is so that $R_Q \times N_Q \leq MAX_{reconciliation}$, with $MAX_{reconciliation}$ being corridor-dependent (min 36 hours, can up to 3-4 weeks or even more).

Polling with HWS/status requests after $MAX_{reconciliation}$ is harmless and may just be sub-optimal.

This sample design assumes that all HWS requests are implemented using the local retransmission mechanism suggested in [Request-level Retransmission Mechanism](#).

This workflow is quite generic and may be sub-optimal in some cases. Implementers are free to use it as is, or to adapt it at their convenience.

5.1.5 Indicative End-User Feedbacks

This section provides some guidance to implementors with providing proper feedbacks to sending end-users depending on a request outcome, as well as the expected sending service provider's behavior, depending on the error codes. The request outcome is considered after possible local retransmissions.

Implementors are free to use it as is, or to adapt it at their convenience.

Additional custom error codes can be introduced by the HomeSend platform provider on demand, to handle specific cases.

5.1.5.1 Advice of Charge

For an Advice of Charge, an inconclusive request outcome could be assimilated to a transient failure. A transient failure is a temporary error, and initiating a new transaction a few moments after (several seconds) has a good chance to work correctly. As a consequence, the end-user can be suggested to retry later.

A final failure, on the contrary, means that a new attempt is unlikely to succeed without specific actions. That could be something to reconfigure, or waiting for more time, for instance until AML counters allow the transaction.

Request outcome	Suggested response to the sending end-user	Suggested action on sending Service Provider
OK response received	<i>Service- specific.</i> <i>For instance:</i> <i>You'll be charged XXX EUR, your beneficiary will received YYY PHP. This proposal is valid until Oct, 10, 12:30. Do you accept it ?</i>	The SP should present the received proposals (1 or 2) to the end-user, or directly select one of them, depending on its service logic.
Inconclusive	We are temporarily unable to process your request, please retry later.	Mark the local transaction as failed. Do not reuse its transactionId anymore for new transactions.
Request not processed (any reason)		
Request rejected: Transient failure (any error code: ER0035, ER0036) - for ER0034, see below instead.		

Request rejected: Transient failure: ER0034	N/A	This error is due to a protocol error, as a response to an HWS/<i>adviceOfCharge</i> request for a transaction that already processed successfully a HWS/<i>adviceOfCharge</i> and was already confirmed with a HWS/<i>remittance</i>. This is typically an implementation error on the HWS client, not following HomeSend's HWS call flow and sending requests in an incorrect order. Such a result shall not be pushed toward the end-user, as a previous result was already displayed (the one of the previous HWS/<i>adviceOfCharge</i> request for the transaction). The state of the transaction cannot be deducted when receiving this error. Instead, it is recommended that an alarm is raised to the sending service provider's operation team, and that HomeSend customer care is contacted to know the actual status of the transaction before considering it rejected, successful, or still pending. Sending a HWS/<i>status</i> request would also enable to retrieve the current state of the transaction.
<i>Request rejected: Final failure: specific message per error code</i>		
ER0001	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0002	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0005	We are unable to serve your beneficiary, please contact our customer care for further details.	Mark the local transaction as failed.
ER0007	We are unable to serve your beneficiary, please contact our customer care for further details.	Mark the local transaction as failed.
ER0008	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0010	We cannot serve the beneficiary as the customer is not registered at the chosen destination service.	Mark the local transaction as failed.
ER0011	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0014	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. This should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is highly advised the SP raises an alarm and HomeSend teams are contacted so that correct credentials can be provisioned on both sides.
ER0016	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. It is highly advised the SP raises a major alarm and HomeSend teams are contacted to make sure that there are no configuration mistakes.

ER0020	You cannot send this amount to the chosen destination.	Mark the local transaction as failed. No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0037	You have not provided all the mandatory details, please contact the customer care for further details.	Mark the local transaction as failed. This MAY be a critical error, but typically only affects a subset of the corridors you take part to. However, it is advised the SP raises an alarm and HomeSend teams are contacted to review the technical requirements for the various corridors you are involved in.
ER0040	You are not allowed to perform this transaction for regulatory reasons.	Mark the local transaction as failed. No special action required. Transaction has been blocked due to Sanction Listing screening against the sender and/or beneficiary.
ER0045	The beneficiary's account does not support this currency.	Mark the local transaction as failed. No special action required. This is for reverse advice of charge only.
Any other error	We are unable to process your request due to an internal error, please contact our customer care for further details.	Mark the local transaction as failed. The SP should raise a critical alarm and contact HomeSend to sort out the problem.

Additional custom final failure error codes may be provided on demand to meet specific purposes.

5.1.5.2 Remittance (after Advice of Charge)

A correct handling of a Remittance request outcome is critical for the transaction's consistency across the involved party (the sending system, HomeSend, the receiving system). This is the request whose implementation requires the most attention, and this is where the inconclusive case detection matters.

Request outcome	Suggested response to the sending end-user	Suggested action on sending Service Provider
OK, non-provisional response received	<i>Service-specific.</i> <i>For instance:</i> <i>Your beneficiary has been credited YYY PHP and you have been charged XXX EUR.</i>	The SP should commit/confirm the reserved amount on the sender's balance (if applicable). Close the transaction as successful. What is displayed to the end-user highly depends on the SP's service logic.
OK, provisional response received	<i>Service specific.</i> <i>For instance:</i> <i>Your remittance has been correctly submitted. You have been charged XXX EUR, and your beneficiary will be credited YYY PHP by Oct, 12.</i> <i>For cash-out destinations, a secret code will be provided and thus must be given to the sending user:</i> <i>Your remittance has been correctly submitted. You have been charged XXX EUR, and your beneficiary will be credited YYY PHP by Oct, 12. The secret code for this transaction is ZZZ. It must be used by your beneficiary to claim his/her credit.</i>	The SP should commit/confirm the reserved amount on the sending account/balance (if applicable). Mark the transaction as still open, in a dedicated queue for regular polling. Start polling for an updated transaction's status using HWS/status requests. What is displayed to the end-user highly depends on the service logic. However, displaying the maxCompletionDate, in a way or another, is highly recommended. In addition, when a secret code (VSF 2390.500) is returned, which is the case for cash-out destinations, it should be communicated to the sending user. However, on some provisional responses even for cash-out destinations, the

		secret code may be missing. In that case, it will be made available in a next HWS/status response. The sending SP should then capture it at that time and send it back to the sending user (or update a transaction status) so that the user can get aware of it.
Inconclusive	Your remittance has been submitted. Its status will be provided soon.	The SP should commit/confirm the reserved amount on the sending account/balance (if applicable). Mark the transaction as still open, in a dedicated queue for regular polling. Start polling for an updated transaction's status using HWS/status requests.
Request not processed (any reason)	We are temporarily unable to process your request, please retry later.	The simpler option is to close the transaction as failed, and release the reserved amount on the sending account/balance (if applicable).
Request rejected: Transient failure (any error code: ER0034, ER0035, ER0036)		Note: an enhanced option would be displaying a "We are temporarily unable to confirm your remittance, please retry later" to the end-user, giving him or her an opportunity to reuse the Advice of charge proposal already made, without making him or her another one (that could yield different values).
Request rejected: Final failure: specific message per error code		
ER0001	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0002	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0004	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm and involves the HomeSend support team.
ER0005	We are unable to serve your beneficiary, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served through the selected destination channel.
ER0007	We are unable to serve your beneficiary, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served through the selected destination channel.

ER0008	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0010	We cannot serve the beneficiary as the customer is not registered at the chosen destination service.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served (not-registered) through the selected destination channel.
ER0011	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0012	We are unable to process your request due to an internal error, please contact our customer care for further details.	This error only occurs if reservationRequired is not set to False. This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0013	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No action required, beneficiary can not be served through the selected destination channel.
ER0014	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is highly advised the SP raises an alarm and HomeSend teams are contacted so that correct credentials can be provisioned on both sides.
ER0016	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). It is highly advised the SP raises a major alarm and HomeSend teams are contacted to make sure that there are no configuration mistakes.
ER0017	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). Major alert should be raised at SP side. Either the SP is sending a remittance after more than 36 hours after the initial adviceOfCharge, the HomeSend transaction ID is invalid, or a transaction with the same HomeSend transaction ID was found but with different transaction ID, amount, source or destination URIs. In that case,

		this should be considered as a problem in the sending system implementation.
ER0020	You cannot send this amount to the chosen destination.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0021	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0022	You have reached maximum sent amount or transactions and cannot send further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0023	You have reached maximum sent amount or transactions and cannot send further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0024	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0025	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0026	You have reached maximum sent amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0027	You have reached maximum sent amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). Raise a critical alert as the prefunding account on the HomeSend system may be depleted! Involve the HomeSend support team if needed.
ER0028	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0029	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0031	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to

		this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0032	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0033	Advice of Charge proposal has been expired. Please re-try.	The SP releases the reserved amount on the sending account/balance (if applicable). The user should be suggested to retry a new transaction (starting with the Advice of Charge).
ER0034	We are unable to confirm your proposal at this moment, please try to confirm it later.	The remittance request comes at a moment when HomeSend is not able to process it yet. The Advice of Charge proposal, i.e. the transaction, is still valid and might be confirmed a few seconds after. The SP might alternatively release the funds and close this transaction - HomeSend will expire it after the AoC timeout on its side without further intervention. In that case, the SP should adapt the message given to the end-user as he/she will to do a brand new transaction instead of confirming an existing one.
ER0037	You have not provided all the mandatory details, please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This MAY be a critical error, but typically only affects a subset of the corridors you take part to. However, it is advised the SP raises an alarm and HomeSend teams are contacted to review the technical requirements for the various corridors you are involved in.
ER0040	You are not allowed to perform this transaction for regulatory reasons.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. Transaction has been blocked due to Sanction Listing screening against the sender and/or beneficiary.
ER0042	Your transaction has been expired.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. As stated the transaction has been expired on the receiving SP side ((i.e. the beneficiary did not withdraw the money on time, or did not complete a registration on time)).
Any other error	Your remittance has been submitted. Its status will be provided soon.	The SP should raise a critical alarm and contact HomeSend to get the transaction's status before releasing or confirming the reserved amount.

Additional custom final failure error codes may be provided on demand to meet specific purposes.

5.1.5.3 Cancellation

The following table describes a typical error processing when the cancellation request is sent after a HWS/remittance. Cancelling a transaction after a HWS/adviceOfCharge but before sending a

HWS/remittance, while the transaction is waiting for a user confirmation, does not need to be implemented and would follow different error handling rules.

Request outcome	Suggested response to the sending end-user	Suggested action on sending Service Provider
OK response received	Your cancellation request has been taken into account. The transaction status will be updated when it was confirmed to be cancelled.	The SP shall keep polling HomeSend to get the final transaction status, and shall not assume that the transaction has been cancelled, as some destination systems may not be able to cancel it anymore.
Inconclusive	We are temporarily unable to process your request, please retry later.	This is a pessimistic assumption. The cancellation request may have been taken into account correctly, but the end-user may ask for another cancellation without any problem. The SP shall keep polling HomeSend to get the final transaction status, and shall not assume that the transaction has been cancelled, as some destination systems may not be able to cancel it anymore.
Request not processed (any reason)	We are temporarily unable to process your request, please retry later.	No special action. The transaction is still open and must not be considered cancelled.
Request rejected: Transient failure (any error code: ER0034, ER0036)		
Request rejected: Final failure: specific message per error code		
ER0001	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0002	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0008	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0009	We are temporarily unable to process your request, please retry later.	This is a pessimistic assumption. The cancellation request may have been taken into account correctly, but the end-user may ask for another cancellation without any problem.

		The SP shall keep polling HomeSend to get the final transaction status, and shall not assume that the transaction has been cancelled, as some destination systems may not be able to cancel it anymore.
ER0014	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0017	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0018	This transaction has already been cancelled.	If the transaction was not closed already, The SP releases the reserved amount on the sending account/balance (if applicable), and can close the transaction as cancelled. Alternatively, the SP can wait for the usual status polling to close the transaction.
ER0031	This transaction cannot be cancelled.	The transaction is still open and must not be considered cancelled.
ER0034	This transaction cannot be cancelled at that time.	The transaction is still open and must not be considered cancelled.
ER0044	This transaction cannot be cancelled.	The transaction is still open and must not be considered cancelled.
Any other error	Your cancellation request has been taken into account. The transaction status will be updated when it was confirmed to be cancelled.	The transaction is still open and must not be considered cancelled. This event should be considered as a critical, technical error as any other cancellation attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.

Additional custom final failure error codes may be provided on demand to meet specific purposes.

5.1.5.4 Amend

The following table describes a typical error processing when on a HWS/*amend* request that would be initiated by a sending end-user:

Request outcome	Suggested response to the sending end-user	Suggested action on sending Service Provider
OK response received	Your amend request has been taken into account.	The transaction is still open.
Inconclusive	We are temporarily unable to process your request, please retry later.	The transaction is still open.

		This is a pessimistic assumption: the amending request may have worked correctly. But it may be re-attempted another time by the end-user without any risks.
Request not processed (any reason)	We are temporarily unable to process your request, please retry later.	The transaction is still open.
Request rejected: Transient failure (any error code: ER0034, ER0036)		
Request rejected: Final failure: specific message per error code		
ER0001	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This event should be considered as a critical, technical error as any other amend attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0002	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This event should be considered as a critical, technical error as any other amend attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0008	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This event should be considered as a critical, technical error as any other amend attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0009	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This is a pessimistic assumption: the amending request may have worked correctly. But it may be re-attempted another time by the end-user without any risks.
ER0014	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This event should be considered as a critical, technical error as any other amend attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0017	We are unable to process your request due to an internal error, please contact our customer care for further details.	The transaction is still open. This event should be considered as a critical, technical error as any other amend attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0031	This transaction cannot be amended.	The transaction is still open.
ER0034	This transaction cannot be amended at that time.	The transaction is still open.

ER0037	This transaction cannot be amended.	The transaction is still open.
ER0044	This transaction cannot be amended.	The transaction is still open.
Any other error	Your amend request has been taken into account. The transaction status will be updated when it was confirmed to be cancelled.	The transaction is still open. This event should be considered as a critical, technical error as any other amending attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.

Additional custom final failure error codes may be provided on demand to meet specific purposes.

5.1.5.5 Remittance (One-shot)

The main difference with a Remittance after an Advice of Charge is that some permanent failure error codes are not possible anymore.

Inconclusive responses must be managed with the same care.

Request outcome	Suggested response to the sending end-user	Suggested action on sending Service Provider
OK, non-provisional response received	Service-specific. For instance: Your beneficiary has been credited YYY PHP and you have been charged XXX EUR.	The SP should commit/confirm the reserved amount on the sender's balance (if applicable). Close the transaction as successful. What is displayed to the end-user highly depends on the SP's service logic.
OK, provisional response received	Service specific. For instance: Your remittance has been correctly submitted. You have been charged XXX EUR, and your beneficiary will be credited YYY PHP by Oct, 12. For cash-out destinations, a secret code will be provided and thus must be given to the sending user: Your remittance has been correctly submitted. You have been charged XXX EUR, and your beneficiary will be credited YYY PHP by Oct, 12. The secret code for this transaction is ZZZ. It must be used by your beneficiary to claim his/her credit.	The SP should commit/confirm the reserved amount on the sending account/balance (if applicable). Mark the transaction as still open, in a dedicated queue for regular polling. Start polling for an updated transaction's status using HWS/status requests. What is displayed to the end-user highly depends on the service logic. However, displaying the maxCompletionDate, in a way or another, is highly recommended. In addition, when a secret code (VSF 2390.500) is returned, which is the case for cash-out destinations, it should be communicated to the sending user. However, on some provisional responses even for cash-out destinations, the secret code may be missing. In that case, it will be made available in a next HWS/status response. The sending SP should then capture it at that time and send it back to the sending user (or update a transaction status) so that the user can get aware of it.
Inconclusive	Your remittance has been submitted. Its status will be provided soon.	The SP should commit/confirm the reserved amount on the sending account/balance (if applicable). Mark the transaction as still open, in a dedicated queue for regular polling. Start polling for an updated transaction's status using HWS/status requests.

Request not processed (any reason)	We are temporarily unable to process your request, please retry later.	The simpler option is to close the transaction as failed, and release the reserved amount on the sending account/balance (if applicable).
Request rejected: Transient failure (any error code: ER0034, ER0035, ER0036)		Note: an enhanced option would be displaying a “We are temporarily unable to confirm your remittance, please retry later” to the end-user, giving him or her an opportunity to reuse the Advice of charge proposal already made, without making him or her another one (that could yield different values).
Request rejected: Final failure: specific message per error code		
ER0001	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0002	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0004	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm and involves the HomeSend support team.
ER0005	We are unable to serve your beneficiary, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served through the selected destination channel.
ER0007	We are unable to serve your beneficiary, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served through the selected destination channel.
ER0008	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0010	We cannot serve the beneficiary as the customer is not registered at the chosen destination service.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required, beneficiary can not be served (not-registered) through the selected destination channel.

ER0011	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is advised the SP raises an alarm or makes sure a corrective action is planned. HomeSend support team may be involved in such corrective actions.
ER0012	We are unable to process your request due to an internal error, please contact our customer care for further details.	This error only occurs if reservationRequired is not set to False. This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0013	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No action required, beneficiary can not be served through the selected destination channel.
ER0014	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This should be considered as a critical, technical error as any other transaction attempts are likely to fail too. It is highly advised the SP raises an alarm and HomeSend teams are contacted so that correct credentials can be provisioned on both sides.
ER0016	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). It is highly advised the SP raises a major alarm and HomeSend teams are contacted to make sure that there are no configuration mistakes.
ER0017	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). Major alert should be raised at SP side. This occurs if the SP is reusing the same transactionID as an existing, uncorrelated transaction.
ER0020	You cannot send this amount to the chosen destination.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0021	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0022	You have reached maximum sent amount or transactions and cannot send further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)

ER0023	You have reached maximum sent amount or transactions and cannot send further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0024	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0025	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0026	You have reached maximum sent amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). No action required. (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0027	You have reached maximum sent amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). Raise a critical alert as the prefunding account on the HomeSend system may be depleted! Involve the HomeSend support team if needed.
ER0028	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0029	Beneficiary has reached maximum receivable amount or transactions and cannot be sent further remittances. Please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). (Please check with HomeSend team for the actual corridor limits in case of doubts.)
ER0031	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0032	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is persistent HomeSend support team needs to be involved.
ER0034	We are unable to process your request due to an internal error, please contact our customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This event should be considered as a critical, technical error as any other transaction attempts to this destination channel are likely to fail too. It is advised the SP raises an alarm. If the issue is

		persistent HomeSend support team needs to be involved.
ER0037	You have not provided all the mandatory details, please contact the customer care for further details.	The SP releases the reserved amount on the sending account/balance (if applicable). This MAY be a critical error, but typically only affects a subset of the corridors you take part to. However, it is advised the SP raises an alarm and HomeSend teams are contacted to review the technical requirements for the various corridors you are involved in.
ER0040	You are not allowed to perform this transaction for regulatory reasons.	The SP releases the reserved amount on the sending account/balance (if applicable). No special action required. Transaction has been blocked due to Sanction Listing screening against the sender and/or beneficiary.
Any other error	Your remittance has been submitted. Its status will be provided soon.	The SP should raise a critical alarm and contact HomeSend to get the transaction's status before releasing or confirming the reserved amount.

Additional custom final failure error codes may be provided on demand to meet specific purposes.

5.2 Receiving Side

5.2.1 Operations to Implement

The following table summarizes the operations a receiving system should implement, depending on the kind of destination accounts it is able to manage:

Operation	E/M-Wallet (Synchronous)	E/M-Wallet with Registration support (Asynchronous)	Cash-out (Asynchronous)	Bank (Asynchronous)
TEWS/accountInformation	M	M	M	M
TEWS/credit	M	M	M	M
TEWS/status	Not used	M	M	M
TEWS/statusReport	O (Only if AML verifications are delegated to the Platform Provider)	O (Only if AML verifications are delegated to the Platform Provider)	O (Only if AML verifications are delegated to the Platform Provider)	O (Only if AML verifications are delegated to the Platform Provider)
TEWS/amend	O (Service-dependent)	O (Service-dependent)	O (Service-dependent)	O (Service-dependent)
TEWS/cancellation	Not used	O (Service-dependent)	O (Service-dependent)	O (Service-dependent)
TEWS/deviceWatchdog	M	M	M	M
R-TEWS/statusReport	Not used	M	M	M
R-TEWS/deviceWatchdog	Not used	O	O	O

M: Supporting the operation is mandatory

O: Implementing the operation is optional – may depend on the service the receiving service provider wants to offer

Not used: The operation is not used in this context, no need to implement it (except when told otherwise during integration process).

6 Namespaces

The services (HWS, OEWS, and TEWS) described in this document use data types and elements that are defined in the following namespaces:

Namespace	Alias used in this document	Description
http://www.w3.org/2001/XMLSchema	xsd	The XML Schema standard namespace.
http://www.eservglobal.com/homesend/types/2.0	hos	Common data types that are used in all services.
http://www.eservglobal.com/homesend/types/2.1	hos21	Common data types that are used in all services, introduced in API Suite 2.1.
http://www.eservglobal.com/homesend/types/2.2	hos22	Common data types that are used in all services, introduced in API Suite 2.2.
http://www.eservglobal.com/homesend/types/2.3	hos23	Common data types that are used in all services, introduced in API Suite 2.3.
http://www.eservglobal.com/homesend/hws/2.0	hws	Types and operations that are specific to HWS.
http://www.eservglobal.com/homesend/hws/2.1	hws21	Types and operations that are specific to HWS, introduced in API Suite 2.1.
http://www.eservglobal.com/homesend/hws/2.2	hws22	Types and operations that are specific to HWS, introduced in API Suite 2.2.
http://www.eservglobal.com/homesend/hws/2.3	hws23	Types and operations that are specific to HWS, introduced in API Suite 2.3.
http://www.eservglobal.com/homesend/oews/2.0	oews	Types and operations that are specific to OEWS.
http://www.eservglobal.com/homesend/oews/2.1	oews21	Types and operations that are specific to OEWS, introduced in API Suite 2.1.
http://www.eservglobal.com/homesend/oews/2.2	oews22	Types and operations that are specific to OEWS, introduced in API Suite 2.2.
http://www.eservglobal.com/homesend/tews/2.0	tews	Types and operations that are specific to TEWS and R-TEWS.
http://www.eservglobal.com/homesend/tews/2.1	tews21	Types and operations that are specific to TEWS, introduced in API Suite 2.1.
http://www.eservglobal.com/homesend/tews/2.2	tews22	Types and operations that are specific to TEWS, introduced in API Suite 2.2.
http://www.eservglobal.com/homesend/tews/2.3	tews23	Types and operations that are specific to TEWS, introduced in API Suite 2.3.

7 Common Data Types Definition



Warning :

SOAP element tags are case sensitive; to avoid confusion, consider that only the WSDL contains the official XML element names. The names provided in that document are for explanations only.

7.1 Simple Types

Several restricted string types are defined in this namespace. They are:

Type Name	Description
hos:login	Used to define a login identifier on different systems. String from 1 to 32 characters. ASCII characters only.
hos:password	Used to define the password associated to the login above. String from 0 to 255 characters. ASCII characters only.
hos:uri	Used to define the different URIs identifying the source and destination end-users. String up to 200 characters. The format should match the one defined in End-Users URIs . May contain Unicode characters.
hos:transactionId	Used to define the various transaction identifiers. String from 0 to 30 characters. ASCII characters only.
hos:routingTag	Used to define the routingTag element, identifying a pre-selected corridor, if present in transaction-opening requests, or identifying the selected corridor in TEWS/accountInformation. It may also help route to the proper sub-system in TEWS/credit and OEWS requests. String from 0 to 300 characters, through HomeSend guarantees not to send more than 30 characters in OEWS and TEWS messages. Notice that valid routing tags match the regular expression [a-zA-Z0-9_-]+. This element may also contain a coma-separated list of routing tags in some particular contexts. Its exact format is described when it applies.
hos:senderName	Used to define a free string, identifying the sender with a friendly label / or a legal identification, according to cross-border regulations. String from 0 to 200 characters. Usually contain Unicode characters.
hos:senderAddress	Used to define a free string, identifying the sender's address with a friendly label / or a legal identification, according to cross-border regulations. String from 0 to 200 characters. Usually contain Unicode characters.
hos:description	Used to define a free string, labeling the transaction, usually with a user-provided value. String from 0 to 200 characters. May contain Unicode characters.
hos21:vendorSpecificValue	Used to define a Vendor Specific Field value. String from 0 to 1000 characters. May contain Unicode characters.
hos22:transactionState	Used to identify the current state of a pending transaction. String from 1 to 100 characters in [a-zA-Z0-9]. Valid states depend on the operation and are described later.

hos22:transactionStateLabel	Used to provide a friendly, human-understandable label associated to the state above. String from 0 to 100 characters. Usually ASCII as English is preferred.
-----------------------------	---

7.2 SenderInformation Structure

The `hos:SenderInformation` is a data structure that helps identify the sender with friendly label or a legal identification (such as a verified firstname/lastname and a verified postal address).

It contains the following elements:

Element Name	Type	Optional	Description
name	hos:senderName	No	A friendly or legal identification of the sending end-user.
address	hos:senderAddress	No	A friendly or legal address identification of the sending end-user.

If this structure is present in a message, both elements must be filled. However, filling them with an empty string is still possible.

7.3 AmountInformation Structure

The `hos:AmountInformation` is a data structure that embeds an amount with an associated currency.

Element Name	Type	Optional	Description
currency	xsd:string	No	A currency identifier as defined in ISO 4217 [ISO4217] (3 uppercase characters).
amount	xsd:decimal	No	Amount in the currency above. HomeSend usually rounds this value to 2 to 3 decimal digits, depending on its configuration.
grace	xsd:int	Yes	Deprecated. Should be set to nil/null.
validity	xsd:int	Yes	Deprecated. Should be set to nil/null.

7.4 Security Structure

The `hos:Security` structure is used to provide a couple of technical login/password in all requests between the systems. This is a basic way to authenticate the SOAP request sender, and the only way to do so when not using additional SOAP-level security mechanisms.

Element Name	Type	Optional	Description
login	hos:login	No	The technical login to authenticate
password	hos:password	No	The associated password to the login above. It may be empty.

7.5 VendorSpecificField Structure [2.1]

The `hos21:VendorSpecificField` structure is used to transport a Vendor Specific Field (VSF) in all messages between the systems. Such a structure is then used as an item in the `hos21:VendorSpecificFields` list.

VSFs are globally uniquely identified by a pair of identifiers (Vendor ID, Field ID), where the Vendor ID is a Private Enterprise Number assigned by IANA to the vendor that wants to use such a field, and the Field ID is locally managed and assigned by this vendor. The meaning of its value depends on this identifier,

and is opaque to HomeSend, which only transports it transparently (from message to message, retaining the associated identifiers).

The value is a string (Unicode characters accepted) and may be casted to any object by the destination system if needed. Binary values, when required, are usually encoded to ascii using Base64 or any other method prior sending.

Element Name	Type	Optional	Description
vendorId	xsd:unsignedInt	No	The Vendor Identifier, as a IANA-assigned Private Enterprise Number.
fieldId	xsd:unsignedInt	No	The Field Identifier, as assigned and managed by the Vendor.
value	hos21:vendorSpecificValue	No	The field value. May contain Unicode characters, may be empty.

7.6 VendorSpecificFields Structure [2.1]

The `hos21:VendorSpecificFields` structure basically encapsulates a list of VSFs in each request/response.

If multiple VSFs with the same Vendor ID, Field ID are sent, HomeSend will only consider the last received value for it. HomeSend won't send duplicated VSFs.

Element Name	Type	Cardinality	Description
vsf	hos21:vendorSpecificField	0..50	A list of Vendor Specific Fields.



Note:

While the XSD definition indicates that this is an unbounded list due to development tools limitations (most of them do not support an numeric value for this limit, outside 0 and 1), it is functionally limited to 50 VSFs at most. VSFs after the 50 first ones may not be considered by HomeSend. In return, HomeSend guarantees to never send more than 50 VSFs in a message.

7.7 BaseException Structure

The `hos:BaseException` structure is used in case of an error, and serves as a base to define the specialized types that are exclusively included in the application-level exceptions generated as SOAP Faults: `HomeSendServiceException`, `OriginatingServiceException`, and `TerminatingServiceException`.

It contains an error code, an error message, and optional additional details about the error that occurred. These message and details are for diagnostics only and are not meant to be presented to an end-user.

The error code is significant and is the only element that must be taken into account when analyzing an error. The error message is provided as an additional, human-readable indication for quick diagnostics, but it may vary with the implementation and/or the context the error is generated from.

For instance, the `ER0012` is usually associated to a "Sender charging error". The same error code `ER0012` is also used in case of a "Reservation error", and the message associated to it should be different in both cases to distinguish them.

Element Name	Type	Optional	Description
--------------	------	----------	-------------

code	xsd:string	No	The error code, as described later. Formatted as a string like "ER0000".
message	xsd:string	No	A message associated to the error code. This message is in plain English and is not meant to be displayed to the end-user. Typical error messages associated to a code are provided in the next sections (actual messages may vary).
parameters	xsd:string	Yes	Zero or more of these elements may be present. They contain additional information related to the error, as described later. Depending on the error, one or more parameters must be present. In all cases, additional, non-required parameters are optional and may be present. Such non-required parameters must just be ignored by the entity receiving the error.

8 HomeSend Web Service (HWS)

This section describes types and operations that are defined into the following namespaces:

- `http://www.eservglobal.com/homesend/hws/2.0`, aliases by `hws`,
- `http://www.eservglobal.com/homesend/hws/2.1`, aliases by `hws21`,
- `http://www.eservglobal.com/homesend/hws/2.2`, aliases by `hws22`,
- `http://www.eservglobal.com/homesend/hws/2.3`, aliases by `hws23`.

Common types are referenced with the `hos/hos21/hos22/hos23` prefix, and defined in section [Common Data Types Definition](#).

8.1 Specific Types

The following types are specific to the different HWS messages.

8.1.1 Payer Indicator

This indicator is defined as the `hws:payer` enumeration, containing the following values:

Enumeration Value	Description
RECEIVER	Transaction fees will be paid by the receiver, i.e. implicitly deduced from the transferred amount.
SENDER	Transaction fees will be paid by the sender, i.e. the sender will be charged with the transferred amount + fees.

Such an indication is used in `HWS/remittance` requests.

8.1.2 RemittanceCbsType

The type `hws:remittanceCbsType` is a free string, limited to 30 characters. It is used to determine the remittance's service type, used by HomeSend to apply relevant routing constraints, velocity checks, transaction limiters or pricing models.

Possible values depend on the service configuration, and are documented in the Use Cases documentation. Typical values look like P2P, B2B, B2P, etc.

The default value is set to "EMoneyToEMoneyTransfer" to remain compatible with older client implementations.

8.1.3 ERechargeCbsType Enumeration

This enumeration is deprecated and shall not be used anymore. It was used for E-Recharge operation only, which is also deprecated.

8.1.4 AocValidity

The `hws:aocValidity` is a simple type to define the advice of charge expiration delay, in seconds, in `HWS/adviceOfCharge` response.

It corresponds to an integer > 0.

8.1.5 TransactionSummary Structure

This `hws:TransactionSummary` structure contains the different final amounts involved in a transfer, from the end-user point of view. It is used in HWS/*adviceOfCharge* responses to provide the sender with different transfer proposals, as well as in the final HWS/*remittance* responses to summarize the actually transferred amounts:

- How much the sender was charged/would be charged,
- How much the receiver received/would receive,
- What was/is the base amount used to do service charges calculations (this is also the amount to send in case of reverse quote advice of charge)
- The sum of the service charges that were/would be applied for this transaction

Element Name	Type	Optional	Description
commissionAmount	hos:AmountInformation	No	Contains the transaction fees amount for the transaction (actual, or proposed)
chargedAmount	hos:AmountInformation	No	Contains the debited amount (actual or proposed) in sender currency.
creditedAmount	hos:AmountInformation	No	Contains the received amount (actual or proposed) in receiver currency.
initialAmount	hos:AmountInformation	No	Contains the abse amount used to do all computations. This corresponds to a "net" sent amount, not including any service charges.

8.1.6 ProvisionalResponse Structure [2.2]

The `hws22:ProvisionalResponse` structure is used to taint a HWS/*remittance* response as "provisional", i.e. non-final, indicating that the request was taken into account but its processing may take some time.

It contains the following elements:

Element Name	Type	Optional	Description
maxCompletionDate	xsd:dateTime	No	An indication of the maximum expected completion date for the transaction. While HomeSend cannot guarantee that the transaction will be completed (successfully or rejected) at that time as manual actions may be required in some cases, this is a good indication to provide to the sending end-user to make him/her wait while the transaction is processed. This date/time is always provided with a UTC or a timezone indication, with regards to the standard dateTime format.
state	hos22:transactionState	No	A formal identifier that gives the current transaction's pending state (if the transaction is finished, the <code>complete</code> or <code>rejected</code> choice is used instead). Possible states are provided below.
stateLabel	hos22:transactionStateLabel	No	A human-readable, friendly label associated to the state. Used for diagnostic and logging; only the <code>state</code> field must be used for service logic. This label may be left empty.

stateSpecificFields	hos21:VendorSpecificFields	Yes	Optional fields, encoded as vendor specific fields, that qualify the state further. For instance, if the state is "AmendRequested", this element may contain the list of the fields that the sending user needs to amend, with their current value.
---------------------	----------------------------	-----	---

8.1.7 TransactionStatus Choice [2.2]

The `hws22:TransactionStatus` choice is designed to "tunnel" the information elements HomeSend would normally return in a *HWS/remittance* response or fault.

This structure is used to transport the current status of a remittance that was submitted to HomeSend, but whose final status is not yet known by the sending systems.

This choice contains one of the following elements:

Element Name	Type	Description
pending	hws22:PendingTransactionStatus	This element is used to transport the status of a transaction that is still pending/open, i.e. not complete successfully or explicitly rejected. It is close to the <code>provisionalResponse</code> part of a provisional <i>HWS/remittance</i> response.
complete	hws22:CompleteTransactionStatus	This element is used when the transaction has been successfully completed from the sender's point of view. It is similar to a final (non-provisional) <i>HWS/remittance</i> response.
rejected	hws22:RejectedTransactionStatus	This element is used when the transaction has been rejected, i.e. completed on failure, from the sender's point of view. It is similar to a <i>HWS/remittance</i> fault.

8.1.8 PendingTransactionStatus Structure [2.2]

The `hws22:PendingTransactionStatus` structure is used as a possible choice in a `hws22:TransactionStatus`, and contains the following elements:

Element Name	Type	Optional	Description
maxCompletionDate	xsd:dateTime	No	An indication of the maximum expected completion date for the transaction.
state	hos22:transactionState	No	A formal identifier that gives the current transaction's pending state (if the transaction is finished, the <code>complete</code> or <code>rejected</code> choice is used instead). Possible states are provided below.
stateLabel	hos22:transactionStateLabel	No	A human-readable, friendly label associated to the state. Used for diagnostic and logging; only the <code>state</code> field must be used for service logic. This label may be left empty.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields associated with this provisional response. Convenient to transport, for instance, a transaction secret code that would be required to complete the transaction.
stateSpecificFields	hos21:VendorSpecificFields	Yes	Optional fields, encoded as vendor specific fields, that qualify the state further. For instance, if the state is "AmendRequested", this element may contain the list of the fields that the sending user needs to amend, with their current value.

As explained above, this is just another way to transport the provisional response elements that were available in a HWS/*remittance* provisional response. These elements functionally map as follows:

- `PendingTransactionStatus.maxCompletionDate` corresponds to the HWS/*remittance* response `provisionalResponse.maxCompletionDate` element,
- `PendingTransactionStatus.state` corresponds to the HWS/*remittance* response `provisionalResponse.state` element,
- `PendingTransactionStatus.stateLabel` corresponds to the HWS/*remittance* response `provisionalResponse.stateLabel` element,
- `PendingTransactionStatus.vendorSpecificFields` corresponds to the HWS/*remittance* response `vendorSpecificFields` element.

However, since this `PendingTransactionStatus` is filled after the HWS/*remittance* provisional response was generated, the transaction's state may have evolved, and the `state/stateLabel` elements may have updated values.

As the transaction evolves, Vendor Specific Fields may also be different, and may need to be taken into account by the sending systems to update its local transaction view or notify the end-user. Such cases are discussed when designing custom call flows with the HomeSend platform provider prior to implementing a HWS connector.

Notice that the `maxCompletionDate` value won't change over time.

8.1.9 CompleteTransactionStatus Structure [2.2]

The `hws22:CompleteTransactionStatus` structure is another possible choice in a `hws22:TransactionStatus`, transporting the same information elements as in a final HWS/*remittance* response. It contains the following elements:

Element Name	Type	Optional	Description
transactionSummary	<code>hws:TransactionSummary</code>	No	The transaction summary that would be returned in a final HWS/ <i>remittance</i> response, containing the charged and credited amounts, as well as the transaction fees.
vendorSpecificFields	<code>hos21:VendorSpecificFields</code>	Yes	Vendor Specific Fields that would be returned in a final HWS/ <i>remittance</i> response.

8.1.10 RejectedTransactionStatus Structure [2.2]

The `hws22:RejectedTransactionStatus` structure is the last possible choice in a `hws22:TransactionStatus`, transporting the same information elements as in a final HWS/*remittance* fault. It contains the following elements:

Element Name	Type	Optional	Description
code	<code>xsd:string</code>	No	The error code that would be returned in a HWS/ <i>remittance</i> fault. Formatted as a string like "ER0000".
message	<code>xsd:string</code>	Yes	The message associated to the error code, as it would be returned in a HWS/ <i>remittance</i> fault.

Possible values for those `code` and `message` elements are exactly the same as in a HWS/*remittance* fault.

8.1.11 HomeSendServiceException Structure

The `hws:HomeSendServiceException` structure extends the `hos:BaseException` and is used to implement the single SOAP Fault that HomeSend can generate.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:string	No	Inherited from <code>hos:BaseException</code> . The error code, as described later. Formatted as a string like ER0000.
message	xsd:string	No	Inherited from <code>hos:BaseException</code> . A message associated to the error code. This message is in plain English and is not meant to be displayed to the end-user.
parameters	xsd:string	Yes	Inherited from <code>hos:BaseException</code> . Zero or more of these elements may be present. They contain additional information related to the error, as described later.
transactionId	hos:transactionId	No	The leg transaction ID, from the request the fault is a "response" to. This parameter is always provided by HomeSend, and enables to correlate the SOAP Fault with the request, independently from the tcp transport layer. Useful for traceability and non-repudiation.
hsTransactionId	hos:transactionId	Yes	The HomeSend transaction ID, if it was available when generating the fault. In case of low-level errors (such as application-level parameters validation), this ID may not be provided. Even if its technical type allows for more characters, <code>hsTransactionIds</code> are guaranteed to only contain <code>[A-Za-z0-9_-]</code> only.

8.2 Operations

This section describes the operations HomeSend makes available to a sender E-Wallet system that needs to trigger a remittance (one-shot or with advice of charge) or an e-recharge through the hub.

Each operation is made of one request, as a single *operationRequest* element whose type is *OperationRequest*, and a response, taking a single *operationResponse* element whose type is *OperationResponse*.

These types are described below.

8.2.1 HWS/adviceOfCharge

8.2.1.1 Request

The *HWS/adviceOfCharge* request takes a `hws:AdviceOfChargeRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password the sender E-Wallet system should use to connect onto HomeSend.

transactionId	hos:transactionId	No	The originating transaction ID, generated by the E-Wallet system to reference the transaction on its side. Must be unique per E-Wallet system.
sourceUri	hos:uri	No	The sender end-user's URI.
destinationUri	hos:uri	No	The receiver end-user's URI.
amountInformation	hos:AmountInformation	No	If this is a standard advice of charge, this amount to transfer, with its associated currency. This amount is considered to include the sender-payable service charges for the "receiver payer" proposal, or is considered as the principal, net amount to send, not including any service charges, for the "sender payer" proposal. For a reverse quote, this is the amount the sender wants the beneficiary to receive, with the (beneficiary's) associated currency. Grace and validity, if provided, are ignored in this context.
cbsType	hws:remittanceCbsType	Yes	The type of service this transfer refers to (B2B, P2P, etc). Check the Use Cases documentation for additional details about possible values.
routingTag	hos:routingTag	Yes	A routing tag to use to pre-select the corridor(s) to use. Unless you were explicitly told to use one by your HomeSend platform provider, you should not set it. See the introduction to HomeSend's routing for details. If not provided, no pre-selection occurs.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields generated by the sending E-Wallet system. They are forwarded in next messages according to the HomeSend configuration.
reverseQuote	hws23:ReverseQuote	Yes	If provided, this adviceOfCharge request will be interpreted as a "reverse quote" request, i.e. the sending end-users wants to know how much he/she will be charged so that the beneficiary receives the amount indicating in amountInformation. In this case, this reverseQuote element contains the charging/sending currency as a subelement.

Notice that no `hsTransactionId` is provided, as this request opens a new transaction. The HomeSend transaction ID will be generated by HomeSend itself.

8.2.1.2 Response

The `HWS/adviceOfCharge` response takes a `hws:AdviceOfChargeResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, generated by HomeSend. Should be reused in next operations that part of the same transaction (in particular, the associated <code>HWS/remittance</code> request). Even if its technical type allows for more characters, <code>hsTransactionIds</code> are guaranteed to only contain <code>[A-Za-z0-9_-]</code> only.
senderPayerProposal	hws:TransactionSummary	No	The advice of charge information in case of the sender pays for the service charges. This proposal is always provided.

receiverPayerProposal	hws:TransactionSummary	No	The advice of charge information in case of the receiver pays for the service charges. While the element is always provided, its value may be nil/null if the received amount in this case is negative (≤ 0). It avoids proposing a meaningless option to the end-user.
expirationDelay	hws:aocValidity	No	The advice of charge's validity period, in seconds. (integer, > 0). Instead of considering this field, the expirationDate may be preferred, as it provides a more absolute way to determine the advice of charge proposal expiration date.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Forwarded Vendor Specific Fields, if any.
expirationDate	xsd:dateTime	No	The date/time of the advice of charge end of validity period. This completes the expirationDelay element with a more absolute way to determine when the advice of charge proposal expires. Implementers are advised to discard the expirationDelay element and prefer this one, at their convenience.

The source E-Wallet system vendor is free to implement a way to notify the end-user when the advice of charge's validity has expired, and/or to prevent the end-user from accepting a proposal once it has expired.

In any case, HomeSend will reject associated HWS/remittance requests, if received after this timeout.

8.2.1.3 Faults

In case of an error, HomeSend replies with a `hws:HomeSendServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Error Type	Typical Message	Specific Description (in this context)
ER0001	Final failure	Service not authorized	The Remittance (thus advice of charge) service is not supported on this platform.
ER0002	Final failure	Invalid input value	An application-level validator detected at least one syntactically invalid value. This error comes in addition to possible SOAP faults that may be generated by XML schema validators.
ER0003	Inconclusive	Request already in progress	This advice of charge has already been received and has not responded yet. This error usually occurs when retransmitting a request (real-time retry).
ER0004	Final failure	Service request license reached	The Remittance service is supported on this platform, but due to licensing limitations the system was unable to process the request.
ER0005	Final failure	Destination system route error	Unable to find a receiver E-Wallet system to serve the destination end-user (but a corridor was found).
ER0007	Final failure	Destination service provider not allowed	Unable to find a corridor to route the transaction (service routing error); the destination service provider cannot be reached from this sender.
ER0008	Final failure	Internal error	An unspecified, internal error occurred.
ER0010	Final failure	Unknown receiver	No routing table for the provided destination, or unable to find a receiver E-Wallet that servers this destination.
ER0011	Final failure	Exchange rate error	The system was unable to compute an amount, due to a missing exchange rate or an unsupported currency.

ER0014	Final failure	Authentication error	Invalid login or password.
ER0016	Final failure	Service charge computation error	HomeSend was unable to compute a service charge, computed a negative service charge, or computed a negative transferred amount due to high service charges.
ER0020	Final failure	Maximum sending amount exceeded Minimum sending amount not reached	The sender is not allowed to send this much/ little amount of money to this destination.
ER0034	Transient failure	Unexpected request	Receiving this error is an abnormal case, as it would mean that the HWS/ <i>adviceOfCharge</i> request has been received at a time when the transaction's state machine, as managed by HomeSend, does not expect it. It usually means that the transaction call flow has not been followed by the sending service provider, i.e. requests are sent in an incorrect order for a given transaction. Cases of ER0034 being returned here include a HWS/ <i>adviceOfCharge</i> after a HWS/ <i>remittance</i> , which is illegal protocol-wise and is likely due to a faulty HWS client implementation.
ER0035	Transient failure	Destination temporarily unavailable	A non-permanent routing error occurred; please try again later. This could happen when the MNP resolution for a ported-out destination failed, or when the destination system is temporarily unreachable.
ER0036	Transient failure	Service temporarily unavailable	HomeSend is temporarily unavailable to handle the request (typical reasons: overload protection, maintenance)
ER0037	Final failure	Missing required fields	At least one vendor specific field (VSF) is missing, preventing one of the E-Wallet system from working correctly. Various VSFs may be required depending on the corridor and/or the service providers involved for the transaction, and are documented in the agreements between parties that are interconnected through HomeSend.
ER0040	Final failure	Transaction not authorized	The transaction was barred for administrative or regulatory reasons.
ER0045	Final failure	Unsupported destination currency	When requesting a "reverse" advice of charge, the provided destination currency is not supported by the beneficiary's account.

8.2.2 HWS/remittance

8.2.2.1 Request

The HWS/*remittance* request takes a `hws:RemittanceRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password the sender E-Wallet system should use to connect onto HomeSend.
transactionId	hos:transactionId	No	The originating transaction ID, generated by the E-Wallet system to reference the transaction on its side. Must be unique per E-Wallet system. If the remittance follows an advice of charge, the same originating transaction ID shall be used.

hsTransactionId	hos:transactionId	Yes	Only provided in remittances following an advice of charge. The HomeSend transaction ID, as provided in the advice of charge response this remittance is associated with.
sourceUri	hos:uri	Yes	The sender end-user's URI. For a remittance associated with an advice of charge, can be omitted, or, if provided, must be the same as in the advice of charge request.
destinationUri	hos:uri	Yes	The receiver end-user's URI. For a remittance associated with an advice of charge, can be omitted, or, if provided, must be the same as in the advice of charge request.
amountInformation	hos:AmountInformation	No	The amount to transfer, with its associated currency. This amount is considered to include the sender-payable service charges if the payer element is set to RECEIVER, or is considered as the principal, net amount to send if the payer element is set to SENDER (i.e. additional service charges may be paid by the sender). Grace and validity, if provided, are ignored in this context. For remittance associated with a standard advice of charge, must be the same as in the advice of charge request. If not, the ER0017 will be replied (see below). For remittance associated with a reverse advice of charge, must be the same as the initialAmount returned in the advice of charge response. If not, the ER0017 will be replied (see below).
payer	hos:payer	Yes	This flag indicates two things: whether the provided amount to send includes the service charges (RECEIVER) or not (SENDER), and who should pay the charges for charges that can be paid by any party (some charges can only be paid by a side, in this case this indicator has no impact on them). This indicator is usually explicitly set when the remittance is associated with an advice of charge. If not provided, it is defaulted to RECEIVER.
cbsType	hws:remittanceCbsType	Yes	The type of service this transfer refers to (B2B, P2P, etc). Check the Use Cases documentation for additional details about possible values. If not provided, defaulted to EMoneyToEMoneyTransfer. For remittance associated with an advice of charge, this field is ignored and may be discarded. The value provided with the advice of charge takes precedence.
routingTag	hos:routingTag	Yes	A routing tag to use to pre-select the corridor(s) to use. Unless you were explicitly told to use one by your HomeSend platform provider, you should not set it. See the introduction to HomeSend's routing for details. If not provided, no pre-selection occurs. For remittance associated with an advice of charge, this field is ignored and may be discarded. The value provided with the advice of charge takes precedence.
reservationRequired	xsd:Boolean	Yes	This flag indicates if the OEWS call flow (reserve + charge) should be used for this remittance. If set to false, the remittance is executed without calling the sender E-Wallet system for end-user balance reservation. If not provided, defaulted to true.
senderInformation	hos:senderInformation	Yes	The verified sender information (usually a name and a postal address), as it may be required by regulations.

description	hos:description	Yes	An optional label describing the transaction purpose (for instance). Such information is usually user-provided and is forwarded as is to the receiver E-Wallet system that could display it to the receiver end-user, and to the sender E-Wallet system for accounting during reservation/charge cycle.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields generated by the sending E-Wallet system. They are forwarded in next messages according to the HomeSend configuration.

Notice that no `hsTransactionId` is provided for single-shot remittance only, as such a request opens a new transaction in this case. When the remittance correlates an advice of charge, this identifier is mandatory (otherwise, HomeSend will consider a new remittance, possibly using different foreign exchange rates from the one used for the advice of charge).

8.2.2.2 Response

The `HWS/remittance` response takes a `hws:RemittanceResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID. Even if its technical type allows for more characters, <code>hsTransactionIds</code> are guaranteed to only contain <code>[A-Za-z0-9_-]</code> only.
transactionSummary	hws:TransactionSummary	No	The summary of the (successful) remittance. Includes the charged amount in the sender's currency, the credited amount in the destination's currency, as well as the transfer fees in the sender's currency. If the <code>provisionalResponse</code> element is provided, this element is still present, but explicitly nulled (<code>xsi:nil</code> is set to true). Otherwise, this element always contains an actual transaction summary.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Forwarded Vendor Specific Fields, if any.
provisionalResponse	hws22:ProvisionalResponse	Yes	If this element is present, the response must be considered "provisional", indicating that the transaction has been taken into account and will take some time to process. Further client-side polling (using <code>HWS/status</code> requests) will be needed to get the transaction's final status. This element contains a deadline date that is meant to notify the sending end-user to not "panic" if no final response is received before this date, as it may take several days or weeks depending on the receiving systems & business (cash-out, banks, etc). However, if the transaction's status did not change after it, it is likely that a manual transaction repairing action must be performed on destination side, and the sending service provider should contact the HomeSend platform provider.

8.2.2.3 Faults

In case of an error, HomeSend replies with a `hws:HomeSendServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Error Type	Typical Message	Specific Description (in this context)
ER0001	Final failure	Service not authorized	The Remittance service is not supported on this platform.
ER0002	Final failure	Invalid input value	An application-level validator detected at least one syntactically invalid value. This error comes in addition to possible SOAP faults that may be generated by XML schema validators.
ER0004	Final failure	Service request license reached	The Remittance service is supported on this platform, but due to licensing limitations the system was unable to process the request.
ER0005	Final failure	Destination system route error	Unable to find a receiver E-Wallet system to serve the destination end-user (but a corridor was found).
ER0007	Final failure	Destination service provider not allowed	Unable to find a corridor to route the transaction (service routing error); the destination service provider cannot be reached from this sender.
ER0008	Final failure	Internal error	An unspecified, internal error occurred. The remittance request has not been processed.
ER0010	Final failure	Unknown receiver	No routing table for the provided destination, or unable to find a receiver E-Wallet that serves this destination.
ER0011	Final failure	Exchange rate error	The system was unable to compute an amount, due to a missing exchange rate or an unsupported currency.
ER0012	Final failure	Sender reservation error Sender charging error	The sender E-Wallet system was unable to reserve or charge the requested amount (high-level error).
ER0013	Final failure	Receiver crediting error	The receiver E-Wallet system was unable to credit the user (high-level error).
ER0014	Final failure	Authentication error	Invalid login or password.
ER0016	Final failure	Service charge computation error	HomeSend was unable to compute a service charge, computed a negative service charge, or computed a negative transferred amount due to high service charges.
ER0017	Final failure	Correlation error	The HomeSend transaction ID was not found (the transaction does not exist), or at least one of the transactionId, sourceUri, destinationUri, amountInformation fields mismatched an existing transaction with the same HomeSend transaction ID.
ER0020	Final failure	Maximum sending amount exceeded Minimum sending amount not reached	The sender is not allowed to send this much/little money, either due to a control on the sender E-Wallet system, or on HomeSend itself (corridor configuration). This fault contains a single argument in parameters[0], which is the maximum/minimum allowed amount. If provided by HomeSend, this amount is rounded to N decimal digits, as configured for the sending service provider. If provided by a sender E-Wallet system (upon reservation), the amount is transparently forwarded here.
ER0021	Final failure	Maximum reception amount exceeded	The receiver is not allowed to receive as much (controlled by the receiver E-Wallet system) in a single transaction.

ER0022	Final failure	Maximum number of transactions exceeded by sender	The sender is not allowed to perform a new transaction, either due to a control on the sender E-Wallet system, or on HomeSend itself (corridor configuration). This fault contains two additional arguments: parameter[0]: the maximum authorized number of transactions, parameter[1]: the considered period, in number of days, for such a check (the limit is a max number of transactions per period).
ER0023	Final failure	Maximum cumulative amount of transactions exceeded by sender	The sender is not allowed to perform a new transaction, either due to a cumulated amount control on the sender E-Wallet system, or on HomeSend itself (corridor configuration). This fault contains two additional arguments: parameter[0]: the maximum cumulated authorized amount, in sender's currency (rounded to N significant digits, according to HomeSend configuration, or transmitted as is from the sender E-Wallet system), parameter[1]: the considered period, in number of days, for such a check (the limit is a max number of transactions per period).
ER0024	Final failure	Maximum number of transactions exceeded by receiver	The receiver is not allowed to receive a new transaction, either due to a control performed by the receiver E-Wallet system, or by HomeSend itself (corridor configuration).
ER0025	Final failure	Maximum cumulative amount of transactions exceeded by receiver	The receiver is not allowed to receive a new transaction, either due to a cumulated amount control performed by the receiver E-Wallet system, or by HomeSend itself (corridor configuration).
ER0026	Final failure	Maximum number of transactions exceeded by sending network	The sender could not perform the transaction due to his/her service provider having reached its maximum number of transactions per period, for the selected corridor.
ER0027	Final failure	Maximum cumulative amount of transactions exceeded by sending network	The sender could not perform the transaction due to his/her service provider having reached its maximum cumulated transfers per period, for the selected corridor. Also sent if the sending service provider's balance is depleted. In both cases, this is a critical error as the service provider is unlikely to perform any other transactions through the hub (for this corridor, or for any corridor).
ER0028	Final failure	Maximum number of transactions exceeded by receiving network	HomeSend could not perform the transaction due to the selected receiving service provider having reached its maximum number of transactions per period, for the selected corridor.
ER0029	Final failure	Maximum cumulative amount of transactions exceeded by receiving network	HomeSend could not perform the transaction due to the selected receiving service provider having reached its maximum cumulated transfers per period, for the selected corridor.
ER0031	Final failure	Incompatible destination service	Due to different network APIs with different capabilities, HomeSend was unable to credit the destination.
ER0032	Final failure	Incompatible source service	Due to different network APIs with different capabilities, HomeSend was unable to perform a reservation.

ER0033	Final failure	Transaction cancelled	Remittance with advice of charge only: the associated advice of charge has expired.
ER0034	Transient failure	Unexpected request	The current transaction's state does not allow such a request at this moment (even to retransmit the latest known response).
ER0035	Transient failure	Destination temporarily unavailable	A non-permanent routing error occurred; please try again later. This could happen when the MNP resolution for a ported-out destination failed, or when the destination system is temporarily unreachable.
ER0036	Transient failure	Service temporarily unavailable	HomeSend is temporarily unavailable to handle the request (typical reasons: overload protection, maintenance)
ER0037	Final failure	Missing required fields	At least one vendor specific field (VSF) is missing, preventing one of the E-Wallet system from working correctly. Various VSFs may be required depending on the corridor and/or the service providers involved for the transaction, and are documented in the agreements between parties that are interconnected through HomeSend.
ER0040	Final failure	Transaction not authorized	The transaction was barred for administrative or regulatory reasons.
ER0042	Final failure	Transaction Expired	The transaction expired on the receiving side, i.e. the receiving end-user was requested to perform some actions that were not done on time (registration, cash-out, etc). This error code is typically only returned as status query responses (as a response to a retransmitted HWS/remittance request, or in a HWS/status response).

8.2.3 HWS/eRecharge

This operation is deprecated and shall not be used anymore.

8.2.4 HWS/cancellation

Use this request to explicitly cancel a running advice of charge. Sending it to cancel an unknown or already expired advice of charge after its lingering period will result in an error (fault).

8.2.4.1 Request

The HWS/cancellation request takes a `hws:CancellationRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password the sender E-Wallet system should use to connect onto HomeSend.
transactionId	hos:transactionId	No	The originating transaction ID, as used in the advice of charge request/response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as provided in the advice of charge response.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields generated by the sending E-Wallet system. They are forwarded in next messages according to the HomeSend configuration.

reason	hws23:CancellationReason	Yes	An optional cancellation reason. If not provided, defaulted to 1 / Unspecified.
--------	--------------------------	-----	---

Cancellation reasons are made of a numeric code and a human-readable message. While there are no formal constraints regarding their code, the following reasons are suggested:

Code	Typical Message	Specific Description (in this context)
1	Unspecified	No particular reason was specified.
2	End-user cancellation	The end-user decided to cancel the transaction.
3	Administrative cancellation	The transaction was cancelled administratively.

Other reasons may be added depending on the business needs, and should be notified to the platform provider before being used.

8.2.4.2 Response

The HWS/*cancellation* response takes a `hws:CancellationResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Forwarded Vendor Specific Fields, if any.

When sending a HWS/*cancellation* request to cancel a transaction whose credit is in progress (HWS/*remittance* provisional response received or equivalent), receiving such a response only guarantees that the cancellation request was submitted successfully to the destination system. However, this system may not be able to process this cancellation immediately (if not at all), and you should then continue polling HomeSend until the transaction is completed. Even if you asked for a cancellation, the request may eventually be accepted.

In all other contexts (no provisional response received yet, for instance after an advice of charge response but before a remittance request), receiving this response is enough to guarantee that the advice of charge was cancelled.

8.2.4.3 Faults

In case of an error, HomeSend replies with a `hws:HomeSendServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Error Type	Typical Message	Specific Description (in this context)
ER0001	Final failure	Service not authorized	The Remittance service is not supported on this platform.
ER0002	Final failure	Invalid input value	An application-level validator detected at least one syntactically invalid value. This error comes in addition to possible SOAP faults that may be generated by XML schema validators.
ER0008	Final failure	Internal error	An internal error occurred.
ER0009	Inconclusive	Inconclusive request	No response was received from the destination system after forwarding the cancellation order. The transaction may or may not be cancelled. The sending system may retry to send a cancellation request again, but this is not mandatory.
ER0014	Final failure	Authentication error	Invalid login/password in request.
ER0017	Final failure	Correlation error	Invalid HomeSend transaction ID (the transaction was not found) or invalid originating transaction ID for this transaction.
ER0018	Final failure	Transaction already cancelled	The cancellation was already performed / the advice of charge has already expired.
ER0031	Final failure	Incompatible destination service	The destination system does not support transaction cancellations.
ER0034	Transient failure	Unexpected Request	The transaction state does not allow a cancellation (for instance, HomeSend is actively processing the transaction).
ER0036	Transient failure	Service temporarily unavailable	HomeSend is temporarily unavailable to handle the request (typical reasons: overload protection, maintenance)
ER0037	Final failure	Missing required fields	At least one vendor specific field (VSF) is missing, preventing one of the E-Wallet system from working correctly. Various VSFs may be required depending on the corridor and/or the service providers involved for the transaction, and are documented in the agreements between parties that are interconnected through HomeSend.

ER0044	Final failure	Unable to complete operation	The destination system does not support transaction cancellations.
--------	---------------	------------------------------	--

8.2.5 HWS/status [2.2]

This operation is used to get the status of a remittance the HWS client received a provisional response for. It can also be used safely after a (synchronous or asynchronous) transaction completed to get its status if it was missed for whatever reason, even days or weeks after completion

However, HomeSend may not be able to retrieve old transactions by this mean, i.e. transactions that are typically closed for 2 weeks or more. Should you need to get such transactions' status, other interfaces are available to query them, such as the HomeSend Web Customer Care.

8.2.5.1 Request

The *HWS/status* request takes a `hws22:StatusRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password the sender E-Wallet system should use to connect onto HomeSend.
transactionId	hos:transactionId	No	The originating transaction ID of the transaction to query, as used in the advice of charge request/response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID of the transaction to query, as provided in the advice of charge response. As this field is mandatory, you cannot query the status of a transaction for which you didn't have any response yet. In such cases, just replay your initial request as is within <i>MAX_{reconciliation}</i> after the first attempt.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields generated by the sending E-Wallet system, and related to the status operation itself. Reserved for future usage, currently ignored if provided.

8.2.5.2 Response

The *HWS/status* response takes a `hws22:StatusResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
status	hws22:TransactionStatus	No	The status of the transaction, as a choice between three structures: pending, rejected, complete.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	HomeSend-generated Vendor Specific Fields. Reserved for future usage, not provided for now.

The `status` choice is designed to “tunnel” the information elements you would normally have in a corresponding final *HWS/remittance* response:

- If the `complete` choice is selected, it contains a `hws22:CompleteTransactionStatus` structure that transports the `transactionSummary` and `vendorSpecificFields` fields that you would have in a successful *HWS/remittance* response,

- If the `rejected` choice is selected, it contains a `hws22:RejectedTransactionStatus` structure that transports the `code` and `message` fields that you would have in a `HWS/remittance` fault. The code field can be parsed and used as if it was extracted from such a fault for your client-side error handling.
- If the `pending` choice is selected, it contains a `hws22:PendingTransactionStatus` structure that transports the current transaction's state, an associated label, a recall of the maximum completion date, and possibly updated vendor specific fields.

When the transaction is still pending, the following transaction states may be returned:

State (Identifier, case sensitive)	Typical State Label	Description
Processing		HomeSend is actively processing the transaction.
InProgress		The transaction has been submitted to the destination system, we are now waiting for it to process it. This is a common state when connected to a destination party that uses an asynchronous processing model.
AcceptedInProgress		The outcome of the transaction is already known and it will be OK, however some additional processing is still required to close the transaction. The transaction summary is not available at this time, and you should wait for a final Complete status to announce the sending en-user that the transaction was successful.
RejectedInProgress		The outcome of the transaction is already known and it will be NOK, however some additional processing is still required to close the transaction. The error code is not available at this time, and you should wait for a final Rejected status to announce the sending en-user that the transaction was failed.
WaitingForSenderInput		An action on the source side is required to continue the transaction, i.e. an advice of charge was provided, waiting for user confirmation.
AmendRequested		The destination system requested the sender to update some transaction details, as it is unable to complete it with the current data. A typical case would be requesting an update on a misspelled beneficiary's name or address. In this case, the state specific fields contains the fields to update, with their current values.
VerificationRequired		The transaction is blocked due to an inconclusive sanction list screening operation. The sending service provider's compliance officer is invited to resolve the case with the sanction list filtering system provider directly. The transaction will be resumed automatically when the case is resolved. Such a state is only possible if the sending service provider delegates sanction list filtering tasks to the platform provider.

(any other value)		A compliant HWS client MUST be able to accept any other value here. Even if such a value can't be explicitly understood, the sending system must consider it as if it was a "InProgress" state. HWS clients are not required to be able to understand every possible state value.
-------------------	--	---

Keep in mind that the state label may change without notice and is provided for information only. The state value, however, can be used in the sending system's business logic.

8.2.5.3 Faults

In case of an error, HomeSend replies with a `hws:HomeSendServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Error Type	Typical Message	Specific Description (in this context)
ER0001	Final failure	Service not authorized	The Remittance service is not supported on this platform.
ER0002	Final failure	Invalid input value	An application-level validator detected at least one syntactically invalid value. This error comes in addition to possible SOAP faults that may be generated by XML schema validators.
ER0008	Final failure	Internal error	An internal error occurred.
ER0014	Final failure	Authentication error	Invalid login/password in request.
ER0017	Final failure	Correlation error	Invalid HomeSend transaction ID (the transaction was not found) or invalid originating transaction ID for this transaction.
ER0036	Transient failure	Service temporarily unavailable	HomeSend is temporarily unavailable to handle the request (typical reasons: overload protection, maintenance)
ER0037	Final failure	Missing required fields	At least one vendor specific field (VSF) is missing, preventing one of the E-Wallet system from working correctly. Various VSFs may be required depending on the corridor and/or the service providers involved for the transaction, and are documented in the agreements between parties that are interconnected through HomeSend.

8.2.6 HWS/amend [2.3]

This operation is used to update one or several user-information-related fields for asynchronous, in-progress transactions.

8.2.6.1 Request

Such a request is typically sent once the sending system is aware that the transaction is in a "AmendRequested" substate. However, unsolicited amend requests are also supported, but are more likely to be rejected by the destination system if it does not expect them.

The `HWS/status` request takes a `hws22:StatusRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
--------------	------	----------	-------------

security	hos:Security	No	The login/password the sender E-Wallet system should use to connect onto HomeSend.
transactionId	hos:transactionId	No	The originating transaction ID of the transaction to query, as used in the advice of charge request/response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID of the transaction to query, as provided in the advice of charge response. As this field is mandatory, you cannot query the status of a transaction for which you didn't have any response yet. In such cases, just replay your initial request as is within <i>MAX_{reconciliation}</i> after the first attempt.
amendedFields	hos21:VendorSpecificFields	No	The list of fields to amend. For each amended field, a new value is provided. You should amend at least one field per request. You can amend multiple fields at once. If the same field (same vendorId, fieldId) is provided multiple times, only its last provided value is taken into account.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields generated by the sending E-Wallet system, and related to the status operation itself. Reserved for future usage, currently ignored if provided.

8.2.6.2 Response

The HWS/amend response takes a `hws23:AmendResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	HomeSend-generated Vendor Specific Fields. Reserved for future usage, not provided for now.

Receiving such a response means that the destination system accepted the update. The transaction is then continued as usual, and you should still keep polling HomeSend for its final status.

8.2.6.3 Faults

In case of an error, HomeSend replies with a `hws:HomeSendServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Error Type	Typical Message	Specific Description (in this context)
ER0001	Final failure	Service not authorized	The Remittance service is not supported on this platform.
ER0002	Final failure	Invalid input value	An application-level validator detected at least one syntactically invalid value. This error comes in addition to possible SOAP faults that may be generated by XML schema validators.
ER0008	Final failure	Internal error	An internal error occurred.
ER0009	Inconclusive	Inconclusive request	No response was received from the destination system after forwarding the update order. The transaction may or may not be amended. The sending system may retry to send an amend request again, but this is not mandatory. Checking the transaction's state will help take a decision.

ER0014	Final failure	Authentication error	Invalid login/password in request.
ER0017	Final failure	Correlation error	Invalid HomeSend transaction ID (the transaction was not found) or invalid originating transaction ID for this transaction.
ER0018	Final failure	Transaction already cancelled	The cancellation was already performed / the advice of charge has already expired.
ER0031	Final failure	Incompatible destination service	The destination system does not support transaction updates.
ER0034	Transient failure	Unexpected Request	The transaction state does not allow amending at this time(for instance, the transaction is not waiting for an asynchronous completion).
ER0036	Transient failure	Service temporarily unavailable	HomeSend is temporarily unavailable to handle the request (typical reasons: overload protection, maintenance)
ER0037	Final failure	Missing required fields	At least one vendor specific field (VSF) is missing, preventing one of the E-Wallet system from working correctly. Various VSFs may be required depending on the corridor and/or the service providers involved for the transaction, and are documented in the agreements between parties that are interconnected through HomeSend.
ER0044	Final failure	Unable to complete operation or Cannot amend at least one of these fields	HomeSend did not accept the amend request due to an attempt to amend fields that are not allowed to be amended, or due to another business processing error.

8.2.7 HWS/deviceWatchdog [2.2]

This operation is used to monitor a HWS link actively. A HWS client may regularly send a device watchdog request to check that the HomeSend service is still reachable and active.

8.2.7.1 Request

The *HWS/deviceWatchdog* request takes a `hws22:DeviceWatchdogRequest` structure as argument, which is an empty structure.

Unlike the other requests, this one should not be digitally signed, even if the HWS link is configured to require digital signatures. If signed, HomeSend won't validate its signature but will still accept it as a valid request, provided its content is valid.

8.2.7.2 Response

The *HWS/deviceWatchdog* response takes a `hws22:DeviceWatchdogResponse` structure as argument, which is an empty structure.

Unlike the other responses, this one will never be signed digitally, even if the HWS link is configured to require digital signatures. The HWS client should be able to accept the response as a valid one even if not signed in this case.

8.2.7.3 Faults

This operation does not return any fault.

9 Originating Endpoint Web Service (OEWS)

This section describes types and operations that are defined into the following namespaces:

- `http://www.eservglobal.com/homesend/oews/2.0`, aliases by `oews`,
- `http://www.eservglobal.com/homesend/oews/2.1`, aliases by `oews21`,
- `http://www.eservglobal.com/homesend/oews/2.2`, aliases by `oews22`.

Common types are referenced with the `hos/hos21/hos22` prefixes, and defined in section [Common Data Types Definition](#).

9.1 Specific Types

The following types are specific to the different OEWS messages.

9.1.1 Reservation Identifier

The `oews:reservationId` simple type defines the way a reservation token/identifier should be transmitted to HomeSend. This is a simple string, from 1 to 30 characters, containing ASCII characters only.

9.1.2 Chargeback Reason

The `oews22:ChargebackReason` structure is used to identify the reason why a chargeback operation is performed.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:int	No	A normative code to formally identify the reason why the chargeback occurs. An OEWS business logic shall only rely on this field.
message	xsd:string	No	A human readable, friendly description associated to the code above. To be used for diagnostic & logging. It may be empty, and may vary for a same code. This field must not be used by the OEWS business logic.

Possible reasons are described with the [chargeback request description](#).

9.1.3 OriginatingServiceException Structure

The `oews:OriginatingServiceException` structure extends the `hos:BaseException` and is used to implement the single SOAP Fault that the sender E-Wallet system can generate.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:string	No	Inherited from <code>hos:BaseException</code> . The error code, as described later. Formatted as a string like <code>ER0000</code> .

message	xsd:string	No	Inherited from <code>hos:BaseException</code> . A message associated to the error code. This message is in plain English and is not meant to be displayed to the end-user.
parameters	xsd:string	Yes	Inherited from <code>hos:BaseException</code> . Zero or more of these elements may be present. They contain additional information related to the error, as described later.
transactionId	hos:transactionId	No	The leg transaction ID, from the request the fault is a "response" to. This ID is always provided in the request and shall be always included in such a fault.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, from the request the fault is a "response" to. This ID is always provided in the request and shall be always included in such a fault. Even if its technical type allows for more characters, <code>hsTransactionIds</code> are guaranteed to only contain <code>[A-Za-z0-9_-.]</code> only.

9.2 Operations

This section describes the operations the sender E-Wallet must publish if it supports the reservation mechanism.



Warning:

To guarantee HomeSend's SLA, 90% of these operations must be replied in less than 500ms.

Each operation is made of one request, as a single *operationRequest* element whose type is *OperationRequest*, and a response, taking a single *operationResponse* element whose type is *OperationResponse*.

These types are described below.

9.2.1 OEWS/reserve

This request is sent by HomeSend when a remittance transaction requires a sender's balance reservation. The targeted sender E-Wallet system shall then reply with a reservation identifier (if the reservation is OK) and should expect an additional OEWS/*charge* request from HomeSend to confirm the reservation, or an OEWS/*release* request to cancel it explicitly.

In some particular cases, called "Ambiguous cases", HomeSend is unable to know if the destination was correctly credited or not and cannot confirm or cancel this sender-side reservation. In this case, the E-Wallet system shall keep the reservation until it has been confirmed or cancelled manually.

To be able to support such cases, **the reservation shall not expire for at least 3 months**. Your HomeSend platform provider may give you a shorter delay, depending on its ambiguous cases resolution policies.

9.2.1.1 Request

The OEWS/*reserve* request takes a `oews:ReserveRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
--------------	------	----------	-------------

security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding HWS/ <i>remittance</i> request. The sender E-Wallet may ignore it and not use it to correlate with a previous sent remittance, as the systems implementing the HWS client and this OEWS server may not be directly connected.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, the unique identifier for the whole HomeSend transaction. May be used, or not, as a correlator by the targeted system. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_.-] only.
uri	hos:uri	No	The URI identifying the account to reserve.
amountInformation	hos:AmountInformation	No	The amount to reserve, with its associated currency. The amount may include additional service charges, if the sender pays them. The associated currency is the one as received in the initial HWS/ <i>remittance</i> request.
routingTag	hos:routingTag	Yes	This routing tag, if provided, may help the sender E-Wallet system identify the service provider the request is intended to, in case of a E-Wallet system hosting multiple service providers.
description	hos:description	Yes	A description labeling the transaction, as provided in the initial HWS/ <i>remittance</i> request, if available.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.

9.2.1.2 Response

The OEWS/*reserve* response takes a `oews:ReserveResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding reserve request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as received in the reserve request.
reservationId	tews:reservationId	No	The reservation identifier/token for this reservation.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.

Sending this response means that the reservation for the whole indicated amount was performed.

9.2.1.3 Faults

In case of an error, the sender E-Wallet system shall reply with a `oews:OriginatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
------------	-----------------	--

ER0001	Service not authorized	Reservation is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a reservation due to licensing issues (depends on the E-Wallet implementation).
ER0008	Internal error	An internal error occurred.
ER0009	Ambiguous case	The reservation system was unable to guarantee that the reservation was correctly recorded.
ER0010	Unknown receiver	No balance can be found for the provided URI, or unsupported URI scheme.
ER0011	Exchange rate error	The reservation currency is unsupported / could not use it to perform the reservation.
ER0012	Reservation error	The reservation could not be performed.
ER0014	Authentication error	Invalid login/password in request.
ER0020	Maximum reservation amount exceeded	The sender is not allowed to reserve as much. This fault contains a single argument in parameters[0], which is the maximum allowed amount.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The sender E-Wallet system should use this error if its reservation service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the sender E-Wallet system. When sending this error, the sender E-Wallet system must make sure that the reservation was not performed.

9.2.2 OEWS/charge

This request is used to confirm/commit a previous reservation.

Notice that this *charge* operation should not require vendor specific fields, contrary to the reserve operation that can be replied with the ER0037 Missing required fields error.

9.2.2.1 Request

The OEWS/charge request takes a `oews:ChargeRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.

transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding HWS/remittance request. The sender E-Wallet may ignore it and not use it to correlate with a previous sent remittance, as the systems implementing the HWS client and this OEWS server may not be directly connected.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, the unique identifier for the whole HomeSend transaction. May be used, or not, as a correlator by the targeted system. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
reservationId	oews:reservationId	No	The reservation ID to confirm, as provided in the associated reserve response.
amountInformation	hos:AmountInformation	No	The confirmed amount. Always equal to the reserved amount (same amount, same currency).
routingTag	hos:routingTag	Yes	This routing tag, if provided, may help the sender E-Wallet system identify the service provider the request is intended to, in case of a E-Wallet system hosting multiple service providers.
description	hos:description	Yes	A description labeling the transaction, as provided in the initial HWS/remittance request, if available.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.

9.2.2.2 Response

The OEWS/charge response takes a `oews:ChargeResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding charge and reserve requests.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as received in the charge and reserve requests.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.

Sending this response means that the reservation was confirmed and the user charged.

9.2.2.3 Faults

In case of an error, the sender E-Wallet system shall reply with a `oews:OriginatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Charging is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).

ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a reservation due to licensing issues (depends on the E-Wallet implementation).
ER0008	Internal error	An internal error occurred.
ER0009	Ambiguous case	The reservation system was unable to guarantee that the reservation was correctly confirmed.
ER0010	Invalid reservation identifier	The reservation ID is invalid on this system.
ER0011	Exchange rate error	The reservation currency is unsupported / could not use it to perform the reservation.
ER0012	Charging error	The user could not be charged.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	The originating ID or HomeSend transaction ID is invalid / not associated to the provided reservation ID.

9.2.3 OEWS/release

This request is used to cancel a previous reservation.

Notice that this *release* operation should not require vendor specific fields, contrary to the *reserve* operation that can be replied with the `ER0037 Missing required fields` error.

9.2.3.1 Request

The *OEWS/release* request takes a `oews:ReleaseRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding <i>HWS/remittance</i> request. The sender E-Wallet may ignore it and not use it to correlate with a previous sent remittance, as the systems implementing the HWS client and this OEWS server may not be directly connected.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, the unique identifier for the whole HomeSend transaction. May be used, or not, as a correlator by the targeted system. Even if its technical type allows for more characters, <code>hsTransactionIds</code> are guaranteed to only contain <code>[A-Za-z0-9_-]</code> only.
reservationId	oews:reservationId	No	The reservation ID to confirm, as provided in the associated <i>reserve</i> response.
routingTag	hos:routingTag	Yes	This routing tag, if provided, may help the sender E-Wallet system identify the service provider the request is intended to, in case of a E-Wallet system hosting multiple service providers.
description	hos:description	Yes	A description labeling the transaction, as provided in the initial <i>HWS/remittance</i> request, if available.

vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.
----------------------	----------------------------	-----	---

9.2.3.2 Response

The OEWS/*release* response takes a `oews:ReleaseResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding charge and reserve requests.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as received in the charge and reserve requests.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.

Sending this response means that the reservation was correctly released.

9.2.3.3 Faults

In case of an error, the sender E-Wallet system shall reply with a `oews:OriginatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Reservation release is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a reservation due to licensing issues (depends on the E-Wallet implementation).
ER0008	Internal error	An internal error occurred.
ER0009	Ambiguous case	The reservation system was unable to guarantee that the reservation was correctly released.
ER0010	Invalid reservation identifier	The reservation ID is invalid on this system.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	The originating ID or HomeSend transaction ID is invalid / not associated to the provided reservation ID / Mismatching amount.

9.2.4 OEWS/chargeback [2.2]

This request is currently used to revert a charge operation in the case where an asynchronous credit was finally rejected.

Notice that this *chargeback* operation should not require vendor specific fields, contrary to the reserve operation that can be replied with the ER0037 Missing required fields error.

9.2.4.1 Request

The OEWS/*chargeback* request takes a `oews22:ChargebackRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding HWS/ <i>remittance</i> request. The sender E-Wallet may ignore it and not use it to correlate with a previous sent remittance, as the systems implementing the HWS client and this OEWS server may not be directly connected.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, the unique identifier for the whole HomeSend transaction. May be used, or not, as a correlator by the targeted system. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
reservationId	oews:reservationId	Yes	The reservation ID corresponding to the charge/reserve operation. Only provided if a reservation was performed for this transaction, which is currently always the case.
amountInformation	hos:AmountInformation	No	The amount to re-credit/revert on the account. Always equal to the charged amount (same amount, same currency).
reason	oews22:ChargebackReason	No	The reason why the chargeback is performed. Possible reasons are described below.
routingTag	hos:routingTag	Yes	This routing tag, if provided, may help the sender E-Wallet system identify the service provider the request is intended to, in case of a E-Wallet system hosting multiple service providers.
description	hos:description	Yes	A description labeling the transaction, as provided in the initial HWS/ <i>remittance</i> request, if available.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.

Current possible chargeback reasons are:

Reason Code	Typical Message	Specific Description (in this context)
1	Transaction rejected (unspecified)	The transaction was rejected by the destination system after an asynchronous processing. The exact reason is unspecified.

New reasons will be introduced in next OEWS API versions, along with new OEWS/*chargeback* operation usages.

9.2.4.2 Response

The OEWS/*chargeback* response takes a `oews22:ChargebackResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
--------------	------	----------	-------------

security	hos:Security	No	The login/password HomeSend should use to connect onto the sender E-Wallet system.
transactionId	hos:transactionId	No	The originating transaction ID, as received in the corresponding charge and reserve requests.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as received in the charge and reserve requests.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.

Sending this response means that the reservation was correctly released.

9.2.4.3 Faults

In case of an error, the sender E-Wallet system shall reply with a `oews:OriginatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Reservation release is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a reservation due to licensing issues (depends on the E-Wallet implementation).
ER0008	Internal error	An internal error occurred.
ER0009	Ambiguous case	The reservation/charging system was unable to guarantee that the charging was correctly reversed.
ER0010	Invalid reservation identifier	The reservation ID is invalid on this system.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	The originating ID or HomeSend transaction ID is invalid / not associated to the provided reservation ID / Mismatching amount.

9.2.5 OEWS/deviceWatchdog [2.2]

This operation is used to monitor a OEWS link actively. HomeSend can be configured to regularly send a device watchdog request to check that your implementation is still reachable, online, and running a OEWS server implementation.

Supporting this operation is mandatory when implementing a OEWS server.

9.2.5.1 Request

The `OEWS/deviceWatchdog` request takes a `oews22:DeviceWatchdogRequest` structure as argument, which is an empty structure.

Unlike the other requests, this one is never signed digitally, even if the OEWS link is configured to require digital signatures.

9.2.5.2 Response

The OEWS/*deviceWatchdog* response takes a `oews22:DeviceWatchdogResponse` structure as argument, which is an empty structure.

Sending anything else is considered as a failed probe by HomeSend, which may then consider your system as down or malfunctioning.

Unlike the other responses, this one should not be digitally signed, even if the OEWS link is configured to require digital signatures. If signed, HomeSend won't validate its signature but will still accept it as a valid probe response, provided its content is valid.

9.2.5.3 Faults

This operation does not return any fault.

10 Terminating Endpoint Web Services (TEWS & R-TEWS)

This section describes types and operations that are defined into the following namespaces:

- `http://www.eservglobal.com/homesend/tews/2.0`, aliases by `tews`,
- `http://www.eservglobal.com/homesend/tews/2.1`, aliases by `tews21`,
- `http://www.eservglobal.com/homesend/tews/2.2`, aliases by `tews22`,
- `http://www.eservglobal.com/homesend/tews/2.3`, aliases by `tews23`.

Common types are referenced with the `hos/hos21/hos22/hos23` prefixes, and defined in section [Common Data Types Definition](#).

10.1 TEWS and R-TEWS Technical Interfaces

The messages that are exchanged between HomeSend and a receiver E-Wallet system are all parts of the same logical API link called TEWS. However, due to the HTTP binding used to implement it, two different web services (from a SOAP point of view) are used:

- One called TEWS, which this is the technical link/HTTP binding from HomeSend to the receiver E-Wallet system (HomeSend implements a TEWS client, the E-Wallet system implements a TEWS server),
- Another one called R-TEWS, for Reverse-TEWS, which is a technical link/HTTP binding from the receiver E-Wallet system to HomeSend (the E-Wallet system implements a R-TEWS client, HomeSend provides a R-TEWS server).

All SOAP messages exchanged on both these technical links share the same TEWS namespaces, as they complement each other from a functional point of view.

10.2 Specific Types

The following types are specific to the different TEWS and R-TEWS messages.

10.2.1 State Enumeration (TEWS only)

The `tews:state` enumeration enables to qualify the destination balance state. It is provided to HomeSend for future usage and only by receiver E-Wallet systems this state makes sense for.

Enumeration Value	Description
ACTIVE	Active account
INACTIVE	Inactive account
DEACTIVATED	Deactivated account
UNKNOWN	Unknown

10.2.2 AccountInformation Structure (TEWS only)

The `tews:AccountInformation` structure is the main element the receiver E-Wallet system usually provides in the `TEWS/accountInformation` response.

Element Name	Type	Optional	Description
balanceCurrency	xsd:string	No	The balance currency, as an ISO 4217 [ISO4217] identifier (3 uppercase characters). If the balance can manage multiple currencies, this is its default or main currency.
state	tews:state	Yes	The account state. Usually not provided.
secondaryCurrencies	tews23:secondaryCurrencies	Yes	A list of secondary currencies (xsd:string), not including the one provided in balanceCurrency. Only set if the balance can manage more than a single currency.

10.2.3 ProvisionalResponseSupport Structure [2.2]

The `tews22:ProvisionalResponseSupport` structure is used in TEWS/*credit* requests to provide the technical information that a destination system needs to report status asynchronously, i.e. an access to the R-TEWS interface.

This resembles an application-level implementation of Ws-Addressing model, but provides a better control and limit SOAP stacks interoperability problems.

This structure contains the following elements:

Element Name	Type	Optional	Description
reportStatusTo	tews22:endpointAddress (xsd:string, max 500 characters)	No	The complete URL of the R-TEWS SOAP endpoint where R-TEWS requests should be sent for this transaction. Network & visibility rules may prevent the destination system from using this URL as is. Actually, only the path part of the URL must be followed. The actual hostname and port may be locally overridden.
token	tews22:token (xsd:string, max 100 characters)	No	A security token to reuse in all subsequent R-TEWS requests for this transaction.

10.2.4 ProvisionalResponse Structure [2.2]

The `tews22:ProvisionalResponse` structure is used to taint a TEWS/*credit* response as “provisional”, i.e. non-final, indicating that the credit request was taken into account but its processing may take some time.

It contains the following elements:

Element Name	Type	Optional	Description
maxCompletionDate	xsd:dateTime	No	An indication of the maximum expected completion date for the transaction. This is typically calculated from the transaction expiration time in the destination system. This date/time must always be provided with a UTC or a timezone indication, with regards to the standard dateTime format.
state	tews22:transactionState	No	A formal identifier that gives the current transaction's pending state. When used in a TEWS/ <i>credit</i> provisional response, should be set to <code>CreditInProgress</code> . Other values may be supported later.

stateLabel	tews22:transactionStateLabel	No	A human-readable, friendly label associated to the state. Used for diagnostic and logging; only the <code>state</code> field will be used for service logic. This label may be left empty.
------------	------------------------------	----	---

10.2.5 TransactionStatus Choice [2.2]

The `tews22:TransactionStatus` choice is designed to “tunnel” the information elements a TEWS server would normally return in a corresponding HWS/*credit* response.

This structure is used to retrieve a credit status asynchronously, either by client-side polling (HomeSend sends TEWS/*status* requests to the TEWS endpoint, expecting a response with this structure in it), or by destination-side status notification/credit completion (sending a R-TEWS/*statusReport* request).

This choice contains one of the following elements:

Element Name	Type	Description
pending	tews22:PendingTransactionStatus	This element is used to transport the status of a transaction that is still pending/open, i.e. not complete successfully or rejected. It is close to the <code>provisionalResponse</code> part of a provisional TEWS/ <i>credit</i> response, but may contain different pending status to indicate the transaction processing progress.
complete	tews22:CompleteTransactionStatus	This element is used when the transaction has been successfully complete and does not require any further processing on the receiving side. It is similar to a final (non-provisional) TEWS/ <i>credit</i> response.
rejected	tews22:RejectedTransactionStatus	This element is used when the transaction has been rejected, i.e. completed on failure. It does not require any further processing on the receiving side. It is similar to a TEWS/ <i>credit</i> fault.

10.2.6 PendingTransactionStatus Structure [2.2]

The `tews22:PendingTransactionStatus` structure is used as a possible choice in a `tews22:TransactionStatus`, and contains the following elements:

Element Name	Type	Optional	Description
maxCompletionDate	xsd:dateTime	No	An indication of the maximum expected completion date for the transaction.
state	hos22:transactionState	No	A formal identifier that gives the current transaction's pending state (if the transaction is finished, the <code>complete</code> or <code>rejected</code> choice must be used instead). Supported states are provided below.
stateLabel	hos22:transactionStateLabel	No	A human-readable, friendly label associated to the state. Used for diagnostic and logging; only the <code>state</code> field is used for service logic.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields associated with this provisional response. Convenient to transport, for instance, a transaction secret code that would be required to complete the transaction.
stateSpecificFields	hos21:VendorSpecificFields	Yes	State Specific Fields associated with this particular pending state. This is used to transport additional attributes according to the state. For instance, if the state is “AmendRequested”, this state specific fields list may contain the list of the fields to amend, with their current values.

As explained above, this is just another way to transport the provisional response elements that were available in a TEWS/*credit* provisional response. These elements functionally map as follows:

- `PendingTransactionStatus.maxCompletionDate` corresponds to the HWS/*credit* response `provisionalResponse.maxCompletionDate` element,
- `PendingTransactionStatus.state` corresponds to the TEWS/*credit* response `provisionalResponse.state` element
- `PendingTransactionStatus.stateLabel` corresponds to the TEWS/*credit* response `provisionalResponse.stateLabel` element
- `PendingTransactionStatus.vendorSpecificFields` corresponds to the TEWS/*credit* response `vendorSpecificFields`.

However, since this `PendingTransactionStatus` is filled after the TEWS/*credit* provisional response was generated, the transaction's state may have evolved, and the `state/stateLabel` elements may have updated values.

As the transaction's state evolves, Vendor Specific Fields may also be different. In this case, HomeSend merge all the VSF lists received in the initial TEWS/*credit* provisional response and in the various `PendingTransactionStatus` elements, and only keeps the last ones for each (VendorId, FieldId) pair for its final reporting.

The `maxCompletionDate`, however, is expected to remain the same. HomeSend will ignore it if updated.

10.2.7 CompleteTransactionStatus Structure [2.2]

The `tews22:CompleteTransactionStatus` structure is another possible choice in a `tews22:TransactionStatus`, transporting the same information elements as in a final TEWS/*credit* response.

It is used in TEWS/*status* response or R-TEWS/*statusReport* request to notify HomeSend that the credit operation was successfully processed and the transaction is now closed. Depending on the receiver systems service logic, it can be after the receiving end-user actually withdrew the funds, or once his/her e-wallet has been successfully created after a registration process, etc.

It contains the following elements:

Element Name	Type	Optional	Description
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that would be returned in a final TEWS/ <i>credit</i> response. Typical usages in this context include sending a transaction receipt back to HomeSend for reporting and reference.

10.2.8 RejectedTransactionStatus Structure [2.2]

The `tews22:RejectedTransactionStatus` structure is the last possible choice in a `tesws22:TransactionStatus`, transporting the same information elements as in a final TEWS/*credit* fault.

It is used in TEWS/*status* response or R-TEWS/*statusReport* request to notify HomeSend that the credit operation was rejected and the transaction has failed and is now closed. This could be after a failed registration process, failed AML/CTF verifications, withdrawal expiration, etc.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:string	No	The error code that would be returned in a TEWS/credit fault. Formatted as a string like "ER0000".
message	xsd:string	Yes	The message associated to the error code, as it would be returned in a TEWS/credit fault.

Possible values for those `code` and `message` elements are:

Error Code	Typical Message	Specific Description (in this context)
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0010	Unknown receiver	The receiver is unknown on this system, the URI scheme is unsupported, the subscriber is ported-out, or cannot be served through this corridor (routing tag).
ER0011	Exchange rate error	The currency of the amount to credit is unsupported on this system/for this user.
ER0013	Receiver crediting error	The receiver could not be credited.
ER0017	Correlation error	Inconsistency in Ids. May not be checked by the receiver E-Wallet system (for instance, it may rely on the HomeSend transaction ID only, generating a fake/unused transaction ID).
ER0021	Maximum reception amount exceeded	The maximum creditable amount in a single operation has been reached.
ER0024	Maximum number of transactions exceeded by receiver	The maximum number of transactions (usually, for a given period) has been reached for the user to credit.
ER0025	Maximum cumulative amount of transactions exceeded by receiver	The maximum cumulated amount of transactions (usually, for a given period) has been reached for the user to credit.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0042	Transaction Expired	The credit was accepted, but the receiving end-user did not perform the requested operations to get the funds (withdrawal, registration, etc).

Notice that some error codes that are available in TEWS/*credit* faults are not available here anymore. They correspond to errors that should be detected synchronously when handling the TEWS/*credit* request (such as authentication or licensing errors).

10.2.9 CancellationReason Structure [2.3]

The `tews23:CancellationReason` structure is used to transport a cancellation reason basically explaining why a transaction cancellation is requested. It is used for information only, and what it contains is not meant to be displayed to the end-user. A local mapping between the cancellation reason code and a friendly label displayed to the receiving end-user, if needed, may be implemented by the receiving system.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:int	No	A formal, short code that can be safely stored in records in the receiving system, for future reference. If a mapping is implemented as explained above, this should be the mapping key.
message	xsd:string	No	A human-readable string associated to the code, providing for improved troubleshooting only. May be left empty.

Usual/well known codes and typical associated messages are described in [TEWS/cancellation \[2.3\]](#).

10.2.10 HsProvidedTransactionStatus Choice [2.3]

The `tews23:HsProvidedTransactionStatus` choice is used to report a transaction status from HomeSend to the receiving system. It is very close to the `TransactionStatus` choice that is used to transport a status update from the receiving system to HomeSend, and due to its reverse use it is designed to evolve independently from it.

This choice contains one of the following elements:

Element Name	Type	Description
pending	<code>tews23:HsProvidedPendingTransactionStatus</code>	This element is used to transport the status of a transaction that is still pending/open, i.e. not complete successfully or rejected yet.
complete	<code>tews23:HsProvidedCompleteTransactionStatus</code>	This element is used when the transaction has been successfully complete and does not require any further processing on the receiving side. For now, HomeSend won't notify the receiving system that a transaction has been successfully complete, through this may be a possible evolution in a future version (same API).
rejected	<code>tews23:HsProvidedRejectedTransactionStatus</code>	This element is used when the transaction has been rejected, i.e. completed on failure. It does not require any further processing on the receiving side. For now, HomeSend won't notify the receiving system that a transaction has been rejected, through this may be a possible evolution in a future version (same API).

10.2.11 HsProvidedPendingTransactionStatus Structure [2.3]

The `tews23:HsProvidedPendingTransactionStatus` structure is used as a possible choice in a `tews23:HsProvidedTransactionStatus`, and contains the following elements:

Element Name	Type	Optional	Description
state	hos22:transactionState	No	A formal identifier that gives the current transaction's pending state (if the transaction is finished, the <code>complete</code> or <code>rejected</code> choice must be used instead). States that can be provided are provided below.
stateLabel	hos22:transactionStateLabel	No	A human-readable, friendly label associated to the state. Used for diagnostic and logging; only the <code>state</code> field is used for service logic.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields associated with this provisional response. Convenient to transport, for instance, a transaction secret code that would be required to complete the transaction.
stateSpecificFields	hos21:VendorSpecificFields	Yes	State Specific Fields associated with the pending state.

The possible state values that HomeSend may push to the destination system are provided in [TEWS/status request](#).

10.2.12 HsProvidedCompleteTransactionStatus Structure [2.3]

The `tews23:HsProvidedCompleteTransactionStatus` structure is used as a possible choice in a `tews23:HsProvidedTransactionStatus`, and contains the following element:

Element Name	Type	Optional	Description
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields associated with this provisional response. Convenient to transport, for instance, a transaction secret code that would be required to complete the transaction.

Notice that as of writing this documentation, HomeSend won't use such a structure as it won't notify a destination system of a transaction failure.

10.2.13 HsProvidedRejectedTransactionStatus Structure [2.3]

The `tews23:HsProvidedRejectedTransactionStatus` structure is used as a possible choice in a `tews23:HsProvidedTransactionStatus`, and contains the following elements:

Element Name	Type	Optional	Description
code	xsd:string	No	The error code indicating the reason why the transaction failed. Formatted as a string like "ER0000". Error codes could be any error code returned by a HWS/remittance operation.
message	xsd:string	No	A human-readable message associated to the error code.

Notice that as of writing this documentation, HomeSend won't use such a structure as it won't notify a destination system of a transaction successful completion.

10.2.14 TerminatingServiceException Structure [2.2]

The `tews:TerminatingServiceException` structure extends the `hos:BaseException` and is used to implement the single SOAP Fault that a receiver E-Wallet system can generate.

It contains the following elements:

Element Name	Type	Optional	Description
code	xsd:string	No	Inherited from <code>hos:BaseException</code> . The error code, as described later. Formatted as a string like <code>ERR0000</code> .
message	xsd:string	No	Inherited from <code>hos:BaseException</code> . A message associated to the error code. This message is in plain English and is not meant to be displayed to the end-user.
parameters	xsd:string	Yes	Inherited from <code>hos:BaseException</code> . Zero or more of these elements may be present. They contain additional information related to the error, as described later.
transactionId	hos:transactionId	Yes	The leg transaction ID, from the request the fault is a "response" to. This ID may not be generated yet by the receiver E-Wallet system – in this case, may not be provided.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID. This ID is always provided in the request and shall be always included in such a fault.

10.3 Operations

This section describes the operations the receiver E-Wallet must publish.



Warning:

To guarantee HomeSend's SLA, 90% of these operations must be replied in less than 500ms.

Each operation is made of one request, as a single *operationRequest* element whose type is *OperationRequest*, and a response, taking a single *operationResponse* element whose type is *OperationResponse*.

These types are described below.

10.3.1 TEWS/accountInformation

This request is used by HomeSend to identify the receiver E-Wallet system that will serve the receiver end-user for this transaction.

Such a system should only reply a normal response if it is ready to serve the designated destination URI for the invoked service/corridor, indirectly identified by the routingTag element.

10.3.1.1 Request

The *TEWS/accountInformation* request takes a `tews:AccountInformationRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.

hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
destinationUri	hos:uri	No	The destination URI HomeSend is looking for a serving receiver E-Wallet system.
routingTag	hos:routingTag	Yes	This routing tag, if provided, helps the receiver E-Wallet system identify the kind of invoked service (corridor) and/or the service provider the request is intended to, in case of a E-Wallet system hosting multiple service providers. This routing tag is the one the receiver E-Wallet integrator configured locally on the E-Wallet, and transmitted to the HomeSend platform provider so that it can provision it into HomeSend.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.

10.3.1.2 Response

The TEWS/*accountInformation* response takes a `tews:AccountInformationResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The terminating transaction ID, as locally generated by the receiver E-Wallet system to identify the transaction on its side.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as provided in the incoming request.
accountInformation	tews:AccountInformation	No	The information HomeSend requires performing the credit. Usually only contains the destination's balance currency.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.

Providing this response means that the destination E-Wallet system is willing to serve this destination.

HomeSend will send a TEWS/*credit* request to it afterwards, with the same transactionId and hsTransactionId values as provided here.

10.3.1.3 Faults

In case of an error, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	AccountInformation requests are not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).

ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform the operation due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0010	Unknown receiver	The destination URI cannot be served by this system. This includes unsupported URI schemes. In case of a ported-out subscriber, the destination system MUST return this error code.
ER0014	Authentication error	Invalid login/password in request.
ER0030	Unsupported service for this user	The destination user is known and can be served by this system, but not through the requested service (indicated in the routing tag). In this case, depending on HomeSend's routing configuration, the receiver E-Wallet system should usually expect a new TEWS/accountInformation with a new routingTag to try another corridor/service for the same user.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system.

10.3.2 TEWS/credit

10.3.2.1 Request

The TEWS/credit request takes a `tews:CreditRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.
transactionId	hos:transactionId	No	The transaction ID as provided in the account information response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_.-] only.

sourceUri	hos:uri	No	The sender's identifier. If it is a tel URI, may be directly presented to the end-user, as it will be human-understandable. Other URI schemes, such as ewallet or IBAN, may be considered as more technical and the E-Wallet system implementers may prefer present senderInformation (if available).
destinationUri	hos:uri	No	The receiver's identifier. HomeSend sends the same URI as in the preliminary account information request.
amountInformation	hos:AmountInformation	No	The amount to credit, with the associated currency (the balance currency, as returned in the TEWS/accountInformation response).
routingTag	hos:routingTag	Yes	If a routing tag was provided in the previous TEWS/accountInformation request, it is also provided here for convenience. It may help the E-Wallet system route the credit operation to the correct sub-system, in particular if it acts as a load-balancer or a gateway toward actual serving E-Wallet systems.
senderInformation	hos:SenderInformation	Yes	If available, this structure contains free text containing human-readable sender's name and address information. Unicode; may be displayed to the end-user in a notification to help him/her identifying the sender.
description	hos:description	Yes	A description labeling the transaction, as provided in the initial HWS/remittance request, if available.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields forwarded as is (optionally filtered, according to the HomeSend configuration) from the previous exchanged messages, or locally generated by HomeSend.
provisionalResponseSupport	tews22:ProvisionalResponseSupport	Yes	This field is used to advertise the support of the asynchronous model for this transaction, and includes a "reply address" as well as a unique token to use when sending further state updates notification as R-TEWS/statusReport requests. While this field is technically optional, HomeSend guarantees to always provide it in TEWS/credit requests.

10.3.2.2 Response

The TEWS/credit response takes a `tews:CreditResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The terminating transaction ID, as locally generated by the receiver E-Wallet system to identify the transaction on its side.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID, as provided in the incoming request.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields the E-Wallet system wants to send back to HomeSend, if any.
provisionalResponse	tews22:ProvisionalResponse	Yes	If present, HomeSend considers that the credit operation has been accepted, but not yet processed. The transaction is called to be "asynchronous". The transaction's final status will be provided later, either via a destination-pushed notification (R-TEWS/statusReport), or via

Sending this response without the `provisionalResponse` element means that the credit was fully executed.

10.3.2.3 Faults

In case of an error, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a credit due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0009	Ambiguous case	This error code must be returned in case of doubt regarding subscriber balance crediting has been performed.
ER0010	Unknown receiver	The receiver is unknown on this system, the URI scheme is unsupported, the subscriber is ported-out, or cannot be served through this corridor (routing tag).
ER0011	Exchange rate error	The currency of the amount to credit is unsupported on this system/for this user.
ER0013	Receiver crediting error	The receiver could not be credited.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids. May not be checked by the receiver E-Wallet system (for instance, it may rely on the HomeSend transaction ID only, generating a fake/unused transaction ID).
ER0021	Maximum reception amount exceeded	The maximum creditable amount in a single operation has been reached.
ER0024	Maximum number of transactions exceeded by receiver	The maximum number of transactions (usually, for a given period) has been reached for the user to credit.
ER0025	Maximum cumulative amount of transactions exceeded by receiver	The maximum cumulated amount of transactions (usually, for a given period) has been reached for the user to credit.

ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0042	Transaction Expired	The credit was accepted, but the receiving end-user did not perform the requested operations to get the funds (withdrawal, registration, etc). This error code is typically only returned as status query responses (as a response to a retransmitted HWS/remittance request, or in a HWS/status response).

10.3.3 TEWS/status

This operation is used to get the status of a credit operation HomeSend received a provisional response for.

Such TEWS/status requests are typically sent regularly after the deadline provided in the TEWS/credit provisional response expired, for an amount of time that is configured by the HomeSend platform provider. It is used to recover from a TEWS/statusReport request HomeSend never received for the transaction; as a consequence, its response contains the same level of information that would have been expected in such a TEWS/statusReport.

10.3.3.1 Request

The TEWS/status request takes a `tews22:StatusRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.
transactionId	hos:transactionId	No	The transaction ID as provided in the account information response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
routingTag	hos:routingTag	Yes	If a routing tag was provided in the previous TEWS/accountInformation request, it is also provided here for convenience. It may help the E-Wallet system route the credit operation to the correct sub-system, in particular if it acts as a load-balancer or a gateway toward actual serving E-Wallet systems.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

10.3.3.2 Response

The TEWS/status response takes a `tews22:StatusResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
status	tews22:TransactionStatus	No	The status of the transaction, as a choice between three structures: <code>pending</code> , <code>rejected</code> , <code>complete</code> . The same structure can be pushed to HomeSend in a TEWS/statusReport request. The TEWS/status request is just another way for HomeSend to get it (client-side polling instead of asynchronous notification).
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

The `status` choice is designed to “tunnel” the information elements you would normally have in a corresponding final TEWS/credit response:

- If the `complete` choice is selected, it contains a `tews22:CompleteTransactionStatus` structure that transports an optional `vendorSpecificFields` field that would have been set in a successful TEWS/credit response – using the `complete` choice is sufficient to instruct HomeSend the transaction was successful and its credit was committed.
- If the `rejected` choice is selected, it contains a `tews22:RejectedTransactionStatus` structure that transports the `code` and `message` fields that would have been set in a TEWS/credit fault. The same code and message values should be used.
- If the transaction is still being processed by the receiving systems, the `pending` choice shall be used, containing a `tews22:PendingTransactionStatus` structure transporting the current transaction’s state, an associated label, a recall of the maximum completion date, and possibly updated vendor specific fields.

The state is an open state identifier that will be ignored by HomeSend for now – future versions may take them into account to be able to track the transaction’s progress more accurately. For now, any pending status is enough to consider that the credit operation is not fully processed yet:

State (Identifier, case sensitive)	Typical State Label	Description
CreditInProgress	Credit taken into account, processing	The destination system is still processing the credit operation.

Other states may be formalized after discussing with the HomeSend service provider (for instance, specific states requesting the sending end-user to amend the transaction, indicating that a withdrawal is being prepared, etc).

The same states can also be pushed by the destination system via a R-TEWS/statusReport request.

10.3.3.3 Faults

In case of an error, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a credit due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0041	Transaction not found	There is no such transaction on this system, or the transactions status cannot be queried by this interface anymore. Some TEWS implementations may only be able to retrieve a transaction's status for recent transactions. Depending on the implementation, returning ER0017 in this case is also possible.

10.3.4 TEWS/cancellation [2.3]

This operation is used to request a pending transaction cancellation. The receiving system should reply OK if the cancellation request has been successfully submitted, even if the actual transaction cancellation is not done yet. Then it should complete the transaction as usual – that is by rejecting the transaction, or even accepting it if the cancellation attempt could not be taken into account.

A *TEWS/cancellation* request can only be received after the receiving system sent a *TEWS/credit* provisional response. If the *TEWS/credit* was not sent, was definitively accepted (synchronous transaction), or rejected, then it is not need to be ready to accept such a request.

10.3.4.1 Request

The *TEWS/cancellation* request takes a `tews22:CancellationRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.
transactionId	hos:transactionId	No	The transaction ID as provided in the account information response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_.-] only.
routingTag	hos:routingTag	Yes	If a routing tag was provided in the previous TEWS/ <i>accountInformation</i> request, it is also provided here for convenience. It may help the E-Wallet system route the credit operation to the correct sub-system, in particular if it acts as a load-balancer or a gateway toward actual serving E-Wallet systems.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

10.3.4.2 Response

The TEWS/*cancellation* response takes a `tews23:CancellationResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

Sending such a response only means that the cancellation request was submitted successfully, i.e. the receiving system can basically try to cancel the transaction. It does not mean that the transaction was actually cancelled yet.

If the receiving system knows that the transaction cannot be cancelled (unsupported by its back-office systems, invalid transaction state, etc), it must reply with a fault instead.

10.3.4.3 Faults

In case of an error or when it is not possible to attempt to cancel the transaction, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).

ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a credit due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids.
ER0018	Transaction already cancelled	The transaction was already cancelled, no need to cancel it again.
ER0034	Unexpected request	The transaction is not in a valid state to accept a cancellation attempt.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0041	Transaction not found	There is no such transaction on this system, or the transactions status cannot be queried by this interface anymore. Some TEWS implementations may only be able to retrieve a transaction's state for recent transactions. Depending on the implementation, returning ER0017 in this case is also possible.
ER0044	Operation not supported	This operation was not implemented on this system.

10.3.5 TEWS/amend [2.3]

10.3.5.1 Request

The TEWS/amend request takes a `tews23:AmendRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.
transactionId	hos:transactionId	No	The transaction ID as provided in the account information response.

hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_.-] only.
routingTag	hos:routingTag	Yes	If a routing tag was provided in the previous TEWS/accountInformation request, it is also provided here for convenience. It may help the E-Wallet system route the credit operation to the correct sub-system, in particular if it acts as a load-balancer or a gateway toward actual serving E-Wallet systems.
amendedFields	hos21:VendorSpecificFields	No	The list of fields to amend, all using the standard VSF format. Fields are not ordered. Fields are not duplicated.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

When receiving such a request, the receiving system should first check if it is authorized to update all the fields provided with their new values in `amendedFields`. If any of them is not updatable, the request must be rejected with `ER0044 Cannot amend at least one of these fields and no fields must be updated`, even if some others could have been updated.

10.3.5.2 Response

The TEWS/amend response takes a `tews23:AmendResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

Sending this response means that the provided fields were successfully amended.

If the transaction Credit in progress substate was set to "AmendRequested" before receiving this response (solicited amend case), HomeSend automatically switch the transaction state to the previous known Credit in progress substate when receiving it. In other words, it is not needed to notify HomeSend that the Credit in progress substate is no longer "AmendRequested".

10.3.5.3 Faults

In case of an error or when it is not possible to attempt to cancel the transaction, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).

ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a credit due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids.
ER0034	Unexpected request	The transaction is not in a valid state to accept an amend attempt.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0041	Transaction not found	There is no such transaction on this system, or the transactions status cannot be queried by this interface anymore. Some TEWS implementations may only be able to retrieve a transaction's state for recent transactions. Depending on the implementation, returning ER0017 in this case is also possible.
ER0044	Operation not supported Cannot amend at least one of these fields	This operation was not implemented on this system. An attempt was made to amend at least one non-amendable field.

10.3.6 TEWS/statusReport [2.3]

This operation enables HomeSend to notify a receiving system of specific transaction events, involving a manual action from the receiving service provider.

It must not be mixed up with the R-TEWS/statusReport operation:

- TEWS/statusReport: used by HomeSend to push a HomeSend-provided status to the receiving system,
- R-TEWS/statusReport: used by the receiving system to push a terminating-provided status to HomeSend.

10.3.6.1 Request

The TEWS/statusReport request takes a `tews23:TewsStatusReportRequest` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password HomeSend should use to connect onto the receiver E-Wallet system.
transactionId	hos:transactionId	No	The transaction ID as provided in the account information response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
routingTag	hos:routingTag	Yes	If a routing tag was provided in the previous TEWS/accountInformation request, it is also provided here for convenience. It may help the E-Wallet system route the credit operation to the correct sub-system, in particular if it acts as a load-balancer or a gateway toward actual serving E-Wallet systems.
status	tews23:HsProvidedTransactionStatus	No	The list of fields to amend, all using the standard VSF format. Fields are not ordered. Fields are not duplicated.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

At the time this documentation is being written, HomeSend can only report a pending transaction status, and the following states may be reported in this case:

Reported State	Typical Label	When	Description
TerminatingVerificationRequired	A transaction needs a compliance officer's attention.	Between TEWS/accountInformation response and TEWS/credit request	Only possible if the receiving service provider delegated a sanction list screening service to HomeSend. A transaction that was about to be submitted to the receiving system needs further investigations before being submitted. A compliance officer should connect onto the partner's sanction list screening system to resolve the case. There is no state specific fields associated with this state.

Additional states may also be reported.

When getting such a notification, the receiving system may discard or capture the reported state for reference, explicitly reject the notification, or handle it explicitly depending on the decided service logic to implement.

States that must be explicitly handled are discussed when integrating a receiving system with HomeSend.

10.3.6.2 Response

The TEWS/statusReport response takes a `tews23:TewsStatusReportResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The originating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.

vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.
----------------------	----------------------------	-----	--

Sending such a response means that the status update was taken into account.

10.3.6.3 Faults

In case of an error or when receiving a TEWS/*statusReport* request, the receiver E-Wallet system shall reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to perform a credit due to licensing issues (depends on the E-Wallet implementation).
ER0005	Destination system route error	Unable to route this request to the correct sub-system (for instance, when acting as a load-balancer or a multiple service provider E-Wallet system).
ER0008	Internal error	An internal error occurred.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids.
ER0034	Unexpected request	The transaction is not in a valid state to accept a status report.
ER0037	Missing required fields	At least one vendor specific field (VSF) is missing. The receiver E-Wallet system should use this error if its service logic expected a particular VSF to be provided by HomeSend. Such mandatory VSFs should be communicated to the HomeSend platform provider so that it can make sure they will be provided to the receiver E-Wallet system. When sending this error, the receiver E-Wallet system must make sure that no credit operation was performed.
ER0041	Transaction not found	There is no such transaction on this system, or the transactions status cannot be queried by this interface anymore. Some TEWS implementations may only be able to retrieve a transaction's state for recent transactions. Depending on the implementation, returning ER0017 in this case is also possible.
ER0044	Operation not supported	This operation was not implemented on this system.

10.3.7 TEWS/deviceWatchdog [2.2]

This operation is used to monitor a TEWS link actively. HomeSend can be configured to regularly send a device watchdog request to check that your implementation is still reachable, online, and running a TEWS server implementation.

Supporting this operation is mandatory when implementing a TEWS server.

10.3.7.1 Request

The *TEWS/deviceWatchdog* request takes a `tews22:DeviceWatchdogRequest` structure as argument, which is an empty structure.

Unlike the other requests, this one is never signed digitally, even if the TEWS link is configured to require digital signatures.

10.3.7.2 Response

The *TEWS/deviceWatchdog* response takes a `tews22:DeviceWatchdogResponse` structure as argument, which is an empty structure.

Sending anything else is considered as a failed probe by HomeSend, which may then consider your system as down or malfunctioning.

Unlike the other responses, this one should not be digitally signed, even if the TEWS link is configured to require digital signatures. If signed, HomeSend won't validate its signature but will still accept it as a valid probe response, provided its content is valid.

10.3.7.3 Faults

This operation does not return any fault.

10.3.8 R-TEWS/statusReport

This operation is a "server-pushed" variant of the *TEWS/status* operation. Instead of waiting for HomeSend to ask an updated status for a transaction, the destination system may asynchronously push an updated status to HomeSend via this operation.

While its usage is currently limited to notify of the actual transaction completion (success or failure), intermediate states may be considered in next HomeSend versions, such as "Awaiting registration", "Preparing for withdrawal", etc.

Due to HTTP binding limitations, however, this is implemented as a request toward a dedicated "reverse" TEWS service.

This request **MUST** be sent to an URL that is dynamically constructed from the `provisionalResponseSupport.reportStatusTo` value. While the host and port provided in this value may be discarded and replaced with locally-configured values (due to local network rules & host visibility), the path part of the provided URL must be used, as it may change from one transaction to another for technical and maintenance reasons.

For instance, if the `reportStatusTo` is valued to "https://homesend.com:4443/rtews/2.2", the R-TEWS client must make sure that it will send requests that will arrive exactly there. NAT or VPN gateways may make this URL visible as "https://my-local-gw:6732/rtews/2.2" from the R-TEWS client, but the `/rtews/2.2` part must not be overridden locally.

Notice that you may ask the HomeSend platform provider to advertise in `provisionalResponseSupport` an URL that is only meaningful for you, with regards to your visibility and network rules. This avoids a local configuration setting.

10.3.8.1 Request

The R-TEWS/*statusReport* request takes a `tews22:StatusReportRequest` structure as argument.

Basically, a R-TEWS/*statusReport* request contains the same functional elements and the same level of information as in a TEWS/*status* response:

Element Name	Type	Optional	Description
security	hos:Security	No	The login/password the receiver E-Wallet system should use to connect onto HomeSend's R-TEWS interface.
token	tews:token	No	The correlation token that was provided in the TEWS/ <i>credit</i> the status report is related to.
transactionId	hos:transactionId	No	The terminating transaction ID identifying the transaction, as provided in the account information response.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID that should be kept for the all the subsequent message exchanges related to this transaction. Even if its technical type allows for more characters, hsTransactionIds are guaranteed to only contain [A-Za-z0-9_-] only.
status	tews22:TransactionStatus	No	The status of the transaction, as a choice between three structures: pending, rejected, complete. Typically only rejected (transaction NOK) or complete (transaction OK) should be used. Reporting a pending status will be ignored by HomeSend for now.
vendorSpecificFields	tews21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically unused, unless specified otherwise on a project basis.

The use of the status element is exactly the same as in a TEWS/*status* [Response](#).

10.3.8.2 Response

The R-TEWS/*statusReport* response takes a `tews22:StatusReportResponse` structure as argument, containing the following elements:

Element Name	Type	Optional	Description
transactionId	hos:transactionId	No	The terminating transaction ID, as provided into the replied request.
hsTransactionId	hos:transactionId	No	The HomeSend transaction ID.
vendorSpecificFields	hos21:VendorSpecificFields	Yes	Vendor Specific Fields that are related to the status operation itself. This is typically left empty or not provided, unless specified otherwise on a project basis.

Getting such a response means that HomeSend took the status update into account.

10.3.8.3 Faults

In case of an error, HomeSend will reply with a `tews:TerminatingServiceException` SOAP Fault, as described in [ServiceException Structure](#).

The following error codes may be generated:

Error Code	Typical Message	Specific Description (in this context)
ER0001	Service not authorized	Crediting is not supported on this system (or from HomeSend, with the provided login).
ER0002	Invalid input value	The request contained at least one invalid field (complements possible SOAP faults generated by XML schema validators).
ER0003	Request already in progress	The request has already been received, and is currently being processed.
ER0004	Service request license reached	Unable to handle the request due to request throttling from this receiving system.
ER0008	Internal error	An internal error occurred.
ER0014	Authentication error	Invalid login/password in request.
ER0017	Correlation error	Inconsistency in Ids, invalid token, or the transaction was not found.
ER0040	Unsupported status update	A pending status update is not understood (unknown state name), or the transaction is not in a state that can accept such an update (for instance, when a success is reported while the transaction was already failed)

10.3.9 R-TEWS/deviceWatchdog

This operation is used to monitor a R-TEWS link actively. A destination system implementing a R-TEWS client may regularly send a device watchdog request to check that HomeSend is still reachable and active.

Supporting this operation is optional when implementing a R-TEWS client.

10.3.9.1 Request

The R-TEWS/deviceWatchdog request takes a `tews22:DeviceWatchdogRequest` structure as argument, which is an empty structure.

Unlike the other requests, this one should not be digitally signed, even if the TEWS (thus R-TEWS) link is configured to require digital signatures. If signed, HomeSend won't validate its signature but will still accept it as a valid request, provided its content is valid.

10.3.9.2 Response

The R-TEWS/deviceWatchdog response takes a `tews22:DeviceWatchdogResponse` structure as argument, which is an empty structure.

Unlike the other responses, this one will never be signed digitally, even if the TEWS (thus R-TEWS) link is configured to require digital signatures. The R-TEWS client should be able to accept the response as a valid one even if not signed in this case.

10.3.9.3 Faults

This operation does not return any fault.

11 Annexes

11.1 Call Flows

11.1.1 Simple Remittance Call Flows

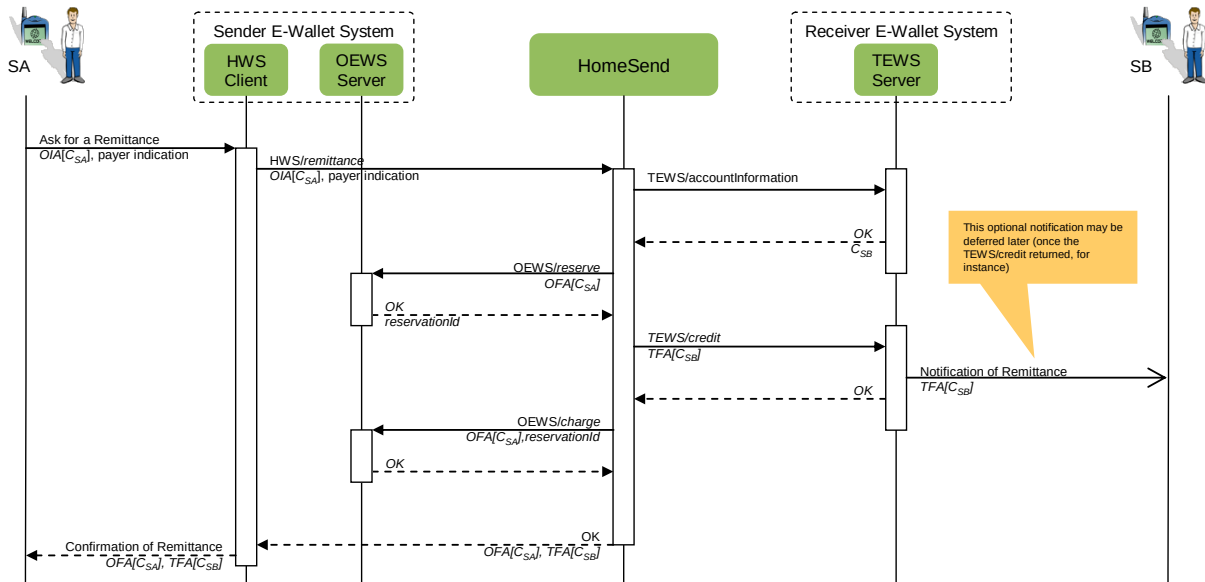


Figure 28 - One-shot remittance (OK)

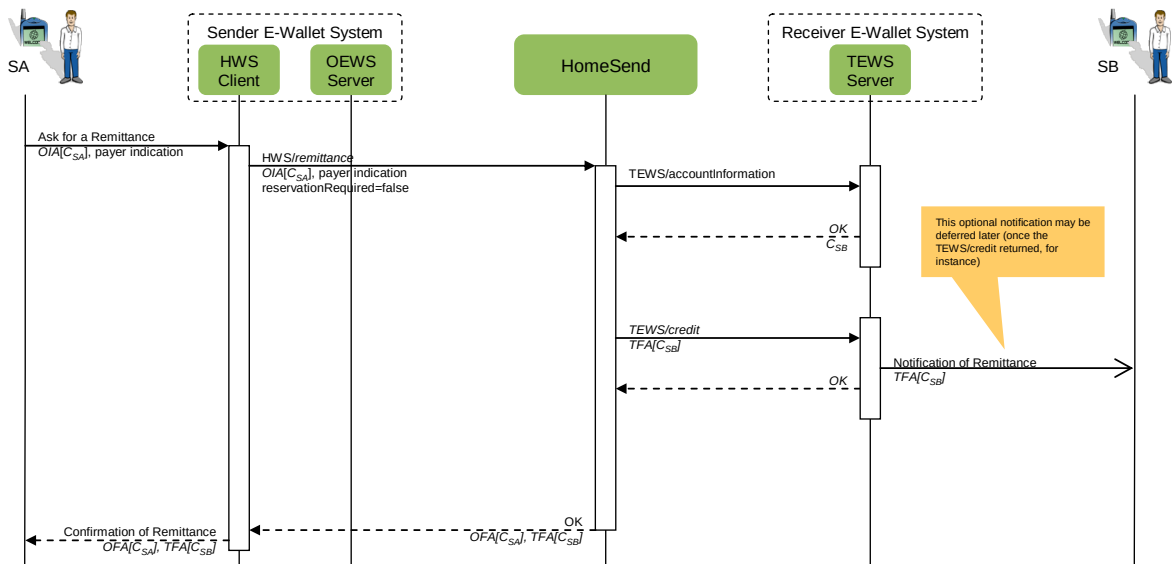


Figure 29 - One-shot remittance with reservationRequired=False (OK)

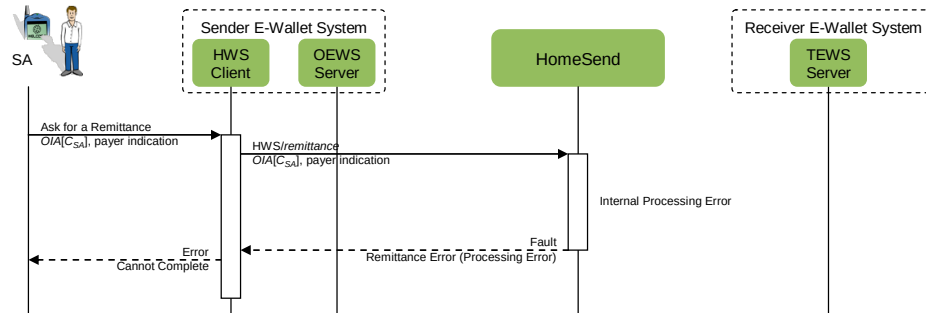


Figure 30 - One-shot remittance, processing error

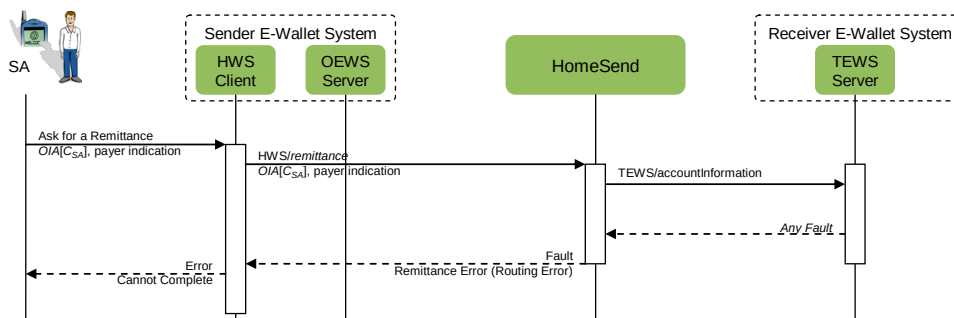


Figure 31 - One-shot remittance, typical routing error

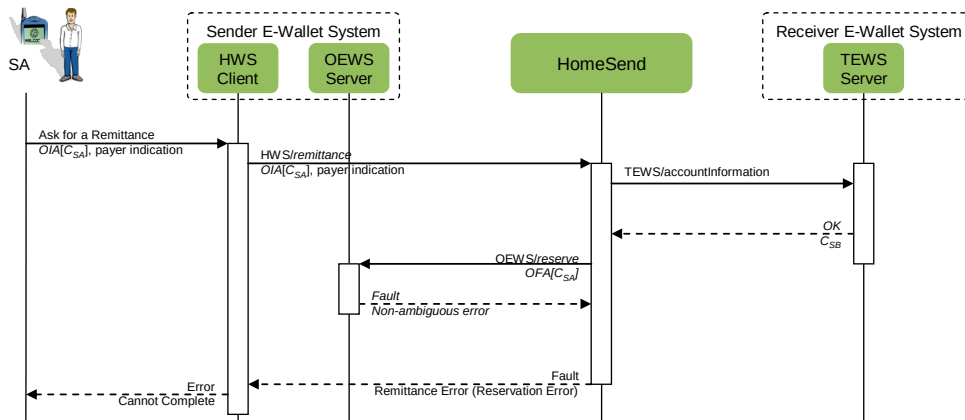


Figure 32 - One-shot remittance, reservation error

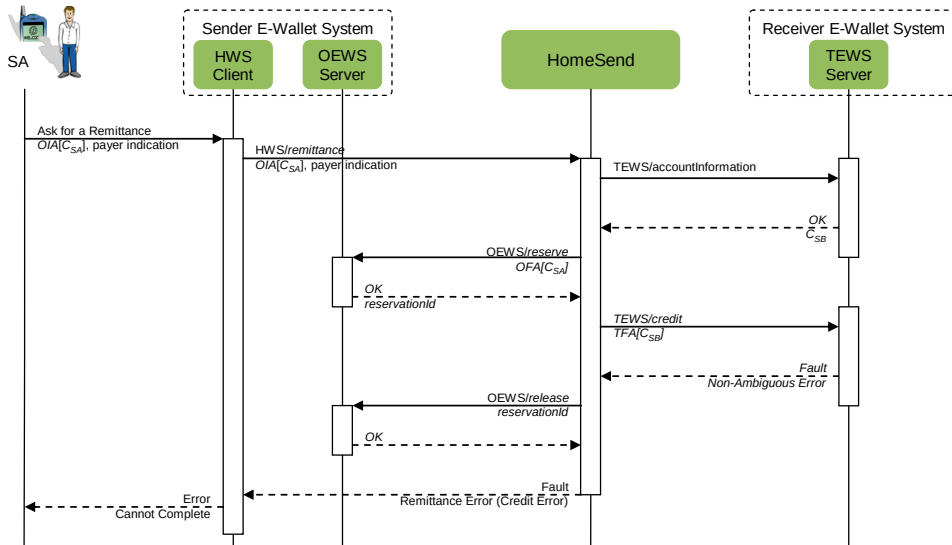


Figure 33 - One-shot remittance, non-ambiguous credit error

11.1.2 Advice of Charge Call Flows

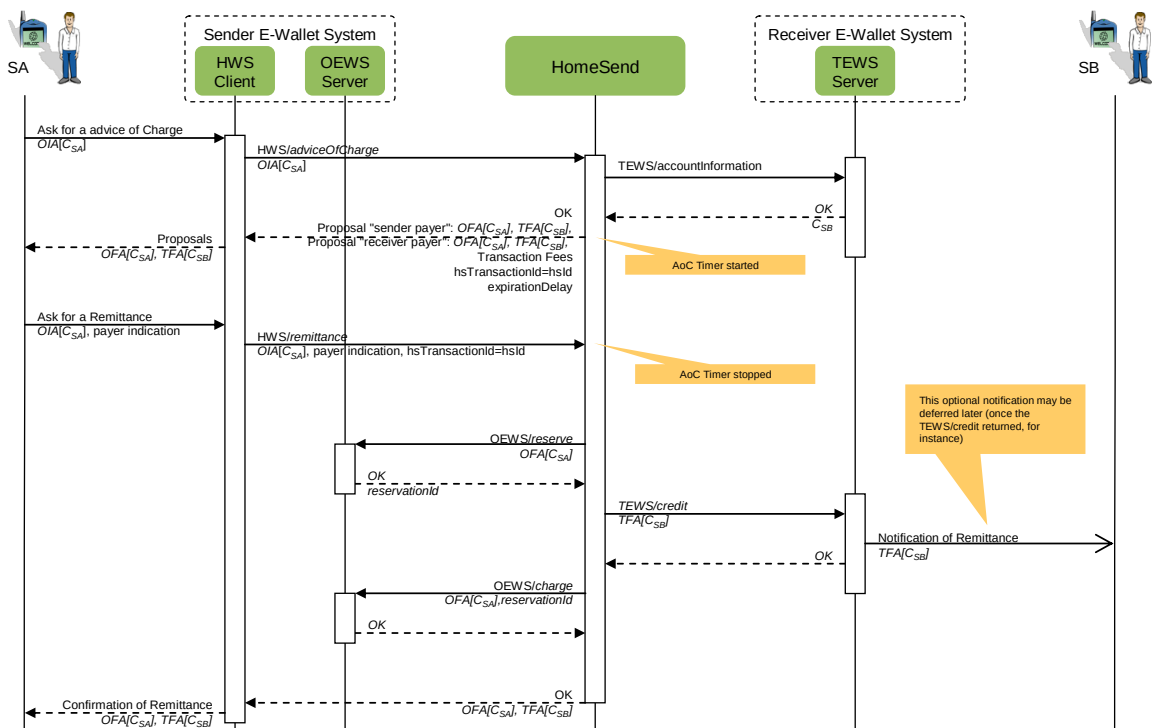


Figure 34 - Remittance with advice of charge (OK)

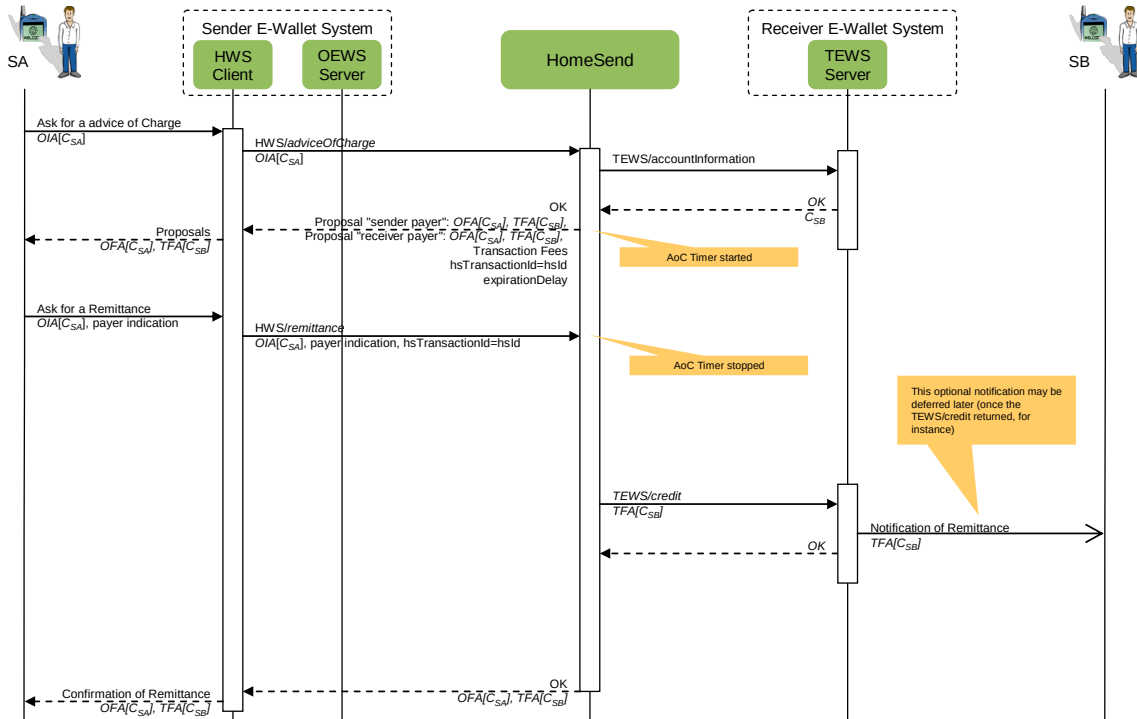


Figure 35 - Remittance with advice of charge, with reservationRequired=False (OK)

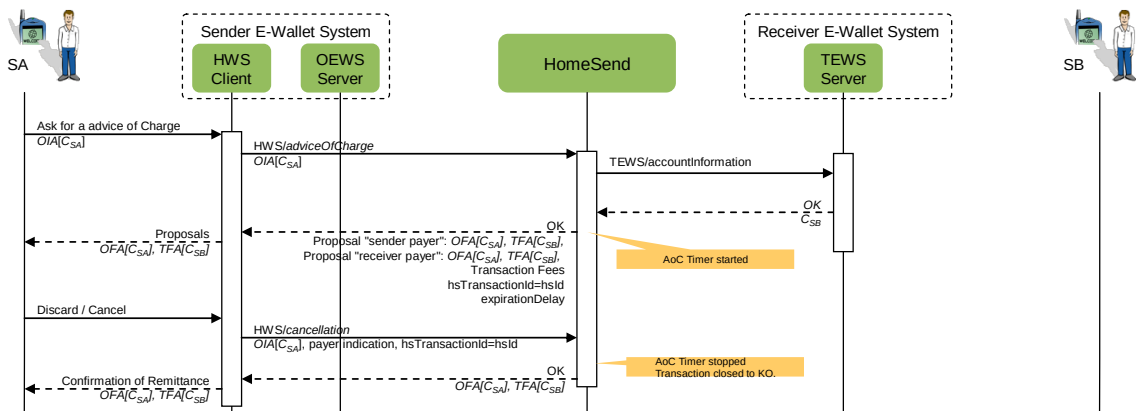


Figure 36 - Advice of charge, explicit cancellation

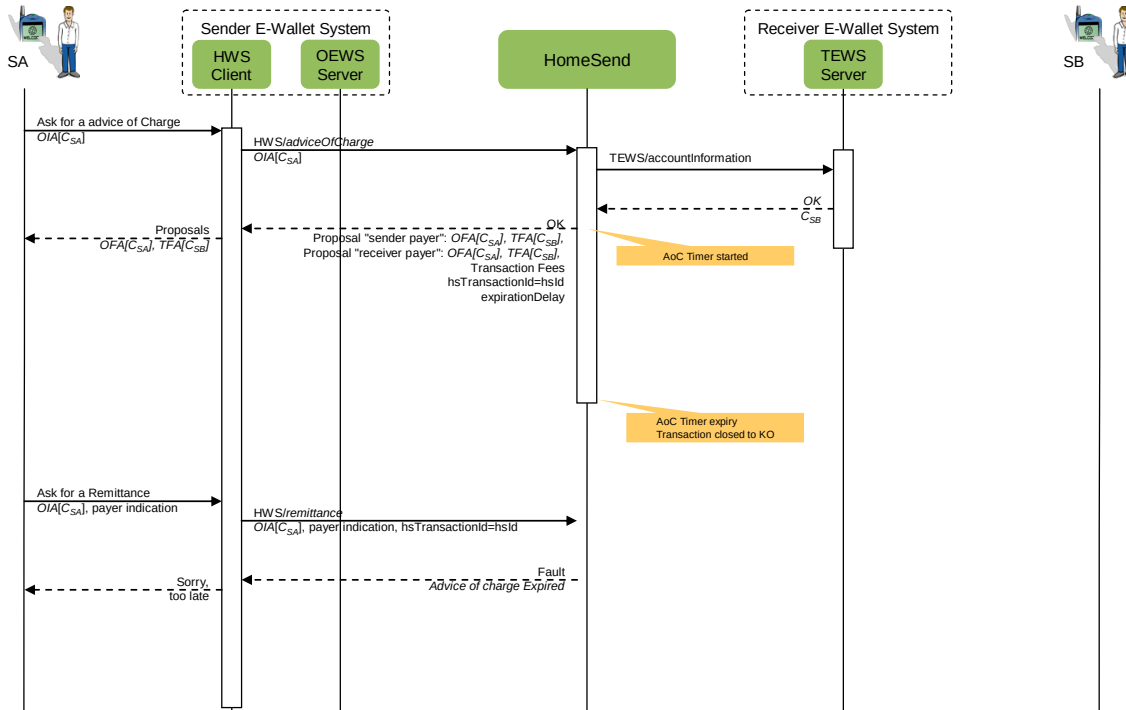


Figure 37 - Remittance with advice of charge, late remittance

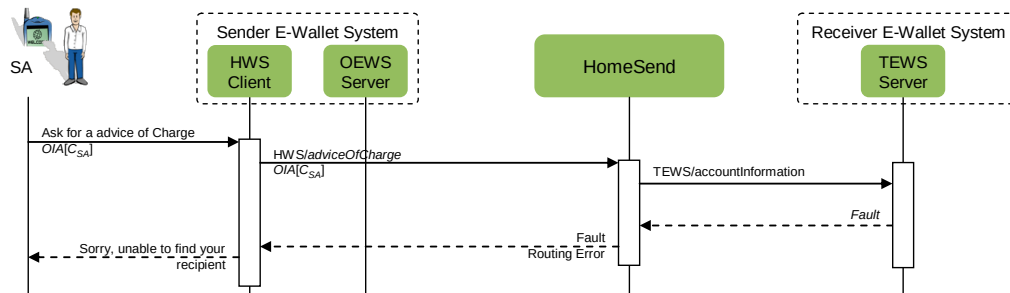


Figure 38 - Advice of charge, typical routing error

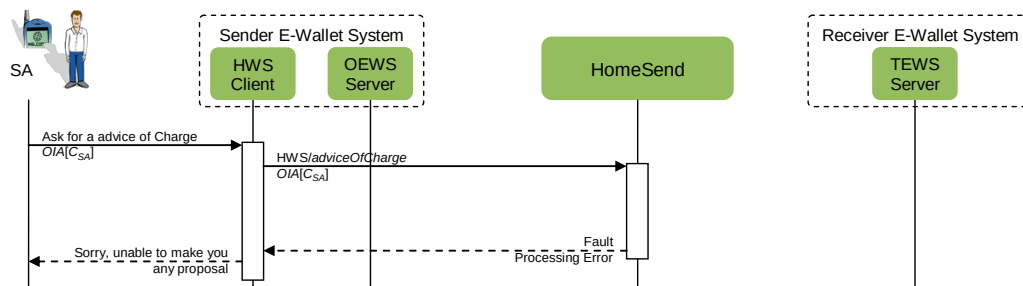


Figure 39 - Advice of charge, processing error

11.1.3 Automatic Reconciliation Call Flows

This mechanism applies to one-shot remittances, remittance with advice of charge, and e-recharge transactions.

These call flows are provided for information, to help understand the automatic reconciliation mechanism that tries to recover or complete a transaction once a first response has been provided.

This procedure is completely driven by HomeSend and does not require any particular support from interconnected E-Wallet systems, excepting the basic support for incoming request retransmissions for $MAX_{reconciliation}$ time (all cases) and the capability of the HWS client to re-send the initial request to get an updated status (ambiguous credit case).

11.1.3.1 Completing the Transaction after an Ambiguous Reservation Request

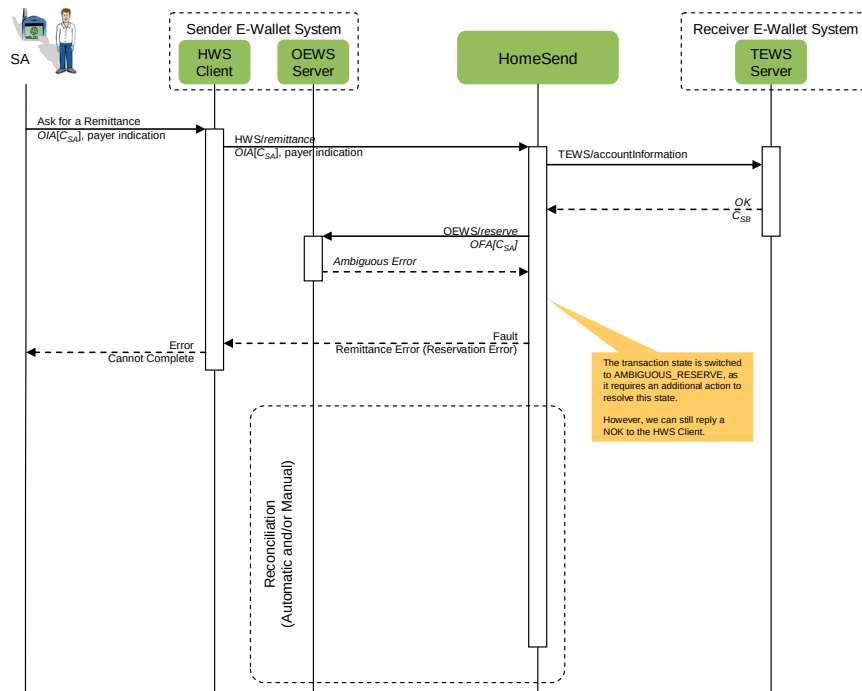


Figure 40 - Ambiguous OEWS/reserve request

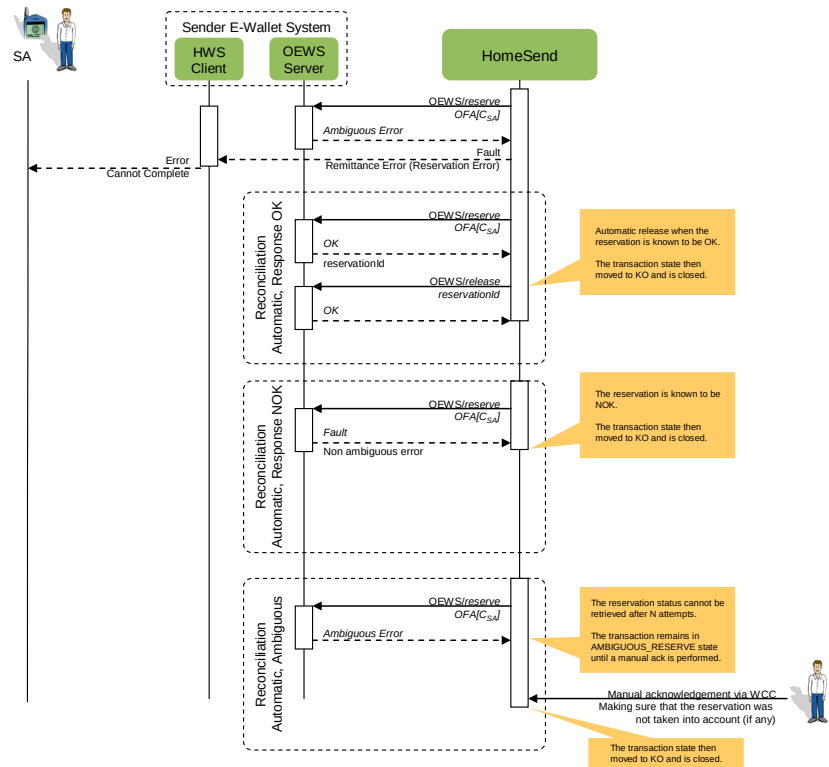


Figure 41 - Reconciliation after an ambiguous OEWS/reserve request

11.1.3.2 Completing the Transaction after an Ambiguous/Error Charge Request

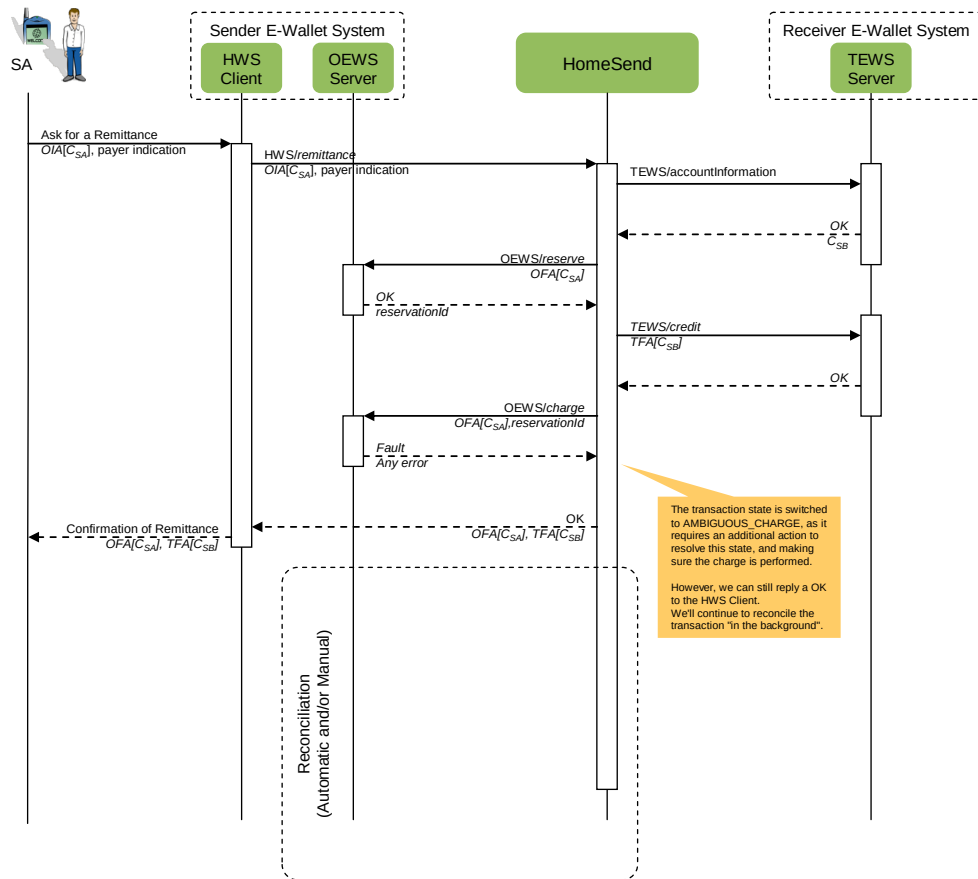


Figure 42 – OEWS/charge error, requesting a reconciliation to complete the transaction

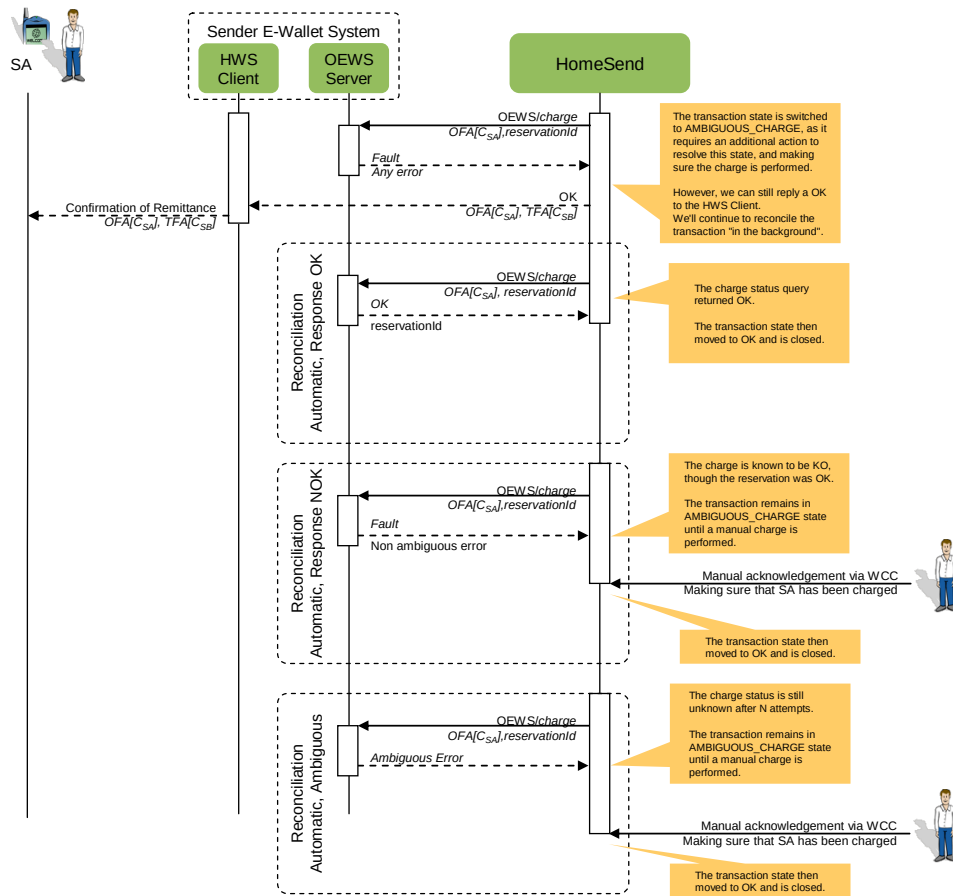


Figure 43 - Automatic reconciliation after an OEWS/charge error

11.1.3.3 Completing the Transaction after an Ambiguous/Error Release Request

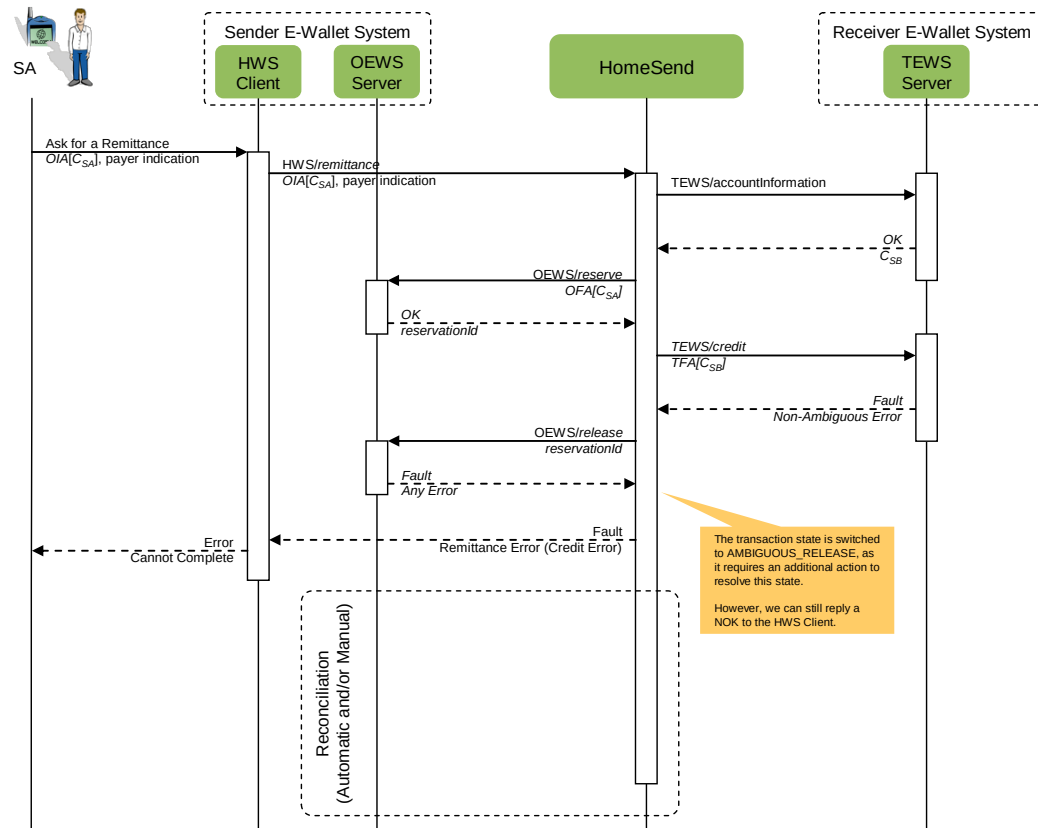


Figure 44 - OEWS/release error, requesting a reconciliation to complete the transaction

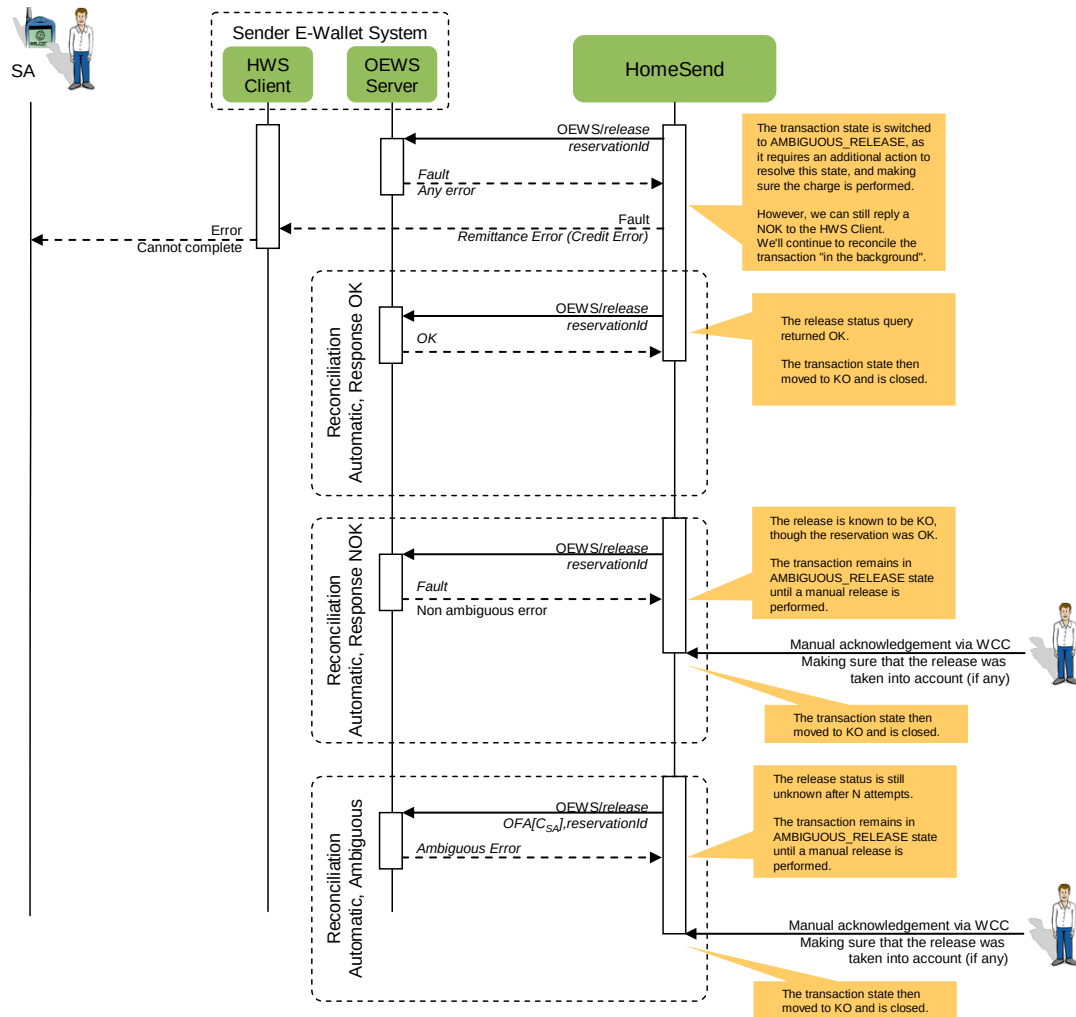


Figure 45 - Automatic reconciliation after an OEWS/release error

11.1.3.4 Recovering from Ambiguous Credit Request

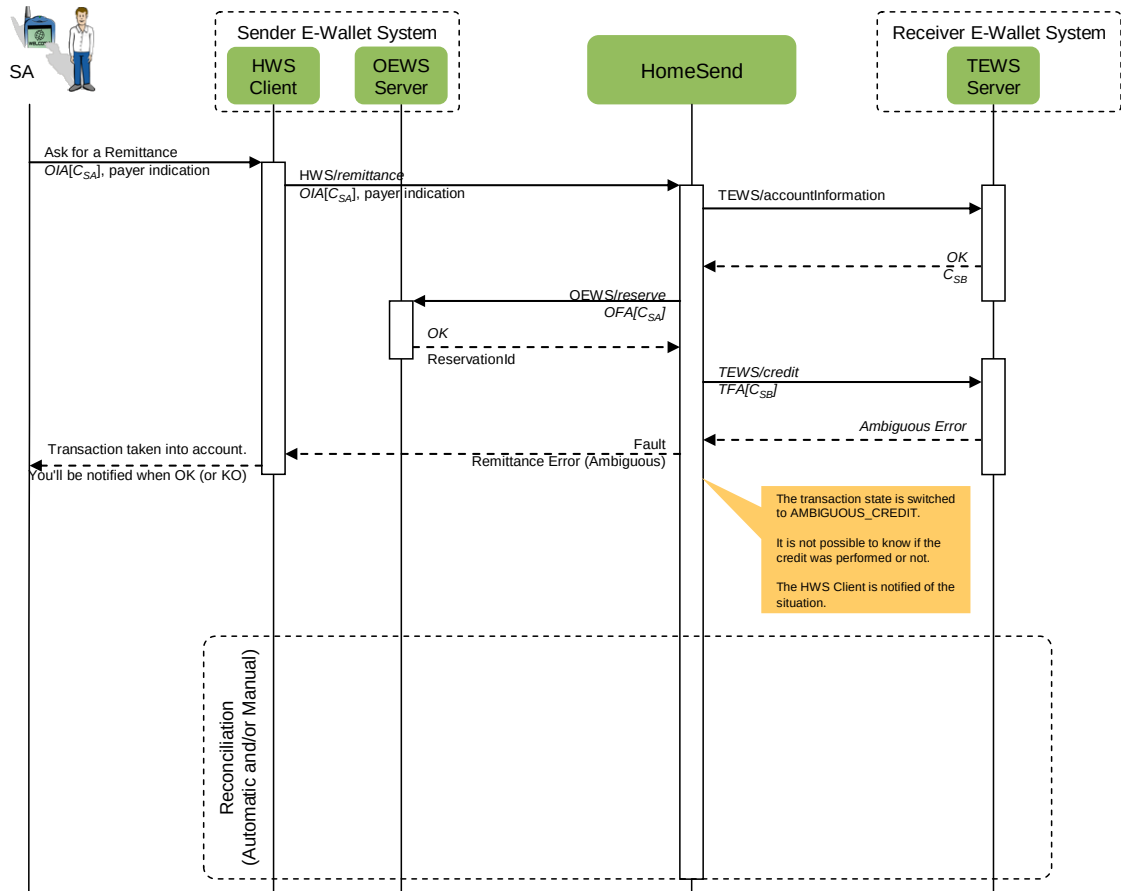


Figure 46 - Ambiguous TEWS/credit, requesting a reconciliation

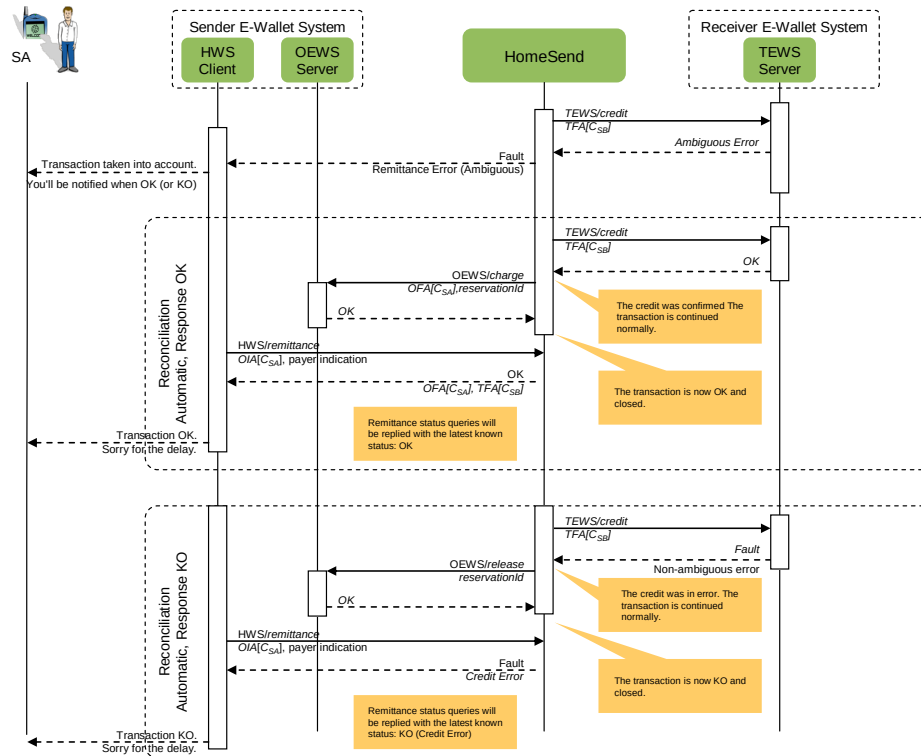


Figure 47 - Reconciliation after an ambiguous TEWS/credit (1)

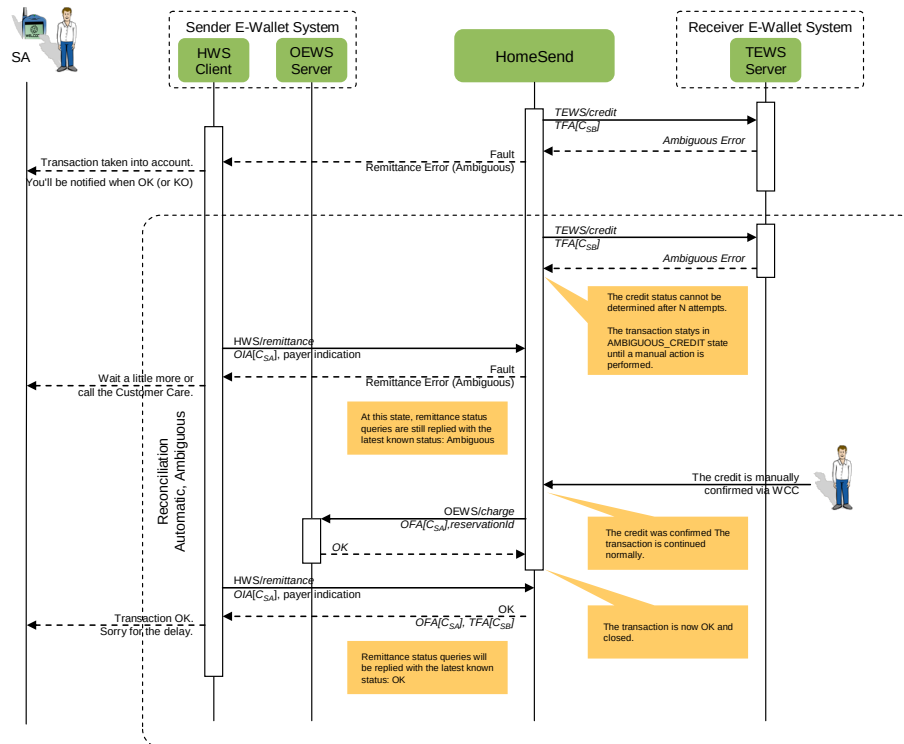


Figure 48 - Reconciliation after an ambiguous TEWS/credit (2)

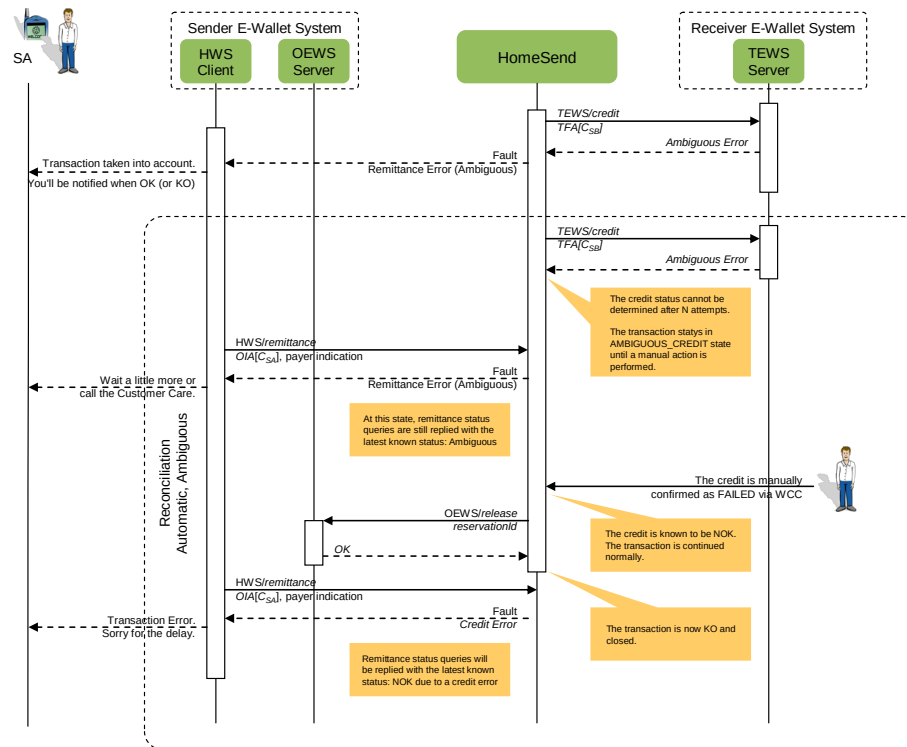


Figure 49 - Reconciliation after an ambiguous TEWS/credit (3)

11.2 WSDL Files

The WSDL representation of these interface specifications are provided as a references when developing a HWS client, a OEWS server, or a TEWS server/R-TEWS client.



Tip:

These WSDL files are also embedded into this PDF document, as attached files.

11.2.1 HomeSend Web Service (HWS) 2.3

The current HWS 2.3 version is HWS 2.3.2.

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- HomeSend Web Service (HWS) 2.3.1 -->

<definitions name="homesend_hws_2_3_1" targetNamespace="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns="http://schemas.xmlsoap.org/wsdl/" xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
  xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/hws/2.0" xmlns:hws="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:hws21="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:hws22="http://www.eservglobal.com/homesend/types/2.2"
  xmlns:hws23="http://www.eservglobal.com/homesend/types/2.3"
  xmlns:hws21="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:hws22="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:hws23="http://www.eservglobal.com/homesend/hws/2.3">

  <documentation>Version 2.3, extending 2.2.
  Minor revisions:
  - 2.3.1: relaxed requirements on sourceUri and destinationUri in remittanceRequest.
  - 2.3.2: relaxed enumeration requirement on cbsType, which is now a string (0..30).
  </documentation>

  <types>

    <!-- HomeSend API Shared Types -->
```

```
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.0" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.0">

  <xsd:simpleType name="login">
    <xsd:restriction base="xsd:string">
      <xsd:minLength value="1" />
      <xsd:maxLength value="32" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="password">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="255" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="uri">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="200" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="transactionId">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="30" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="routingTag">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="300" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="senderName">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="200" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="senderAddress">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="200" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="description">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="200" />
    </xsd:restriction>
  </xsd:simpleType>

  <!-- Common Complex Types -->
  <xsd:complexType name="SenderInformation">
    <xsd:sequence>
      <xsd:element name="name" type="tns:senderName" minOccurs="1"
        nillable="false" />
      <xsd:element name="address" type="tns:senderAddress"
        minOccurs="1" nillable="false" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="AmountInformation">
    <xsd:sequence>
      <xsd:element name="currency" type="xsd:string" minOccurs="1"
        nillable="false" />
      <xsd:element name="amount" type="xsd:decimal" minOccurs="1"
        nillable="false" />
      <xsd:element name="grace" type="xsd:int" minOccurs="0"
        nillable="true" />
      <xsd:element name="validity" type="xsd:int" minOccurs="0"
        nillable="true" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="Security">
    <xsd:sequence>
      <xsd:element name="login" type="tns:login" minOccurs="1"
        nillable="false" />
      <xsd:element name="password" type="tns:password"
        minOccurs="1" nillable="false" />
    </xsd:sequence>
  </xsd:complexType>

  <!-- Exception Types -->
  <xsd:complexType name="BaseException">
    <xsd:sequence>
      <xsd:element name="code" type="xsd:string" minOccurs="1"
        nillable="false" />
      <xsd:element name="message" type="xsd:string" minOccurs="1"
        nillable="false" />
      <xsd:element name="parameters" type="xsd:string"
        minOccurs="0" maxOccurs="unbounded" nillable="true" />
    </xsd:sequence>
  </xsd:complexType>
```

```

</xsd:schema>

<!-- HomeSend API 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.1" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.1">

  <xsd:simpleType name="vendorSpecificValue">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="1000" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:complexType name="VendorSpecificField">
    <xsd:sequence>
      <xsd:element name="vendorId" type="xsd:unsignedInt"
        minOccurs="1" maxOccurs="1" />
      <xsd:element name="fieldId" type="xsd:unsignedInt"
        minOccurs="1" maxOccurs="1" />
      <xsd:element name="value" type="tns:vendorSpecificValue"
        minOccurs="1" maxOccurs="1" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="VendorSpecificFields">
    <xsd:sequence>
      <!--
        maxOccurs should normally be set to 50. However, most development
        tools do not support an integer value as maxOccurs, outside 0 or 1.
      -->
      <xsd:element name="vsf" type="tns:VendorSpecificField"
        minOccurs="0" maxOccurs="unbounded" />
    </xsd:sequence>
  </xsd:complexType>

</xsd:schema>

<!-- HWS 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/hws/2.1">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
  <xsd:element name="vendorSpecificFields" type="tns:VendorSpecificFields"
    nillable="false" />

</xsd:schema>

<!-- HomeSend API 2.2 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.2" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.2">

  <xsd:simpleType name="transactionState">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="100" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="transactionStateLabel">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="100" />
    </xsd:restriction>
  </xsd:simpleType>

</xsd:schema>

<!-- HWS 2.2 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:tns="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:hws="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:hos22="http://www.eservglobal.com/homesend/types/2.2">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.1" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.3" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.3" />

  <xsd:complexType name="CompleteTransactionStatus">
    <xsd:sequence>
      <xsd:element name="transactionSummary" type="hws:TransactionSummary" />
      <xsd:element ref="hos21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="RejectedTransactionStatus">
    <xsd:sequence>
      <xsd:element name="code" type="xsd:string" />
      <xsd:element name="message" type="xsd:string" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="PendingTransactionStatus">
    <xsd:sequence>
      <xsd:element name="maxCompletionDate" type="xsd:dateTime" />

```

```

        <xsd:element name="state" type="hos22:transactionState" />
        <xsd:element name="label" type="hos22:transactionStateLabel" />
        <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
        <xsd:element ref="hws23:stateSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="TransactionStatus">
    <xsd:choice>
        <xsd:element name="complete" type="tns:CompleteTransactionStatus" />
        <xsd:element name="rejected" type="tns:RejectedTransactionStatus" />
        <xsd:element name="pending" type="tns:PendingTransactionStatus" />
    </xsd:choice>
</xsd:complexType>

<xsd:complexType name="ProvisionalResponse">
    <xsd:sequence>
        <xsd:element name="maxCompletionDate" type="xsd:dateTime" minOccurs="1" nillable="false" />
        <xsd:element name="state" type="hos22:transactionState" minOccurs="1" nillable="false" />
        <xsd:element name="label" type="hos22:transactionStateLabel" minOccurs="1" nillable="false" />
        <xsd:element ref="hws23:stateSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

<xsd:element name="provisionalResponse" type="tns:ProvisionalResponse"
    nillable="false" />

<!-- Status -->
<xsd:element name="statusRequest" type="tns:StatusRequest" />
<xsd:complexType name="StatusRequest">
    <xsd:sequence>
        <xsd:element name="security" type="hos:Security"
            minOccurs="1" nillable="false" />
        <xsd:element name="transactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="hsTransactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

<xsd:element name="statusResponse" type="tns:StatusResponse" />
<xsd:complexType name="StatusResponse">
    <xsd:sequence>
        <xsd:element name="transactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="hsTransactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="status" type="tns:TransactionStatus" minOccurs="1" nillable="false" />
        <!-- Request-level VSFs -->
        <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

<!-- Device Watchdog -->
<xsd:complexType name="DeviceWatchdogRequest" />
<xsd:complexType name="DeviceWatchdogResponse" />

<xsd:element name="deviceWatchdogRequest" type="tns:DeviceWatchdogRequest" />
<xsd:element name="deviceWatchdogResponse" type="tns:DeviceWatchdogResponse" />

</xsd:schema>

<!-- HomeSend API 2.3 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
    xmlns:tns="http://www.eservglobal.com/homesend/types/2.3" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
    targetNamespace="http://www.eservglobal.com/homesend/types/2.3">

    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />

    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
    <xsd:complexType name="StateSpecificFields">
        <xsd:sequence>
            <!--
                maxOccurs should normally be set to 50. However, most development
                tools do not support an integer value as maxOccurs, outside 0 or 1.
            -->
            <xsd:element name="ssf" type="hos21:VendorSpecificField"
                minOccurs="0" maxOccurs="unbounded" />
        </xsd:sequence>
    </xsd:complexType>

</xsd:schema>

<!-- HWS 2.3 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
    targetNamespace="http://www.eservglobal.com/homesend/hws/2.3"
    xmlns:tns="http://www.eservglobal.com/homesend/hws/2.3"
    xmlns:hws="http://www.eservglobal.com/homesend/hws/2.0"
    xmlns:hos22="http://www.eservglobal.com/homesend/types/2.2">

    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />

```

```

<xsd:import namespace="http://www.eservglobal.com/homesend/types/2.3" />
<xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.0" />
<xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.1" />

<xsd:import namespace="http://www.eservglobal.com/homesend/types/2.3" />
<xsd:element name="stateSpecificFields" type="hos23:StateSpecificFields"
  nillable="false" />

  <xsd:complexType name="ReverseQuote">
    <xsd:sequence>
      <xsd:element name="sendingCurrency" type="xsd:string" minOccurs="1" />
    </xsd:sequence>
  </xsd:complexType>

<xsd:element name="reverseQuote" type="tns:ReverseQuote" nillable="false" />

<xsd:element name="initialAmount" type="hos:AmountInformation" nillable="false" />

<xsd:complexType name="CancellationReason">
  <xsd:sequence>
    <xsd:element name="code" type="xsd:int" minOccurs="1" />
    <xsd:element name="message" type="xsd:string" minOccurs="1" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="reason" type="tns:CancellationReason" nillable="false" />

  <xsd:element name="expirationDate" type="xsd:dateTime" nillable="false" />

<!-- Amend Operation -->
<xsd:element name="amendRequest" type="tns:AmendRequest" />
<xsd:complexType name="AmendRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="amendedFields" minOccurs="1" type="hos21:VendorSpecificFields" />
    <!-- Request-level VSFs -->
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="amendResponse" type="tns:AmendResponse" />
<xsd:complexType name="AmendResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <!-- Request-level VSFs -->
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

</xsd:schema>

  <!-- HWS Definitions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/hws/2.0" version="2.3">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.1" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.2" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/hws/2.3" />

  <xsd:simpleType name="payer">
    <xsd:restriction base="xsd:string">
      <xsd:enumeration value="RECEIVER" />
      <xsd:enumeration value="SENDER" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="remittanceCbsType">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="30" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="eRechargeCbsType">
    <xsd:restriction base="xsd:string">
      <xsd:enumeration value="ERetailRoamingRecharge" />
      <xsd:enumeration value="ATMRoamingRecharge" />
      <xsd:enumeration value="ElectronicRoamingRecharge" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="aocValidity">
    <!-- AoC Validity period in seconds. 0 is a reserved value for "unlimited" -->
    <xsd:restriction base="xsd:int">
      <xsd:minInclusive value="0" />
    </xsd:restriction>
  </xsd:simpleType>

```



```

<xsd:complexType name="TransactionSummary">
  <xsd:sequence>
    <xsd:element name="commissionAmount" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="chargedAmount" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="creditedAmount" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element ref="hws23:initialAmount" minOccurs="1" />
  </xsd:sequence>
</xsd:complexType>

<!-- Request and Response Types -->

<!-- AdviceOfCharge Operation -->
<xsd:element name="adviceOfChargeRequest" type="tns:AdviceOfChargeRequest"
  nillable="true" />
<xsd:complexType name="AdviceOfChargeRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="sourceUri" type="hos:uri" minOccurs="1"
      nillable="false" />
    <xsd:element name="destinationUri" type="hos:uri"
      minOccurs="1" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="cbsType" type="tns:remittanceCbsType"
      minOccurs="0" nillable="true" default="EMoneyToEMoneyTransfer" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="hws23:reverseQuote" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="adviceOfChargeResponse" type="tns:AdviceOfChargeResponse"
  nillable="true" />
<xsd:complexType name="AdviceOfChargeResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="senderPayerProposal" type="tns:TransactionSummary"
      minOccurs="1" nillable="true" />
    <xsd:element name="receiverPayerProposal" type="tns:TransactionSummary"
      minOccurs="1" nillable="true" />
    <xsd:element name="expirationDelay" type="tns:aocValidity"
      minOccurs="1" nillable="false" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="hws23:expirationDate" minOccurs="1" />
  </xsd:sequence>
</xsd:complexType>

<!-- Remittance Operation -->
<xsd:element name="remittanceRequest" type="tns:RemittanceRequest"
  nillable="true" />
<xsd:complexType name="RemittanceRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="0" maxOccurs="1" nillable="true" />
    <xsd:element name="sourceUri" type="hos:uri" minOccurs="0"
      nillable="false" />
    <xsd:element name="destinationUri" type="hos:uri"
      minOccurs="0" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="payer" type="tns:payer" default="RECEIVER"
      minOccurs="0" nillable="true" />
    <xsd:element name="cbsType" type="tns:remittanceCbsType"
      minOccurs="0" nillable="true" default="EMoneyToEMoneyTransfer" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="reservationRequired" type="xsd:boolean"
      minOccurs="0" nillable="false" default="true" />
    <xsd:element name="senderInformation" type="hos:SenderInformation"
      minOccurs="0" nillable="false" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="remittanceResponse" type="tns:RemittanceResponse"
  nillable="true" />
<xsd:complexType name="RemittanceResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />

```

```

    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionSummary" type="tns:TransactionSummary"
      minOccurs="1" nillable="true" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="hws22:provisionalResponse" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- eRecharge Operation -->
<xsd:element name="eRechargeRequest" type="tns:ERechargeRequest"
  nillable="true" />
<xsd:complexType name="ERechargeRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="sourceUri" type="hos:uri" minOccurs="1"
      nillable="false" />
    <xsd:element name="destinationUri" type="hos:uri"
      minOccurs="1" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="cbsType" type="tns:eRechargeCbsType"
      minOccurs="0" nillable="true" default="ElectronicRoamingRecharge" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="senderInformation" type="hos:SenderInformation"
      minOccurs="0" nillable="false" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="eRechargeResponse" type="tns:ERechargeResponse"
  nillable="true" />
<xsd:complexType name="ERechargeResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionSummary" type="tns:TransactionSummary"
      minOccurs="1" nillable="true" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Cancellation Operation -->
<xsd:element name="cancellationRequest" type="tns:CancellationRequest"
  nillable="true" />
<xsd:complexType name="CancellationRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="hws23:reason" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="cancellationResponse" type="tns:CancellationResponse"
  nillable="true" />
<xsd:complexType name="CancellationResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="hws21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Exceptions -->
<xsd:complexType name="HomeSendServiceException">
  <xsd:complexContent>
    <xsd:extension base="hos:BaseException">
      <xsd:sequence>
        <!-- The originating transaction ID is always provided -->
        <xsd:element name="transactionId" type="hos:transactionId"
          minOccurs="1" nillable="false" />
        <!-- The HomeSend transaction ID may not have been generated yet -->
        <xsd:element name="hsTransactionId" type="hos:transactionId"
          minOccurs="1" nillable="true" />
      </xsd:sequence>
    </xsd:extension>
  </xsd:complexContent>
</xsd:complexType>

<xsd:element name="homeSendServiceException" type="tns:HomeSendServiceException"
  nillable="false" />

```

```
</xsd:schema>
</types>

<!-- HWS Messages Definitions -->
<message name="remittance">
  <part name="remittanceRequest" element="tns:remittanceRequest" />
</message>

<message name="remittanceResponse">
  <part name="remittanceResponse" element="tns:remittanceResponse" />
</message>

<message name="eRecharge">
  <part name="eRechargeRequest" element="tns:eRechargeRequest" />
</message>

<message name="eRechargeResponse">
  <part name="eRechargeResponse" element="tns:eRechargeResponse" />
</message>

<message name="cancellation">
  <part name="cancellationRequest" element="tns:cancellationRequest" />
</message>

<message name="cancellationResponse">
  <part name="cancellationResponse" element="tns:cancellationResponse" />
</message>

<message name="adviceOfCharge">
  <part name="adviceOfChargeRequest" element="tns:adviceOfChargeRequest" />
</message>

<message name="adviceOfChargeResponse">
  <part name="adviceOfChargeResponse" element="tns:adviceOfChargeResponse" />
</message>

<message name="status">
  <part name="statusRequest" element="hws22:statusRequest" />
</message>

<message name="statusResponse">
  <part name="statusResponse" element="hws22:statusResponse" />
</message>

<message name="deviceWatchdog">
  <part name="deviceWatchdogRequest" element="hws22:deviceWatchdogRequest" />
</message>

<message name="deviceWatchdogResponse">
  <part name="deviceWatchdogResponse" element="hws22:deviceWatchdogResponse" />
</message>

<message name="amend">
  <part name="amendRequest" element="hws23:amendRequest" />
</message>

<message name="amendResponse">
  <part name="amendResponse" element="hws23:amendResponse" />
</message>

<!-- Exception Messages Definitions -->
<message name="homeSendServiceException">
  <part name="homeSendServiceException" element="tns:homeSendServiceException" />
</message>

<!-- HWS Port Definition -->
<portType name="HWSPortType_2_3">
  <operation name="eRecharge">
    <input message="tns:eRecharge" />
    <output message="tns:eRechargeResponse" />
    <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
  </operation>
  <operation name="remittance">
    <input message="tns:remittance" />
    <output message="tns:remittanceResponse" />
    <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
  </operation>
  <operation name="cancellation">
    <input message="tns:cancellation" />
    <output message="tns:cancellationResponse" />
    <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
  </operation>
  <operation name="adviceOfCharge">
    <input message="tns:adviceOfCharge" />
    <output message="tns:adviceOfChargeResponse" />
    <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
  </operation>
  <operation name="status">
    <input message="tns:status" />
    <output message="tns:statusResponse" />
    <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
  </operation>
  <operation name="deviceWatchdog">
    <input message="tns:deviceWatchdog" />
```

```
<output message="tns:deviceWatchdogResponse" />
</operation>
<operation name="amend">
  <input message="tns:amend" />
  <output message="tns:amendResponse" />
  <fault name="HomeSendServiceException" message="tns:homeSendServiceException" />
</operation>
</portType>

<!-- HWS Binding (SOAP/http, document/literal style) -->
<binding name="HWSBinding_2_3" type="tns:HWSPortType_2_3">
  <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http" />
  <operation name="remittance">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="remittanceRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="remittanceResponse" />
    </output>
    <fault name="HomeSendServiceException">
      <soap:fault name="HomeSendServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>

  <operation name="eRecharge">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="eRechargeRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="eRechargeResponse" />
    </output>
    <fault name="HomeSendServiceException">
      <soap:fault name="HomeSendServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>

  <operation name="cancellation">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="cancellationRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="cancellationResponse" />
    </output>
    <fault name="HomeSendServiceException">
      <soap:fault name="HomeSendServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>

  <operation name="adviceOfCharge">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="adviceOfChargeRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="adviceOfChargeResponse" />
    </output>
    <fault name="HomeSendServiceException">
      <soap:fault name="HomeSendServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>

  <operation name="status">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="statusRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="statusResponse" />
    </output>
    <fault name="HomeSendServiceException">
      <soap:fault name="HomeSendServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>

  <operation name="deviceWatchdog">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="deviceWatchdogRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="deviceWatchdogResponse" />
    </output>
  </operation>

  <operation name="amend">
    <soap:operation soapAction="" />
    <input>
```

```

        <soap:body use="literal" parts="amendRequest" />
    </input>
    <output>
        <soap:body use="literal" parts="amendResponse" />
    </output>
    <fault name="HomeSendServiceException">
        <soap:fault name="HomeSendServiceException" use="literal"
            encodingStyle="" />
    </fault>
</operation>

</binding>

<!-- HWS Service Definition -->
<service name="HomeSend HWS_2_3">
    <port name="HWSPort_2_3" binding="tns:HWSBinding_2_3">
        <soap:address location="http://localhost/hws/2.3" />
    </port>
</service>
</definitions>

```

11.2.2 Originating Endpoint Web Service (OEWS) 2.3

```

<?xml version="1.0" encoding="UTF-8"?>
<!-- HomeSend Originating Endpoint Web Service (OEWS) 2.3 -->

<definitions name="homesend_oews_2_3" targetNamespace="http://www.eservglobal.com/homesend/oews/2.0"
    xmlns="http://schemas.xmlsoap.org/wsdl/" xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
    xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
    xmlns:tns="http://www.eservglobal.com/homesend/oews/2.0" xmlns:hos="http://www.eservglobal.com/homesend/types/2.0"
    xmlns:hos21="http://www.eservglobal.com/homesend/types/2.1" xmlns:oews21="http://www.eservglobal.com/homesend/oews/2.1"
    xmlns:oews22="http://www.eservglobal.com/homesend/oews/2.2">

    <documentation>Version 2.3, extending 2.2</documentation>

    <types>

        <!-- HomeSend API Shared Types -->
        <xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
            xmlns:tns="http://www.eservglobal.com/homesend/types/2.0" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
            targetNamespace="http://www.eservglobal.com/homesend/types/2.0">

            <xsd:simpleType name="login">
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="32" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="password">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="255" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="uri">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="200" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="transactionId">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="30" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="routingTag">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="300" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="senderName">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="200" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="senderAddress">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="200" />
                </xsd:restriction>
            </xsd:simpleType>

            <xsd:simpleType name="description">
                <xsd:restriction base="xsd:string">
                    <xsd:maxLength value="200" />
                </xsd:restriction>
            </xsd:simpleType>

        <!-- Common Complex Types -->
        <xsd:complexType name="SenderInformation">

```

```

<xsd:sequence>
  <xsd:element name="name" type="tns:senderName" minOccurs="1"
    nillable="false" />
  <xsd:element name="address" type="tns:senderAddress"
    minOccurs="1" nillable="false" />
</xsd:sequence>
</xsd:complexType>

<xsd:complexType name="AmountInformation">
  <xsd:sequence>
    <xsd:element name="currency" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="amount" type="xsd:decimal" minOccurs="1"
      nillable="false" />
    <xsd:element name="grace" type="xsd:int" minOccurs="0"
      nillable="true" />
    <xsd:element name="validity" type="xsd:int" minOccurs="0"
      nillable="true" />
  </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="Security">
  <xsd:sequence>
    <xsd:element name="login" type="tns:login" minOccurs="1"
      nillable="false" />
    <xsd:element name="password" type="tns:password"
      minOccurs="1" nillable="false" />
  </xsd:sequence>
</xsd:complexType>

<!-- Exception Types -->
<xsd:complexType name="BaseException">
  <xsd:sequence>
    <xsd:element name="code" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="message" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="parameters" type="xsd:string"
      minOccurs="0" maxOccurs="unbounded" nillable="true" />
  </xsd:sequence>
</xsd:complexType>

</xsd:schema>

<!-- HomeSend API 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.1" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.1">

  <xsd:simpleType name="VendorSpecificValue">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="1000" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:complexType name="VendorSpecificField">
    <xsd:sequence>
      <xsd:element name="vendorId" type="xsd:unsignedInt"
        minOccurs="1" maxOccurs="1" />
      <xsd:element name="fieldId" type="xsd:unsignedInt"
        minOccurs="1" maxOccurs="1" />
      <xsd:element name="value" type="tns:VendorSpecificValue"
        minOccurs="1" maxOccurs="1" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="VendorSpecificFields">
    <xsd:sequence>
      <!--
        maxOccurs should normally be set to 50. However, most development
        tools do not support an integer value as maxOccurs, outside 0 or 1.
      -->
      <xsd:element name="vsf" type="tns:VendorSpecificField"
        minOccurs="0" maxOccurs="unbounded" />
    </xsd:sequence>
  </xsd:complexType>

</xsd:schema>

<!-- OEWS 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/oews/2.1">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
  <xsd:element name="VendorSpecificFields" type="tns:VendorSpecificFields"
    nillable="false" />

</xsd:schema>

<!-- OEWS 2.2 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:tns="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:oews20="http://www.eservglobal.com/homesend/oews/2.0">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />

```

```

<xsd:import namespace="http://www.eservglobal.com/homesend/oews/2.0" />
<xsd:import namespace="http://www.eservglobal.com/homesend/oews/2.1" />
<xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />

<xsd:complexType name="ChargebackReason">
  <xsd:sequence>
    <xsd:element name="code" type="xsd:int" minOccurs="1" />
    <xsd:element name="message" type="xsd:string" minOccurs="1" />
  </xsd:sequence>
</xsd:complexType>

<!-- Chargeback -->
<xsd:element name="chargebackRequest" type="tns:ChargebackRequest" />
<xsd:complexType name="ChargebackRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="reservationId" type="oews20:reservationId"
      minOccurs="1" nillable="true" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="reason" type="tns:ChargebackReason"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="false" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="chargebackResponse" type="tns:ChargebackResponse" />
<xsd:complexType name="ChargebackResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Device Watchdog -->
<xsd:complexType name="DeviceWatchdogRequest" />
<xsd:complexType name="DeviceWatchdogResponse" />

<xsd:element name="deviceWatchdogRequest" type="tns:DeviceWatchdogRequest" />
<xsd:element name="deviceWatchdogResponse" type="tns:DeviceWatchdogResponse" />

</xsd:schema>

<!-- OEWS Definitions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/oews/2.0" version="2.3">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/oews/2.1" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/oews/2.2" />

  <xsd:simpleType name="reservationId">
    <xsd:restriction base="xsd:string">
      <xsd:minLength value="1" />
      <xsd:maxLength value="30" />
    </xsd:restriction>
  </xsd:simpleType>

  <!-- Request and Response Types -->

  <!-- Reserve Operation -->
  <xsd:element name="reserveRequest" type="tns:ReserveRequest"
    nillable="true" />
  <xsd:complexType name="ReserveRequest">
    <xsd:sequence>
      <xsd:element name="security" type="hos:Security"
        minOccurs="1" nillable="false" />
      <xsd:element name="transactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element name="hsTransactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element name="uri" type="hos:uri" minOccurs="1"
        nillable="false" />
      <xsd:element name="amountInformation" type="hos:AmountInformation"
        minOccurs="1" nillable="false" />
      <xsd:element name="routingTag" type="hos:routingTag"
        minOccurs="0" nillable="true" />
      <xsd:element name="description" type="hos:description"
        minOccurs="0" nillable="false" />
      <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:element name="reserveResponse" type="tns:ReserveResponse"

```

```

        nillable="true" />
<xsd:complexType name="ReserveResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="reservationId" type="tns:reservationId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Release Operation -->
<xsd:element name="releaseRequest" type="tns:ReleaseRequest"
  nillable="true" />
<xsd:complexType name="ReleaseRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="reservationId" type="tns:reservationId"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="releaseResponse" type="tns:ReleaseResponse"
  nillable="true" />
<xsd:complexType name="ReleaseResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Charge Operation -->
<xsd:element name="chargeRequest" type="tns:ChargeRequest"
  nillable="true" />
<xsd:complexType name="ChargeRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="reservationId" type="tns:reservationId"
      minOccurs="1" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="chargeResponse" type="tns:ChargeResponse"
  nillable="true" />
<xsd:complexType name="ChargeResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="oews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Exceptions -->
<xsd:complexType name="OriginatingServiceException">
  <xsd:complexContent>
    <xsd:extension base="hos:BaseException">
      <xsd:sequence>
        <!-- The originating transaction ID should always be provided -->
        <xsd:element name="transactionId" type="hos:transactionId"
          minOccurs="1" nillable="false" />
        <!-- The HomeSend transaction ID should always be provided, too -->
        <xsd:element name="hsTransactionId" type="hos:transactionId"
          minOccurs="1" nillable="false" />
      </xsd:sequence>
    </xsd:extension>
  </xsd:complexContent>
</xsd:complexType>

```



```
<xsd:element name="originatingServiceException" type="tns:OriginatingServiceException"
  nillable="false" />

</xsd:schema>
</types>

<!-- OEWS Messages Definitions -->
<message name="reserve">
  <part name="reserveRequest" element="tns:reserveRequest" />
</message>

<message name="reserveResponse">
  <part name="reserveResponse" element="tns:reserveResponse" />
</message>

<message name="release">
  <part name="releaseRequest" element="tns:releaseRequest" />
</message>

<message name="releaseResponse">
  <part name="releaseResponse" element="tns:releaseResponse" />
</message>

<message name="charge">
  <part name="chargeRequest" element="tns:chargeRequest" />
</message>

<message name="chargeResponse">
  <part name="chargeResponse" element="tns:chargeResponse" />
</message>

<message name="chargeback">
  <part name="chargebackRequest" element="tns:chargebackRequest" />
</message>

<message name="chargebackResponse">
  <part name="chargebackResponse" element="tns:chargebackResponse" />
</message>

<message name="deviceWatchdog">
  <part name="deviceWatchdogRequest" element="tns:deviceWatchdogRequest" />
</message>

<message name="deviceWatchdogResponse">
  <part name="deviceWatchdogResponse" element="tns:deviceWatchdogResponse" />
</message>

<!-- Exception Messages Definitions -->
<message name="originatingServiceException">
  <part name="originatingServiceException" element="tns:originatingServiceException" />
</message>

<!-- OEWS Port Definition -->
<portType name="OEWSPortType_2_3">
  <operation name="reserve">
    <input message="tns:reserve" />
    <output message="tns:reserveResponse" />
    <fault name="OriginatingServiceException" message="tns:originatingServiceException" />
  </operation>
  <operation name="release">
    <input message="tns:release" />
    <output message="tns:releaseResponse" />
    <fault name="OriginatingServiceException" message="tns:originatingServiceException" />
  </operation>
  <operation name="charge">
    <input message="tns:charge" />
    <output message="tns:chargeResponse" />
    <fault name="OriginatingServiceException" message="tns:originatingServiceException" />
  </operation>
  <operation name="chargeback">
    <input message="tns:chargeback" />
    <output message="tns:chargebackResponse" />
    <fault name="OriginatingServiceException" message="tns:originatingServiceException" />
  </operation>
  <operation name="deviceWatchdog">
    <input message="tns:deviceWatchdog" />
    <output message="tns:deviceWatchdogResponse" />
  </operation>
</portType>

<binding name="OEWSBinding_2_3" type="tns:OEWSPortType_2_3">
  <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http" />
  <operation name="reserve">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="reserveRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="reserveResponse" />
    </output>
    <fault name="OriginatingServiceException">
      <soap:fault name="OriginatingServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>
```

```

<operation name="release">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="releaseRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="releaseResponse" />
  </output>
  <fault name="OriginatingServiceException">
    <soap:fault name="OriginatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="charge">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="chargeRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="chargeResponse" />
  </output>
  <fault name="OriginatingServiceException">
    <soap:fault name="OriginatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="chargeback">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="chargebackRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="chargebackResponse" />
  </output>
  <fault name="OriginatingServiceException">
    <soap:fault name="OriginatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="deviceWatchdog">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="deviceWatchdogRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="deviceWatchdogResponse" />
  </output>
</operation>
</binding>

<service name="HomeSend_OEWS_2_3">
  <port name="OEWSPort_2_3" binding="tns:OEWSBinding_2_3">
    <soap:address location="http://localhost/oews/2.3" />
  </port>
</service>
</definitions>

```

11.2.3 Terminating Endpoint Web Service (TEWS and R-TEWS) 2.3

```

<?xml version="1.0" encoding="UTF-8"?>
<!-- HomeSend Terminating Endpoint Web Service (TEWS) and associated Reverse Service (R-TEWS) 2.3 -->

<definitions name="homesend tews 2_2" targetNamespace="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns="http://schemas.xmlsoap.org/wsdl/" xmlns:wsdl="http://schemas.xmlsoap.org/wsdl/"
  xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/tews/2.0" xmlns:hos="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:hos21="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:hos22="http://www.eservglobal.com/homesend/types/2.2"
  xmlns:hos23="http://www.eservglobal.com/homesend/types/2.3"
  xmlns:tews21="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:tews22="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:tews23="http://www.eservglobal.com/homesend/tews/2.3">

  <documentation>Version 2.3, extending 2.2</documentation>

  <types>

    <!-- HomeSend API Shared Types -->
    <xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
      xmlns:tns="http://www.eservglobal.com/homesend/types/2.0" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
      targetNamespace="http://www.eservglobal.com/homesend/types/2.0">

      <xsd:simpleType name="login">
        <xsd:restriction base="xsd:string">
          <xsd:minLength value="1" />
          <xsd:maxLength value="32" />
        </xsd:restriction>
      </xsd:simpleType>

      <xsd:simpleType name="password">
        <xsd:restriction base="xsd:string">
          <xsd:maxLength value="255" />
        </xsd:restriction>

```

```

</xsd:simpleType>

<xsd:simpleType name="uri">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="200" />
  </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="transactionId">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="30" />
  </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="routingTag">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="300" />
  </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="senderName">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="200" />
  </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="senderAddress">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="200" />
  </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="description">
  <xsd:restriction base="xsd:string">
    <xsd:maxLength value="200" />
  </xsd:restriction>
</xsd:simpleType>

<!-- Common Complex Types -->
<xsd:complexType name="SenderInformation">
  <xsd:sequence>
    <xsd:element name="name" type="tns:senderName" minOccurs="1"
      nillable="false" />
    <xsd:element name="address" type="tns:senderAddress"
      minOccurs="1" nillable="false" />
  </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="AmountInformation">
  <xsd:sequence>
    <xsd:element name="currency" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="amount" type="xsd:decimal" minOccurs="1"
      nillable="false" />
    <xsd:element name="grace" type="xsd:int" minOccurs="0"
      nillable="true" />
    <xsd:element name="validity" type="xsd:int" minOccurs="0"
      nillable="true" />
  </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="Security">
  <xsd:sequence>
    <xsd:element name="login" type="tns:login" minOccurs="1"
      nillable="false" />
    <xsd:element name="password" type="tns:password"
      minOccurs="1" nillable="false" />
  </xsd:sequence>
</xsd:complexType>

<!-- Exception Types -->
<xsd:complexType name="BaseException">
  <xsd:sequence>
    <xsd:element name="code" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="message" type="xsd:string" minOccurs="1"
      nillable="false" />
    <xsd:element name="parameters" type="xsd:string"
      minOccurs="0" maxOccurs="unbounded" nillable="true" />
  </xsd:sequence>
</xsd:complexType>

</xsd:schema>

<!-- HomeSend API 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.1" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.1">

  <xsd:simpleType name="vendorSpecificValue">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="1000" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:complexType name="VendorSpecificField">

```

```

<xsd:sequence>
  <xsd:element name="vendorId" type="xsd:unsignedInt"
    minOccurs="1" maxOccurs="1" />
  <xsd:element name="fieldId" type="xsd:unsignedInt"
    minOccurs="1" maxOccurs="1" />
  <xsd:element name="value" type="tns:vendorSpecificValue"
    minOccurs="1" maxOccurs="1" />
</xsd:sequence>
</xsd:complexType>

<xsd:complexType name="VendorSpecificFields">
  <xsd:sequence>
    <!--
      maxOccurs should normally be set to 50. However, most development
      tools do not support an integer value as maxOccurs, outside 0 or 1.
    -->
    <xsd:element name="vsf" type="tns:VendorSpecificField"
      minOccurs="0" maxOccurs="unbounded" />
  </xsd:sequence>
</xsd:complexType>

</xsd:schema>

<!-- TEWS 2.1 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/tews/2.1">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
  <xsd:element name="vendorSpecificFields" type="hos21:VendorSpecificFields"
    nillable="false" />

</xsd:schema>

<!-- HomeSend API 2.2 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.2" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.2">

  <xsd:simpleType name="transactionState">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="100" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="transactionStateLabel">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="100" />
    </xsd:restriction>
  </xsd:simpleType>

</xsd:schema>

<!-- TEWS/R-TEWS 2.2 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:tns="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:tews20="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:hos22="http://www.eservglobal.com/homesend/types/2.2">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.1" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.3" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />

  <xsd:simpleType name="token">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="100" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:simpleType name="endpointAddress">
    <xsd:restriction base="xsd:string">
      <xsd:maxLength value="500" />
    </xsd:restriction>
  </xsd:simpleType>

  <xsd:complexType name="CompleteTransactionStatus">
    <xsd:sequence>
      <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="RejectedTransactionStatus">
    <xsd:sequence>
      <xsd:element name="code" type="xsd:string" />
      <xsd:element name="message" type="xsd:string" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="PendingTransactionStatus">
    <xsd:sequence>
      <xsd:element name="maxCompletionDate" type="xsd:dateTime" />
      <xsd:element name="state" type="hos22:transactionState" />
      <xsd:element name="stateLabel" type="hos22:transactionStateLabel" />
      <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
      <xsd:element ref="tews23:stateSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

```

```

</xsd:complexType>

<xsd:complexType name="TransactionStatus">
  <xsd:choice>
    <xsd:element name="complete" type="tns:CompleteTransactionStatus" />
    <xsd:element name="rejected" type="tns:RejectedTransactionStatus" />
    <xsd:element name="pending" type="tns:PendingTransactionStatus" />
  </xsd:choice>
</xsd:complexType>

<xsd:complexType name="ProvisionalResponse">
  <xsd:sequence>
    <xsd:element name="maxCompletionDate" type="xsd:dateTime" minOccurs="1" nillable="false" />
    <xsd:element name="state" type="hos22:transactionState" minOccurs="1" nillable="false" />
    <xsd:element name="label" type="hos22:transactionStateLabel" minOccurs="1" nillable="false" />
    <xsd:element ref="tews23:stateSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="provisionalResponse" type="tns:ProvisionalResponse"
  nillable="false" />

<xsd:complexType name="ProvisionalResponseSupport">
  <xsd:sequence>
    <xsd:element name="reportStatusTo" type="tns:endpointAddress" minOccurs="1" nillable="false" />
    <xsd:element name="token" type="tns:token" minOccurs="1" nillable="false" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="provisionalResponseSupport" type="tns:ProvisionalResponseSupport"
  nillable="false" />

<!-- Status -->
<xsd:element name="statusRequest" type="tns:StatusRequest" />
<xsd:complexType name="StatusRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="statusResponse" type="tns:StatusResponse" />
<xsd:complexType name="StatusResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="status" type="tns:TransactionStatus" minOccurs="1" nillable="false" />
    <!-- Request-level VSFs -->
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Status Report (R-TEWS) -->
<xsd:element name="statusReportRequest" type="tns:StatusReportRequest" />
<xsd:complexType name="StatusReportRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="token" type="tns:token"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="status" type="tns:TransactionStatus" minOccurs="1" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="statusReportResponse" type="tns:StatusReportResponse" />
<xsd:complexType name="StatusReportResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Device Watchdog -->
<xsd:complexType name="DeviceWatchdogRequest" />
<xsd:complexType name="DeviceWatchdogResponse" />

<xsd:element name="deviceWatchdogRequest" type="tns:DeviceWatchdogRequest" />
<xsd:element name="deviceWatchdogResponse" type="tns:DeviceWatchdogResponse" />

```

```

</xsd:schema>

<!-- HomeSend API 2.3 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/types/2.3" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://www.eservglobal.com/homesend/types/2.3">

  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.1" />
  <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.2" />

  <xsd:complexType name="StateSpecificFields">
    <xsd:sequence>
      <!--
        maxOccurs should normally be set to 50. However, most development
        tools do not support an integer value as maxOccurs, outside 0 or 1.
      -->
      <xsd:element name="ssf" type="hos21:VendorSpecificField"
        minOccurs="0" maxOccurs="unbounded" />
    </xsd:sequence>
  </xsd:complexType>

</xsd:schema>

<!-- TEWS 2.3 Extensions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:tns="http://www.eservglobal.com/homesend/tews/2.3"
  xmlns:tns2="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:tns1="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:tns2_2="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:tns2_1="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:tns2_2_2="http://www.eservglobal.com/homesend/types/2.2"
  xmlns:tns2_3="http://www.eservglobal.com/homesend/types/2.3" />

  <xsd:complexType name="Currencies">
    <xsd:sequence>
      <xsd:element name="currency" type="xsd:string" minOccurs="0" maxOccurs="unbounded" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:element name="secondaryCurrencies" type="tns:Currencies"
    nillable="false" />

  <xsd:element name="stateSpecificFields" type="hos23:StateSpecificFields"
    nillable="false" />

  <xsd:complexType name="CancellationReason">
    <xsd:sequence>
      <xsd:element name="code" type="xsd:int" minOccurs="1" />
      <xsd:element name="message" type="xsd:string" minOccurs="1" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:complexType name="ReversalReason">
    <xsd:sequence>
      <xsd:element name="code" type="xsd:int" minOccurs="1" />
      <xsd:element name="message" type="xsd:string" minOccurs="1" />
    </xsd:sequence>
  </xsd:complexType>

  <!-- Cancellation Operation -->
  <xsd:element name="CancellationRequest" type="tns:CancellationRequest" />
  <xsd:complexType name="CancellationRequest">
    <xsd:sequence>
      <xsd:element name="security" type="hos:Security"
        minOccurs="1" nillable="false" />
      <xsd:element name="transactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element name="hsTransactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element name="routingTag" type="hos:routingTag"
        minOccurs="0" nillable="true" />
      <xsd:element name="reason" minOccurs="1" type="tns:CancellationReason" />
      <!-- Request-level VSFs -->
      <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

  <xsd:element name="CancellationResponse" type="tns:CancellationResponse" />
  <xsd:complexType name="CancellationResponse">
    <xsd:sequence>
      <xsd:element name="transactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element name="hsTransactionId" type="hos:transactionId"
        minOccurs="1" nillable="false" />
      <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
  </xsd:complexType>

  <!-- Amend Operation -->
  <xsd:element name="amendRequest" type="tns:AmendRequest" />
  <xsd:complexType name="AmendRequest">
    <xsd:sequence>

```

```

    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="amendedFields" minOccurs="1" type="hos21:VendorSpecificFields" />
    <!-- Request-level VSFs -->
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="amendResponse" type="tns:AmendResponse" />
<xsd:complexType name="AmendResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!-- Reversal Operation -->
<xsd:element name="reversalRequest" type="tns:ReversalRequest" />
<xsd:complexType name="ReversalRequest">
  <xsd:sequence>
    <xsd:element name="security" type="hos:Security"
      minOccurs="1" nillable="false" />
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="reason" type="tns:ReversalReason"
      minOccurs="1" nillable="false" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <!-- Request-level VSFs -->
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="reversalResponse" type="tns:ReversalResponse" />
<xsd:complexType name="ReversalResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<!--
      These structures are used to transport a transaction status as provided by HomeSend.
      They are logically different from the transactions status as provided by the destination system,
      and may evolve differently in the next API versions.
-->
<xsd:complexType name="HsProvidedCompleteTransactionStatus">
  <xsd:sequence>
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>
<xsd:complexType name="HsProvidedRejectedTransactionStatus">
  <xsd:sequence>
    <xsd:element name="code" type="xsd:string" />
    <xsd:element name="message" type="xsd:string" />
  </xsd:sequence>
</xsd:complexType>
<xsd:complexType name="HsProvidedPendingTransactionStatus">
  <xsd:sequence>
    <xsd:element name="state" type="hos22:transactionState" />
    <xsd:element name="stateLabel" type="hos22:transactionStateLabel" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="tews23:stateSpecificFields" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="HsProvidedTransactionStatus">
  <xsd:choice>
    <xsd:element name="complete" type="tns:HsProvidedCompleteTransactionStatus" />
    <xsd:element name="rejected" type="tns:HsProvidedRejectedTransactionStatus" />
    <xsd:element name="pending" type="tns:HsProvidedPendingTransactionStatus" />
  </xsd:choice>
</xsd:complexType>

<!-- Status Report (TEWS) -->
<xsd:element name="statusReportRequest" type="tns:TewsStatusReportRequest" />
<xsd:complexType name="TewsStatusReportRequest">
  <xsd:sequence>

```

```

        <xsd:element name="security" type="hos:Security"
            minOccurs="1" nillable="false" />
        <xsd:element name="transactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="hsTransactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="routingTag" type="hos:routingTag"
            minOccurs="0" nillable="false" />
        <xsd:element name="status" type="tns:HsProvidedTransactionStatus" minOccurs="1" nillable="false" />
        <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

<xsd:element name="statusReportResponse" type="tns:TewsStatusReportResponse" />
<xsd:complexType name="TewsStatusReportResponse">
    <xsd:sequence>
        <xsd:element name="transactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element name="hsTransactionId" type="hos:transactionId"
            minOccurs="1" nillable="false" />
        <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>

</xsd:schema>

<!-- TEWS Definitions -->
<xsd:schema elementFormDefault="qualified" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
    targetNamespace="http://www.eservglobal.com/homesend/tews/2.0" version="2.3">

    <xsd:import namespace="http://www.eservglobal.com/homesend/types/2.0" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.1" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.2" />
    <xsd:import namespace="http://www.eservglobal.com/homesend/tews/2.3" />

    <xsd:simpleType name="state">
        <xsd:restriction base="xsd:string">
            <xsd:enumeration value="ACTIVE" />
            <xsd:enumeration value="INACTIVE" />
            <xsd:enumeration value="DEACTIVATED" />
            <xsd:enumeration value="UNKNOWN" />
        </xsd:restriction>
    </xsd:simpleType>

    <xsd:complexType name="AccountInformation">
        <xsd:sequence>
            <xsd:element name="balanceCurrency" type="xsd:string"
                minOccurs="1" nillable="false" />
            <xsd:element name="state" type="tns:state" minOccurs="0"
                nillable="false" />
            <xsd:element ref="tews23:secondaryCurrencies" minOccurs="0" />
        </xsd:sequence>
    </xsd:complexType>

    <!-- Request and Response Types -->

    <!-- accountInformation Operation -->
    <xsd:element name="accountInformationRequest" type="tns:AccountInformationRequest"
        nillable="true" />
    <xsd:complexType name="AccountInformationRequest">
        <xsd:sequence>
            <xsd:element name="security" type="hos:Security"
                minOccurs="1" nillable="false" />
            <xsd:element name="hsTransactionId" type="hos:transactionId"
                minOccurs="1" nillable="false" />
            <xsd:element name="destinationUri" type="hos:uri"
                minOccurs="1" nillable="false" />
            <xsd:element name="routingTag" type="hos:routingTag"
                minOccurs="0" nillable="true" />
            <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
        </xsd:sequence>
    </xsd:complexType>

    <xsd:element name="accountInformationResponse" type="tns:AccountInformationResponse"
        nillable="true" />
    <xsd:complexType name="AccountInformationResponse">
        <xsd:sequence>
            <xsd:element name="transactionId" type="hos:transactionId"
                minOccurs="1" nillable="false" />
            <xsd:element name="hsTransactionId" type="hos:transactionId"
                minOccurs="1" nillable="false" />
            <xsd:element name="accountInformation" type="tns:AccountInformation"
                minOccurs="1" nillable="false" />
            <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
        </xsd:sequence>
    </xsd:complexType>

    <!-- Credit Operation -->
    <xsd:element name="creditRequest" type="tns:CreditRequest"
        nillable="true" />
    <xsd:complexType name="CreditRequest">
        <xsd:sequence>
            <xsd:element name="security" type="hos:Security"
                minOccurs="1" nillable="false" />
            <xsd:element name="transactionId" type="hos:transactionId"
                minOccurs="1" nillable="false" />

```



```

    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="sourceUri" type="hos:uri" minOccurs="1"
      nillable="false" />
    <xsd:element name="destinationUri" type="hos:uri"
      minOccurs="1" nillable="false" />
    <xsd:element name="amountInformation" type="hos:AmountInformation"
      minOccurs="1" nillable="false" />
    <xsd:element name="senderInformation" type="hos:SenderInformation"
      minOccurs="0" nillable="false" />
    <xsd:element name="routingTag" type="hos:routingTag"
      minOccurs="0" nillable="true" />
    <xsd:element name="description" type="hos:description"
      minOccurs="0" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="tews22:provisionalResponseSupport" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:element name="creditResponse" type="tns:CreditResponse"
  nillable="true" />
<xsd:complexType name="CreditResponse">
  <xsd:sequence>
    <xsd:element name="transactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element name="hsTransactionId" type="hos:transactionId"
      minOccurs="1" nillable="false" />
    <xsd:element ref="tews21:vendorSpecificFields" minOccurs="0" />
    <xsd:element ref="tews22:provisionalResponse" minOccurs="0" />
  </xsd:sequence>
</xsd:complexType>

<xsd:complexType name="TerminatingServiceException">
  <xsd:complexContent>
    <xsd:extension base="hos:BaseException">
      <xsd:sequence>
        <!-- The terminating transaction ID may not have been generated yet -->
        <xsd:element name="transactionId" type="hos:transactionId"
          minOccurs="1" nillable="true" />
        <!-- The HomeSend transaction ID is mandatory -->
        <xsd:element name="hsTransactionId" type="hos:transactionId"
          minOccurs="1" nillable="false" />
      </xsd:sequence>
    </xsd:extension>
  </xsd:complexContent>
</xsd:complexType>

<xsd:element name="terminatingServiceException" type="tns:TerminatingServiceException"
  nillable="false" />
</xsd:schema>
</types>

<!-- TEWS Messages Definitions -->
<message name="accountInformation">
  <part name="accountInformationRequest" element="tns:accountInformationRequest" />
</message>

<message name="accountInformationResponse">
  <part name="accountInformationResponse" element="tns:accountInformationResponse" />
</message>

<message name="credit">
  <part name="creditRequest" element="tns:creditRequest" />
</message>

<message name="creditResponse">
  <part name="creditResponse" element="tns:creditResponse" />
</message>

<message name="status">
  <part name="statusRequest" element="tews22:statusRequest" />
</message>

<message name="statusResponse">
  <part name="statusResponse" element="tews22:statusResponse" />
</message>

<message name="deviceWatchdog">
  <part name="deviceWatchdogRequest" element="tews22:deviceWatchdogRequest" />
</message>

<message name="deviceWatchdogResponse">
  <part name="deviceWatchdogResponse" element="tews22:deviceWatchdogResponse" />
</message>

<message name="amend">
  <part name="amendRequest" element="tews23:amendRequest" />
</message>

<message name="amendResponse">
  <part name="amendResponse" element="tews23:amendResponse" />
</message>

<message name="cancellation">
  <part name="cancellationRequest" element="tews23:cancellationRequest" />

```

```
</message>

<message name="cancellationResponse">
  <part name="cancellationResponse" element="tews23:cancellationResponse" />
</message>

<message name="reversal">
  <part name="reversalRequest" element="tews23:reversalRequest" />
</message>

<message name="reversalResponse">
  <part name="reversalResponse" element="tews23:reversalResponse" />
</message>

<message name="statusReport">
  <part name="statusReportRequest" element="tews22:statusReportRequest" />
</message>

<message name="statusReportResponse">
  <part name="statusReportResponse" element="tews22:statusReportResponse" />
</message>

<message name="tewsStatusReport">
  <part name="statusReportRequest" element="tews23:statusReportRequest" />
</message>

<message name="tewsStatusReportResponse">
  <part name="statusReportResponse" element="tews23:statusReportResponse" />
</message>

<!-- Exception messages -->
<message name="terminatingServiceException">
  <part name="terminatingServiceException" element="tns:terminatingServiceException" />
</message>

<!-- TEWS Port Definition -->
<portType name="TEWSPortType_2_3">
  <operation name="accountInformation">
    <input message="tns:accountInformation" />
    <output message="tns:accountInformationResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="credit">
    <input message="tns:credit" />
    <output message="tns:creditResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="status">
    <input message="tns:status" />
    <output message="tns:statusResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="deviceWatchdog">
    <input message="tns:deviceWatchdog" />
    <output message="tns:deviceWatchdogResponse" />
  </operation>
  <operation name="amend">
    <input message="tns:amend" />
    <output message="tns:amendResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="cancellation">
    <input message="tns:cancellation" />
    <output message="tns:cancellationResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="reversal">
    <input message="tns:reversal" />
    <output message="tns:reversalResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="statusReport">
    <input message="tns:tewsStatusReport" />
    <output message="tns:tewsStatusReportResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
</portType>

<!-- R-TEWS Port Definition -->
<portType name="RTEWSPortType_2_3">
  <operation name="statusReport">
    <input message="tns:statusReport" />
    <output message="tns:statusReportResponse" />
    <fault name="TerminatingServiceException" message="tns:terminatingServiceException" />
  </operation>
  <operation name="deviceWatchdog">
    <input message="tns:deviceWatchdog" />
    <output message="tns:deviceWatchdogResponse" />
  </operation>
</portType>

<binding name="TEWSBinding_2_3" type="tns:TEWSPortType_2_3">
  <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http" />
  <operation name="credit">
    <soap:operation soapAction="" />
    <input>
```

```
<soap:body use="literal" parts="creditRequest" />
</input>
<output>
  <soap:body use="literal" parts="creditResponse" />
</output>
<fault name="TerminatingServiceException">
  <soap:fault name="TerminatingServiceException" use="literal"
    encodingStyle="" />
</fault>
</operation>
<operation name="accountInformation">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="accountInformationRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="accountInformationResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="status">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="statusRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="statusResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="deviceWatchdog">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="deviceWatchdogRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="deviceWatchdogResponse" />
  </output>
</operation>
<operation name="amend">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="amendRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="amendResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="cancellation">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="cancellationRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="cancellationResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="reversal">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="reversalRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="reversalResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
<operation name="statusReport">
  <soap:operation soapAction="" />
  <input>
    <soap:body use="literal" parts="statusReportRequest" />
  </input>
  <output>
    <soap:body use="literal" parts="statusReportResponse" />
  </output>
  <fault name="TerminatingServiceException">
    <soap:fault name="TerminatingServiceException" use="literal"
      encodingStyle="" />
  </fault>
</operation>
```

```

</operation>
</binding>

<binding name="RTEWSBinding_2_3" type="tns:RTEWSPortType_2_3">
  <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http" />
  <operation name="statusReport">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="statusReportRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="statusReportResponse" />
    </output>
    <fault name="TerminatingServiceException">
      <soap:fault name="TerminatingServiceException" use="literal"
        encodingStyle="" />
    </fault>
  </operation>
  <operation name="deviceWatchdog">
    <soap:operation soapAction="" />
    <input>
      <soap:body use="literal" parts="deviceWatchdogRequest" />
    </input>
    <output>
      <soap:body use="literal" parts="deviceWatchdogResponse" />
    </output>
  </operation>
</binding>

<service name="HomeSend TEWS 2_3">
  <port name="TEWSPort_2_3" binding="tns:TEWSBinding_2_3">
    <soap:address location="http://localhost/tews/2.3" />
  </port>
</service>

<service name="HomeSend RTEWS 2_3">
  <port name="RTEWSPort_2_3" binding="tns:RTEWSBinding_2_3">
    <soap:address location="http://localhost/rtews/2.3" />
  </port>
</service>
</definitions>

```

11.3 Sample Remittance Traces

11.3.1 Synchronous Credit Model

The following HTTP traces describe a typical one-shot remittance from URI tel:+3530000 to URI tel:+2130000, using the synchronous credit model.

The transactions identifiers are:

Originating Transaction ID: 20100430-120342-trid_1, allocated by the Sender M-Wallet System upon the initial transaction request,

HomeSend Transaction ID: 201, allocated by HomeSend when receiving the initial transaction request,

Terminating Transaction ID: 4, allocated by the Receiver M-Wallet System when receiving the TEWS/accountInformation request,

Reservation ID: RRR35300002130000, allocated by the Sender M-Wallet System upon successful reservation.

In this sample, some vendor specific fields were also added in the Remittance request, and forwarded by HomeSend in the Credit request. Some additional info from the Credit response were also forwarded back to the sender system in the Remittance response.

This call flow is very similar to a API 2.1 one. The only difference is the introduction of a provisionalResponseSupport element in the TEWS/credit request, which advertises the asynchronous credit response model support. It is then up to the TEWS endpoint to use it or not – here, it stuck to the synchronous model by responding with a non-provisional response immediately.

Initial HWS/remittance request (Sender E-Wallet System --> HomeSend):

```
POST /hws/2.2 HTTP/1.1
Connection: close
User-Agent: Jakarta Commons-HttpClient/3.1
Host: 172.17.107.56:7777
Content-Length: 1487
Content-Type: text/xml

<?xml version="1.0" encoding="ISO-8859-1"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body xmlns:ns1="http://www.eservglobal.com/homesend/hws/2.0"
    xmlns:ns2="http://www.eservglobal.com/homesend/types/2.0"
    xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
    xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
    <ns1:remittanceRequest>
      <ns1:security>
        <ns2:login>fip2001RTlogin</ns2:login>
        <ns2:password>fip2001RTlogin</ns2:password>
      </ns1:security>
      <ns1:hsTransactionId xmlns:ns3="http://www.w3.org/2001/XMLSchema-instance"
        ns3:nil="true" />
      <ns1:transactionId>20100430-120342-trid_1
      </ns1:transactionId>
      <ns1:sourceUri>tel:+3530000</ns1:sourceUri>
      <ns1:destinationUri>tel:+2130000</ns1:destinationUri>
      <ns1:amountInformation>
        <ns2:currency>GBP</ns2:currency>
        <ns2:amount>54</ns2:amount>
        <ns2:grace>5</ns2:grace>
        <ns2:validity>2</ns2:validity>
      </ns1:amountInformation>
      <ns1:payer>RECEIVER</ns1:payer>
      <ns1:cbsType>EMoneyToEMoneyTransfer</ns1:cbsType>
      <ns1:routingTag>sendmaxCBS20routingTag
      </ns1:routingTag>
      <ns1:reservationRequired>true</ns1:reservationRequired>
      <ns1:senderInformation>
        <ns2:name>User's name</ns2:name>
        <ns2:address>User's Addr</ns2:address>
      </ns1:senderInformation>
      <ns3:vendorSpecificFields>
        <ns4:vsf>
          <ns4:vendorId>1234</ns4:vendorId>
          <ns4:fieldId>1</ns4:fieldId>
          <ns4:value>CODE012345</ns4:value>
        </ns4:vsf>
      </ns3:vendorSpecificFields>
    </ns1:remittanceRequest>
  </soap:Body>
</soap:Envelope>
```

TEWS/accountInformation request (HomeSend --> Receiver E-Wallet System):

```
POST /tews/2.2 HTTP/1.1
Host: 172.17.100.150:8897
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
```

```
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 646

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:accountInformationRequest>
      <ns0:security>
        <ns1:login>fip2002RTlogin</ns1:login>
        <ns1:password>fip2002RTlogin</ns1:password>
      </ns0:security>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:destinationUri>tel:+2130000</ns0:destinationUri>
      <ns0:routingTag>phonebankCBS20routingTag</ns0:routingTag>
    </ns0:accountInformationRequest>
  </env:Body>
</env:Envelope>
```

TEWS/accountInformation response (Receiver E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Date: Fri, 30 Apr 2010 10:03:48 GMT
Server: Oracle Containers for J2EE
Content-Length: 632
Connection: Keep-Alive
Keep-Alive: timeout=15, max=100
Content-Type: text/xml; charset=UTF-8
SOAPAction: ""

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:accountInformationResponse>
      <ns0:transactionId>4</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:accountInformation>
        <ns0:balanceCurrency>EGP</ns0:balanceCurrency>
        <ns0:state>ACTIVE</ns0:state>
      </ns0:accountInformation>
    </ns0:accountInformationResponse>
  </env:Body>
</env:Envelope>
```

OEWS/reserve request (HomeSend --> Sender E-Wallet System):

```
POST /oews/2.2 HTTP/1.1
Host: 172.17.100.150:8897
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
```

```
Content-type: text/xml; charset=UTF-8
Content-length: 816

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:reserveRequest>
      <ns0:security>
        <ns1:login>fip01RTlogin</ns1:login>
        <ns1:password>fip2001RTlogin</ns1:password>
      </ns0:security>
      <ns0:transactionId>20100430-120342-trid_1</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:uri>tel:+3530000</ns0:uri>
      <ns0:amountInformation>
        <ns1:currency>GBP</ns1:currency>
        <ns1:amount>54</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:routingTag>sendmaxCBS20routingTag</ns0:routingTag>
    </ns0:reserveRequest>
  </env:Body>
</env:Envelope>
```

OEWS/reserve response (Sender E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Date: Fri, 30 Apr 2010 10:03:48 GMT
Server: Oracle Containers for J2EE
Content-Length: 563
Connection: Keep-Alive
Keep-Alive: timeout=15, max=100
Content-Type: text/xml; charset=UTF-8
SOAPAction: ""

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:reserveResponse>
      <ns0:transactionId>20100430-120342-trid_1</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:reservationId>RRR35300002130000</ns0:reservationId>
    </ns0:reserveResponse>
  </env:Body>
</env:Envelope>
```

TEWS/credit request (HomeSend --> Receiver E-Wallet System):

```
POST /tews/2.2 HTTP/1.1
Host: 172.17.100.150:8897
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
```

```
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 1023

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns3="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:ns5="http://www.eservglobal.com/homesend/tews/2.2">
  <env:Body>
    <ns0:creditRequest>
      <ns0:security>
        <ns1:login>fip2002RTlogin</ns1:login>
        <ns1:password>fip2002RTlogin</ns1:password>
      </ns0:security>
      <ns0:transactionId>4</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:sourceUri>tel:+3530000</ns0:sourceUri>
      <ns0:destinationUri>tel:+2130000</ns0:destinationUri>
      <ns0:amountInformation>
        <ns1:currency>EGP</ns1:currency>
        <ns1:amount>375.31</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:senderInformation>
        <ns1:name>User's name</ns1:name>
        <ns1:address>User's Addr</ns1:address>
      </ns0:senderInformation>
      <ns0:routingTag>phonebankCBS20routingTag</ns0:routingTag>
      <ns3:vendorSpecificFields>
        <ns4:vsf>
          <ns4:vendorId>1234</ns4:vendorId>
          <ns4:fieldId>1</ns4:fieldId>
          <ns4:value>CODE012345</ns4:value>
        </ns4:vsf>
      </ns3:vendorSpecificFields>
      <ns5:provisionalResponseSupport>

<ns5:reportStatusTo>https://homesend.example.com/rtews/2.2</ns5:reportStatusTo>
      <ns5:token>hqkj23hjk1gsiA289f</ns5:token>
    </ns5:provisionalResponseSupport>
  </ns0:creditRequest>
</env:Body>
</env:Envelope>
```

TEWS/credit response (Receiver E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Date: Fri, 30 Apr 2010 10:03:48 GMT
Server: Oracle Containers for J2EE
Content-Length: 484
Connection: Keep-Alive
Keep-Alive: timeout=15, max=100
Content-Type: text/xml; charset=UTF-8
SOAPAction: ""
```



```
<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns3="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:creditResponse>
      <ns0:transactionId>4</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns3:vendorSpecificFields>
        <ns4:vsf>
          <ns4:vendorId>1234</ns4:vendorId>
          <ns4:fieldId>2</ns4:fieldId>
          <ns4:value>Receiver's name</ns4:value>
        </ns4:vsf>
        <ns4:vsf>
          <ns4:vendorId>1234</ns4:vendorId>
          <ns4:fieldId>3</ns4:fieldId>
          <ns4:value>Receiver's address</ns4:value>
        </ns4:vsf>
      </ns3:vendorSpecificFields>
    </ns0:creditResponse>
  </env:Body>
</env:Envelope>
```

OEWS/charge request (HomeSend --> Sender E-Wallet System):

```
POST /oews/2.2 HTTP/1.1
Host: 172.17.100.150:8897
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 839

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:chargeRequest>
      <ns0:security>
        <ns1:login>fip01RTlogin</ns1:login>
        <ns1:password>fip2001RTlogin</ns1:password>
      </ns0:security>
      <ns0:transactionId>20100430-120342-trid_1
      </ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:reservationId>RRR35300002130000</ns0:reservationId>
      <ns0:amountInformation>
        <ns1:currency>GBP</ns1:currency>
        <ns1:amount>54</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:routingTag>sendmaxCBS20routingTag
```

```
</ns0:routingTag>
</ns0:chargeRequest>
</env:Body>
</env:Envelope>
```

OEWS/charge response (Sender E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Date: Fri, 30 Apr 2010 10:03:48 GMT
Server: Oracle Containers for J2EE
Content-Length: 505
Connection: Keep-Alive
Keep-Alive: timeout=15, max=100
Content-Type: text/xml; charset=UTF-8
SOAPAction: ""

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:chargeResponse>
      <ns0:transactionId>20100430-120342-trid_1</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
    </ns0:chargeResponse>
  </env:Body>
</env:Envelope>
```

Final HWS/remittance response (HomeSend --> Sender M-Wallet System):

```
HTTP/1.1 200 OK
Date: Fri, 30 Apr 2010 10:03:48 GMT
Server: Oracle-Application-Server-10g/10.1.3.4.0 Oracle-HTTP-Server
Content-Length: 983
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:remittanceResponse>
      <ns0:transactionId>20100430-120342-trid_1</ns0:transactionId>
      <ns0:hsTransactionId>201</ns0:hsTransactionId>
      <ns0:transactionSummary>
        <ns0:commissionAmount>
          <ns1:currency>GBP</ns1:currency>
          <ns1:amount>8.46</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:commissionAmount>
        <ns0:chargedAmount>
          <ns1:currency>GBP</ns1:currency>

```

```
<ns1:amount>54</ns1:amount>
<ns1:grace xsi:nil="1" />
<ns1:validity xsi:nil="1" />
</ns0:chargedAmount>
<ns0:creditedAmount>
  <ns1:currency>EGP</ns1:currency>
  <ns1:amount>375.31</ns1:amount>
  <ns1:grace xsi:nil="1" />
  <ns1:validity xsi:nil="1" />
</ns0:creditedAmount>
</ns0:transactionSummary>
<ns3:vendorSpecificFields>
  <ns4:vsf>
    <ns4:vendorId>1234</ns4:vendorId>
    <ns4:fieldId>2</ns4:fieldId>
    <ns4:value>Receiver's name</ns4:value>
  </ns4:vsf>
  <ns4:vsf>
    <ns4:vendorId>1234</ns4:vendorId>
    <ns4:fieldId>3</ns4:fieldId>
    <ns4:value>Receiver's address</ns4:value>
  </ns4:vsf>
</ns3:vendorSpecificFields>
</ns0:remittanceResponse>
</env:Body>
</env:Envelope>
```

11.3.2 Asynchronous Credit Model – Successful Transaction

This sample corresponds to a remittance with advice of charge implemented using the asynchronous credit response mechanism.

The following HTTP traces describe a typical remittance with adviceOfCharge from URI `tel:+123456` to URI `tel:+21212123456`, using the asynchronous credit model.

The transactions identifiers are:

Originating Transaction ID: `otid-1301497055.86`, allocated by the Sender M-Wallet System upon the initial transaction request,

HomeSend Transaction ID: `2004`, allocated by HomeSend when receiving the initial transaction request,

Terminating Transaction ID: `testerman-t-1301497056.12`, allocated by the Receiver M-Wallet System when receiving the *TEWS/accountInformation* request,

Reservation ID: `tm-res-1301497058.42`, allocated by the Sender M-Wallet System upon successful reservation.

The transaction starts exactly as the synchronous one, until the *TEWS/credit* request included. Then, the TEWS endpoint replies with a provisional response, rendering the transaction asynchronous.

Initial HWS/adviceOfCharge request (Sender M-Wallet System --> HomeSend):

```
POST /hws/2.2 HTTP/1.0
content-type: text/xml
Content-Length: 851
```

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body xmlns:ns1="http://www.eservglobal.com/homesend/hws/2.0"
    xmlns:ns2="http://www.eservglobal.com/homesend/types/2.0">
    <ns1:advicesOfChargeRequest>
      <ns1:security>
        <ns2:login>demo01-login</ns2:login>
        <ns2:password>password</ns2:password>
      </ns1:security>
      <ns1:transactionId>otid-1301497055.86
      </ns1:transactionId>
      <ns1:sourceUri>tel:+123456</ns1:sourceUri>
      <ns1:destinationUri>tel:+21212123456</ns1:destinationUri>
      <ns1:amountInformation>
        <ns2:currency>EUR</ns2:currency>
        <ns2:amount>100</ns2:amount>
      </ns1:amountInformation>
      <ns1:cbsType>EMoneyToEMoneyTransfer</ns1:cbsType>
    </ns1:advicesOfChargeRequest>
  </soap:Body>
</soap:Envelope>
```

TEWS/accountInformation request (HomeSend --> Receiver E-Wallet System):

```
POST /tews/2.2 HTTP/1.1
Host: localhost:9097
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 787

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/tews/2.2">
  <env:Body>
    <ns0:accountInformationRequest>
      <ns0:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns0:security>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:destinationUri>tel:+21212123456</ns0:destinationUri>
      <ns0:routingTag xsi:nil="1" />
    </ns0:accountInformationRequest>
  </env:Body>
</env:Envelope>
```

TEWS/accountInformation response (Receiver E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Connection: close
```

```
Content-Type: text/xml
Content-Length: 750

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns2="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:accountInformationResponse>
      <ns0:transactionId>testerman-t-1301497056.12
      </ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:accountInformation>
        <ns0:balanceCurrency>MAD</ns0:balanceCurrency>
        <ns0:state>ACTIVE</ns0:state>
      </ns0:accountInformation>
    </ns0:accountInformationResponse>
  </env:Body>
</env:Envelope>
```

HWS/adviceOfCharge response (HomeSend --> Sender E-Wallet System):

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:57:37 GMT
Server: Oracle-Application-Server-10g
Content-Length: 1723
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:adviceOfChargeResponse>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:senderPayerProposal>
        <ns0:commissionAmount>
          <ns1:currency>EUR</ns1:currency>
          <ns1:amount>0</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:commissionAmount>
        <ns0:chargedAmount>
          <ns1:currency>EUR</ns1:currency>
          <ns1:amount>100.00</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:chargedAmount>
        <ns0:creditedAmount>
          <ns1:currency>MAD</ns1:currency>
          <ns1:amount>1132.55</ns1:amount>
          <ns1:grace xsi:nil="1" />
        </ns0:creditedAmount>
      </ns0:senderPayerProposal>
    </ns0:adviceOfChargeResponse>
  </env:Body>
</env:Envelope>
```

```

        <ns1:validity xsi:nil="1" />
      </ns0:creditedAmount>
    </ns0:senderPayerProposal>
    <ns0:receiverPayerProposal>
      <ns0:commissionAmount>
        <ns1:currency>EUR</ns1:currency>
        <ns1:amount>0</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:commissionAmount>
      <ns0:chargedAmount>
        <ns1:currency>EUR</ns1:currency>
        <ns1:amount>100</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:chargedAmount>
      <ns0:creditedAmount>
        <ns1:currency>MAD</ns1:currency>
        <ns1:amount>1132.55</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:creditedAmount>
    </ns0:receiverPayerProposal>
    <ns0:expirationDelay>120</ns0:expirationDelay>
  </ns0:adviceOfChargeResponse>
</env:Body>
</env:Envelope>

```

After a little while, one of the proposal is selected by the sending end-user. Here he/she selected the “sender payer” option:

HWS/remittance request (Sender E-Wallet System --> HomeSend):

```

POST /hws/2.2 HTTP/1.0
content-type: text/xml
Content-Length: 1046

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body xmlns:ns1="http://www.eservglobal.com/homesend/hws/2.0"
    xmlns:ns2="http://www.eservglobal.com/homesend/types/2.0"
    xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
    xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
    <ns1:remittanceRequest>
      <ns1:security>
        <ns2:login>demo01-login</ns2:login>
        <ns2:password>password</ns2:password>
      </ns1:security>
      <ns1:transactionId>otid-1301497055.86
      </ns1:transactionId>
      <ns1:hsTransactionId>2004</ns1:hsTransactionId>
      <ns1:sourceUri>tel:+123456</ns1:sourceUri>
      <ns1:destinationUri>tel:+21212123456</ns1:destinationUri>
      <ns1:amountInformation>
        <ns2:currency>EUR</ns2:currency>
        <ns2:amount>100</ns2:amount>
      </ns1:amountInformation>
      <ns1:cbsType>EMoneyToEMoneyTransfer</ns1:cbsType>
      <ns1:reservationRequired>true</ns1:reservationRequired>
    </ns1:remittanceRequest>
  </soap:Body>
</soap:Envelope>

```

HomeSend will continue the transaction with the reservation part:

OEWS/reserve request (HomeSend --> Sender E-Wallet System):

```
POST /oews/2.2 HTTP/1.1
Host: localhost:9095
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 953

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/oews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:reserveRequest>
      <ns0:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns0:security>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:uri>tel:+123456</ns0:uri>
      <ns0:amountInformation>
        <ns1:currency>EUR</ns1:currency>
        <ns1:amount>100</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:routingTag xsi:nil="1" />
    </ns0:reserveRequest>
  </env:Body>
</env:Envelope>
```

OEWS/reserve response (Sender E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Connection: close
Content-Type: text/xml
Content-Length: 658

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/oews/2.1"
  xmlns:ns2="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:reserveResponse>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
    </ns0:reserveResponse>
  </env:Body>
</env:Envelope>
```

```
<ns0:reservationId>tm-res-1301497058.42</ns0:reservationId>
</ns0:reserveResponse>
</env:Body>
</env:Envelope>
```

TEWS/credit request (HomeSend --> Receiver E-Wallet System):

```
POST /tews/2.2 HTTP/1.1
Host: localhost:9097
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 1203

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/tews/2.2">
  <env:Body>
    <ns0:creditRequest>
      <ns0:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns0:security>
      <ns0:transactionId>testerman-t-1301497056.12</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:sourceUri>tel:+123456</ns0:sourceUri>
      <ns0:destinationUri>tel:+21212123456</ns0:destinationUri>
      <ns0:amountInformation>
        <ns1:currency>MAD</ns1:currency>
        <ns1:amount>1132.55</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:routingTag xsi:nil="1" />
      <ns4:provisionalResponseSupport>
        <ns4:reportStatusTo>http://localhost:7777/rtews/2.2</ns4:reportStatusTo>
        <ns4:token>ljp9hchxqy6m</ns4:token>
      </ns4:provisionalResponseSupport>
    </ns0:creditRequest>
  </env:Body>
</env:Envelope>
```

Now the receiver E-Wallet system triggers the asynchronous model by responding with a provisional response.

TEWS/credit response (Receiver E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Connection: close
Content-Type: text/xml
Content-Length: 849
```



```
<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/types/2.1"
  xmlns:ns3="http://www.eservglobal.com/homesend/tews/2.2">
  <env:Body>
    <ns0:creditResponse>
      <ns0:transactionId>testerman-t-1301497056.12</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns3:provisionalResponse>
        <ns3:maxCompletionDate>2011-03-30T15:00:38Z</ns3:maxCompletionDate>
        <ns3:state>CreditInProgress</ns3:state>
        <ns3:label>Credit In Progress</ns3:label>
      </ns3:provisionalResponse>
    </ns0:creditResponse>
  </env:Body>
</env:Envelope>
```

HomeSend can then “precharge” the sending account using a OEWS/charge request (HomeSend --> Sender E-Wallet System):

```
POST /oews/2.2 HTTP/1.1
Host: localhost:9095
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 980

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/oews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:chargeRequest>
      <ns0:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns0:security>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:reservationId>tm-res-1301497058.42</ns0:reservationId>
      <ns0:amountInformation>
        <ns1:currency>EUR</ns1:currency>
        <ns1:amount>100</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns0:amountInformation>
      <ns0:routingTag xsi:nil="1" />
    </ns0:chargeRequest>
  </env:Body>
</env:Envelope>
```

OEWS/charge response (Sender E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Connection: close
Content-Type: text/xml
Content-Length: 537
<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:chargeResponse>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
    </ns0:chargeResponse>
  </env:Body>
</env:Envelope>
```

Provisional HWS/remittance response (HomeSend --> Sender M-Wallet System):

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:57:40 GMT
Server: Oracle-Application-Server-10g
Content-Length: 1350
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:remittanceResponse>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:transactionSummary>
        <ns0:commissionAmount>
          <ns1:currency>EUR</ns1:currency>
          <ns1:amount>0</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:commissionAmount>
        <ns0:chargedAmount>
          <ns1:currency>EUR</ns1:currency>
          <ns1:amount>100</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:chargedAmount>
        <ns0:creditedAmount>
          <ns1:currency>MAD</ns1:currency>
          <ns1:amount>1132.55</ns1:amount>
          <ns1:grace xsi:nil="1" />
          <ns1:validity xsi:nil="1" />
        </ns0:creditedAmount>
      </ns0:transactionSummary>
    </ns0:remittanceResponse>
  </env:Body>
</env:Envelope>
```

```
</ns0:creditedAmount>
</ns0:transactionSummary>
<ns2:provisionalResponse>
  <ns2:maxCompletionDate>2011-03-
30T17:00:38.000+00:00</ns2:maxCompletionDate>
  <ns2:state>InProgress</ns2:state>
  <ns2:label>Credit In Progress</ns2:label>
</ns2:provisionalResponse>
</ns0:remittanceResponse>
</env:Body>
</env:Envelope>
```

After a while, the Receiver E-Wallet System completes the transaction, using a R-TEWS/statusReport request (Receiver E-Wallet System --> HomeSend):

```
POST /rtews/2.2 HTTP/1.0
connection: close
content-type: text/xml
Content-Length: 897

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:statusReportRequest>
      <ns0:security>
        <ns1:login>demo01-rlogin</ns1:login>
        <ns1:password>rpassword</ns1:password>
      </ns0:security>
      <ns0:token>ljp9hchxqy6m</ns0:token>
      <ns0:transactionId>testerman-t-1301497056.12</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
      <ns0:status>
        <ns0:complete>
          </ns0:complete>
        </ns0:status>
      </ns0:statusReportRequest>
    </env:Body>
  </env:Envelope>
```

In this sample, no additional completion VSFs were provided.

R-TEWS/statusReport response (HomeSend --> Receiver E-Wallet System):

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:57:45 GMT
Server: Oracle-Application-Server-10g
Content-Length: 693
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
```

```
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns2:statusReportResponse>
      <ns2:transactionId>testerman-t-1301497056.12
      </ns2:transactionId>
      <ns2:hsTransactionId>2004</ns2:hsTransactionId>
    </ns2:statusReportResponse>
  </env:Body>
</env:Envelope>
```

From the Receiver system's or HomeSend's point of view, the transaction has now been successfully completed. However, the Sending system must still retrieve the final's transaction status, using a HWS/status request (Sender E-Wallet System --> HomeSend):

```
POST /hws/2.2 HTTP/1.0
content-type: text/xml
Content-Length: 729

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.2"
    xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
    xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.1"
    xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1">
    <ns0:statusRequest>
      <ns0:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns0:security>
      <ns0:transactionId>otid-1301497055.86</ns0:transactionId>
      <ns0:hsTransactionId>2004</ns0:hsTransactionId>
    </ns0:statusRequest>
  </soap:Body>
</soap:Envelope>
```

As the transaction is completed successfully, HomeSend replies with:

HWS/status response (HomeSend --> Sender E-Wallet System):

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:57:45 GMT
Server: Oracle-Application-Server-10g
Content-Length: 1195
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
```

```

xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.2"
xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
<env:Body>
  <ns2:statusResponse>
    <ns2:transactionId>otid-1301497055.86
  </ns2:transactionId>
    <ns2:hsTransactionId>2004</ns2:hsTransactionId>
    <ns2:status>
      <ns2:complete>
        <ns2:transactionSummary>
          <ns0:commissionAmount>
            <ns1:currency>EUR</ns1:currency>
            <ns1:amount>0</ns1:amount>
            <ns1:grace xsi:nil="1" />
            <ns1:validity xsi:nil="1" />
          </ns0:commissionAmount>
          <ns0:chargedAmount>
            <ns1:currency>EUR</ns1:currency>
            <ns1:amount>100</ns1:amount>
            <ns1:grace xsi:nil="1" />
            <ns1:validity xsi:nil="1" />
          </ns0:chargedAmount>
          <ns0:creditedAmount>
            <ns1:currency>MAD</ns1:currency>
            <ns1:amount>1132.55</ns1:amount>
            <ns1:grace xsi:nil="1" />
            <ns1:validity xsi:nil="1" />
          </ns0:creditedAmount>
        </ns2:transactionSummary>
      </ns2:complete>
    </ns2:status>
  </ns2:statusResponse>
</env:Body>
</env:Envelope>

```

If the transaction was not completed yet, HomeSend would have responded with:

HWS/status response – pending transaction (HomeSend --> Sender E-Wallet System):

```

HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:57:45 GMT
Server: Oracle-Application-Server-10g
Content-Length: 818
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns2:statusResponse>
      <ns2:transactionId>otid-1301497055.86
    </ns2:transactionId>
      <ns2:hsTransactionId>2004</ns2:hsTransactionId>
      <ns2:status>

```

```
<ns2:pending>
  <ns2:maxCompletionDate>2011-03-
30T17:00:38.000+00:00</ns2:maxCompletionDate>
  <ns2:state>InProgress</ns2:state>
  <ns2:label></ns2:label>
</ns2:pending>
</ns2:status>
</ns2:statusResponse>
</env:Body>
</env:Envelope>
```

11.3.3 Asynchronous Credit Model – Failed Transaction

This section contains some additional traces for the messages that are involved in an asynchronous failed transaction.

The message flow is exactly the same as above until the R-TEWS/statusReport request, which transports a rejected element in case of a failure:

```
POST /rtews/2.2 HTTP/1.0
connection: close
content-type: text/xml
Content-Length: 992

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns0:statusReportRequest>
      <ns0:security>
        <ns1:login>demo01-rlogin</ns1:login>
        <ns1:password>rpassword</ns1:password>
      </ns0:security>
      <ns0:token>bz7tptaa71aw</ns0:token>
      <ns0:transactionId>testerman-t-1301497143.45</ns0:transactionId>
      <ns0:hsTransactionId>2012</ns0:hsTransactionId>
      <ns0:status>
        <ns0:rejected>
          <ns0:code>ER0013</ns0:code>
          <ns0:message>Rejected for unspecified reason</ns0:message>
        </ns0:rejected>
      </ns0:status>
    </ns0:statusReportRequest>
  </env:Body>
</env:Envelope>
```

The R-TEWS/statusReport response remains the same:

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:59:10 GMT
Server: Oracle-Application-Server-10g
Content-Length: 693
SOAPAction: ""
Connection: close
```

```
Content-Type: text/xml; charset=UTF-8

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/tews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/tews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns2:statusReportResponse>
      <ns2:transactionId>testerman-t-1301497143.45
      </ns2:transactionId>
      <ns2:hsTransactionId>2012</ns2:hsTransactionId>
    </ns2:statusReportResponse>
  </env:Body>
</env:Envelope>
```

Then HomeSend reverses the charge operation on the sending side using a OEWS/chargeback request (HomeSend --> Sender E-Wallet System):

```
POST /oews/2.2 HTTP/1.1
Host: localhost:9095
Connection: Keep-Alive, TE
TE: trailers, deflate, gzip, compress
User-Agent: Oracle HTTPClient Version 10h
SOAPAction: ""
Accept-Encoding: gzip, x-gzip, compress, x-compress
Content-type: text/xml; charset=UTF-8
Content-length: 1040

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/oews/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns2:chargebackRequest>
      <ns2:security>
        <ns1:login>demo01-login</ns1:login>
        <ns1:password>password</ns1:password>
      </ns2:security>
      <ns2:transactionId>otid-1301497143.18</ns2:transactionId>
      <ns2:hsTransactionId>2012</ns2:hsTransactionId>
      <ns2:reservationId>tm-res-1301497143.47</ns2:reservationId>
      <ns2:amountInformation>
        <ns1:currency>EUR</ns1:currency>
        <ns1:amount>100</ns1:amount>
        <ns1:grace xsi:nil="1" />
        <ns1:validity xsi:nil="1" />
      </ns2:amountInformation>
      <ns2:reason>
        <ns2:code>0</ns2:code>
        <ns2:message>message</ns2:message>
      </ns2:reason>
    </ns2:chargebackRequest>
  </env:Body>
```

```
</env:Envelope>
```

OEWs/chargeback response (Sender E-Wallet System --> HomeSend):

```
HTTP/1.1 200 OK
Connection: close
Content-Type: text/xml
Content-Length: 541

<?xml version="1.0" encoding="UTF-8"?>
<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/oews/2.2"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0">
  <env:Body>
    <ns0:chargebackResponse>
      <ns0:transactionId>otid-1301497143.18
    </ns0:transactionId>
      <ns0:hsTransactionId>2012</ns0:hsTransactionId>
    </ns0:chargeResponse>
  </env:Body>
</env:Envelope>
```

Then, when the Sender E-Wallet System gets an updated status through a HWS/status request, it will get a HWS/status response such as:

```
HTTP/1.1 200 OK
Date: Wed, 30 Mar 2011 14:59:11 GMT
Server: Oracle-Application-Server-10g
Content-Length: 764
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
  xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
  xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.2"
  xmlns:ns3="http://www.eservglobal.com/homesend/hws/2.1"
  xmlns:ns4="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <ns2:statusResponse>
      <ns2:transactionId>otid-1301497143.18</ns2:transactionId>
      <ns2:hsTransactionId>2012</ns2:hsTransactionId>
      <ns2:status>
        <ns2:rejected>
          <ns2:code>ER0013</ns2:code>
          <ns2:message>Receiver crediting error</ns2:message>
        </ns2:rejected>
      </ns2:status>
    </ns2:statusResponse>
  </env:Body>
</env:Envelope>
```


11.3.4 Sample Custom Faults

This section provides some network traces of the various custom SOAP faults, as they are expected to be or are encoded on the network.

11.3.4.1 HomeSendServiceException Fault Sample

A HomeSend Service Exception is encoded as follows:

```
HTTP/1.1 500 Internal Server Error
Date: Tue, 01 Mar 2011 17:02:10 GMT
Server: Oracle-Application-Server-10g
Content-Length: 889
SOAPAction: ""
Connection: close
Content-Type: text/xml; charset=UTF-8

<env:Envelope xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:ns0="http://www.eservglobal.com/homesend/hws/2.0"
xmlns:ns1="http://www.eservglobal.com/homesend/types/2.0"
xmlns:ns2="http://www.eservglobal.com/homesend/hws/2.1"
xmlns:ns3="http://www.eservglobal.com/homesend/types/2.1">
  <env:Body>
    <env:Fault xsi:type="env:Fault">
      <faultcode xmlns="">env:Server</faultcode>
      <faultstring xmlns="">Missing required fields</faultstring>
      <detail xmlns="">
        <ns0:homeSendServiceException
xsi:type="ns0:HomeSendServiceException">
          <ns1:code>ER0037</ns1:code>
          <ns1:message>Missing required fields</ns1:message>
          <ns0:transactionId>originating-
1298998818.32</ns0:transactionId>
          <ns0:hsTransactionId>6781</ns0:hsTransactionId>
        </ns0:homeSendServiceException>
      </detail>
    </env:Fault>
  </env:Body>
</env:Envelope>
```

The `transactionId` element is always provided, and corresponds to the incoming, originating transaction ID provided in the request the fault is sent in response to. The `hsTransactionId` element is always provided and equals to the incoming one (if it was provided), or is optionally provided otherwise, depending on the error code – HomeSend may not have an opportunity to generate it yet (for instance, when dealing with authentication errors).

11.3.4.2 OriginatingServiceException Fault

An Originating Service Exception is expected to be encoded as follows (or any equivalent XML document):

```
HTTP/1.1 500 Internal Server Error
Server: Sun Java System Application Server 9.1_01
Content-Type: text/xml; charset="utf-8"
Content-Length: 846
Date: Wed, 16 Mar 2011 05:23:32 GMT
Connection: close
```

```
<?xml version="1.0" ?>
<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">
  <S:Body>
    <ns2:Fault xmlns:ns2="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns3="http://www.w3.org/2003/05/soap-envelope">
      <faultcode>ns2:Server</faultcode>
      <faultstring>Authentication error.</faultstring>
      <detail>
        <ns2:originatingServiceException
xmlns:ns2="http://www.eservglobal.com/homesend/oews/2.0"
xmlns="http://www.eservglobal.com/homesend/types/2.0">
          <code>ER0014</code>
          <message>Authentication error</message>
          <ns2:transactionId>1269</ns2:transactionId>
          <ns2:hsTransactionId>otid-
1300292834.41</ns2:hsTransactionId>
        </ns2:originatingServiceException>
      </detail>
    </ns2:Fault>
  </S:Body>
</S:Envelope>
```

As the corresponding incoming request always contains a `transactionId` and a `hsTransactionId` fields (containing the originating transaction ID and the HomeSend transaction ID, respectively), such a fault must always contain these two elements, taken as is from the request.

11.3.4.3 TerminatingServiceException Fault Sample

A Terminating Service Exception is expected to be encoded as follows (or any equivalent XML document):

```
HTTP/1.1 500 Internal Server Error
Server: Sun Java System Application Server 9.1_01
Content-Type: text/xml; charset="utf-8"
Content-Length: 846
Date: Wed, 16 Mar 2011 05:24:36 GMT
Connection: close

<?xml version="1.0" ?>
<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">
  <S:Body>
    <ns2:Fault xmlns:ns2="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:ns3="http://www.w3.org/2003/05/soap-envelope">
      <faultcode>ns2:Server</faultcode>
      <faultstring>Authentication error.</faultstring>
      <detail>
        <ns2:terminatingServiceException
xmlns:ns2="http://www.eservglobal.com/homesend/tews/2.0"
xmlns="http://www.eservglobal.com/homesend/types/2.0">
          <code>ER0014</code>
          <message>Authentication error</message>
          <ns2:transactionId>1269</ns2:transactionId>
          <ns2:hsTransactionId>otid-
1300292834.41</ns2:hsTransactionId>
        </ns2:terminatingServiceException>
      </detail>
    </ns2:Fault>
  </S:Body>
</S:Envelope>
```

A fault to a *TEWS/credit* or *TEWS/status* request must always contain a `transactionId` and a `hsTransactionId` fields (containing the terminating transaction ID and the HomeSend transaction ID, respectively), taken as is from the request it replies to.

TEWS/accountInformation faults, however, may not contain a `transactionId`, as the terminating transaction ID may not have been generated by the destination system at the time the error was detected. Providing the `hsTransactionId` (from the incoming request), however, remains mandatory.

HomeSend may also return `TerminatingServiceException` faults on *R-TEWS/statusReport* requests. Both `transactionId` and `hsTransactionId` fields will be provided, with the value from the incoming requests.

---000---